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Model DRS

Unit 3,4 Systems

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Criteria 1

Design Brief

Formula 1 is the pinnacle of motorsport, representing the best of complex automotive engineering. Due to the complexity and competitive nature of the sport it is nearly impossible to gain a clear and in depth understanding of all the facets of engineering behind the scenes. As some of the most incredible feats of engineering happen behind closed doors, prospective future engineers are not presented with the opportunities to be fully immersed within the design, development, engineering, and fabrication of Formula 1 cars. The non-existent platform for testing and evaluating inhibits an influx of future development within the sport. To combat this issue, a realistic and industry spec teaching aid is required. A suitable and innovative solution to this problem is the Model Drag Reduction System (Model DRS), the system would work to educate engineering students who are unfamiliar with the more technical aspects of the sport.

The Model DRS will have three versions that progress based on test results acquired through calculations, digital model testing and by running multiple tests on each iteration. For the system to achieve the user's requirements of being an educational aid, the controls need to be logically laid out and be easy to use to enhance the interactivity of the system.

The DRS model will include a rear wing actuator which will efficiently open and close the DRS at the press of a button. As stated in the FIA Sporting Regulations, the DRS cannot be activated unless the car enters the DRS zone and is within a one second gap of the car in front, this includes cars being lapped. To simulate this the system must gain input from a variety of sensors to determine if the button can be used to activate the DRS. An extension on this could be to include failsafe backup sensors to simulate having extra reliability in the hypothetical failure of previous systems. In keeping with the same FIA regulations, it should be considered to include a delay for the activation of DRS if the button is pressed before the sensors have indicated the DRS zone has been entered. (Autosport, 2023)

To ensure that the no one is harmed during manufacturing, testing or operation of the system, all wires must be properly covered and secured in the system. To achieve a clean, safe, and easy to modify electronic portion of the system proper wire mounting, casing and management should be considered. For the purpose of safety and keeping within the legal requirements for the unit 3/4 Systems Engineering course construction of the Model DRS must not include processes that use 240V.

As the Model DRS is intended to be used as a teaching tool for complex engineering in motorsport it must exist within the FIA's regulations. This aids to replicate the constraints that teams must adhere to when producing their cars for the following season. Included in these regulations are the dimensions for the rear wing, which are maximum 1010mm width with 50mm additional each side for endplates and a maximum depth of 220mm. (FIA Formula One World Championship, 2023) Another FIA regulated constraint is the types of material that the system can be made out of. The DRS model must adhere to this regulation in order to properly replicate and educate on the engineering design, processes and innovation that is required in order to maximise performance in accordance with the regulations. Not only do the materials need to adhere to the FIA regulations, but also must be suitable for the systems needs and must be assembled to maximise strength. This ties in the DRS Model simulating the rear wing requirements of a F1 car. In conjunction with previously stated constraints for the system to function at its highest capacity and adequately

represent realistic parts of an F1 car, the DRS model must be durable and strong. F1 cars drive at up wards of 360kph and can accelerate from 0 to 100kph in 2.4 seconds exerting high aerodynamic forces onto the car components (Formula One car, 2023). With the 2023 race calendar including 24 races, all parts of the car must be long lasting and strong to minimise the cost of replacing broken parts and to reduce modifications needed to be made throughout the season.

The purpose of the DRS on an F1 car is to reduce drag in DRS zones, allowing the cars to reach top speeds. For the DRS model to simulate its real-world functionality and efficiency of fulfilling its purpose, it must be aerodynamic and work to reduce drag. In addition to these simulator factors of the DRS model, the inclusion of a brake pedal could be considered. The brake pedal would simulate the FIA Regulations regarding the deactivation of the DRS after drivers press the brakes, while this feature isn't necessary for the functionality of the system it makes it more interactive and realistic. (FIA Formula One World Championship, 2023) Another thing to consider with the system is to make it visually appealing, this will not only improve the aesthetic finish of the Model DRS but also make it a more appealing education tool.

As Formula 1 has committed to having the championship's carbon footprint at net zero by 2030 (Formula 1 , 2023)it is vital that this system utilises sustainable processes in order to demonstrate possible developments within the Formula 1 industry regarding wastage and environmental impact. This includes the use of sustainably sourced materials and minimising waste.

To simulate the FIA's regulations on the cost cap this project must be completed within $500. In order to replicate the engineering processes active in the Formula 1 teams the project must be completed by the 1st of September 2023, this mimics the timeline in place in Formula 1 for cars to be ready well before the start of the next season so they can go through testing and modification. Based on the time frame allowed, it should be considered to utilise the most efficient production methods to save time during production of the Model DRS. This allows for testing, development and modifications to the Model DRS simulating the engineering processes followed by Formula 1 teams.

Constraints and Considerations

All constraint and considerations referring to the FIA 2023 Technical Regulations Issue are further clarified in the 'FIA Technical Regulations' and 'FIA Sporting Regulations' section of this folio.

Constraints	Justification
Must only use materials permitted in the FIA 2023 technical regulations issue. (Appendix A)	To maximise the complexity of the engineering processes to simulate the progressions of designing and developing a F1 grade rear wing with DRS, the project is required to adhere to the FIA 2023 technical regulations regarding materials. This enhances the purpose of the model to educate and enhance engineering students understanding of engineering in motorsport.
Must include efficient usage of the rear wing actuator to activate and deactivate the DRS.	To function to the highest standard the electronics need to be efficient to replicate F1's stress on performance. This helps to produce a model that is realistic with the value placed of precision and the engineering processes required to achieve this.
Must gain input from sensors to then allow a button to be used to activate DRS.	In keeping with the FIA regulations on DRS, the model must take input from sensors which then allow a button to be pressed to activate DRS. This follows the change in 2011's FIA regulations (Autosport, 2023) regarding the usage of DRS only in specific zones. By following the FIA regulations students can be educated on the flexibility and ingenuity of engineering processes to adapt to regulation changes.
All wiring for the system must be properly covered and safe.	The wiring of the system must be safe for anyone working on, interacting with or observing the Model DRS. This constraint is in place to keep everyone as safe as possible to minimise serious injury.
The system must have logically laid out and easily understandable controls.	To increase the interactivity with the Model DRS and to enhance the learning experience it provides, the end result must be logical and easy to use. This helps to ensure that the users have the best experience with the system and that it functions well and with ease.
Must fit within the FIA's dimension rules of maximum 1010mm width with 50mm additional each side for endplates and a maximum depth of 220mm. (or a scaled alternative) (Appendix 1)	By following constraints placed on the wings dimensions the model's construction and design will have to adhere to the FIA regulations. As previously stated these regulations force the design to be creative whist still remaining within the specifications followed by all F1 teams. This provides a space for learning techniques and methods for incorporating creativity within the set margins. These dimensions also aid in aligning the rear wing with a Formula 1 car creating a more realistic process and end result.

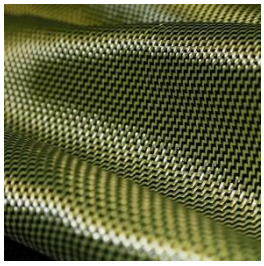
Must be completed within a budget of $500.	The budget for this project is in place to ensure that it is completed within the allocated allowance. The budget also has the added benefit of simulating the cost regulations that are put on F1 teams, achieving a more realistic system develop process.
Must be completed by the 1st of September.	Having a set date for the completion of the DRS Model is crucial to keep designing, testing and production of track. It also compares to the time pressure placed on F1 engineers to test, modify, and present a car at the start of the season.
Must maximise sustainable processes and minimise environmental impact.	Sustainable processes must be maximised, and environmental impact minimised in the design, testing, production, and modification of the DRS Model. With ever growing climate and environmental concerns, it is vital to take a step in the right direction by reducing negative impacts on the environment. Social sustainability also must be maximised by sourcing materials that are ethically obtained as well as focusing on economic sustainability. Not only does this work to create an economically, environmentally, and socially sustainable final product but also represents Formula 1's current concerns regarding their own environmental impact.
Must be able to withstand high aerodynamic forces through strength and durability.	To achieve the intended purpose of being a model version of a Formula 1 rear wing with DRS, the final system needs to be produced with durability in mind. F1 cars reach speeds of upwards of 300 km/h and accelerate from 0-100 km/h in around 2.9 seconds, (Formula One car, 2023) this requires all aspects of the cars to be durable and equipped to handle high aerodynamic forces. While the system being constructed is only a model that will not be fitted onto an F1 car, it is important for educational purposes that the Model DRS follows the process that all parts of F1 cars go through to ensure they are able to properly function when handling extreme forces.
Must be aerodynamic and reduce drag.	In keeping with the purpose of the Model DRS to educate engineering students on automotive engineering, the final project must be aerodynamic creating high levels of downforce whilst reducing drag. Proper engineering process, testing and modification must be made in order to achieve this which in turn educates on the steps to follow to achieve this constraint.
Must use strong and suitable materials, that are assembled/constructed to maximise strength.	To further enhance the comparison to a rear wing designed, engineered, and constructed to endure the forces of a Formula 1 car, the materials need to be suitable for the stresses they would be placed under. This further progresses the educational purpose of the Model DRS by constructing it through the lens of engineering the F1 rear wings with DRS. The various iterations of the Model DRS will be tested to make changes and improve the strength of the material joins or weak points to achieve a strong end result.

Construction of the Model DRS must not include processes that use 240V. "In Victoria, the relevant legislation for electrical safety is the Electricity Safety Act 1998 and associated regulations. Only persons who hold an appropriate current electrical licence are permitted to carry out electrical work on products or equipment that require voltage greater than 50 volts AC or 120 volts ripple-free DC. This requirement means that students are not permitted to carry out any electrical work on electrical products or equipment that operate above 50 volts AC or 120 volts ripple-free DC" **(Victorian Certificate of Education , 2023)**	For the purpose of safety and keeping within the legal requirements for the unit 3/4 systems engineering course 240V cannot be tampered with or altered during construction.

Considerations	Justification
Adequate component organisation, managed properly to allow for easy access and maintenance.	Organised wire management should be considered in order to minimise possible wiring mistakes and to make troubleshooting an easier and more straight forward process. This speeds up the wiring processes as well as creates a higher quality project.
The inclusion of failsafe backup sensors.	The inclusion of failsafe and backup sensors would mimic the inclusion of these on F1 cars. This added feature would increase reliability and functionality. However, it is not required for the system to function properly. The extra sensors would enhance the system's functionality but is not required.
Have a delayed activation of the DRS if it is attempted to be activated before the sensors indicate it is able to do so.	The addition of delayed activation of DRS would be more similar to the rules and regulations currently surrounding the DRS in F1. Enhancing the educational comparisons that could be attained by interacting with the Model DRS. However, it is not a requirement to add this to the model.
Have visually appealing and aesthetic qualities.	Aesthetics should be considered for the Model DRS as it enhances the visual quality of the end product. Not only does it draw people in to interact with the system, but this consideration ties in with the final result having a high-quality finish.
Include a brake pedal to deactivate the DRS.	The inclusion of a brake pedal into the Model DRS should be considered as it would create a more interactive model and would be a more realistic representation of a Formula 1 car. It fits withing the FIA regulations that the brakes will deactivate the DRS. Whilst also, creating a more realistic method of deactivating the DRS.
The most efficient methods of production being utilised.	By considering the most efficient methods of producing the Model DRS, time will be saved resulting in a finished product. This also simulates the prioritisation of the most important aspects of a F1 car during its developmental stages. With the inclusion of efficient production methods, testing and development of the Model DRS can be achieved.

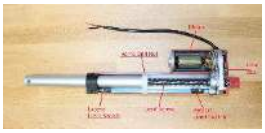
Initial Research

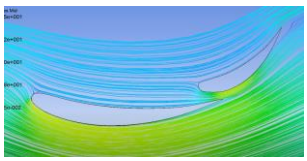
Material	Applicability	Properties	Price	Image
3D printer filaments	During the initial iterations of the Model DRS 3D printing will be a great method for rapid prototyping, testing, and modifying various aspects of the wings. The speed of 3D printing and allowance for rapid changes to be made and tested from CAD makes it a very accessible initial prototyping tool. While carbon fibre filament has been included for comparison, it has its own restrictions in terms for requiring specifics with printers.	PLA: Strength: medium Print temperature: 180 – 230 degrees Celsius Warping: minimal ABS: Strength: high Print temperature: 210 – 250 degrees Celsius Warping: High Carbon Fibre (additive to base filament): Strength: very high Print temperature: Wide range Warping: moderate Possible base filaments: nylon, ABS, PLA. The most commonly used base filament is nylon due to its compatible properties with carbon fibre. Adding the carbon fibre increases the strength, durability, and heat resistance of the nylon. (Best 3D Printer Filament, 2023)	PLA: $20-$40 per Kg ABS: $30-$50 per Kg Carbon Fibre: $70 upwards per Kg (Best 3D Printer Filament, 2023)	

Carbon fibre	Carbon fibre is applicable in the later iteration of the system, this material is used on today's Formula 1 cars and furthers the aim to make the model as close FIA spec as possible. Carbon fibre is a strong choice of material due to its excellent tensile properties.	Tensile properties for carbon fibre include, low density, good thermal and electrical conductivity, high quality creep resistance and high thermal and chemical stabilities. Carbon fibre is a low-density material with a high strength to weight ratio. (Carbon Fibre Plain, 2023)	Approximately $50 per metre	
Kevlar	Kevlar is an alternative material to carbon fibre as it fits with the same purpose within the final iteration. Both materials are comparative in their properties and would perform well with the constraints put on the Model DRS.	High tensile strength, high toughness, high chemical stability at high temperatures and low density. (Kevlar Aramid Fiber Technical Guide, 2023) While both Kevlar and carbon fibre are synthetic fibres with high strength they have some minute differences. Kevlar is lighter but not as strong as carbon fibre, they also share some thermal and impact resistance differences. For enhanced composites carbon Kevlar is used, where both materials are weaved together as seen in 'images' .	Approximately $50 per metre	

Aluminium	For the last iteration aluminium will be useful in the load baring point of the Model DRS which are often weak point. By adding this material, the strength and durability increases as well as the ease of joining components together.	Low density, high thermal conductivity, easily cast and machined.	Approximately $2-$3 per kg	
High Performance foam cores (ROHACELL)	ROHACELL structural foam is often used as the core for lightweight designs that have high shear and pressure resistance (even at high temperatures). The application of this material is to be the core in the carbon fibre or Kevlar components for extra weight reduction. This structural foam is used in Formula 1 as a main method of reducing weight.	The polymethacrylimide based lightweight structural foam comes in a variety of densities making it possible to meet various weight and mechanical requirements. The density margin is 32 to 200 kg/m^2. (High-performance foam cores for sandwich structures ROHACELL, 2023) While this exact specific material may not be feasible due to cost, supply and access, a high-performance foam alternative will be further researched.	Wide range depending on foam grade and thickness.	

Electronic	Applicability	Pros	Cons	Image
Raspberry pi 3, Model A+	This microcontroller will act as the central controller for the system, using the information received from the sensors and buttons. Following the code, the Raspberry pi will have a main output of activating the linear actuator. The main specs of the microcontroller are; Power: Input voltage: 5 volts GPIO pins: 40 (Raspberry Pi 3 Model A+, 2023)	Having all 17 of its GPIO pins are digital, this improves the efficiency of organising the wiring along with allowing for more inputs/outputs. Can read multiple scripts at once, allowing for more layered programming.	Small database of information with limited range of resources. The microcontroller doesn't include storage and would require an external storage method. Extremely limited stock.	
Arduino Uno Rev3	The Arduino would be the central controller for the system, taking input from the sensors and using the information to follow the code. The main output of the Arduino will be activating the linear actuator. The main specs of the microcontroller are; Power: 2.7-5.5 volts Input voltage: 7-17 volts (recommended) Analog input pins: 6 Digital input/output pins: 14 (Arduino Uno Rev3, 2023)	Uses C++ which is a widely accessible and familiar coding language. Has a wide range of resources with an extensive database of information.	Reduced memory storage capacity, with only 2kb of SRAM and 32kb of flash memory. (What are the advantages and disadvantages of ardunio , 2023) Not optimised for performance, this Arduino board is augmented for beginners meaning optimising performance comes with the cost of reducing the microcontrollers overall power ability. Can only read one script at a time is an	

			issue for the microcontroller.	
Linear actuator	The linear actuator will be a mechanical component that controls the opening and closing of the DRS on the rear wing. A linear actuator is applicable due to only needing to move the wing linearly.	Linear actuators are very accurate and reliable at repeatedly moving a payload. Linear actuators are versatile and reliable making it easy to incorporate them into designs. Safe and accurate motion control that is provided from the linear actuator. (Actuator as a keystone of motion, 2023)	With the three types of actuators (hydraulic, pneumatic, and electric) the cons all vary. However, there are some overall cons. (Pros & cons of hydrolic, pneumatic and electric linear actuators, 2023) High quality and efficient linear actuators can be very expensive. The actuators can be highly sensitive to vibrations which has the potential to impact efficiency.	
Motion sensor	This PIR (Passive infrared detection) will be incorporated into the system to simulate the detection of the DRS zone. The sensor will provide an input to the system to allow for the DRS activation.	Low false alarm rate in comparison to microwave and ultrasonic sensors. Large lens version provides a wider frame for motion detection. this creates a larger area to use as the DRS detection zone. (Advantages and disadvantages of PIRs, 2023)	Large changes of environmental temperature around the sensor can impact the number of false alarms.	

Rear wing (with DRS) for a Lancer Evo	Overview	Pros	Cons
   (Making a carbon fiber rear wing with DRS, 2023)	The rear wing was initially constructed with foam moulds utilising a foam core to reduce weight in the wings. Inside the wings are 4 aluminium inserts to assist in attaching the wing to the supports and the end plates are constructed with aluminium moulds. The foam moulds were shaped using CNC then filled with plaster and fiberglass to create a second stage mould. The second stage moulds were used to shape the carbon fibre and make the foam core of the wings. The aluminium inserts were machined using CNC to achieve exact clean cuts. The end plates were created using a clamshell type mould produced on the CNC machine. The method of bolting together the two sides of the mould have the intended purpose of squeezing out any air or resin between the carbon fibre. To enhance the structural strength in the end plates, small aluminium inserts were added. These inserts also assist with the assembling and aligning of the rear wing's pieces. The DRS is controlled with the use of an electric motor with is attached to one of the aluminium supports. The open and closing of the wing is achieved through the linkage between the motor and the wing.	The supports are top mounted which aids aerodynamics, as they don't disturb the airflow underneath the wing. The shaping of the wings performs to a level that provides 400 kg of downforce at 200 km/h, The stress points of the wing have aluminium underneath the carbon fibre to provide extra strength. The process of clamshell moulding the end plates produces a higher quality result than vacuum bagging due to the clamping force being much higher. The activation of the DRS is controlled by a button on the steering wheel and the brake pedal. To open the rear wing a button is pushed and to close it the brake pedal is linked into the system.	As the supports were designed to bolt onto the same location on the Lancer Evo as the factory wing they are far apart. This has some downside with no accounting for the optimal support position for durability and airflow. To reduce the weight of supports, pockets could have been drilled out of the main frame. However, this creates the possibly of interfering with the airflow underneath the wing. The motor and wing linkage add extra weight to one side of the wing due to the mounting location. This has minute impacts on the balance of the wing. The placement of the components could impact the airflow due to them obstructing the air before it reaches the wing.

Scaled Formula 1 Rear Wing with DRS	Overview
 (How Does DRS work? F1 Tech Talk, 2023) 	The rear wing is a scale model of the one fitted to the Alpine Formula 1 cars. The wing is scaled down to 50% of its original size. With modern Formula 1 cars, DRS is vital to increasing overtake opportunities by increasing speed from the impact on downforce. Rules outlined: - Two race laps must be completed before DRS can be used. - In the case of a safety car, DRS cannot be enabled until two laps after the safety car has been removed. - Can only be used in DRS zones. - To use DRS the gap to the car in front has to be one second or less. These rules are also in (Appendix B) The DRS has an adjustable element (open and closing of the DRS) for the aerodynamics which is not allowed anywhere else but on rear wing DRS. When the wing is open (DRS enabled) the gap is 85mm. This measurement is the same for every team in every race of the season. The DRS is deactivated when the drivers press the brakes, it is vital that this is done at the end of the DRS zone especially if the zone is followed by a corner. With the DRS on the back of the car loses roughly 30% of downforce, while this increases speed on straights. The downforce is needed for the corners. The top flap of the rear wing is moved by a piston to open and close the gap in the wing. The piston on the scaled model is pneumatic however the F1 cars use a hydraulics system which are similar to the hydraulics in the steering wheel which create a faster opening of the wing. The curved shape of the closed rear wing creates downforce on the car due to the air traveling up based on the angle of the curve. By activating the DRS which opens the gap in the wings the airflow changes to reduce this down force.

Evaluation Criteria

	Constraints
	Considerations

Criteria	Description	How will this be achieved?	How will this be tested?/ Kept on track?	Importance (1-10)
Functionality	Does the system include an efficient usage of the rear wing actuator to activate and deactivate the DRS?	To ensure that the system activates and deactivates DRS using the rear wing efficiently, time will be taken to ensure a high-quality instalment of the actuator along with quality coding and wiring. More specifically to the functionality of the actuator, trails and calculations will be carried out in regard to the pitch of the lead screw, gearing of the motor and applied voltage.	This will be tested by monitoring the speed and precision of the actuator under different settings before attaining requirements for the most efficient one. This will be kept on track by allowing for the testing in the scheduled production plan and carrying these tests out on all iterations improving each time.	10

		Is the system aerodynamic and reduce drag?	Producing an aerodynamic Model DRS that reduces drag will be achieved by researching the best methods of directing airflow to reduce downforce. This includes the shape, material, and other attributes of the rear wing. As well as carrying out extensive research, iterative models will be created and tested in conjunction with CAD testing specifically CFD to ensure that the system is aerodynamic and reduces drag.	This will be tested on not only the final version of the system but also on any previous iterations or CAD models. The CAD models can be tested using the appropriate computer software to look at the airflow around the model. With the physical iterations they will be placed on scaled and air will be projected at the Model DRS when it is open and then when it is closed. If the weight of the system decreases when the DRS is activated then the system is reducing the downforce. This will either be tested using a wind tunnel or by projecting water mixed with air at the system.	9
		Does the Model DRS include delayed activation of the DRS if it is attempted to be activated before the sensors indicated it is able to do so?	The inclusion of delayed DRS activation could be achieved through developing a code that allows for this. This requires the sensors to indicate the incorrect DRS activation attempt has occurred and to place a time delay on the actuator.	This will be assessed by running tests on the sensors' readings based on the location of the Model DRS. This will allow an understanding to be gained based on the accuracy of the sensors to then be fed into the code of the system. Multiple tests will be carried out on the time it takes from the failed DRS activation to the delayed activation will be recorded and compared.	5

		Does the system include a brake pedal to deactivate DRS?	Producing a Model DRS with a brake pedal to simulate the method of deactivation could be achieved by allowing for a brake pedal in the design of the system and factoring it in to the wiring and componentry.	This will be kept on track by including the components into the design of the system and planning for the connection of the brake pedal to the design. Through different iterations of the design, the time between pressing the brake and the deactivation of the DRS will be recorded and methods of speeding up the deactivation will be researched and implemented into the nest versions of the system.	3
User needs and requirements		Does the Model DRS gain input from a sensor to then allow a button to activate DRS?	To achieve the utilisation of a button to activate DRS, the design of not only the physical components but also the schematic plans for the electronic components and the code will incorporate the sensors. Allowing for them to connect to the DRS activation button to have it open and close the wing. To achieve this to a high quality, extensive research will be carried out to gain a good understanding of the functionality of the sensors and the systems requirements to allow them to perform the best. As well as research, a scheduled production plan will be created to allow adequate time to construct, connect and test the sensors.	To test the inputs from the various sensors, not only will time related efficiency tests be carried out but also more focused tests in regards to the electronics will be done. These electronic tests could include current draw recordings. To test the efficiency of sensors, the time between the Model DRS entering the simulated DRS zone and the button being able to be pressed for the activation of DRS will be recorded and analysed. This will be assessed over various iterations to make improvements on the location and sensitivity of the sensors.	10

		Is the system completed by the 1st of September?	The completion of the system by the date allocated will be achieved by creating a detailed production plan allowing for interruptions and delays. This will also be achieved by utilising the most efficient production methods.	This will be kept on track by planning out a production timeline in the form of a Gantt chart to schedule each step in production and testing. By doing this the Model DRS's construction will be kept on track and at the end of the timeframe there will be a completed and tested system.	8
		Does the system have logically laid out and easily understandable controls?	This will be achieved by designing and constructing a set of controls that are simplistic and clearly laid out and labelled.	This will be kept on track by gaining feedback on how logical the controls design is allowing for modifications to be made for later versions. On the final system the same tests will be run on how easy people find it to use and how understandable it is. From this qualitative and quantitative feedback will be obtained to give a wide scope of responses.	6
Materials and components		Were only materials permitted in the FIA 2023 technical regulations used?	Keeping withing the FIA 2023 technical regulations requiring materials will be achieved by completing extensive research into the allowed materials including and composite materials. During the production stage if any alterations to the plan are made in regards to the materials, checks will be completed to endure that the alternative material is permitted.	To keep on track with only using FIA permitted materials (Appendix A) during planning and design a summary list of acceptable materials will be regularly checked against. With the final iteration an analysis will be done on the materials used both by doing a physical assessment and by referring to a documented production process.	7
		Are the materials chosen strong and suitable for the system? Are they assembled using methods to	This will be achieved by researching various materials and comparing their properties to determine the most suitable materials for the	The strength and suitability of the chosen materials and production processes will be tested by adding load to the materials and recording any notable strain on potential weak point.	7

		maximise strength?	system. Along with research of materials, research on the most suitable construction techniques and processes will be carried out to gain an understanding of a wide range of methods to produce a strong Model DRS, comparative to the requirement of F1 car's rear wings. When constructing the model time and care will be taken when joining, machining, or carrying out other construction processes to ensure they produce a strong outcome.	This stress testing will be carried out on the materials prior to construction of the project to ensure they are the best choice, they will also be tested during different iterations so that improvements can be made to increase the strength if the system.	
Environmental use (use area)		Is the Model DRS able to withstand high aerodynamic forces through strength and durability?	A High level of strength and durability against aerodynamic forces to simulate the environment of a F1 car will be achieved through researching, planning, testing, and improving iterations of the Model DRS. Not only will the system be planned in terms of materials, construction processes and the minimisation of weak points but quality production will be performed in order to ensure that the Model DRS meets these requirements.	This will be initially tested by running strength tests on a CAD model of the system including CFD testing and calculating the durability and strength of the model. For tests on further physical iterations of the design a combination of aerodynamic tests and strengths will be carried out to ascertain this information. The aerodynamic testing includes observing the airflow around the Model DRS, this test can either be run by projecting air mixed with water at the system. Alternatively, an air tunnel would give more accurate test results. For tests around durability tests similar to the material strength tests will be run to ascertain any weak points and to ensure that the final project will be able to withstand large	8

				forces. Specific zones of the Model DRS for example load baring areas can be put under specific directional load tests to observe the durability.	
Safety		Is all wiring for the system covered and safe?	This will be achieved by taking care and allowing for extra time when wiring the system. Along with planning for wire management processes, which could include specific casings for wiring or simply using heat shrink on exposed parts of wires.	This will be kept of track by continually referring to the design that will includes feature to assist in having safe wiring. To test that the final system is covered and safe a visual check will be completed to analyse if there are any exposed wires. Multiple checks will be completed throughout all iterations of the design to ensure that they are all safe for use, maintenance, and modifications.	10
Cost		Is the system completed within the budget of $500?	Keeping within the budget will be achieved by initially creating a comprehensive Bill of Materials that indicates not only price of the component/material but also the cost of transportation. To further ensure that the system stays with budget is to use materials that are either recycled or are already in possession.	To determine if this project was successful at staying with the budget of $500 the records of all expenses will be calculated and reviewed to ascertain that information. To keep the project on track to achieve this constraint regular checks will be done to ensure that the system is allowing for future expenses.	8
Waste and energy		Did the system maximise sustainable processes and minimise environmental impact?	Maximizing sustainable processes to minimise the impact on the environment will be achieved by initially researching the most effective methods of reducing environmental impact. This includes research into lean manufacturing techniques. Another way to achieve this is to take care when cutting, shaping, joining ect.	Minimising the effects of environmental impact will be kept on track by following a production plan that includes methods of maximising sustainable processes. These processes will be documented and can be assessed after the finalisation of the project.	8

			Materials to reduce the likelihood of making a mistake that could mean wasting materials.		
Quality standards Styling and appearance		Does the system have appealing and aesthetic qualities?	This could be achieved by taking time in all stages of the systems engineering process to ensure the best quality end result is being created. This includes designing, planning for and producing a quality product. To ensure the end result is visually appealing checks could be done throughout design and construction to gain ongoing feedback on the appearance of the system.	This can be tested throughout the systems engineering process and particularly at the end of the project by gaining feedback and conducting surveys in regards to the appearance and quality off the Model DRS.	4
Performance and durability		Does the Model DRS include failsafe backup sensors?	This could be achieved by planning and designing for the worst case scenario and implementing backup sensors to combat any issues. By using the initial sensors as a guide, failsafe sensors could be included with the systems design having them linked to the initial sensors.	To test the functionality of the failsafe sensors and how reliable they are, worst case scenario test will be run. This involves manually shutting off the initial sensors and recording the reaction of the failsafe sensors. This can be recorded as how long it takes for them to initiate and how accurate they are in comparison to the initial sensors.	5
Size		Does the Model DRS fit withing the FIA's dimension rules or a scaled alternative?	This will be achieved by carefully measuring throughout all stages of the systems engineering process, and double checking any measurements. Adding parameters within CAD applications will help to ensure that anything designed using those tools will be with	The dimensions will be tested by taking precise measurements of the final results or any iterations and checking them against the FIA dimension rulers. (Appendix A)	7

			the required dimensions.		
Maintenance		Are the components organised and managed properly to allow for easy access and maintenance? This includes wires, electronics, and power supply.	This could be achieved in conjunction with the constraint regarding wire safety, aiming for planned wire management containers and carefully laid out schematic designs to plan the location of all wires.	This will be kept on track by regularly allowing time to organise the wires, components, or electronics. The method of testing this will occur during production, by recording how easy it was to locate any issues and preform maintenance by swapping out components or fixing broken ones. This can also be tested by visually assessing the organisation of all components, how they are laid out along with how easy it is to see the connections and wiring.	5
Production methods		Did the system utilise the most efficient production methods?	This could be achieved by researching efficient production methods, designing the system to allow for them to be carried out, planning for when and how they will occur with the system construction and by constructing the end result by using them. Time and care should be taken when utilising the efficient production methods ensuring they are also effective.	This will be tested by comparing the production plan/timeline to the actual timeframes for each process. The difference in time will be assessed to achieve an understanding of if the methods of production were efficient.	4
Regulations		Construction of the Model DRS must not include processes that use 240V. “Students are not permitted to carry out any electrical work on electrical products or equipment that operate above 50	This will be achieved by researching and planning for alternate power sources as well as ensure no part of the Model DRS requires 240V to be worked on.	To test that this regulation is adhered to the production journal will be reviewed, ensuring that no 240V altering or utilisation occurred.	10

		volts AC or 120 volts ripple-free DC” (Victorian Certificate of Education , 2023)			

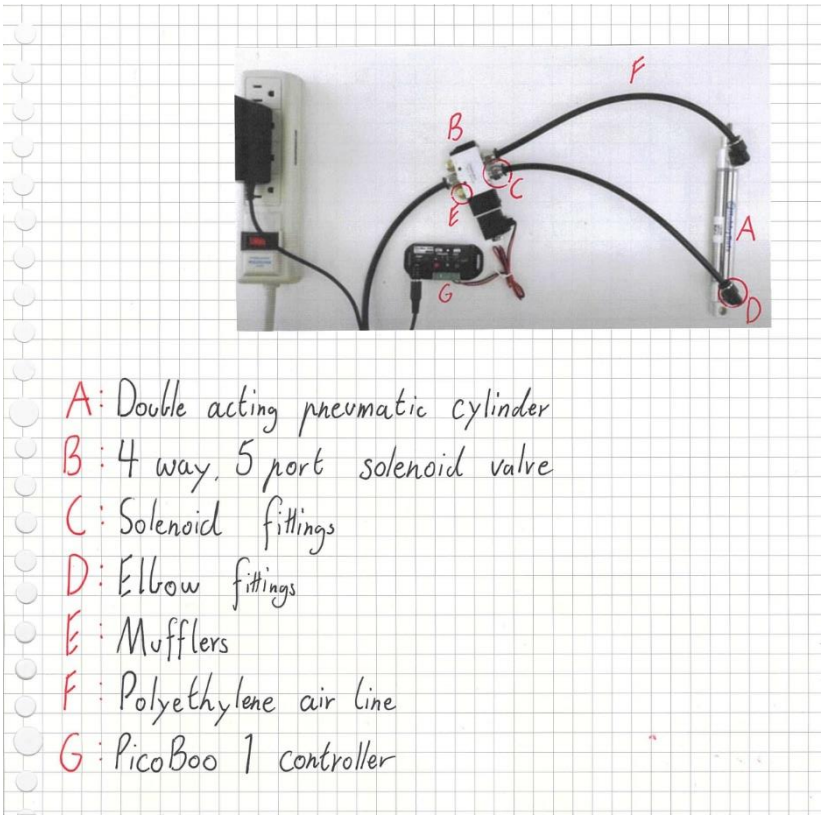
Criteria 2

Existing system analysis

Control system for Pneumatic Cylinder

Precursor:

The main elements of the pneumatic cylinder that are vital for the understanding of the functionality and compatibility of specific cylinders are the; stroke and bore. The stroke is the distance the cylinder rod extends when air is applied. The bore refers to the diameter of the cylinder, a pneumatic cylinder with a larger bore is able to hold more air. This allows the cylinder to be stronger and extend the cylinder rod faster. For this project the most suitable type of cylinder is a double acting cylinder. This is due to it having two air ports which allow air to enter through one to extend the cylinder rod, and air enter through the other to retract it.



A:

The double acting cylinder has two air ports, one for the air to enter through to push the rod out and another port than when air enters pulls the rod back in. As mentioned in the overview for ease of use and construction the cylinder is a universally mounting meaning it has mounting threads at each end of the cylinder.

B:

The purpose of the solenoid value is to direct the flow to the correct port of the pneumatic cylinder to extend and retract the cylinder rod. To summarise the solenoid is a pneumatic switch that controls the flow of air, air enters through the port connected to the compressor then flows to one end of the pneumatic cylinder to extend the rod.

C:

The fittings are sized to match the solenoid threading to minimise any air escaping. On the other end of the fittings is a plastic ring which are sized to connect to the airline. For the 4 way, 5 port solenoid, 3 fittings are needed. Two for the air outputs and one for the air input. The fittings have Teflon rapped around the thread to provide an extra seal, with the aim to reduce the escaping air as much as possible.

D:

The elbow fittings are very similar to the solenoid fittings, with them also have one end threaded and one end with a plastic ring for the air line connection. The main difference is that they bend at 90 degrees, allowing for a cleaner setup of componentry as well as aiding the organisation of the air lines.

E:

On the solenoid there is an exhaust port either side of the input port, these two ports are filled with the mufflers. The mufflers also known as breather vents are threaded on one side with a muffler on the other end of the fitting. The main purpose of the mufflers is to block any air entering the exhaust ports, but they have the added bonus of muffling some of the sound from the exhaust.

F:

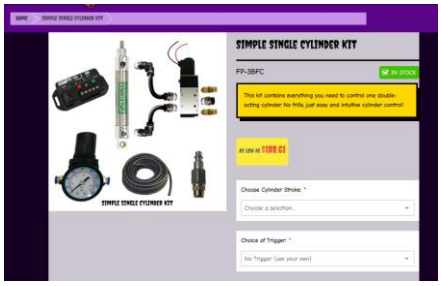
The polyethylene air line is used to connect the air compressor, solenoid, and pneumatic cylinder. The line is easily connected into the fittings, only needing to be pressed in until it is slightly depressed into the fitting.

G:

While the system is functional without the Picoboo 1 controller, the controller increases the systems usability and provides some other functional benefits. The Picoboo 1 controller is created by Fright Props (Prop control, 2023), it is a simplistic controller that has a singular output that is solid state. The biggest functional benefit of the controller is its ability to have a programable sequence of movement. The controller has a record button and a sequencing on/off button, once you hit record you tap out a sequence of various spacings and times for the pneumatic cylinder to be extracted and retracted. Once finished with the sequence you press record again and the controller saves the sequence, meaning once the play button is pressed the controller plays back the latest saved sequence. The controller can also be connected to a trigger, for example a motion sensor which activates the saved sequence.

Analysis:

The Fright Props video demonstration on the set up and controlling of the pneumatic cylinder provided overall context to a system with similar overall componentry and control systems to the initial plan for my Model DRS. While the video simplifies and explains the individual components in the overall system, additional research will need to be carried out to find components best suited for my Model DRS.



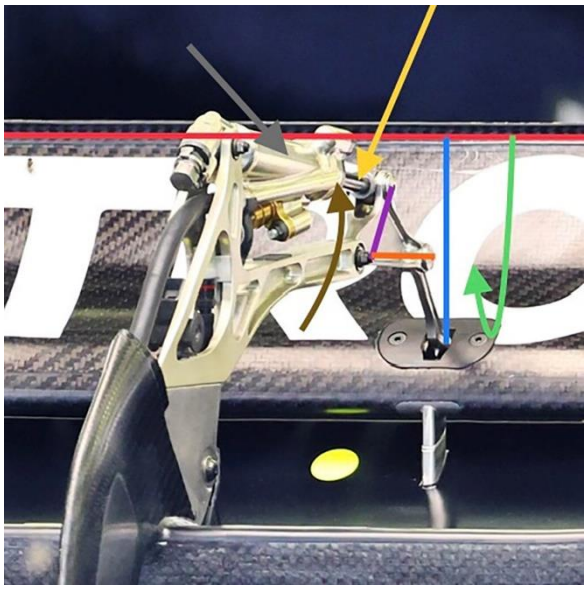
The video explores a purchasable kit from the Fright Props company. While purchasing and using this kit would save time and be a simpler option for my project, I have decided not to based on the reasons listed below.

While the Fright Props simple single cylinder kit has some advantages including; ease of use, simplicity of the programming for the Picoboo 1 controller, and compatibility between all its components. It has some major draw backs that limit its use case for my project. The main ones including;

- Learning, engineering, problem solving, researching and innovation cannot be achieved by using this kit. The kit removes the need for any research on the best componentry and individual components compatibility which negates the purpose of the Model DRS which is to be an educational tool.
- The kit leaves no room for adjustments to be made to customise it for my project. For example, the pneumatic cylinder dimensions, stoke and many other factors that need to be customised for the design of the Model DRS.
- The Picoboo 1 controller cannot save more than one sequence at a time which directly impacts its potential use case to simulate the DRS zones on various F1 circuits.
- This kit removes the fun of discovering, learning, and creating a system from individual parts, through an immense amount of trial and error.

DRS Activation on F1 Cars

DRS activation on Formular 1 cars is achieved through hydraulic fluid being forced into the end of an actuator (hydraulic cylinder) at a high pressure. The fluid fills the internal diameter of the actuator body and the external diameter of the actuator shaft. The pressure created will reduce the length of the actuator shaft, pulling the actuation rocker with it. This opens the rear wing flap, enabling DRS.



To close the actuator the pressure used to open it will be removed and the hydraulic fluid will be forced into the other side of the internal actuator piston, again at high pressure. (Gary Anderson Explain How DRS Actuators Work on F1 Cars, 2023)

- Flap pivot line
- Arc the flap travels when DRS is Activated
- Curve the DRS follows to open the slot gap
- First leg of the actuation rocker
- Second leg of actuation rocker
- Actuator shaft
- Actuator body

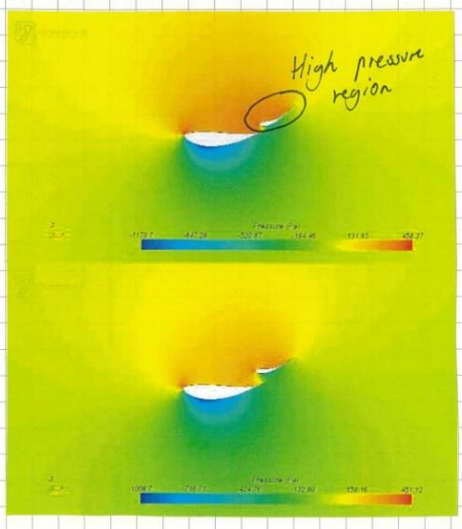
Formula 1 cars use hydraulic systems as they are suitable for moving heavier loads, can function at high speeds and are durable. The hydraulic actuator attached to the rear wing provides greater forces and speeds then electrical or pneumatic systems.

The reason my Model DRS system will not be working off of hydraulics even though it is more efficient, faster, and durable, is because of safety. When not installed, used, or cared for correctly, hydrohalics can be highly dangerous. The Three types of hazards are;

- Burns from hot, high-pressure spray of fluid
- Bruises, cuts, or abrasions from failing hydraulic equipment
- Hydraulic injection of fluid into skin
  (Hyrdaulic Systems Safety, 2023)

Aside from the electrical design of the actuator, its physical design is important to minimise the airflow impact it has on the rear wing. As the actuator and surrounding framework protrude from the centre of the rear wing they are in the direct line of airflow going over the rear wing and through the gap opened with DRS activation. As seen in the diagram above the actuator has not only been machined to reduce its weight, with the cut outs in its framework. It is also curved on majority of the edges to direct airflow around the body structure. Another key design feature that minimises the impact the physical components have on the aerodynamics is that all small features of the actuator are stored inside the body work. This not only stops them from interfering with the aerodynamics but also protects them from high forces, impact and other outside elements that could damage the system.

Model DRS Wing Geometry



- Setting A

- Setting B

- A: Setting A is the off position for the DRS flap. In this position the flap of the wing is responsible for the wing's maximum downforce generation - even with the cost of drag.

  As displayed on the image there is a bigger high pressure region above the wing, as the air has a smaller gap to pass through. Meaning more air is forced above the gap creating more downforce.

- B: Setting B is the on position for the DRS flap. There is a 10 degree difference between Setting A and B, causing more air to flow through the gap in setting B decreasing the downforce. With this model the flap is responsible for 61% of drag generated by the wing assembly in Setting B, whilst maintaining roughly 20% more in downforce.

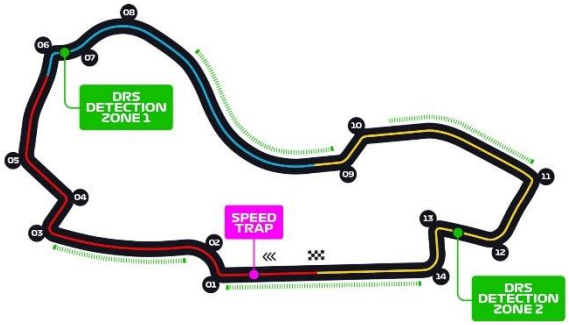
  In this systems results there is a 31% reduction in drag at the 10 degree change in Setting B, which is approximately 3 times more reduction than in an F1 race car.

Time loop System

The intension was to simulate the DRS timings and activation durations. The timings of specific sectors of an F1 track, could have been analysed as well as looking into the timestamps of DRS zones. By ascertaining information on timing, the Model DRS could be coded to simulate a track lap with DRS. This would have enhanced how interactive the Model DRS is and created a more interesting education aid as well as working toward making the model as accurate as possible.

The FIA along with the formula one teams do no release the specific timing points of their car's activation and deactivation of DRS. This results in a less accurate method of gaining that information to be carried out. By looking at the on board fastest laps of qualifying and comparing them to the DRS zones on the track map, it is relatively simple to understand when DRS is activated. The problem arises with when different drivers have different breaking point, which causes a less accurate timing of DRS de-activation. However, this can be estimated when watching the on-board video and listening for changes in the car's noise. While this is an incredibly inaccurate method of timing the deactivation of DRS it is the current method being tested.

2023 Albert Park Max Verstappen Pole Lap



Timestamps for DRS activation:

Sector 1:	Sector 2:	Sector 3:
0.0 -> 3.3	37.1 -> 45.0	49.2 -> 55.2
7.9 -> 13.1		1:11.7 -> 1:16.7

Regulation Annotations

free orientation in space, but must intersect both the stationary and the adjustable part of the profiles for the entire range of adjustment.

d. A fairing that contains a single tyre temperature sensor. The entire fairing and sensor must:

   i. fit in the union between two volumes, a cuboid which is 15mm wide, 60mm long and 50mm high and a circular cylinder which has a base diameter of 30mm and a height of 60mm. The axis of the cylinder must coincide with the major axis of the one of the 15mm x 60mm faces of the cuboid.

   ii. be symmetrical about a plane which is parallel to the 50mm x 60mm faces of the cuboid, and

   iii. intersect either the Front Wing Profiles defined in Article 3.9.1 or the Front Wing Endplate Body defined in Article 3.9.2. A fillet radius no greater than 5mm will be permitted along this intersection.

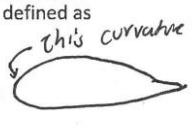
Furthermore, all the components listed above must be arranged such that they are not visible from below and do not lie outboard of the virtual surface defined in 3.9.2.a.

Should there be a requirement for any additional component to be added, the Competitor must write specifically to the FIA with an explanation, design, and calculated aerodynamic effect, in order to get approval. Such a communication will be circulated to rival Competitors if deemed to cover a new aspect that had previously not been considered.

3.9.9 Front Wing Bodywork Group

Once the components defined in Articles 3.9.1 to 3.9.8 have been constructed in accordance to the provisions of those Articles, and any subsequent operation of sub-assembly described within those Articles has been applied, the resulting union of these components is defined as "Front Wing Bodywork".

this curvature



3.10 Rear Wing

3.10.1 Rear Wing Profiles

Bodywork declared as "Rear Wing Profiles" must:

The rear wing flaps curves concave radius >/ 100mm

a. Lie within **RV-RW-PROFILES**.

b. In any Y plane:

   i. Contain exactly two sections.

   ii. The rearmost section, which will be known as the "RW Flap", must have a chord smaller than the chord of the adjacent section.

   iii. With the exception of minimal modifications to permit attachment of the actuator system described in 3.10.10 to the RW Flap, the sections may not contain any concave radius of curvature smaller than 100mm

   iv. The distance between the two sections must lie between 10mm and 15mm at their closest position.

↳ Gap of 2 wing flaps?

Angle of the (top) wing.

Furthermore:

c. Outboard of Y100, the normal to any point of the profiles surface must not subtend an angle greater than 20° to a Y plane.

d. Once the RW Flap is fully defined, a Gurney of up to 20mm may be fitted to the trailing edge. This Gurney is considered to be part of the Rear Wing Profiles and must satisfy the provisions of this Article with the exception of sections b (iii) and (c).

Small tab projecting from the trailing edge of the wing.

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3.10.2 **Pylons**

Bodywork declared as "Rear Wing Pylon" must:

a. Lie within **RV-RW-PYLON**.

b. On any Z plane, when considering both sides of the car, only contain no more than two closed symmetrical sections with a maximum total area of 5000mm^2. This does not apply if the complete section lies within 30mm of the Exhaust Tailpipe defined in Article 3.8.2. For any complete closed section outside of RV-TAIL-EXH, the thickness of each section may not exceed 25mm when measured in the Y direction.

Once the Rear Wing Pylon and Rear Wing Profiles are fully defined the Rear Wing Pylon must be trimmed to the Rear Wing Profiles such that no overlapping surfaces remain. A fillet radius, up to 10mm, may be applied along the intersections between these surfaces.

3.10.3 **Rear Wing Beam**

Bodywork must be declared as "Rear Wing Beam" and must:

a. Lie within **RV-RW-BEAM**.

b. At no point be less than 10mm distant to the **Exhaust Tailpipe** once fully defined in accordance to Article 3.8.2.

c. In any Y plane:

i. Contain no more than two sections.

ii. The section may not contain any concave radius of curvature smaller than 50mm

Furthermore:

d. The Rear Wing Beam must be arranged such that when viewed from above no more than 80,000mm^2 of RS-RW-BEAM may be visible per side of the car.

e. Outboard of Y=175, the normal at any point on the profiles' surface must not subtend an angle greater than 15° to a Y plane. Inboard of Y=175mm, the normal at any point on the profiles surface must not subtend an angle greater than 60° to a Y plane.

Once the profiles have been fully defined:

f. A Gurney of up to 20mm may be fitted to the trailing edge of the rearward most profile. This Gurney is considered to be part of the Rear Wing Beam and must satisfy the provisions of this Article with the exception of sections c (ii) and (e).

g. Provided two closed sections are present as specified in (c), up to two slot gap separator brackets, per side of the car, will be permitted. These brackets must:

i. Be in their entirety within 30mm of both of the two profiles they support in relation to each other.

ii. Be no more than 6mm thick. A fillet radius no greater than 2mm will be permitted where these brackets join the two profiles.

iii. Have no dimension that exceeds 60mm

iv. Not be visible from below with the Rear Wing Beam Profiles present

3.10.4 **Rear Wing Endplate Body**

Bodywork declared as "Rear Wing Endplate Body" must:

a. Lie within **RV-RWEP-BODY**.

b. Be a single volume with no apertures.

c. In any X or Z plane contain only one closed section. Furthermore, any such section, on any X or Z plane, must not contain an external concave radius of curvature smaller than 100mm.

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3.10.5 Rear Wing Tip

Bodywork declared as "Rear Wing Tip" must:

a. Lie within the **RV-RW-TIP**.

b. Be a single volume with no apertures.

No opening, hole or gap.

Furthermore:

c. The following must apply to the intersection between regions of the Rear Wing Tip above Z=670 and any plane that contains the X-aligned axis [Y, Z] = [480, 670]

i. There must be no more than two closed sections.

ii. The distance between adjacent sections must be no more than 15mm at their closest position.

iii. When measured in the X-direction, each section must be larger than 100mm and no more than 40mm of this may overlap, in X, any other section in the same plane.

d. The intersection between the Rear Wing Tip and any Z plane between Z=660mm and Z= 670mm must produce a single closed section.

e. Its complete surface must be tangent continuous, and any concave radius of curvature must be greater than 20mm. Furthermore, any surface visible from the side or above must not contain any concave radius of curvature smaller than 50mm.

→ Continuous curve

f. Curves produced by the intersection of the Rear Wing Tip with any X plane must:

i. Be tangent continuous and not contain any radius less than 20mm forwards of X_R=550.

ii. Not contain any concave radius less than 1000mm if formed from any surface that is visible from the side or any surface forwards of X_R=550 that is visible from above.

Between the wing flaps, endplate(s) and wing tips there should be a smooth connection.

The sole purpose of the external surfaces of the Rear Wing Tip is to create a smooth transition, with no discontinuities, between the Rear Wing Profiles and the Rear Wing Endplate Body. Provided this purpose is observed, and the following are only used at the leading and trailing edges of the sections used to create this transition:

g. Compliance with Section (f) of this Article is not required in the following regions of the Rear Wing Tip:

i. within 10mm of the forward-most point of each individual closed section described in (c) and (d).

ii. within 3mm of the rearward-most point of each individual closed section described in (c) and (d).

h. Compliance with Sections (e) and (f) of this Article are not required in the following regions of the Rear Wing Tip:

i. Regions not in contact with the external air stream after the Rear Wing Assembly is complete.

ii. Within 30mm of a single "Junction Point". This Junction Point must lie above Z= 670.

3.10.6 Rear Wing Separators

Once the Rear Wing Profiles and the Rear Wing Tip are fully defined, the two closed sections described Articles 3.10.1.b and 3.10.5.c must be bridged by three pairs of rigid impervious supports. One pair of supports must be located at Y=0 with the remaining supports arranged symmetrically about the car centre plane and lying in their entirety between Y=+/-470 and Y=+/-490. These pairs of supports must be designed and arranged such that the two closed sections and the relationship between them can only change whilst the car is in motion in accordance to Article 3.10.10. They must:

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a. With the exception of minimal local changes where the two sections are adjacent to each other and with the rearmost element in the closed position, have an inner profile that is coincident to the section they are fixed to, an outer profile that is an offset of the inner profile of no more than 15mm and may not incorporate any radius smaller than 10mm in sideview ('gurney' type trim tabs may however be fitted between the supports).

b. Be aligned as a pair so as to provide a bearing across at least 40mm² when the distance between the two sections is at its closest position.

c. Not be recessed into the Rear Wing Profiles or Rear Wing Tip (where a recess is defined as a reduction in section at a rate greater than 45° with respect to the Y-axis).

d. Be arranged so that any curvature occurs only in a horizontal plane (other than when this bodywork is adjusted in accordance with Article 3.10.10).

e. Not be visible from below when the Rear Wing Profiles and Rear Wing Tip are present.

f. Have no dimension that exceeds 150mm when considering the forward most part of the pair.

g. Have no dimension that exceeds 30mm when considering the rearward most part of the pair.

h. Be between 2mm and 6mm thick.

i. Be rigidly fixed to their respective section. Where they are fixed, a fillet radius of no greater than 2mm will be permitted between the support and the Rear Wing Profiles or the Rear Wing Tip.

j. Be constructed from a material with a modulus greater than 50 GPa.

3.10.7 Rear Wing Endplate

Once the Rear Wing Endplate Body and Rear Wing Tip are fully defined they must be joined to create one unified volume with no remaining overlapping surfaces. The resultant volume, known as the Rear Wing Endplate must be arranged such that, when viewed from the side of the car, no part of **RS-RW-RWEP**, may be visible. Furthermore, the external surfaces at the boundaries between adjacent sections of the Rear Wing Endplate Body and Rear Wing Tip must maintain both continuity and tangency in any X, Y or Z plane.

3.10.8 Continuity

Once the Rear Wing Endplate is fully defined, the external surfaces at the boundaries between adjacent sections of the Rear Wing Endplate, and Rear Wing Profiles must maintain both continuity and tangency in any X, Y or Z plane.

3.10.9 Rear Wing Beam Assembly

Once the Rear Wing Endplate and Rear Wing Beam are fully defined the Rear Wing Beam must be trimmed to the Rear Wing Endplate such that no overlapping surfaces remain. Any Rear Wing Beam bodywork remaining outboard of the Rear Wing Endplate must be discarded. A fillet radius, no greater than 10mm, may be applied along the intersections between these volumes. Furthermore, once the trim is complete, when viewed from the side, no part of the Rear Wing Beam may be visible.

3.10.10 Drag Reduction System (DRS)

The entire RW Flap described in Article 3.10.1, including any gurney if fitted, part of the rearward most section from any regions of the Rear Wing Tip described in Article 3.10.5 where there are exactly two sections in an intersection with any plane that contains the X-aligned axis [Y,Z]= [480, 670], and the portions of the Rear Wing Separators defined in 3.10.6 that are attached to these components, may be rotated about a fixed axis whilst the car is in motion. All bodywork to be rotated shall be known as "DRS Bodywork"

Furthermore:

a. No part of the "DRS Bodywork" may lie outboard of Y=490.

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b. The axis of rotation of the RW Flap must be fixed and located no more than 20mm below the upper extremity and no more than 20mm forward of the rear extremity of **RV-RW-PROFILES** at all times.

c. There must be no relative movement between the constituent parts of the DRS Bodywork.

d. Any such variation of incidence maintains compliance with all of the bodywork regulations with the exception of Article 3.10.1 and 3.10.5.

e. It cannot be used to change the geometry of any duct, either directly or indirectly, other than the change to the distance between adjacent sections permitted by Article 3.10.1 and 3.10.5.

f. The design is such that failure of the system will result in the uppermost closed section returning to the normal high incidence position.

g. Any alteration of the incidence of the uppermost closed section may only be commanded by direct driver input and controlled using the control electronics specified in Article 8.3.

h. At any Y plane, the distance between the two sections of the Rear Wing Profiles at their closest position must lie between 10mm and 85mm.

Maximum open gap is 85mm

Closed gap between 10mm - 85mm

The definition of gap size between stationary and moving flap.

In order to achieve the above adjustment, the actuator mechanism and any associated minimal fairings must lie inboard of Y=25, below Z=940 and not extend beyond **RV-RW-PROFILES** in the forward or rearward direction, when viewed from above. Furthermore, any additional minimal bodywork associated with the adjustment of the DRS Bodywork must be located outboard of Y= 465 and in side-view be below the top surface of **RV-RW-PROFILES,** must not be visible from below and must fit within a cuboid which is 30mm wide, 60mm long and 30mm high. This cuboid may have a free orientation in space, but its width must be aligned with the axis of rotation of the RW flap.

3.10.11 Rear Wing Bodywork Group

Once the components defined in Articles 3.10.1 to 3.10.10 have been constructed in accordance to the provisions of those Articles, and any subsequent operation of sub-assembly described within those Articles has been applied, the resulting union of these components is defined as "Rear Wing Bodywork".

3.11 Final Assembly

This Article gives additional information on the junction between adjacent bodywork groups defined in Articles 3.5 to 3.10.

3.11.1 Front Bodywork to Rear Bodywork

Once both the complete Front Bodywork and the Rear Bodywork are fully defined, the Front Bodywork and the Rear Bodywork must be trimmed to each other, such that no overlapping surfaces remain. A maximum fillet radius, no greater than 50mm, may be applied along the intersections between these surfaces.

3.11.2 Front and Rear Bodywork Assembly to Floor Bodywork

Once both the complete Front and Rear Bodywork assembly defined in Article 3.11.1 and the Floor Bodywork are fully defined, the Front and Rear Bodywork assembly and the Floor Bodywork must be trimmed to each other, such that no overlapping surfaces remain. Additionally, before the trimming is applied, any Mid Chassis or Rear Bodywork remaining below the Floor Bodywork must be discarded. A maximum fillet radius, no greater than 50mm, may be applied along the intersection between these surfaces. Any such fillet must be at least 1mm distant from **RV-FLOOR-EDGE**. Furthermore, once all surfaces are trimmed and filleted, no part of the Mid Chassis, Rear Bodywork, fillet between the Rear Bodywork and the Floor Bodywork, or fillet between the Mid Chassis and the Floor Bodywork, may be visible from below.

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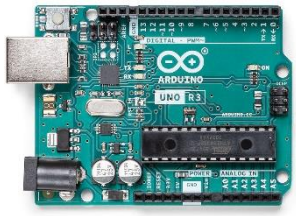
Components

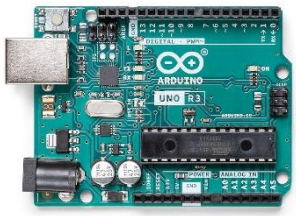
Content of Components

Component Number	Name	
1	Microcontroller	
2	Pneumatic cylinder	
3	Sensors	Temperature
		Pressure
		Load Cell
4	LCD Screen	
5	Solenoid	
6	Regulator	
7	Buttons	

Note: Justifications were completed following the completion of the chosen design option in order to determine how well each component could be integrated into the Model DRS.

Microcontroller

Component	Advantage	Disadvantage	Score (suitability)
 Arduino Uno Rev 3	- Uses familiar coding language - Very accessible, in terms of resources and compatibility - Has an adequate number of pins	- Reduced memory capacity - Can only read one script at a time - Isn't optimised for performance	8/10
 Raspberry pi 3, Model a+	- All 17 GPIO pins are digital - Can read multiple scrips at a time	- Not fully developed community surrounding this microcontroller - Small database of information and example code - Requires an external storage method - Extremely low stock	6/10
 Raspbery pi pico	- Smallest Raspberry pi - Inexpensive - Resources for beginners	- Power limitations - Processing limitations - Hardware and software limitations	4/10

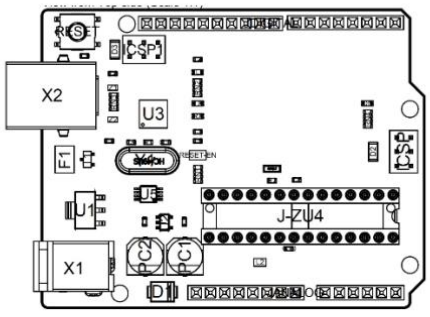
Arduino Uno Rev3	Overview
	The Arduino Uno Rev 3 microcontroller board based on the ATmega328P (ATmega328P Datasheet). The features of the AVR 8-bit microcontroller are; - High performance, low power - 23 programmable I/O lines - Operating voltage of 2.7V to 5.5V - Temperature range of –40°C to 125°C - Speed grade; 0 to 8MHz at 2.7 to 5.5V 0 to 16 MHz at 4.5 to 5.5V - Low power consumption; Active mode: 1.5mA, 3V -4MHz Power-down mode: 1µA at 3V The features of the microcontroller boards are (Arduino Uno Rev3 Datasheet); - 5V operating voltage - 7-12V input voltage - 6-20 input voltage (limit) - 14 digital I/O pins total - 6 PWM output pins - 6 Analog input pins - 20mA of DC current per I/O pin - 50mA of DC current per 3.3V pin - 32 KB of flash memory Dimensions; - 68.6mm length - 53.4 mm width - 25 gram weight (Arduino Uno Rev3, 2023)

Specifications:

For my Model DRS a control board is needed in order to organise and moderate the various electronic inputs and outputs. The Arduino Uno Rev 3 will receive inputs from sensors, LCD screen and any control panel buttons in order to activate the time loop for DRS activation and deactivation.

3.1 Board Topology

Top view



Board topology

Ref.	Description	Ref.	Description
X1	Power jack 2.1x5.5mm	U1	SPX1117M3-L-5 Regulator
X2	USB B Connector	U3	ATMEGA16U2 Module
PC1	EEE-1EA470WP 25V SMD Capacitor	U5	LMV358LIST-A.9 IC
PC2	EEE-1EA470WP 25V SMD Capacitor	F1	Chip Capacitor, High Density
D1	CGRA4007-G Rectifier	ICSP	Pin header connector (through hole 6)
J-ZU4	ATMEGA328P Module	ICSP1	Pin header connector (through hole 6)
Y1	ECS-160-20-4X-DU Oscillator		

Functionality (How it works):

This Arduino board functions with a 5V conventional operating voltage input, receiving input through the 14 I/O pins. The coding language for the Arduino is C++, which the Arduino database provides many prewritten code examples. With the code in place the Arduino will convert the readable inputs into outputs, in the case of the Model DRS the main output is the activation of the pneumatic cylinder.

Explanation (Pro vs Cons):

Pros;

- C++ is a very accessible and familiar coding language, it has a large variety of resources and databases of information.
- Is compatible with familiar programming tools
- Has a sufficient number of pins and functions for my project

Cons;

- Can only read one script at a time.
- Reduced memory capacity.
- Not optimised for performance, even with it being accessible and usable for beginners. This comes with the drawback of reducing the boards overall power ability.

Justification:

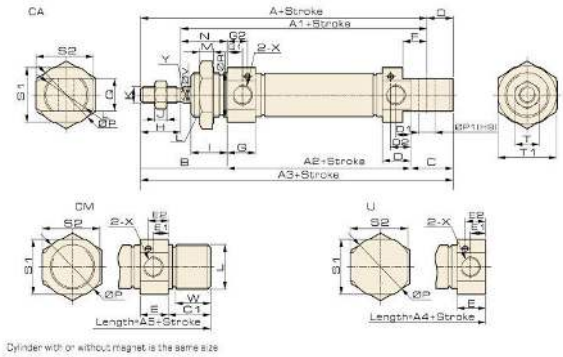
Following the completion of the design of the DRS Model bodywork, but more importantly for this component the circuit diagram. It has been determined that the Arduino Uno Rev3 is the most suitable choice for microcontroller. The board has enough pins to control all components as well as being highly compatible with all the chosen components. The 5V power supply is highly compatible with the components and fits within all their safety and functionality limits. The size of the Uno Rev3 also makes it easy to fit within a systems control box as well as accounting for organisation of the wiring to all components.

Pneumatic Cylinder

Component	Advantage	Disadvantage	Score
MCQI Double Acting ISO Cylinder	- Durable - Wider range of products	- Bore sizing isn't compatible with the size of this project - Very expensive - Not compact	6/10
DSN Double Acting Mini Cylinder	- As the cylinder is double acting there is lots of control over the movement of cylinder. - Fast - Durable - Can be purchased in a variation of stoke and bore sizes	- More expensive than a single acting cylinder - Large in size will require sizable housing	9/10
ISO 21287 Compact Double Acting Cylinder	- Space saving - Requires little maintenance	- Very expensive - Not adjustable - Not suitable for project as doesn't have a single cylinder rod extending. (ISO 21287 Compact Double Acting Cylinder 100-400mm - Magnetic - Non-Rotating, 2023)	2/10

DSN Double Acting Mini Cylinder	Overview
	The double acting mini cylinder from Pneumatic Solutions Australia. The cylinder is double acting as it has two separate ports to extend and retract the cylinder rod. The information on the cylinder is; - Actuation: Double acting - Operating pressure: 14 – 150 PSI - Proof pressure: 217 PSI - Medium: Air (DSN Double Acting Mini Cylinder, 2023) This pneumatic cylinder is available in a wide variety of bore sizes and stroke sizes, which proportionately change the dimensions of the entire cylinder. The bore sizes range from 16 - 32mm, and the stroke sizes range from 25– 150mm. This large versatility in sizing means that a cylinder can be chosen once the designs are finalised, choosing one that is the most efficient and compatible with the design dimensions and shapes.

DSN



Dimension

Bore Symbol	A	A1	A2	A3	A4	A5	B	C	C1	D	D1	D2	E	E1	E2	F	G	G1	G2	H	I
16	98	82	55	111	92	111	37	19	19	10.5	6	7	10.5	6	7	9	10.5	6	7	16	16
20	115	95	64	128	107	128	43	21	21	14	7.5	10	14	7.5	10	9	14	7.5	10	20	18
25	126	104	66	137	116	137	50	21	21	15	8	10.5	15	8	10.5	12	15	8	10.5	22	22
32	135	113	72	147	120	132	48	27	12	17	8.5	12	17	8.5	12	15	17	8.5	12	22	18
40	138	115	72	150	123	136	51	27	14	16	8	11.5	16	8	11.5	15	16	8	11.5	23	20

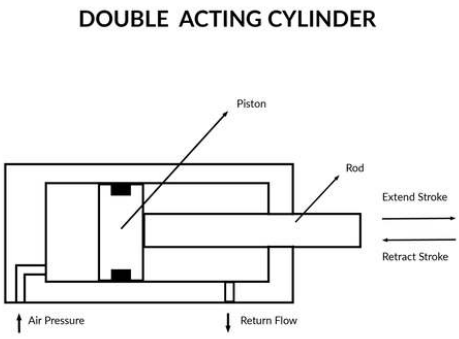
Bore Symbol	J	K	L	M	N	O	P	P1	Q	R	S1	S2	T	T1	X	V	W	Y
16	5	M6×1	M16×1.5	6	21	13	21	6	12	16	19	19	10	24	M5×0.8	6	15.5	5
20	6	M8×1.25	M22×1.5	7	23	13	30	8	16	22	27	28.5	12	29	G1/8	8	17.5	6
25	6	M10×1.25	M22×1.5	7	28	11	30	8	16	22	27	28.5	17	29	G1/8	10	17.5	8
32	6	M10×1.25	M27×2	9	26	12	38	10	20	27	35	36.5	17	36	G1/8	12	8	10
40	8	M14×1.5	M33×2	9	28	12	45	10	20	33	42	43.5	22	45	G1/4	16	10	14

Specifications:

As seen in the diagram and table above, the specifications surrounding the dimensions are dependant of the chosen stroke and bore sizes. The stroke size will be dependent on the rear wing framework and the distance from the cylinder's connection on the rear wing flap to the mounting of the cylinder.

Functionality (How it works):

The cylinder has two ports, one to allow air in to create pressure in the cylinder and extend the rod and another to send air through the other end to retract the rod. The air enters the inner chamber of the cylinder and creates pressure which pushes the piston forward extending the rod. When the rod needs to be retracted airflow into the extension chamber stops, and air is supplied to the other side of the piston. The same principals are used to push the piston back.



For the Model DRS this will be utilised to open and close the rear wing flap, activating and deactivating DRS in a sequence determined by the coding of time loops.

Explanation (Pro vs Cons):

Pros;

- Large amount of control over movement of cylinder rod as pressurised air moves both ways through the cylinder
- Due to the cylinder being double acting it is faster and more durable
- There is an extensive variety of stroke and bore sizes that can be chosen to suit the project

Cons;

- Double acting pneumatic cylinder are more expensive than single acting ones
- The component is large as it requires the sufficient housing for all its parts

Justification:

Following the completion of the calculations for the actuator stroke and bore sizing it can be concluded that this specific product has a selection with the correct dimensions. The actuator will work in conjunction with a fixed size connector to activate and de-activate the DRS by increasing and decreasing the gap size between the two rear wing flaps. The correct stroke size is the main factor for which the dimensions need to be selected around is: The correct stroke size is 25mm, this has a corresponding bore size of 16mm.

This actuator is the perfect choice not only due to its correct dimensions but also for its compatibility with the other pneumatic components. The actuator is durable and fast which also appeals to specific constraints outlined in the evaluation criteria. The final reason to justify the selection of this specific product is its double acting feature. As the cylinder is double acting it allows air in at both ends making it able to be controlled when extending and retracting the rod. Which is required within the context of the project to be able to activate and de-activate the DRS.

Sensors

Component A	Advantage	Disadvantage	Score
 LM335Z Temperature Sensor Linear IC	- Inexpensive - Easy to code - Consumes very little energy	- Non-linear, requires extra calculations - Only measures temperature - Very little support in the Arduino IDE - Calibrated in kelvin	6/10
 DHT11 Temperature and Humidity Sensor	- Very low cost - Very accurate readings - Fast sensor readings - Easily integrated with Arduino	- Not the most durable	9/10
 Thermistor	- High sensitivity - Option for customisation - Easy to use - Accurate	- Only measures temperature - Poor interchangeability - Extremely nonlinear - Not suitable for large temperature range - Susceptible to self-heating errors	7/10

Component B	Advantage	Disadvantage	Score
Gravity BMP388 Barometric Pressure Sensor	- Compatible with Arduino - Inexpensive - Compact - Low power consumption	- Not durable - Sensitive to outside changes	8/10
PiicoDev Pressure Sensor	- Accurate - Low power consumption - Inexpensive	- Not very compatible with Arduino - Sensitive to changes not relevant to measurements	7/10
Thin-Film Pressure Sensor	- Very small and easy to incorporate into design - Compatible with Arduino - Inexpensive	- Does not measure a wide scope of pressures - Fragile	7/10

Component C	Advantage	Disadvantage	Score
TE Connectivity Load Cell	- Compact size - Fast data gathering - Low noise - Durable	- Expensive - Weight range is outside the scope of this project - Designed for industry use	7/10

 Siemens SIWAREX WL Series Single Point Load Cell	- Durable - Accurate	- Way out of price range - Designed for industry use	6/10
 20Kg Beam Load Cell	- Economical - Precise with measurements - Fast feedback on measurements	- Complex integration - Can be damaged from overloading	8/10

		Overview
	A	The DHT11 temperature and relative humidity sensor, is a digital sensor that measures both temperature and humidity. The sensor is easy to use with the Arduino micro controller as it contains a digital chip that does analogue to digital conversion which creates a digital signal with the temperature and humidity. The information for the sensor is; - Communication protocol: One wire - Power supply range: 3 to 5.5 V - Temperature range: 0 to 55 degrees Celsius - Humidity range: 20 to 90% - Sampling period: 1 second (9 Arduino Compatible Temperature Sensors for Your Electronics Projects, 2023) This particular sensor has a large database of information online including; coding libraries, example codes and set up guides.
	B	The Gravity BMP388 Barometric pressure sensor is equipped with the functionality for temperature and pressure measurement. The newer version of the sensor supports Arduino code control. The sensor also has the added feature of measuring altitude and relative floor height, this can be achieved due to the connection between altitude and barometric pressure. The BMP388 enables accurate 3D indoor positioning and navigation through GPS tracking. The information for the sensor is;

		- Operating voltage: 3.3-5.5V - Operating current: 0.5mA - Operating range: 300-1250 hPa - Operating temperature: -40-80 degrees Celsius - Relative accuracy: +_ 8 Pa (Gravity: BMP388 Barometric Pressure Sensors, 2023)
	C	A load cell is used to measure force or weight, accomplished by being an electromechanical sensor. Load cells are a type of transducer which covert mechanical force into a measurable electrical output. Core electronics has Beam Load Cells for various weights, the one that has been chosen to look into is the 20kg Beam Load Cell which can account for the expected weight of the Model DRS. The information for the load cell is; - Comprehensive error: 0.05% F.S - Output sensitivity: 1.0 +_ 0.1 mV / V - Operating temperature: -10-50 degrees Celsius - Recommended excitation voltage: 5-10V - Input impedance: 1000 +_ 50 Ohms - Output impedance: 1000 +_ 50 Ohms (20kg Beam Load Cell, 2023) This load cell is rated for 20Kg, the highest accuracy is at 10Kg (half the rated force).

A: DHT11 Temperature and Relative Humidity Sensor

Specifications:

The temperature and humidity sensor is part of the collection of sensors that will be recording data for the DRS Model. Even though the model will be stationary the sensor is intended to simulate the data Formula 1 teams receive and analysis from their car during a race. The sensor has three pins, (from left to right) signal, VCC, GND. The sensor is very compatible with the Arduino framework.

Functionality (How it works):

The sensor uses a capacitive humidity sensor and a thermistor to measure the surrounding air, relaying the digital signal on the data pin. The sensor will be located on a specifically chosen point on the Model DRS as to ascertain the most relevant information and to minimise its impact on the airflow over and around the wing.

Explanation (Pro vs Cons):

Pros;

- Very low cost
- High accuracy for humidity readings (5%)
- High accuracy for temperature (+_ 2 degrees Celsius)
- Can gain sensor readings every second
- Easy to integrate with Arduino

Cons;

- Made of plastic so isn't the most durable

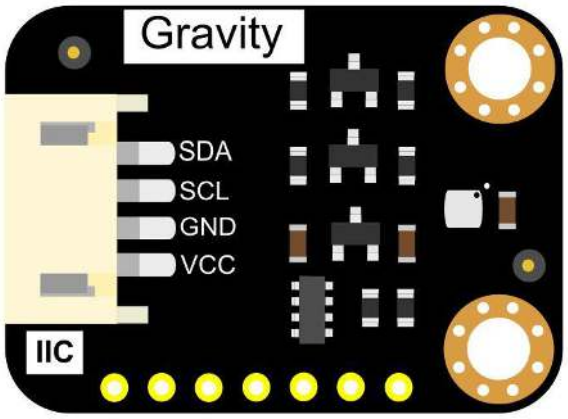
Justification:

This component fits in well with the physical design of the Model DRS as well as the plan for the circuit wiring. Not only does the sensor work incredibly well with the Arduino framework but it will easily be implemented and coded to fit its purpose. The power supply range is within the Arduino's capabilities and the sensor will be attached to the microcontroller in an orderly method. The sensor's size makes it an excellent choice to fit to the exterior of the rear wing body, as it will not interrupt airflow.

B: Gravity BMP388 Barometric Pressure Sensor

Specifications:

The Gravity BMP388 Barometric Pressure Sensor will be used to measure pressure, location and can be used to cross check the temperature readings from the temperature and humidity sensor. The ability for accurate GPS tracking creates the possibility for more advanced functionality of the sensor, this will be outlined in the 'Functionality' section.



Functionality (How it works):

The sensor records pressure through the utilisation of an aneroid cell that expands and contracts when the atmospheric pressure changes (Barometric Sensors 101, 2023). This sensor will be placed on the Model DRS to record data in regards to the atmospheric pressure around the Model. The added bonus of this sensor is its ability to be utilised for elevation detection as well as vertical velocity indication. This means that the sensor can be placed on the moving rear wing flap to determine the change in position, this can help to gain data on the distance the flap moves but can also alert to any failures with the activation/deactivation of DRS.

Explanation (Pro vs Cons):

Pros;

- Is compatible with Arduino framework, with large amount of example code being available
- Is inexpensive
- Has low power consumption
- Is small and easy to fit to the project

Cons;

- Is sensitive and easily influenced by outside activity
- Is not durable

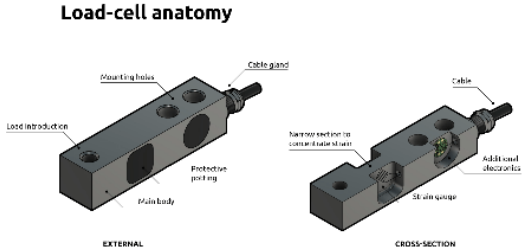
Justification:

This sensor is well within the power supply and current supply capabilities of the chosen microcontroller of the Arduino Uno Rev3. The size and shape of the sensor simplify the process of fitting it to the physical body of the Model DRS, especially the mounting points which aid in the organisation of all componentry. The additional capabilities of the Gravity BMP388 having 3D indoor positioning and GPS tracking, adds a layer of complexity to the information from the sensor which more closely simulates the data Formula 1 teams received from their cars. All these factors justify the selection of this specific sensor.

C: 20Kg Beam Load Cell

Specifications:

The Load cell is used to weigh objects by converting mechanical force into a measurable electrical output. The Load cell will be utilised in determining how well the Model DRS is doing at decreasing downforce.



Functionality (How it works):

The Load cell will be fixed to the Model DRS with the intension of determining how well the model is doing at decreasing downforce. If the Model is succeeding at decreasing downforce when DRS is activated there will be a difference in weight between the activated and deactivated weight (when sprayed with air over the rear wing). The Load cell will be able to ascertain the data regarding the weight which then can be utilised to make improvements to minimise downforce when DRS is activated.

Explanation (Pro vs Cons):

Pros;

- More economical than other sensors
- Precise measurements
- Quick feedback on measurements

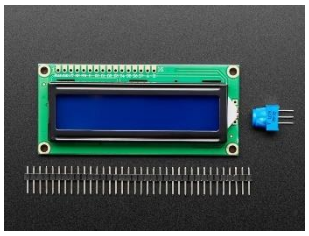
Cons;

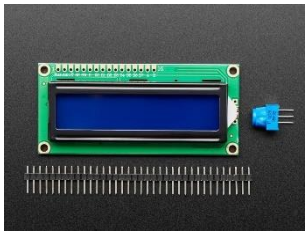
- Complex it integrate into Arduino framework
- Can be damaged from overloading

Justification:

While the additional component of the HX711 Amplifier is required to properly work the Load cell, it is still the most optimal option for determining the change in downforce from the Model DRS. Due to the output from the Load cell being so minute and specific, to enhance the signal for the Arduino to read it the amplifier is required. This is an inexpensive and easy to implement solution that allows the usage of the 20Kg load cell to be justifiable. The wiring and coding of the Load cell is not a straightforward process especially with calibrating the weight. However, there is an excess of resources online to aid with this process. As the decision was made early on to scale the rear wing down to half of the size of an official F1 wing, the 20kg rating is within the estimated weight of the bodywork of the rear wing. All these reasons aid in the justification of this specific component and its validity in the confines of this project.

LCD Screen

Component	Advantage	Disadvantage	Score
 Standard LCD 16x2	- Low power consumption - Relatively easy to program, with lots of resources - Inexpensive	- Limited amount of screen space - Susceptible to weak/stuck pixels - Uneven backlighting	7/10
FeatherWing OLED – 128x64	- High contrast OLED display - 3 mini buttons for a small interface - Only uses 2 I2c pins - Self-admitting light source, meaning higher efficiency	- Requires the Adafruit Feather boards - Complex coding - Not very adaptable to Arduino	5/10
OLED Breakout Board – 16bit colour 1.5”	- Colourful interface - Self-admitting light source, meaning higher efficiency - Colour accuracy	- More expensive - The colourful aspect while more interactive is more complex to code and only acts to enhance aesthetics - Much more complexity for coding	5/10

Standard HD44780 LCD	Overview
	This display screen is the 16x2 standard LCD Screen. This screen is a cost effective and time efficient method of displaying information gained from the sensors. The information of the LCD screen is; - PCB Dimensions: 36mm x 80.6mm x 1mm - Screen Dimensions: 24mm x 69mm - Height (PCB+Screen): 8.4mm - Power Supply Voltage: 5V - Number of Pins to Control: 6 This choice for screen will save time with implementing it into the system as it is familiar, easy to use and has a wide range of sources on how to best code the device. The screen is very compatible with the Arduino framework and fits within the confines of this project.

Specifications:

The LCD screen is a 'Liquid Crystal Display' which is essentially the use of liquid crystals to rotate polarised light (which are backlit), to produce a display. The flat-panel display reads an input from a sensor filtered through a micro controller and displays the received data.

Pin No.	Symbol	Level	Description
1	VSS	0V	Ground.
2	VDD	+5.0V	Power supply for logic operating.
3	V0	--	Adjusting supply voltage for LCD driving.
4	RS	H/L	A signal for selecting registers: 1: Data Register (for read and write) 0: Instruction Register (for write), Busy flag-Address Counter (for read).
5	R/W	H/L	R/W = "H": Read mode. R/W = "L": Write mode.
6	E	H/L	An enable signal for writing or reading data.
7	DB0	H/L	This is an 8-bit bi-directional data bus.
8	DB1	H/L	
9	DB2	H/L	
10	DB3	H/L	
11	DB4	H/L	
12	DB5	H/L	
13	DB6	H/L	
14	DB7	H/L	
15	LED+	+5.0V	Power supply for backlight.
16	LED-	0V	The backlight ground.

Functionality (How it works):

Within the context of the Model DRS the LCD screen will be used to display the data from the sensors that are placed on the rear wing surface. The sensors will be wired to the microcontroller which will then 'pass on' the data to the LCD screen to be displayed on the systems control case. This is intended to simulate the feedback race engineers will be getting from the many sensors on the cars but presented in a more informative and accessible method.

Explanation (Pro vs Cons):

Pros;

- Easily compatible with Arduino framework
- Large volume of sources to aid the usage of this component

- Cost effective
- Low power consumption aids in its compatibility with the other components.
- Sized well to fit within housing of systems control

Cons;

- Limited amount of screen space results in rolling text, which is more difficult to read
- Backlighting can present the issue of uneven light production
- Can have potential issues with weak pixels

Justification:

The standard HD44780 LCD is the best choice for displaying the data gained from the sensors on the Model DRS. The LCD screen's size makes it cohesively fit in the systems control housing, as well as properly display the data. This specific LCD in conjunction with the chosen I2C is highly compatible with the Arduino framework, making coding and assembling this component a time efficient and simplistic process. The power consumption and requirements fit well within the Arduino' capabilities. All these specific factors aid in simplifying the associated processes with this component, justifying its selection.

Solenoid

Component	Advantage	Disadvantage	Score
4V210 5/2 solenoid valve	- Efficient response to inputs - Fits within the scope of this project - Compatible with a large range of other components - Suitable for temperature ranges	- Sensitive to voltage fluctuations - Coil needs replacing with excessive use - Can have valve errors from incorrect set up	8/10
RS PRO 5/2 Pneumatic Solenoid Valve	- Is freestanding, mounting style - Durable - Fits the project's requirements	- Exceeds budget - Valve fluid can be affected by the flow	7/10
JELPC 5/2 Push/Pull Valve (solenoid alternative)	- Less margin for error - Simplifies processes	- Lacks the option for autonomy through coding - Does not fit with the constraints of this project - Is less efficient compared to alternatives	5/10

5/2 Solenoid Valve	Overview
	This solenoid is the 4V210 5/2 solenoid valve from Pneumatic Solutions Australia. This solenoid comes with either 1/8 BSP or 1/4 BSP, and has various voltage options ranging from 24VAC to 220VAC or 12VDC to 24VDC. The information on the solenoid is; - Actuation: Single solenoid/ spring return - Operating pressure: 14 – 150 PSI - Proof pressure: 175 PSI - Temperature: 5 to 50 degrees Celsius - Medium: Air - Function: 5 port, 2 position - Switching frequency: 5 cycles per second (4V210 5/2 Solenoid Valve, 2023) As the solenoid is 5/2 way, it has one input, two outputs, and two exhaust ports. This single coil type of pneumatic valve has a spring or pilot air return, meaning that when the component is de-energised it returns to its starting function. (5/2 Way Single Solenoid Valves, 2023)

Specifications:

For the Model DRS the 5/2 Solenoid valve will be connected to a system running with 1/8 or 1/4 BSP, due to this the Solenoid also has to fit to this sizing. Although the higher voltage solenoid could offer better performance and faster activation of the DRS, the 24VDC will work in combination with the microcontroller board as well as other electronic components. This voltage is also safe and is well within the voltage limits.

Functionality (How it works):

The purpose of the solenoid is to control and alter the flow of air to the pneumatic cylinder, in order to extend and retract the cylinder rod. With the context of the Model DRS this will activate and deactivate the DRS. The solenoid has one input for the air to enter from the air compressor, two outputs supplying air to the pneumatic cylinder and two exhaust ports that will be covered by mufflers. The solenoid basically works as a pneumatic switch, the two outputs have the purpose of supplying air to the pneumatic cylinder, through the cylinder ports. To extend and retract the rod.

Explanation (Pro vs Cons):

Pros;

- Fast response time to inputs (electrical and pneumatic)
- Is programable to fit within the requirements of this project

- Can be functional in conjunction with a variety of components
- Is suitable at a wide range of temperatures

Cons;

- Sensitive to voltage fluctuations
- Coil may need replacing if overused
- If magnetic field isn't correctly set up there can be errors in valve operations (Advantages and Disadvantages to Solenoid Valves, 2023)

Justification:

This specific solenoid was chosen for a variety of reasons, all of which are related to its compatibility with the system as well as its ease of use. This solenoid is extremely compatible with the pneumatic cylinder, they both have the same manufacturer which aids in their cohesion as a subsection of the larger system. The 24VDC works well with the Arduino but did require the inclusion of a 5V Relay. While this was an extra component that needed to be accounted for, not only is it essential for the solenoid's functionality but also aids in the organisation of wires and overall safety. The flexibility of the solenoid's specifics, for example the voltage options and pipe sizing, was a major factor in its selection. Being able to adjust these aspects to fit with the circuit design and the physical rear wing body design., simplified the design process and creates a more fluid system. These reasons justify the selection of this solenoid.

Regulator

Component	Advantage	Disadvantage	Score
 Blackridge Air regulator	- Available locally for easy purchasing - Includes auto drain for ease of use	- Expensive - Only available in 1/4” BSP	7/10
 Ozito 1/4” Mini Filter Compressor Regulator	- Includes locking ring feature to limit accidental regulator adjustments - Reasonably priced	- Only available in 1/4” BSP - Many negative customer reviews regarding easily damaged dial window	7/10
 Airline Regulator (Process Systems)	- Is available in a wide range of port sizes with various flow rate limits - Range of size from 1/4” to 2” BSP allows this product to be very compatible with whatever choices are made with the design - Durable, and reliable - Reasonably priced	- Heavier compared to other products	10/10

Airline Regulator (Process Systems)	Overview
	This airline regulator is from Process Solutions which is an Australian company, this regulator comes in various sizes making it compatible with decisions made in the design stage of the system. The information on the regulator is; - Size: 1/4" to 2” BSP - Body: Die cast Aluminium - Pressure range: 0.5 to 8.5 BAR - Flow rate: 550 to 8000 L/min - Medium temperature: 0 to 60 degrees (Airline Regulator, 2023)

Specifications:

For the pneumatic system to work for the Model DRS the air being fed into the system needs to be regulated in order for it to function safely and efficiently. The data sheet for the airline regulator gives a large overview on the specifics of the regulator and the various changes that different variations have.

Specification	REG-02	REG-03	REG-04	REG-05	REG-06	REG-08	REG-09
Port Size (BSPP)	1/4"	3/8"	1/2"	3/4"	1"	1 ½"	2"
Flow Rate L/min	550	2500	6000	8000	8000		
Pressure Range	0.5 BAR to 8.5 BAR					0.5-16 BAR	
Max Inlet Pressure	10 BAR					16 BAR	
Medium Temperature	0 to 60°C						
Relief Pressure	Set Pressure + 0.5 BAR (relief flow rate of 0.1 L/min)						
Construction	Relief Type, Die Cast Aluminium						
Gauge Port Size	1/8"		1/4"			1/4"	

Functionality (How it works):

In principal air pressure regualtors reduce the supply of air (inlet) pressure to a lower output pressure, they work to maintain a constant outlet pressure even when experiencing fluxtuations in inlet pressure (The Basics of Pressure Regulators, 2023). Within the specifics of this system the air regulator will be providing a constant output pressure to the solenoid when gaining input from a source of air, most likely a compressed air tank of some form.

This aspect of the system in vital as it ensures that no components are damaged by taking on higher pressure air than built for, it also works to keep a consistant flow of air. This adds reliability to the system and works towards guaranteeing a high quality system.

Explanation (Pro vs Cons):

Pros;

- Extensive compatibility with decisions being made regarding the system based on the design. For example, the size of pipe being used. As this regulator has a variety of different variations this ensures its accessibility for the system.

- The product is from an Australian business resulting in better reliability with shipping and reduced costs.
- The product in itself is well priced, benefiting the budget constraints of the project.
- The materials used, while having the minute downside of being heavier than other products make this regulator durable and long lasting.

Cons;

- Heavier and slightly more difficult to manoeuvrer compared to other products.

Justification:

A major factor in the selection of this airline regular was its interchangeability and flexibility with its specifics. This allowed the design of the circuit for the pneumatics to be created independent of the regulator. For example, the pipe sizing could be determined with the priority being the solenoid and pneumatic cylinder instead of needing to account for the regulator. This simplified the process and aids in creating a cohesive system that works together efficiently. With such a complex system with various subsystems, having a simple and reliable airline regulator relives some pressure off the project and allows the more complex sections to be prioritised. This is the major points in the justification of this component.

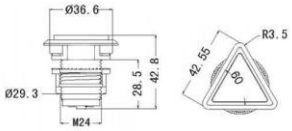
Buttons

Components	Advantage	Disadvantage	Score
 LED Illuminated Triangle Push Button	- Visually appealing, and is similar in aesthetic to other variation of buttons - Comes in a range of colours	- More expensive than alternatives	9/10
 Mini LED Arcade Button	- Lower in price in comparison to alternative choices - Comes in a range of colours	- While on its own is visually appealing does not work in conjunction with other buttons	8/10
 LED Illuminated Pushbutton	- The black body plastic makes this button work visually in conjunction with other buttons	- More expensive than alternatives - Have to change the colour manually, by changing out the LED	8/10

LED Illuminated Push Button Triangle	Overview
	The Triangle LED Illuminated Push Button, inspired by the aesthetics of 1970's Sci-Fi control buttons. This button comes in white, blue, red, green, and yellow. The information on the button is; - Forward voltage: 2.2V - Two connections for the button - Two contacts for the LED (one +, one -) (LED Illuminated Triangle Pushbutton A.K.A 1960s Sci-Fi Buttons - White, 2023)

Specifications:

For the Model DRS to function it requires a control board to activate, deactivate and select the settings for the system. The controls will need buttons in order to facilitate these settings, while other minute electronics including smaller buttons will be incorporated into the design later in the process. The large more significant buttons need to be analysed now. This button appropriately fits within the purpose of the project to be an education aid, as it draws attention to the controls of the system, guiding people through the control process.



Functionality (How it works):

The body of the button has a LED built inside, with there being two contacts for the button and two for the LED. The forward voltage of the LED is roughly 2.2V, so the LED would need to be connected to a resistor. The button is conneted to the microcontroller board, providing it with an input.

In the context of the Model DRS the button will be a part of the control board, connected to the Arduino and coded to control the DRS.

Explanation (Pro vs Cons):

Pros;

- Visually appealing
- Easy to implement into the control board and microcontroller system
- Wide variety of colours to represent various purposes of individual buttons

Cons;

- Expensive in comparison to other choices for buttons.
- Small additional complexity with needing resistors for the buttons

Justification:

The arcade buttons fit well within the design for the systems control case in terms aesthetics and dimensions. The aesthetic of the buttons aids the interactive aspect of the Model DRS as well as clearly labelling the activation and deactivation of the DRS. The colour and shape of the button are also simply appealing and fun. The shape and design of the button makes it easy to insert it into the systems control housing, as well as creating an easy way to fix it to the housing. The buttons are simplistic to wire and organise, creating a more efficient construction process. These points are the reasons for the justification of this component.

Materials (including construction)

Carbon Fibre 3D Printing

3D printing with carbon fibre nylon is intended to be implemented for the initial iterations of the Model DRS as well as additional housing for components. The purpose for using this material is to create a base model that has very similar properties to the final iterations. Carbon fibre nylon is praised for its high modulus of rigidity, improved chemical, and thermal resistance as well as its ease of printing. The recommendations for printer bed settings are as follows;

- Extruder temp: 240-270 degrees Celsius
- Bed temp: 80-110 degrees Celsius
- Nozzle specs: minimum 0.4mm diameter
- Layer height: minimum 0.25mm

The CarbonX PA6+CF Gen 3 from 3DxTech, outlines a set of technical and safety data:

TDS Rev 3.0

Technical Data Sheet: CarbonX™ Carbon Fiber Nylon (Gen3) 3D Printing Filament

Safety Data Sheet

Physical Properties	Standard	Unit	Typical Value
Density	ISO 1183	g/cc	1.17

Mechanical Properties	Standard	Unit	Typical Value
Tensile Strength, Break	ISO 527	MPa	63
Tensile Modulus	ISO 527	MPa	3800
Tensile Elongation, Break	ISO 527	%	3
Flexural Strength	ISO 178	MPa	84
Flexural Modulus	ISO 178	MPa	3750

Thermal Properties	Standard	Unit	Typical Value
Glass Transition Temperature (Tg)	DSC	°C	70
Deflection Temperature at 0.45 MPa (66psi)	ISO 75	°C	147

Electrical Property	Standard	Unit	Typical Value
Surface Resistance	ASTM D257	Ohm/sq	$>10^9$

Printed Specimen Conditions
Printer: Open Source FDM/FFF
Nozzle: 0.4mm
Layer Height: 0.25mm
Infill: 100%, +/- 45°
Extrusion Temp: 275°C
Bed Temp: 80°C
Specimen Orientation: XY Flat

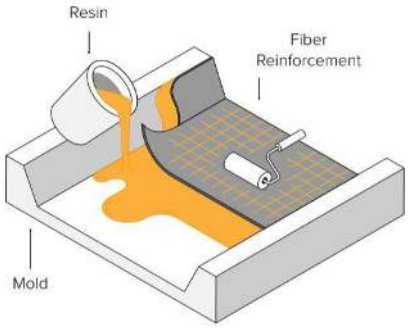
(CARBONX PA6+CF GEN 3, 2023)

Carbon Fibre/ Carbon Kevlar

There are three main processes to creating parts using carbon fibre; Wet lay-up, Prepreg Lamination and Resin Transfer Moulding. Even though the main properties of carbon fibre are attributed to the material itself, a large portion of its overall performance is a result of the laminating process. The way that the fibres are incorporated has a large impact on the parts properties. To achieve the best results the resins are shaped together with the reinforcement in the form of a tool or mould, then cured to form a strong product (How to Manufacture Carbon Fibre Parts, 2023). The laminating techniques are as follows;

Wet Lay-Up:

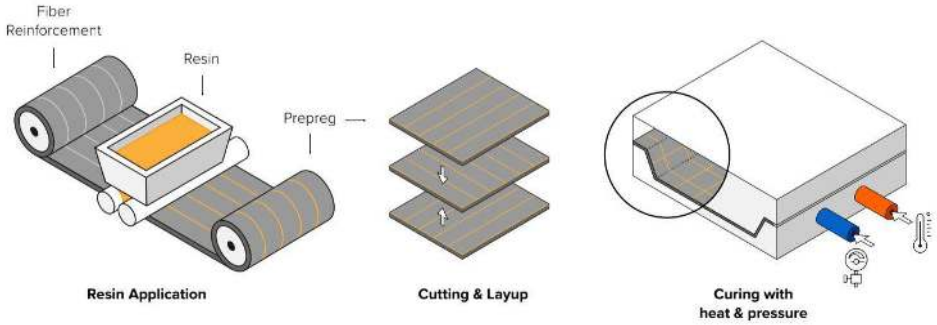
With this technique the fibre is cut and laid into the mold, before the resin is applied. This method is the most technically difficult in order to achive high quality parts, however it is also the least expensive and requires the least amount of specific tools.



In the video provided by Formlabs, they utilise 3D printed moulds and follow the wet lay-up technique. They take the first side of the mould and layer the carbon fibre over it before applying the resin and adding more carbon fibre. Once the carbon fibre sheets are layered over the first side of the mould, the second section of the mould is taken, and the insides coated with resin. This is then placed over the carbon fibre and pressed to secure the two moulds together. Any excess carbon fibre is cut away and the remaining carbon fibre is left to cure.

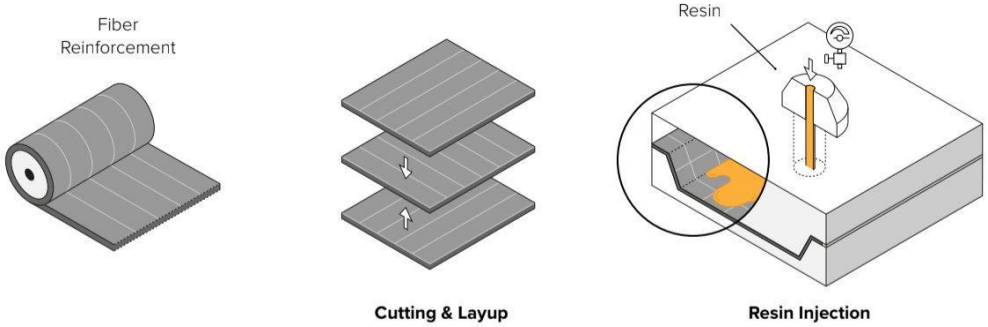
Prepreg Lamination:

With this technique already prepaired sheets of carbon fibre infused with resin are used. These sheets are carefully stored in the cold to stop them from curing. Pieces are curing in the mold under heat and pressure in an autoclave. While this is a very precise and repeatable process, it is also very expensive and reuires specialist machinery. This technique is typically only used for high-performance applications.



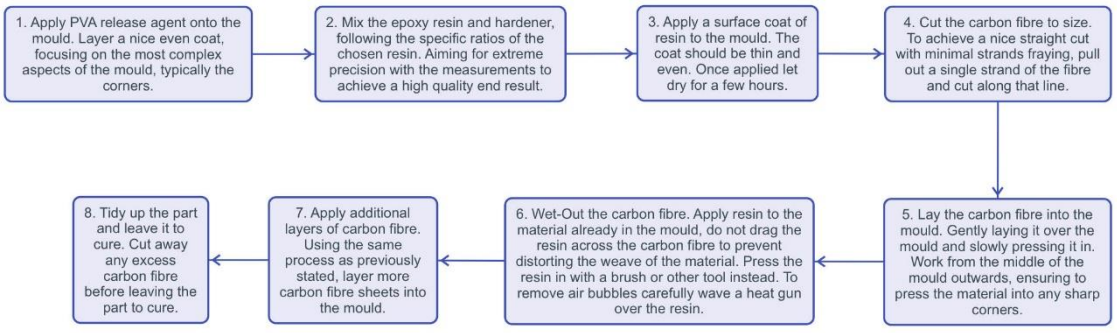
Resin Transfer Moulding:

With RTM moulding the set up includes dry fibre being laid in a two part mould. The mould is secured shut before resin is forced into the mould's cavity at a high pressure. This method is typically automated and is used for producing parts at a high volume.



Following Carbon Mods' video tutorial on creating carbon fibre parts has solidified my understanding of the process along with confirmed that the best method for the Model DRS will be Wet Lay-Up. While this method is more hands on and a longer process it has the potential to create really high-quality parts. It is also less expensive and doesn't require any specific machinery that make the other methods unviable options.

Below is a step by step guide I created whilst following the video tutorial;



TIMELINE

F1 REAR WING

1950s and 1960s

Alfa Romeo 158 Alfetta

In the early stages of Formula 1 the cars didn't have any wings. The cars were very streamlined and had a focus on straight line speed. Due to this wings weren't a major factor in the designs.

1968

Lotus 49

For the 1968 Monaco Grand Prix Colin Chapman's Lotus team were looking to achieve balance and higher downforce to maximise the teams outcome for the specifics of the circuit. The solution was to fit the car with small winglets to the front of the car and curved sheet metal to the rear.

1969

Lotus 49 B

The years following the introduction of rear wing concepts during the 1969 season, saw various developments in the rear wing design. Most notable were the high stilted rear wings run by most teams during 1969. The wings were mounted directly to the suspension, this resulted in the downforce generated by the wing being pushed directly into the rear tyres. This mounting location was ideal for these wings, however the wings durability and safety were majorly lacking. Dangerous crashes caused by failures in the rear wings resulted in an end to the high suspended rear wing era.

1970

Lotus 72

Following the end of the high wing era and with teams gaining a better understanding of the principals of downforce, came the era of wide low wings or 'lunch tray' wings. As teams started to see how the airflow behaved with these new wings, endplates were introduced to reduce drag by stopping high pressure air from being sucked over the edge of the wing and into the low pressure area below it. This made the wings more efficient. Teams also began to plat around with gurney flaps, to change the exit angle of the airflow

1977

Lotus 78

The Lotus team's rear wing developments were a game changer in the philosophy of the wing's purpose. The designers focused less on the downforce the wing produced and more on the ground effect of the cars in relation to the wing. The new design worked to push air under the car through a series of tunnels in order to create a larger low pressure zone under the car. This low pressure zone worked to suction the car to the ground, allowing the cars to go faster through corners.

1974

McLaren M23

The next big development in the design o the rear wings came as a result of trying to solve the issue of the mounting point of the rear wings. Previously the rear wings has been mounted over the gear box and that position wasn't good of aerodynamics. The solution was to raise the wing and push them back as well as to mount them on a single strut, that was away from the hot air and the body work. Mclaren's M23 also aimed to be as slim as possible, working to maximise its effectiveness. This was also the first time that the regulations started to put in place rules for the size of the wings.

1982

Ferrari 126C2 (US GP)

For the 1982 US Grand Prix Ferrari's F1 car was modified to attempt to work around the regulations regarding the rear wings. The rules had set dimensions for the width of the wing, which left a gap in the regulations for Ferrari to interpret as not disallowing the incorporation of two rear wings. The intension being that double the amount of wings would double the amount of down force. The double rear wing was only seen at the US Grand Prix as it was disqualified after Villeneuve placed 3rd.

1984

Toleman TG184

Throughout the early 80s there was an absence of clear, specific and volume of regulations surrounding the rear wings. This allowed for some of the most unique and technically ambitious rear wing designs in Formula 1. Some of the designs include Ferrari's 1983 car with small winglets on the outside of the wing, Tyrrell's 1983 boomerang shaped rear wing, and Toleman's 1984 double rear wing as shown below. The Toleman car danced around the regulations by having the second wing mounted infront of the rear tyres.

1992

Williams FW14

With a greater access to sophisticated technology and more regulations being implemented regarding the rear wings, the rear wings became more uniform and more complexly detailed. Regulations became more specific in terms of the space the rear wings could occupy and the size of that space. This resulted in rear wings becoming smaller but also prompted teams to add more elements within the space provided. This period saw the introduction of steep, two sectioned rear wings. The slot between the upper and lower rear wing flap worked to inject air from the high pressure area on top of the wing, into the airflow under the wing. Aiming to keep the airflow balanced and attached to the lower surface of the wing, allowing the steep wing to be efficient in its downforce production.

1996

Jordan 196 (Monaco)

As the regulations clamped down on the number of elements a rear wing was allowed to have (3), teams had to get creative in order to find other ways to produce more downforce. This was especially important for tracks that required the extra downforce to improve performance such as Monaco. The Jordan F1 team designed an extra rear wing attached to the engine cover, after concluding that there were no specific regulations that inhibited from using that space. Unfortunately the extra rear wing was not very effective.

1997

Tyrrell 025

Similarly to the 1996 Jordan F1 team, in 1997 the Tyrrell F1 team discovered that the regulations did not limit the use of the space to the side of the cockpit. The team utilised this space by adding high wings to either side of the driver, with the intension of increasing downforce over the centre of the car. While the Tyrrell F1 team had an unsuccessful year, they had produced a design that increased downforce by 5%. The high wings were later run towards the end of the 1997 season and throughout the 1998 season by Ferrari, Sauber, Jordan and Prost. However following the 1998 season the wings were banned for safety reasons as they claimed to have restricted the drivers view.

2004

F2004

During the early 2000s the size, shape and location of rear wings was dictated as a result of the regulations which intended on creating closer racing. Lots of focus was placed on teams designing wings that were shaped for the best performance without compromising the strength of the wing. The wing's angle was steeper with the intension of creating a deeper and steep inner section with a shallower outer section. This increased downforce while also reducing drag as vortices were not being created at the edges of the wing. The wider wings also meant there was lots of air hitting the centre of the wing, effectively pushing the car into the ground. These wing designs did however create lots of dirty air for the cars behind.

2009

McLaren MP4-23 vs MP4-24

2009 saw massive regulation changes for the rear wings, making the wings smaller, taller and much more thin. These regulation changes reduced the downforce created in order to reduce the amount of dirty air. Some major design changes as a result of the regulations can bee seen on the differences between the McLaren cars from 2008 and 2009. Most notably the rear wing sizes are vastly different but some more minute details can been seen. The MP4-24 has a rear wing flap that merges with the endplate as well as having slots at the top of the endplates to reduce drag.

2010

McLaren MP4-25 (F-Duct)

For the 2010 season Mclaren engineered a new mechanism called the F-duct. The system consisted of a number of pipework channels and was operated by the drivers on the straights in order to effectively stall the rear wing. The complex system worked through the drivers covering and uncovered a hole in the cockpit. When the hole was uncovered airflow taking in by the chassis snorkel would flow through to the cockpit, and airflow taken in at the airbox would flow through a fluidic switch and then out of the car. When the hole was covered the airflow from the chassis snorkel would bypass the cockpit and go straight through the signal pipework directly to the fluidic switch chamber. Once there it would divert the airflow from the airbox through the upper pipework to the rear wing. Once there the airflow would be fed through a slot in the rear face of the wing, interrupting the airflow on the wings suction surface which then produced the stall. Even though other teams quickly copied the design for the 2010 season, the 'f-duct' was banned for 2011.

2011

Redbull RB7

2011 was the year that DRS was introduced into Formula 1, the use of DRS was classified as an exception to the regulations barring teams from implementing any moving parts whose primary purpose is to impact aerodynamics. DRS aimed to create higher top speeds and increase overtaking, but the inclusion of the system required various changes to the designs of the rear wings. The major changes have to do with the durability of the wings along with the supports housing the associated componentry. While rear wings have always needed to be durable to withstand the forces acting on them, with the inclusion of DRS the moving parts of the system needed to be able to withstand the increased forces aswell. Rear wing pylons needed to be added to store the hydraulics for the DRS activation and deactivation, the pylons' impact on airflow also needed to be considered and designed around.

2013

Lotus E21

2013 saw more regulation changes in terms of both DRS and rear wing physical designs. The changes for DRS restricted the activation of it during practise and qualifying to the DRS zones, just like in a race. The major restriction for the physical design of the rear wing is in relation to the transport of air around the car. While seemingly not a major regulation change banning the use of rear wing endplates to transport air around the car, stunts a lot of airflow concepts and reduces the efficiency of airflow around the wing, which in turn reduces drag.

2022

Ferrari F1-75

2022 Formula 1 cars had an entirely different looking rear wing, seeming to have abandoned endplates altogether. The major change for the rear wing is within the purpose for its curved shape, the rear wing was to collect the wake from the rear wheels and rotate it into the airflow from the diffuser. This sends the dirty air up at an angle that should minimise the impact on the car. This intended to have the result of creating closer racing as the minimisation of dirty air should make it easier to follow the car in front. Statistically the 2021 cars only performed at roughly 53% of their peak performance when trailing the car behind, while the 2022 cars are closer to 83%.

2017

Mercedes AMG F1 W08

Another major regulation over hall saw the rear wings return to being low and wide. The height of the rear wing was reduced to a maximum of 800mm (From 950mm), this created similar looking rear wings the the wings prior to the 2009 regulation changes. This era of F1 rear wings also saw a trend in the shaping of the rear wing edges. The trailing edges are sharp and angular while the front edges are more curved to direct airflow.

Sources:

(When Formula 1 Used Aeroplane Wings, 2023)

(Explaining the New Look 2022 F1 Rear Wing, 2023)

(Seven of the Craziest Wing Designs in F1 History, 2023)

(F1 History Toleman Motorsports, 2023)

(Banned: The 2010 Formula 1 Season's F-duct, 2023)

(Banned: Why McLaren's F-duct was Outlawed, 2023)

(Formula 1's 2021 Regulations: What Will the Cars Look Like in 2022?, 2023)

(F1 2013: The Rule Changes Explained, 2023)

Examples of DRS Issues

Sauber C37

In 2018 at the Monza circuit the Sauber F1 team had many issues with the deactivation of their DRS. Marcus Ericsson had a big crash in FP2 when his rear wing's DRS flap did not reattach (close) with breaking into the first chicane. Ericsson's teammate Charles Leclerc also had similar issue with the rear wing flap not closing correctly when breaking for a corner but continued his practise session in order to run tests on the system for the team. It was discovered that the automatic close of DRS during breaking was the centre of the problem, as manually closing the flap by pressing a button on the steering wheel worked fine.

The team concluded that the issue was caused by a new rear wing being fitted to the cars for the low downforce needs of the Monza circuit, with the rear wing being incorrectly adjusted for the changed downforce. (Sauber Thinks it has Found DRS Fix After Ericsson's Monza Crash, 2023)

Take aways: The major thing to learn from this specific DRS issue is the importance of having a failsafe backup method of closing the rear wing flap, in order to deactivate DRS. Without this failsafe it's not only dangerous but it limits options for testing the rear wing's functionality in case of any issues.

Ferrari F138

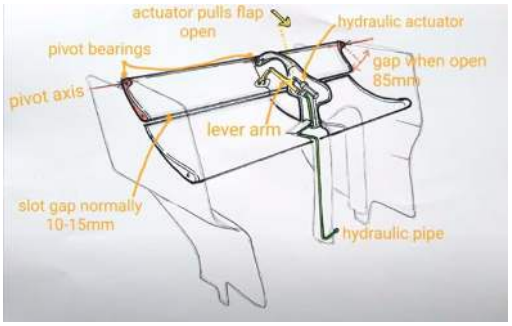
In 2013 Fernando Alonso experienced DRS failure during the Bahrain Grand Prix. On lap 5 of the race the car's DRS jammed open after it was activated to complete an overtake, the car was pitted to have the issue resolved but the fix was only temporary. A few laps later the DRS failed again, and the car was once again returned to the pits, but this time the rear wing flap was permanently close. Stopping Alonso from using DRS for the rest of the race. (Alonso's DRS Failure the First of its Kind - Ferrari, 2023)

Ferrari concluded that the problem had been caused by the breakage of a mechanical component within the system, but the issue raised many questions on the reliability of the car.

Take aways: This specific issue with DRS highlights the importance of reliability to achieve peak performance, as well as solidifying the need to complete in depth calculations to ensure the componentry can handle the forces acting upon it.

Red Bull RB18

At the 2021 Barcelona Grand Prix Max Verstappen's Red Bull car experienced some problems with DRS. The car had minor issues during Q3 with the DRS not opening during the final laps, but more majorly there were some issues during the race. The rear wing flap was acting somewhat unpredictable, DRS would sometimes open on the straights sometimes not. Even activating and deactivating DRS multiple times on one straight.

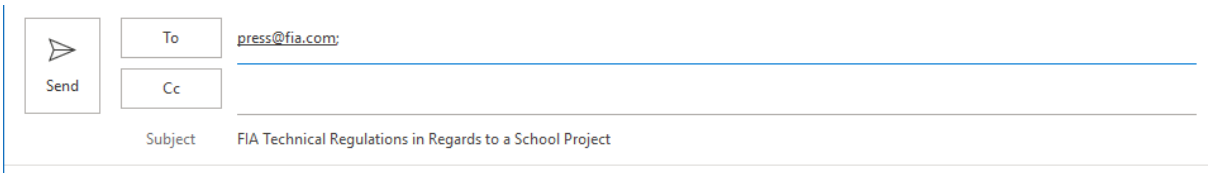


The assumption made in regard to the DRS issue was that it was not an actuator problem but instead an issue with the system binding at the pivot points. The Red Bull rear wing flap has extra pivot points to stop the wing from bowing when the aerodynamic forces are acting on it. Simply that there was so much binding occurring at the three pivot points on the trailing edge causing the flap to not open correctly.

The general understanding for why this issue randomly occurred is due to the weight saving that Red Bull had been changing the designs to achieve. With Red Bull commenting that they had lost 10kg of weight from their cars, the issue lies with the redesigns of the rear wings. (What Caused Max Verstappen's DRS Issues?, 2023)

Take aways: The main thing to learn from this DRS issue is the balance of maximising speed by losing weight from the car and compromising durability. While redesigning some components to minimise their weight is a perfectly logical solutions to increase performance it has the downside of challenging durability of the componentry.

Correspondence



Dear FIA member,

I am writing this email in regards to a school project I am completing, and am requesting assistance in order to produce the best outcome possible. For my Year 12 VCE Systems Engineering project I am designing, constructing, and testing a model Formula 1 rear wing with DRS. I have chosen to constrain my project within the regulations outlined in the 2023 Formula 1 Technical Regulations, to better simulate the engineering processes followed by the Formula 1 teams.

I have completed my initial stages of research and am in the midst of designing and digitally modelling the rear wing. Whilst attempting to attain the regulated dimensions for the rear wing and associated componentry I have run into problems with comprehending the dimensions of the rear wing profiles (3.10.1 of the regulations) and applying the information available to my designs.

Whist going through the regulations searching for any relevant information that could assist me in creating parameters in my CAD model for the rear wing, I came across a point in Appendix 1: Regulation Volumes. In 1.6 of said Appendix it states that;

“The CAD models of all the volumes and surfaces listed in this Appendix are available for Teams to download with a standard range of parameters. The FIA will provide a Team with a specific model on request.”

Is there a possibility of gaining access to these models with associated parameters, or any other useful resources? If you are unable to assist me could you please direct me to the suitable pathway to gain this information, any assistance would be greatly appreciated. I am also more than happy to send through the initial stages of my folio which contains the relevant information for my project if that would be helpful.

Thank you for taking the time to read my email.

Kind regards,
Isabella Handbury

Pre-Season Testing

Pre-season testing occurs before the commencement of the Formula 1 season, in 2023 it is held over three days. Each day has two testing sessions, one in the morning and one in the afternoon. Pre-season testing is the first chance for the teams to gain data on the performance of the car, with the main focus being comparing on track and off-track data.

Aero Rakes

Formula 1 aero rakes are a tool for registering and accessing the airflow pressure and direction around the car on track. They are constructed from special alloys including titanium and have multiple Kiel probes attached to the frame. The sensors send instant feedback to engineers whilst the cars are out on track. (What is an f1 areo rake, 2023)

The Kiel probes measure the flow of air over each data point, the probes are used instead of pilot tubes as they are less sensitive to changes in yaw angle.

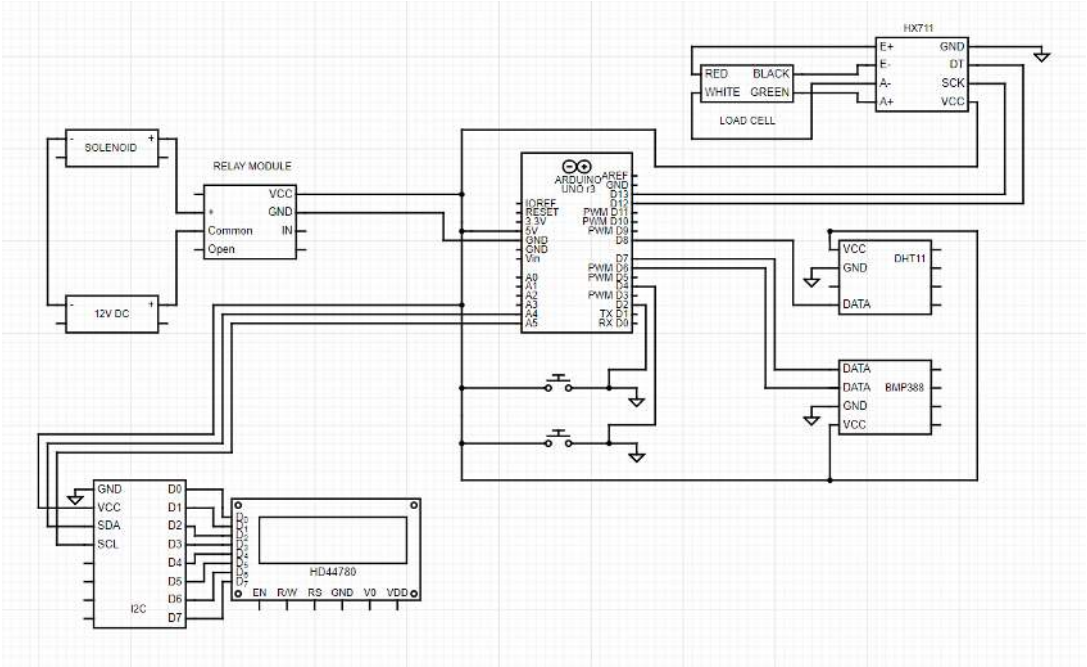
Flow-vis Paint

When investigating the aerodynamics and what the flow structures are doing on the car, F1 teams use flow visualisation paint. Flow-vis is a mixture of fluorescent powder and a light oil, typically paraffin. The mixture is painted onto the car before it goes out on track, as the car gets up to speed and completes a few testing laps the paint dries as the oil evaporates. This allows a visualisation of the flow structures. Analysing the Flow-vis can provide evidence of separation in the airflow, where there isn't a decent flow across the surface of the car. (Testing explained: Rob Smedley on flow-vis paint, 2023)

Aesthetics

Stickers on formula one cars play a large role in not only their economic income in terms of sponsors but also overall aesthetics and appearance. Even surfaces that are vital for airflow and aerodynamics are covered in stickers, this includes front and rear wings. F1 teams do their best to minimise the thickness of the decals to minimise their impact (Do stickers affect the performance of F1 cars?, 2023).

Circuit Diagram



Justification and Explanation

This circuit diagram includes all the electronics for the Model DRS, as well as the ports they need to be connected to. Both for the purpose of functionality but also in terms of organisation.

After further research in regard to all required componentry and their associated electronics three additions components have been added;

The HX711 Load Cell Amplifier is a small breakout board that was added to enhance the functionality of the load cell. With the output from the load cell being relatively difficult to read accurately the amplifier assists in reading the data from the load cell. The amplifier reads the changes in the resistance of the load cell and once calibrated can be very accurate with the weight measurements.

The HX711 utilises a two-wire interface for communication and a connection to any GPIO pin will work. There are also an abundance of libraries and sources for the amplifier making it a time effective solution. The Load cell will use a four wire configuration to connect to the amplifier, with the colour coding of the connections corresponding to the conventional load cell colour system.

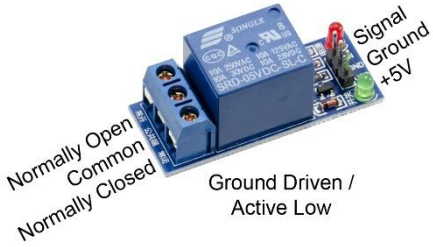
- Red = VCC
- Black = GND
- White = Signal/ Output
- Green = A-, S-/O-
- Yellow = Shield

Features;
Operation Current: <1.5mA
Operation Voltage: 2.7V-5V
Output Data Rate: 10SPS or 80SPS
(SparkFun Load Cell Amplifier Hx711, 2023)

The 5V Single Channel Relay Module 10A, is being implemented to aid the switching on and off of the solenoid due to the large current, essentially the relay acts as a switch. The Relay is required when using a low voltage micro controller such as the Arduino, when controlling components such as motors or solenoids. Within the context of the Model DRS the Relay will be used to help control the solenoid in order to properly work the pneumatic portion of the system.

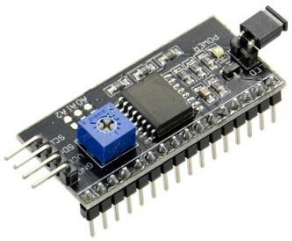
The Relay channel can drive up to 10A at 28VDC which is within the chosen voltage for the pneumatic system.

Features;
Supply Voltage and Input Signal: 5V (can function at 3.3V)
Relay Current: 10A
Max Switching Voltage: 250VAC
Max Switching Current: 15A
Max Switching Power: 3750VA
(5V Single Channel Relay Module 10A, 2023)



The I2C LCD Backpack for 1602 to 2004 LCDs is a connector that reduces the amount of data pins needed to control an LCD screen down to 2, with the exclusion of power and ground pins. This component is being implemented to save pins on the Arduino microcontroller but also to simplify and create better organisation with the wiring. The I2C works as an adapter that converts the 8 outputs pins from the LCD into the GND, VCC, SDA and SCL pins. The I2C is incredibly compatible with the Arduino framework, having an easy to access Arduino library.

Features;
Size: 40mm x 18mm
Voltage: 5V
LCD size: 16x2 or 20x4
(I2C LCD Packpack for 1602 to 2004 LCDs, 2023)



Calculations

Actuator Force Calculations + Downforce Calculations

- Equations

- Pneumatic actuator force

$F = PA$

F = Force (N)

P = Air pressure (Pa, 1 Newton per m^2)

A = Pneumatic piston's usable area (m^2)

- Downforce calculation

$F_d = \frac{1}{2}(W_w \times H_w \times \emptyset_w \times C_L \times \rho \times v)$

F = Downforce (N)

W_w = Wing width (m)

H_w = Wing height (m)

$\emptyset_w$ = Angle of wing (degrees)

~~ρ = Lift coefficient~~

C_L = Lift coefficient $\longrightarrow$ $C_L = \frac{L}{qS}$

ρ = Air density (Kg/m^3)

v = Velocity (ms^{-1})

L = Lift force (N)

q = Fluid dynamic pressure (P_a)

S = Surface area (m^2)

- Force

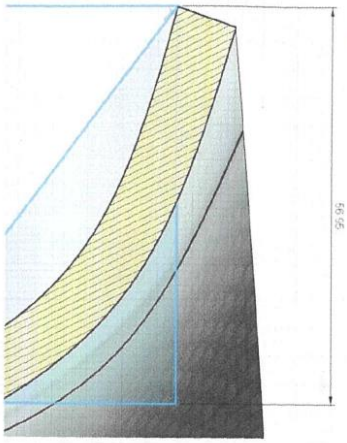
$F = ma$

F = Force (N)

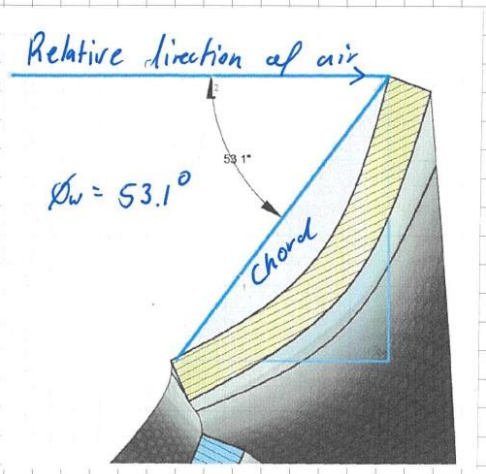
m = mass (Kg)

a = Acceleration (ms^{-2})

Downforce Calculation (Rear Wing Flap (Moving))



H_w (wing height)
= 56.95mm
= 0.05695m



ρ = Air density

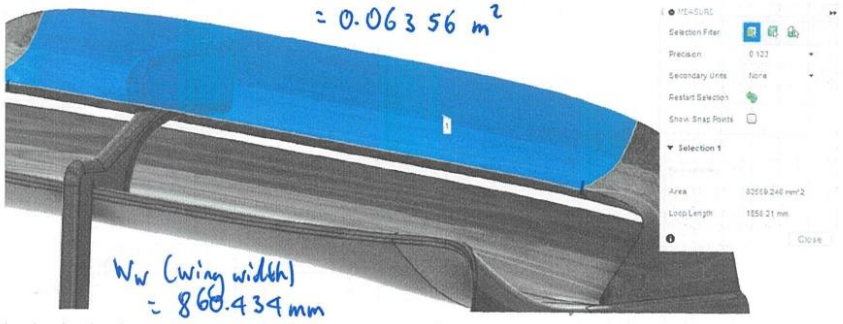
- ~~AS~~ Australian GP 2 Apr, 3:00pm
  -> Temp: 17.8°c
  -> Density: 98.65%

  [$\rho = 1.213\ kg/m^3$]

v = velocity

- Albert park 2023 average speed with DRS
  = 328.9 km/h
  = 330 km/h = 0.278 ms^{-1}

S (surface area) = 63559.246 mm^2
= 0.06356 m^2



W_w (wing width)
= 860.434 mm
= 0.860 m

Lift Coefficient

Coefficient of Lift

In the above calculations, I chose a Cl value of 1.0. However, some airfoils can achieve over twice that much downforce, and you can read about that in Airfoil Comparisons, where I review the best airfoils for car wings.

But don't get too excited about the high Cl numbers. There is always something creating turbulence in front of the wing, which creates downstream losses. This could be hood and fender vents, canards, sharp angles at your roofline, the wake of other cars in front of you, crosswinds, or all of these. Anything that creates turbulence destroys lift.

I saw this firsthand when I changed only the roofline shape on my car and back-to-back tested them: removing the top resulted in only 40% of the downforce as an OEM hard top; A fastback increased rear downforce by 130%. That's a dramatic example, but you can see that it's possible to lose or gain over three times the downforce by changing only the shape of the roof.

So this is a very long-winded way of saying that a coefficient of lift of 1.0 is fine for rough calculations, on a single element wing, on a Miata. For a properly designed dual element wing, you can use an equally rough value of Cl 1.5. My original dual element wing measured more like 1.3, but I've made more modifications since, and it's probably closer to 1.5 now.

- Within the formula for calculating downforce the C_L value is included. This value equates to lift force divided by fluid dynamic pressure multiplied by surface area.

  Lift coefficient is a complex value as it is dependent on wing shape, velocity, angle and turbulence.

  To simplify the calculations a standard value of 1.0 can be used however with the specifics of the design of the Model DRS a more suitable value is 1.5.

  Using a value of 1.5 accounts for the wing being dual element as well as masses that would create turbulence infront of the wing.

  The creation of downforce from the shape and elements of the wing, is altered due to the turbulence infront of the wing. This equals out to produce an estimated C_L value of 1.5.

$F_d = \frac{1}{2}(0.860 \times 0.05695 \times 53.1 \times C_L \times 1.213 \times 0.278)$

Lift coefficient : 1.5

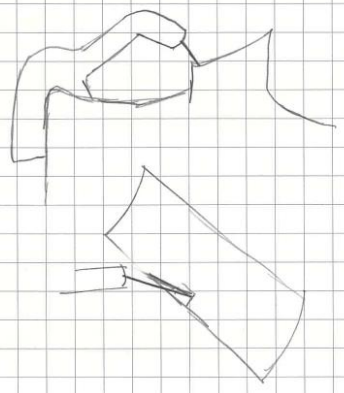
$F_d = \frac{1}{2}(0.860 \times 0.05695 \times 53.1 \times 1.5 \times 1.213 \times 0.278)$

$= \frac{1}{2}(0.87699)$

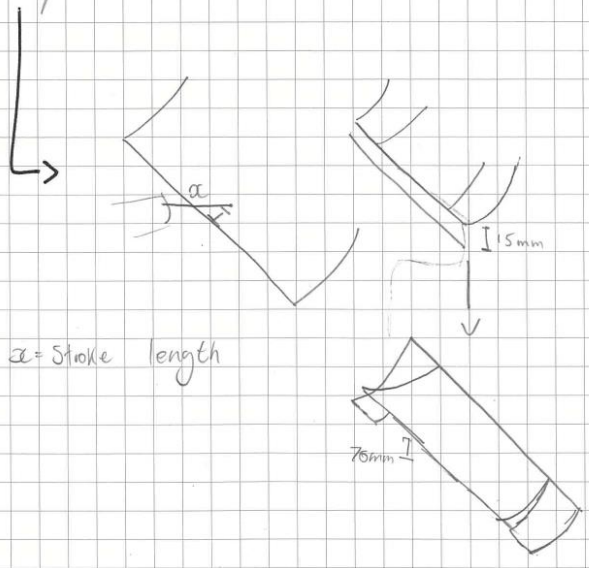
$= 0.4385\,N$ of downforce from the upper rear wing flap.

Actuator Setups and Calculations

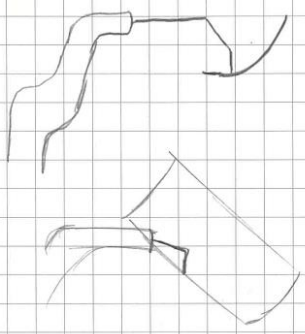
1. Direct line actuator without rocker.



- The stroke of the actuator is directly proportional to the gap with DRS activation
- Simple solution to activation of DRS, however the positioning of the actuator becomes much more important.

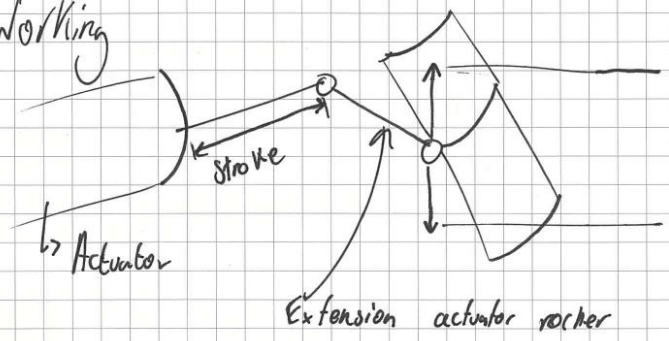


2. Horizontal actuator with rocker.

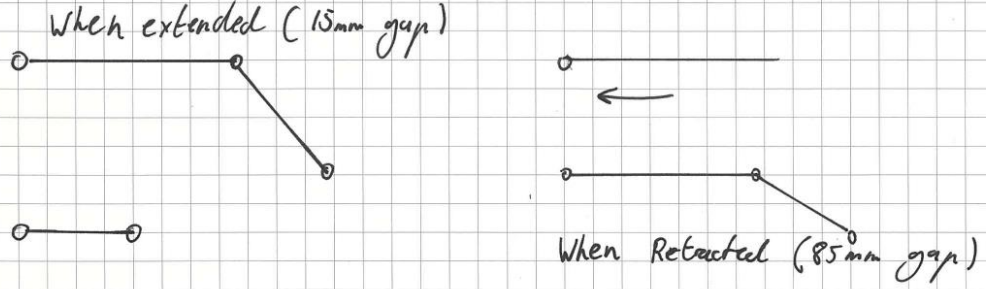


- The calculation of the stroke is much more complex due to the extra componentry.
- The actuator is horizontal which means there needs to be connectors in order to have the wing move vertically.

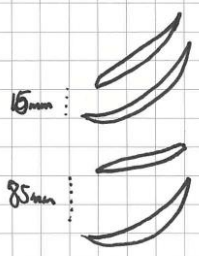
Rough Working

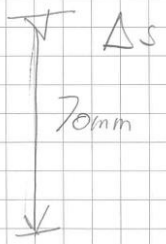


Move from 10mm/15mm gap to 85mm. In 0.5s



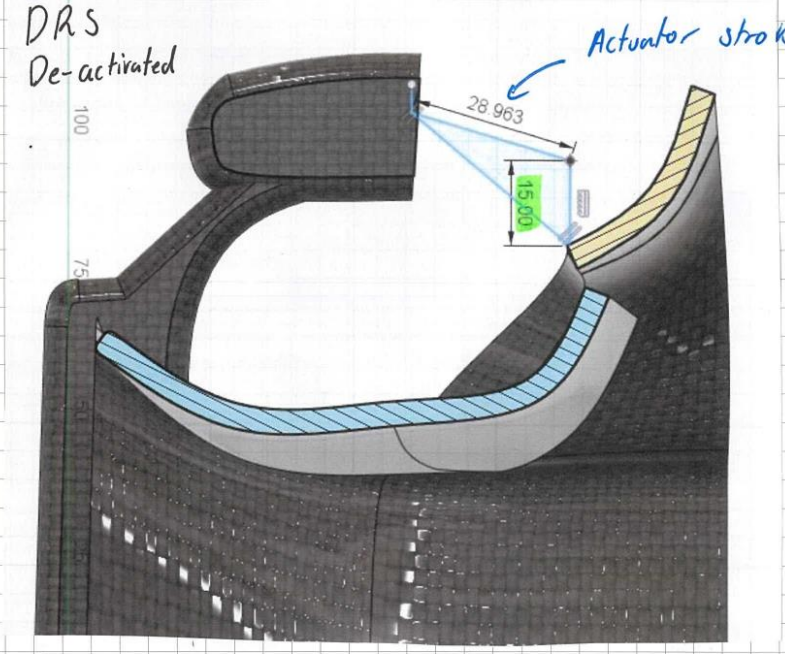
- How long does the stroke need to be?
- How far does it need to retract?

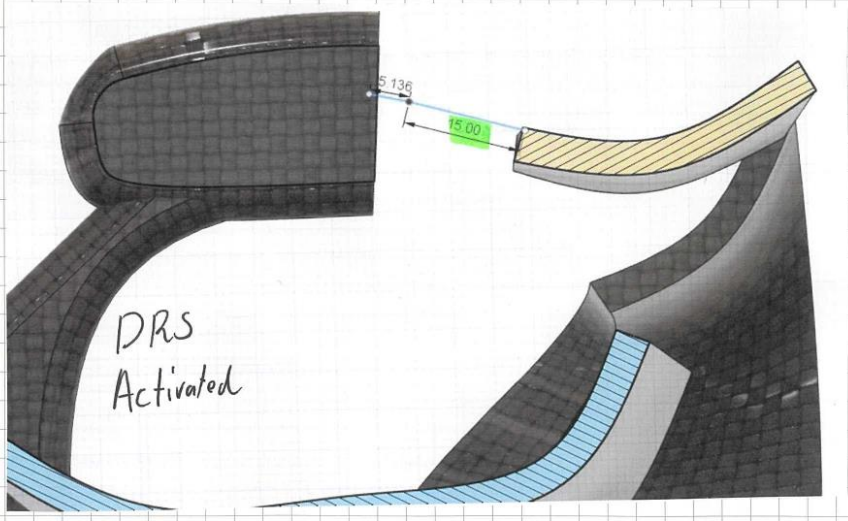




Actuator Calculations · (Rough Estimations)

DRS
De-activated

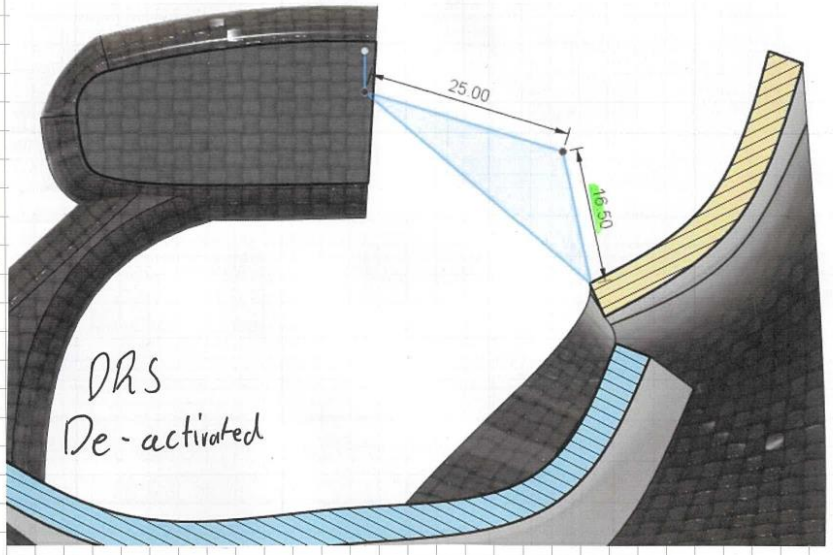




- Fixed connector = 15mm
  - ↳ Used to convert the actuators 'horizontal' motion into moving the wing 'vertically'.
- Δ in stroke length 28.963 − 5.136 (mm)
  = Δ23.827mm

Final Actuator Calculations

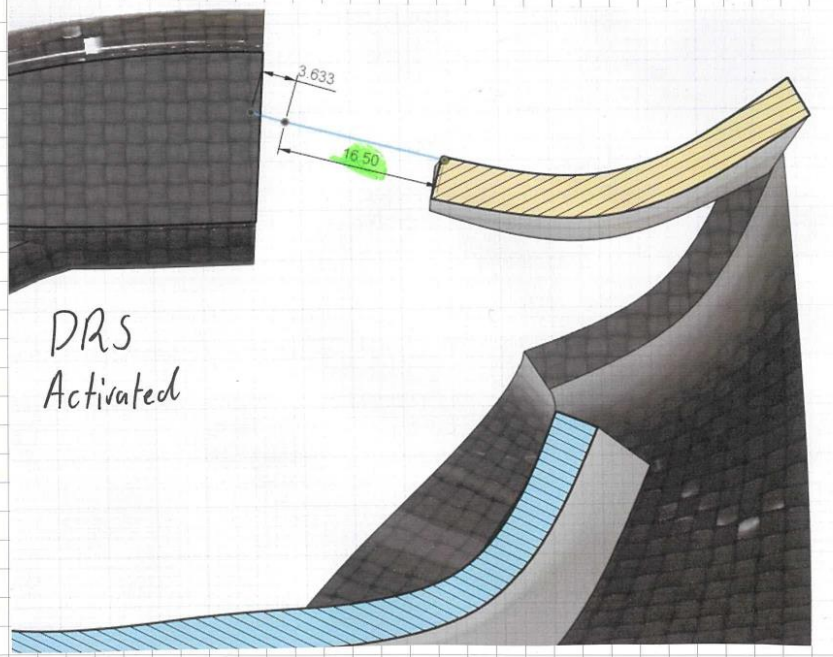
- The DSN Double Acting Mini cylinder has an available sized actuator of bore = 16mm, stroke = 25mm.
- This is the closest dimensions to the estimated fixed connector size's effect on the actuator stroke.
- However some dimension changes need to be made, as seen below.



- Fixed connector size = 16.5mm
- Δ in stroke length is

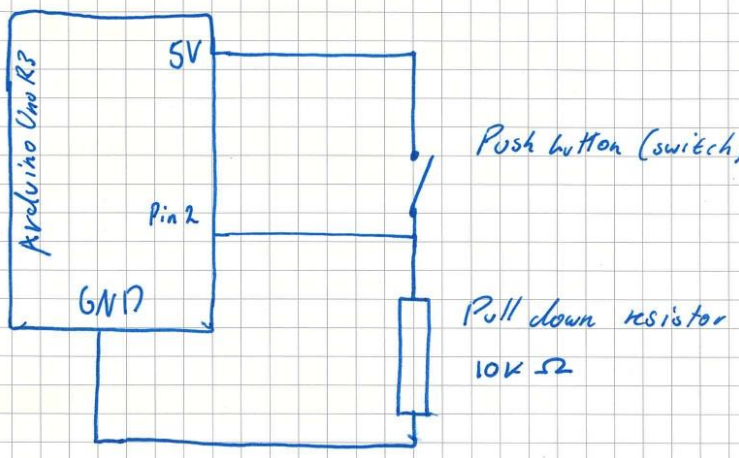
  25 - 3.633mm

∴ The pneumatics solution's actuator is suitable as; Bore: 16mm Stroke: 25mm



Resistor Calculations

- Recommended resistor size of 10K-ohm for push-button to Arduino.



- $\frac{V}{R \times I} = \frac{5}{10 \times 10^3} = 0.5\,mA$.

- The large resistor value produces a smaller current flowing from 5V to GND. This is important as there is less power lost and less heat produced through out the entire circuit.

Eg.

For a 220Ω resistor. VS For a 10K-Ω resistor

$\frac{5}{220} = 23\,mA$

$P_{Loss} = I^2R$

$= (23 \times 10^{-3})^2 \times 220$

$= 0.116\,W$

$\frac{5}{10 \times 10^3} = 0.5\,mA$

$P_{Loss} = I^2R$

$= (0.5 \times 10^{-3})^2 \times 10 \times 10^3$

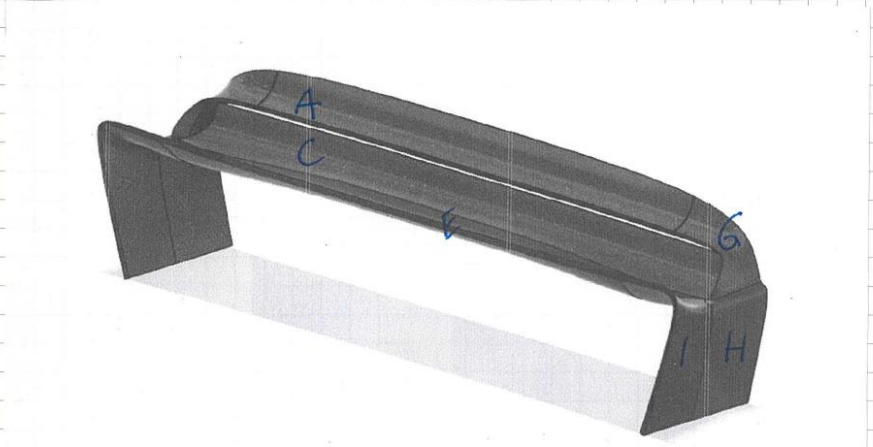
$= 0.0025\,W$

$0.116 > 0.0025$

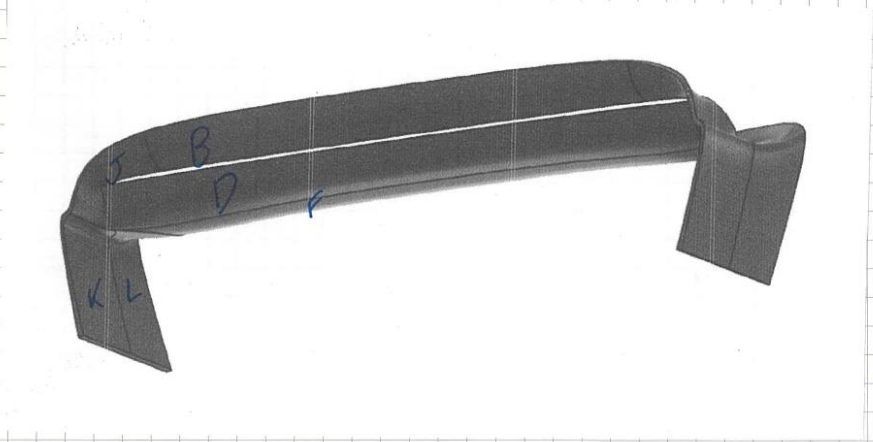
∴ The larger resistance reduces power loss.

Surface Area Calculations

- To determain the amount of carbon fibre sheets required.



- G -> L x2 for both sides of the rear wing.



A = 15889.811 mm^2

B = 17231.093 mm^2

C = 24649.53 mm^2

D = 25592.723 mm^2

E = 26016.366 mm^2

F = 28467.963 mm^2

X2

G = 3027.59 mm^2

H = 6494.894 mm^2

I = 8839.185 mm^2

J = 2719.762 mm^2

K = 6235.84 mm^2

L = 9094.468 mm^2

Total surface area = A + B + C + D + E + F + 2(G + H + I + J + K + L)

= 137847.486 + 2(36411.739)

= 137847.486 + 72823.478

= 210670.964 mm^2) $\times 10^{-6}$

= 0.211 m^2

• Currently available carbon fibre sheet

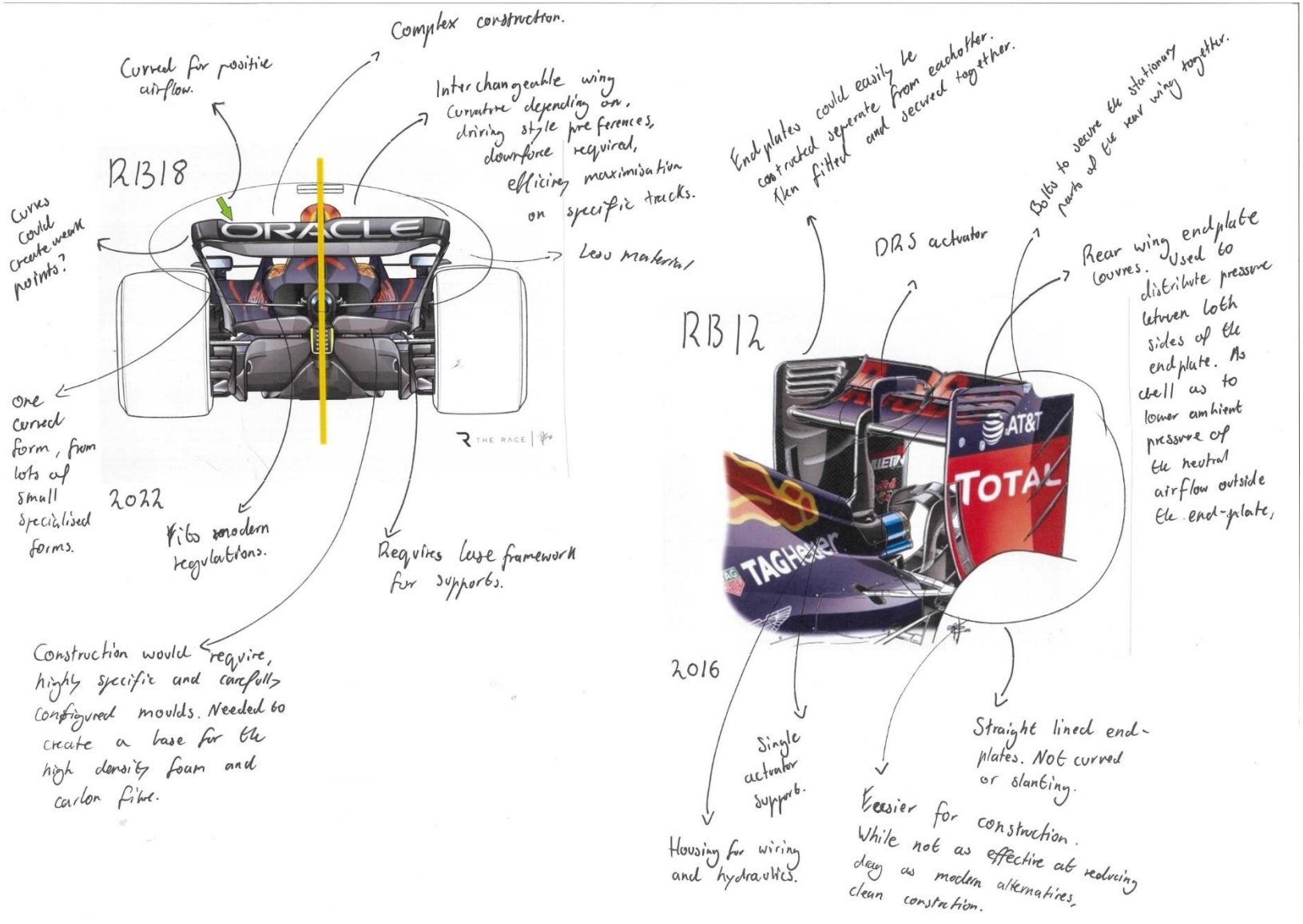
↳ 1.17 × 2.6 m

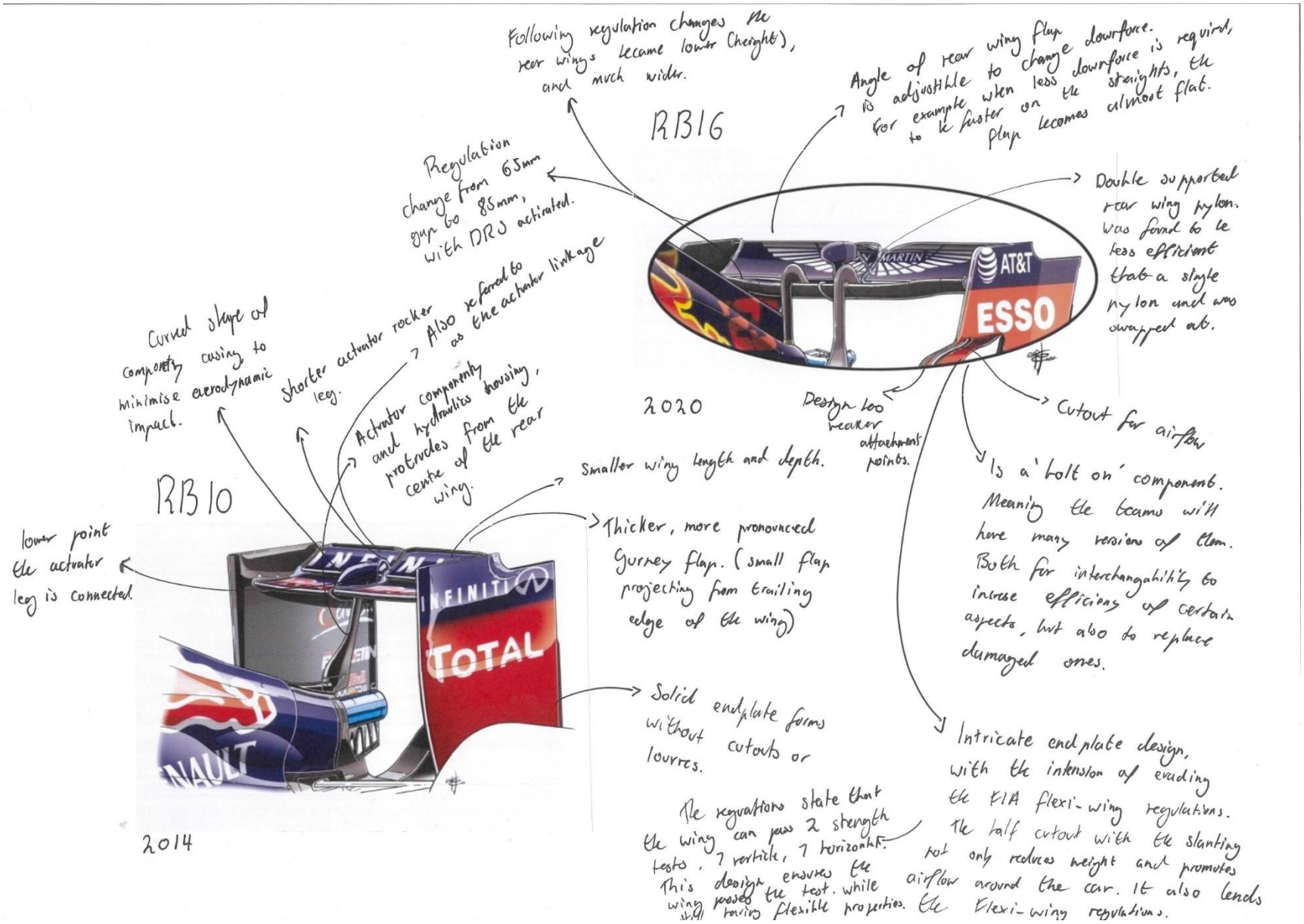
= 3.042 m^2

Each piece requires 2 layers of carbon fibre sheet.

→ 0.211 × 2 ~~= 4.~~ = 0.422 m^2

∴ The currently ~~available~~ available carbon fibre will cover the required material.


RB18
ORACLE
THE RACE
2022
Curved for positive airflow.
Complex construction.
Interchangeable wing curvature depending on, driving style preferences, downforce required, efficiency maximisation on specific tracks.
Curves could create weak points?
Less material
One curved form, from lots of small specialised forms.
Fits to modern regulations.
Requires base framework for supports.
Construction would require, highly specific and carefully configured moulds. Needed to create a base for the high density foam and carbon fibre.
RB12
AT&T
TOTAL
TAGHeuer
2016
End plates could easily be constructed seperate from eachother. Then fitted and secured together.
Bolts to secure the stationary parts of the rear wing together.
DRS actuator
Rear wing endplate louvres. Used to distribute pressure between both sides of the endplate. As well as to lower ambient pressure of the neutral airflow outside the end-plate.
Single actuator support.
Straight lined end-plates. Not curved or slanting.
Housing for wiring and hydraulics.
Easier for construction. While not as effective at reducing drag as modern alternatives, clean construction.


Following regulation changes the rear wings became lower (height), and much wider.
RB16
Angle of rear wing flap is adjustible to change downforce. For example when less downforce is required, to be faster on the straights, the flap becomes almost flat.
Regulation change from 65mm gap to 85mm, with DRS activated.
Double supported rear wing pylon. Was found to be less efficient than a single pylon and was swapped out.
AT&T
ESSO
2020
Design too weaker attachment points.
Cutout for airflow
Is a 'bolt on' component. Meaning the teams will have many versions of them. Both for interchangability to increase efficiency of certain aspects, but also to replace damaged ones.
Curved shape of component casing to minimise aerodynamic impact.
Shorter actuator rocker leg.
Also referred to as the actuator linkage
Actuator components and hydraulics housing, protrudes from the centre of the rear wing.
RB10
Smaller wing length and depth.
lower point the actuator leg is connected.
Thicker, more pronounced Gurney flap. (small flap projecting from trailing edge of the wing)
INFINITI
TOTAL
Solid endplate forms without cutouts or louvres.
Intricate endplate design, with the intension of evading the FIA flexi-wing regulations. The half cutout with the slanting not only reduces weight and promotes airflow around the car. It also bends the flexi-wing regulations.
The regulations state that the wing can pass 2 strength tests, 1 vertical, 1 horizontal. This design ensures the wing passes the test. while having flexible properties.
2014

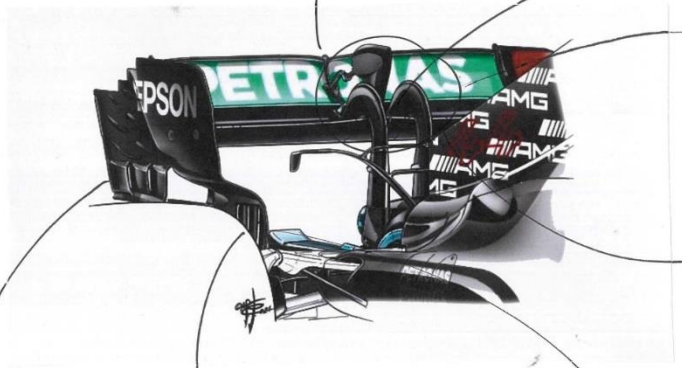
W11

Top edge of upper wing, sweeps downwards towards the centre of the DRS mechanism.

> Double pylon wing mount.

> Interchanged for the more effective single support pillar

> Pillar is merged with the DRS actuator pod, creating a bigger central wing frame.



2020

Endplate slots for airflow and managment of air pressure.

Upgraded to 3 'blowholes' to manage the vortices coming from the rotation of the rear tyres.

The endplate vertical slots and extensions, aid in extending the effective width of the car's rear wing. This increases airflow expansion around this area, generating more downforce.

The 2014 regulation changes prompted large alterations in the wing designs. 3 rule changes with the greatest impact were, Larger stroke DRS, Smaller overall rear wing depth (plus removal of the beam wing), and beamwing in the rear for the main plane to be supported by central pillars.

F14

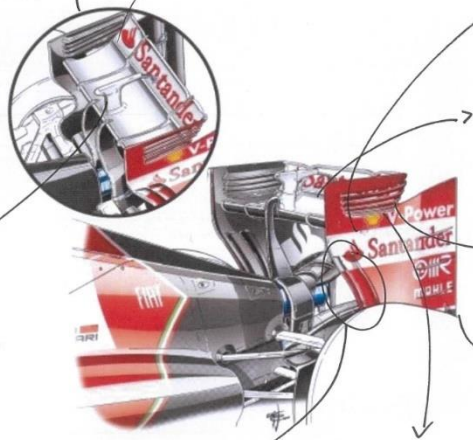
20mm shallower due to new regulations

Main plate gradient slots, inspired by the RB10. Efficient solution to pressure issues.

> Increased opening window of DRS flap

> Thin, closely spaced endplate louvres.

> Sharp angle of trailing edge strakes.



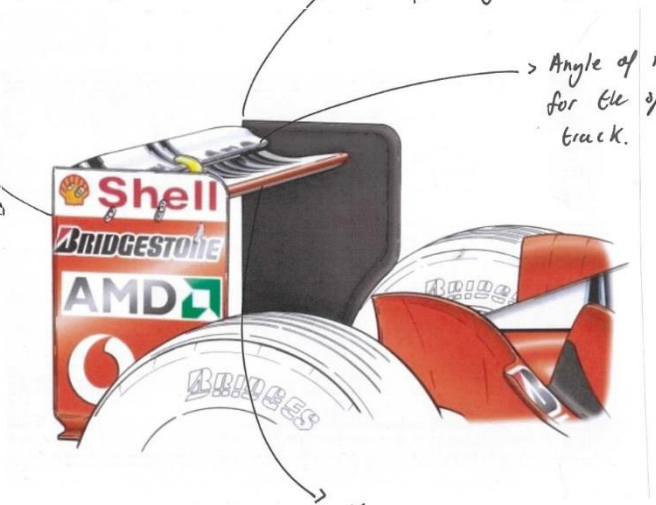
Central housing of DRS actuator pod. Protruding from the centre of the stationary base wing.

2014

Leading edge tyre wake slots. Angle directs turbulent air from rear tyres inside the endplates to balance the pressure.

Number of louvres was interchanged throughout the season to best fit the requirements of certain tracks.

F 2003-GA



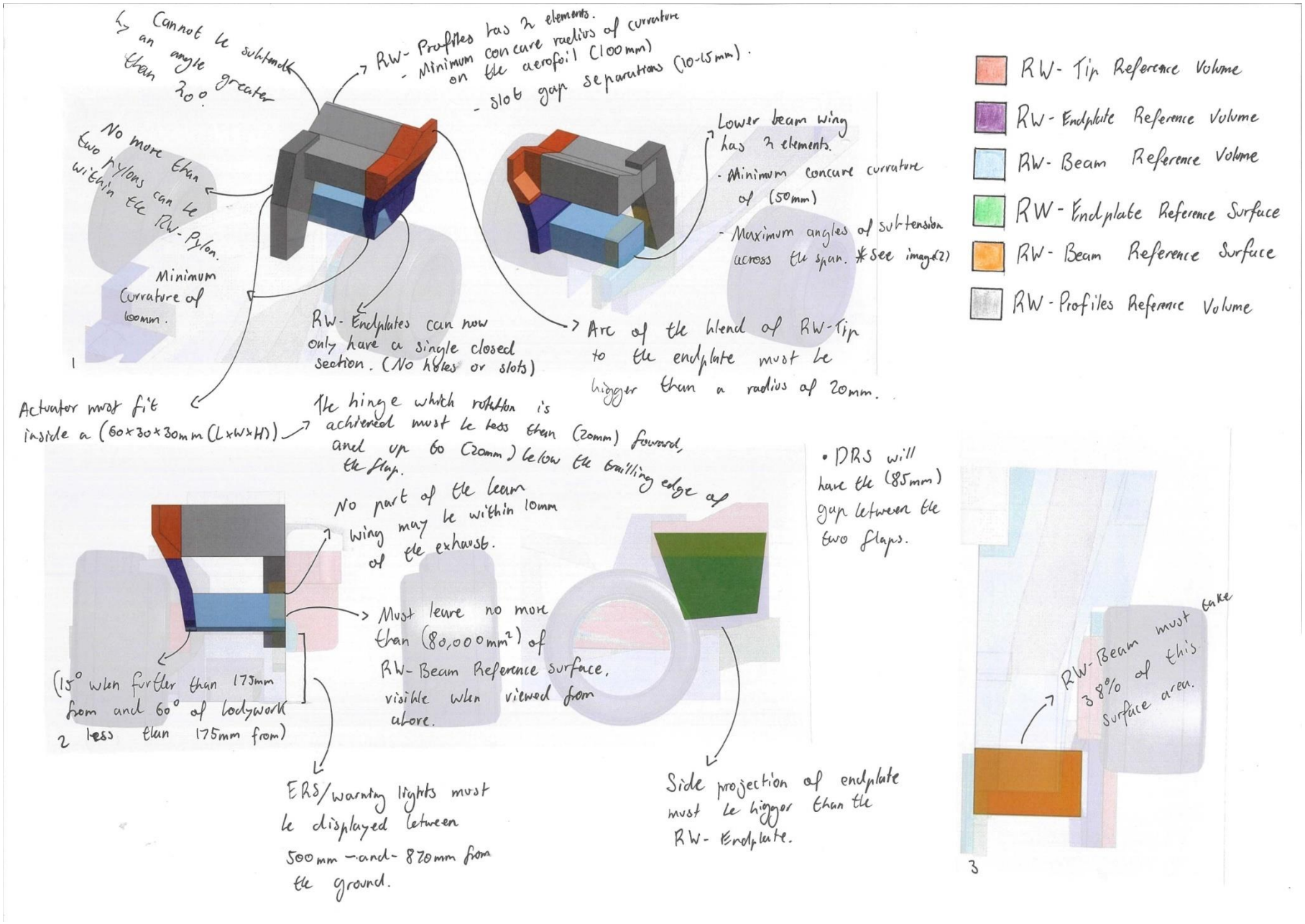
2003

Competed prior to the introduction of DRS. Minimal prioritising extravagant rear wing design. Prioritising maximising a speed / downforce balance.

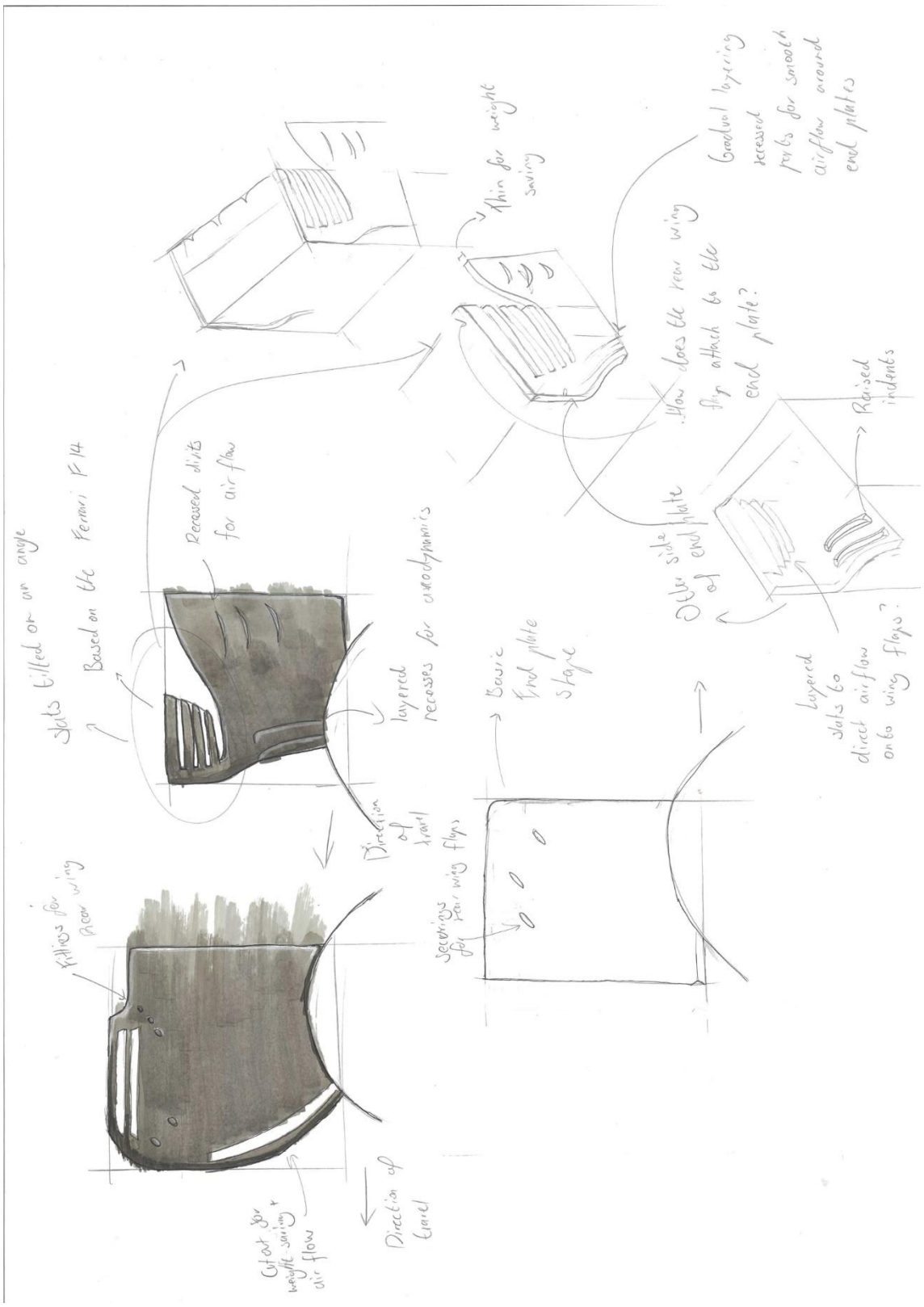
> Angle of rear wing changable for the specifics of each track.

No endplate louvres, pressure within wing components was less complex.

No support pillars required, meaning airflow through the wing was maximised

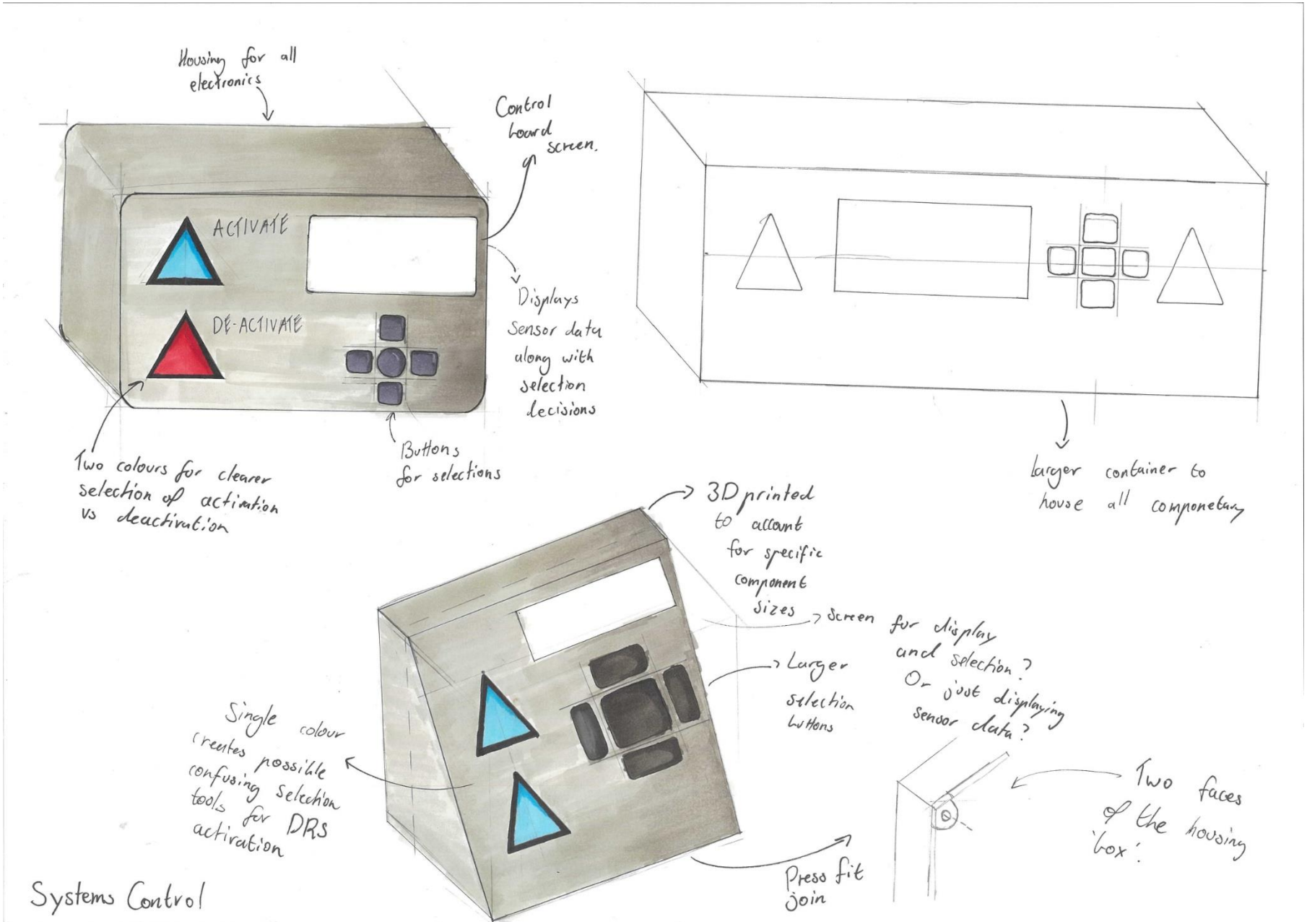

Cannot be subtended at an angle greater than 20°.
RW- Profiles has 2 elements.
- Minimum concave radius of curvature on the aerofoil (100mm)
- Slot gap separations (10-15mm).
No more than two pylons can be within the RW- Pylon.
Minimum Curvature of 600mm.
Lower beam wing has 2 elements.
- Minimum concave curvature of (50mm)
- Maximum angles of subtension across the span. *see image(2)
RW- Endplates can now only have a single closed section. (No holes or slots)
Arc of the blend of RW-Tip to the endplate must be bigger than a radius of 20mm.
RW- Tip Reference Volume
RW- Endplate Reference Volume
RW- Beam Reference Volume
RW- Endplate Reference Surface
RW- Beam Reference Surface
RW- Profiles Reference Volume
1
Actuator must fit inside a (60×30×30mm (L×W×H))
The hinge which rotation is achieved must be less than (20mm) forward, and up to (20mm) below the trailing edge of the flap.
No part of the beam wing may be within 10mm of the exhaust.
Must leave no more than (80,000mm²) of RW- Beam Reference surface, visible when viewed from above.
(15° when further than 175mm from and 60° of bodywork less than 175mm from)
2
ERS/warning lights must be displayed between 500mm -and- 870mm from the ground.
Side projection of endplate must be bigger than the RW- Endplate.
• DRS will have the (85mm) gap between the two flaps.
RW- Beam must take 38% of this surface area.
3

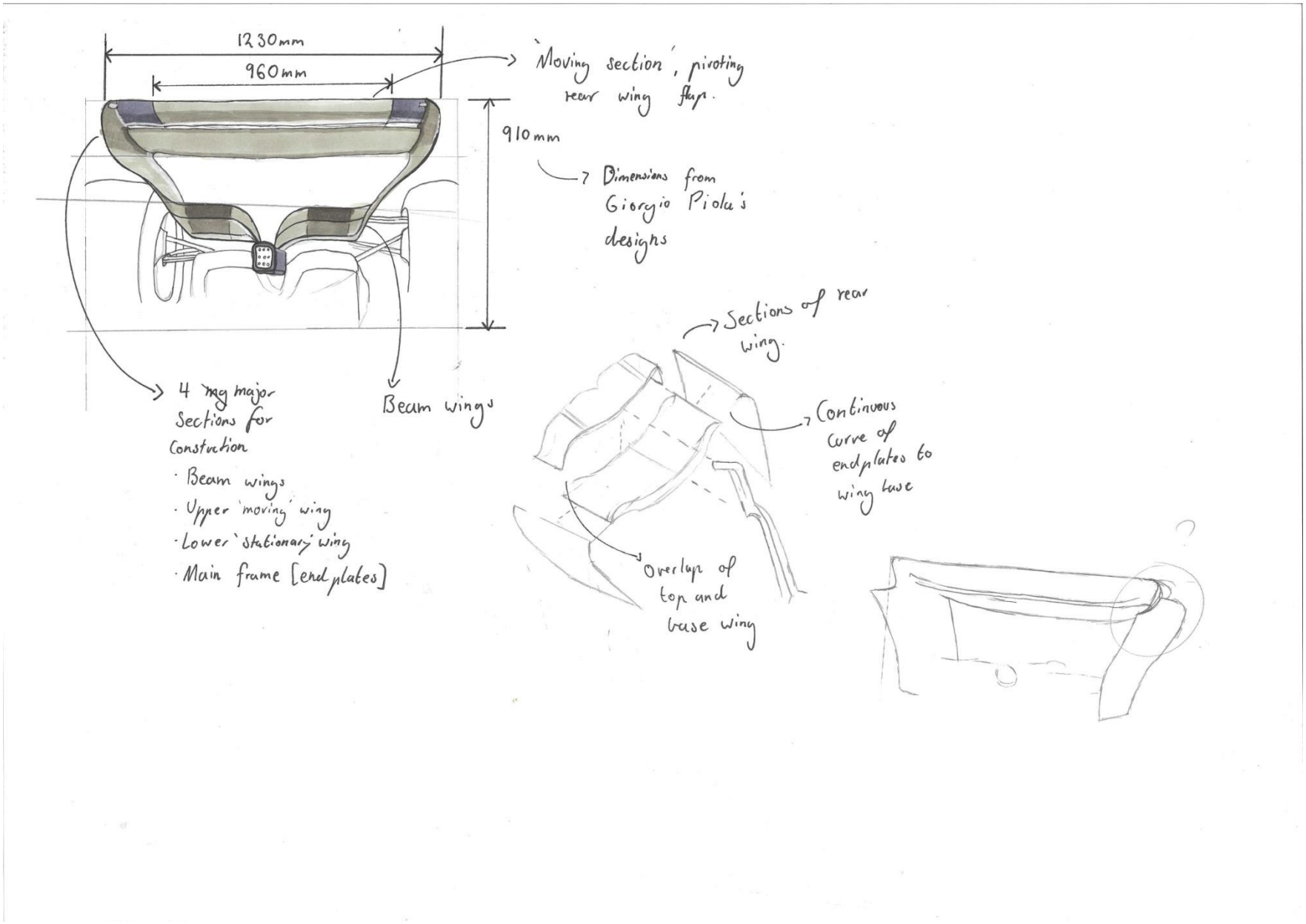
Design Option Ideation

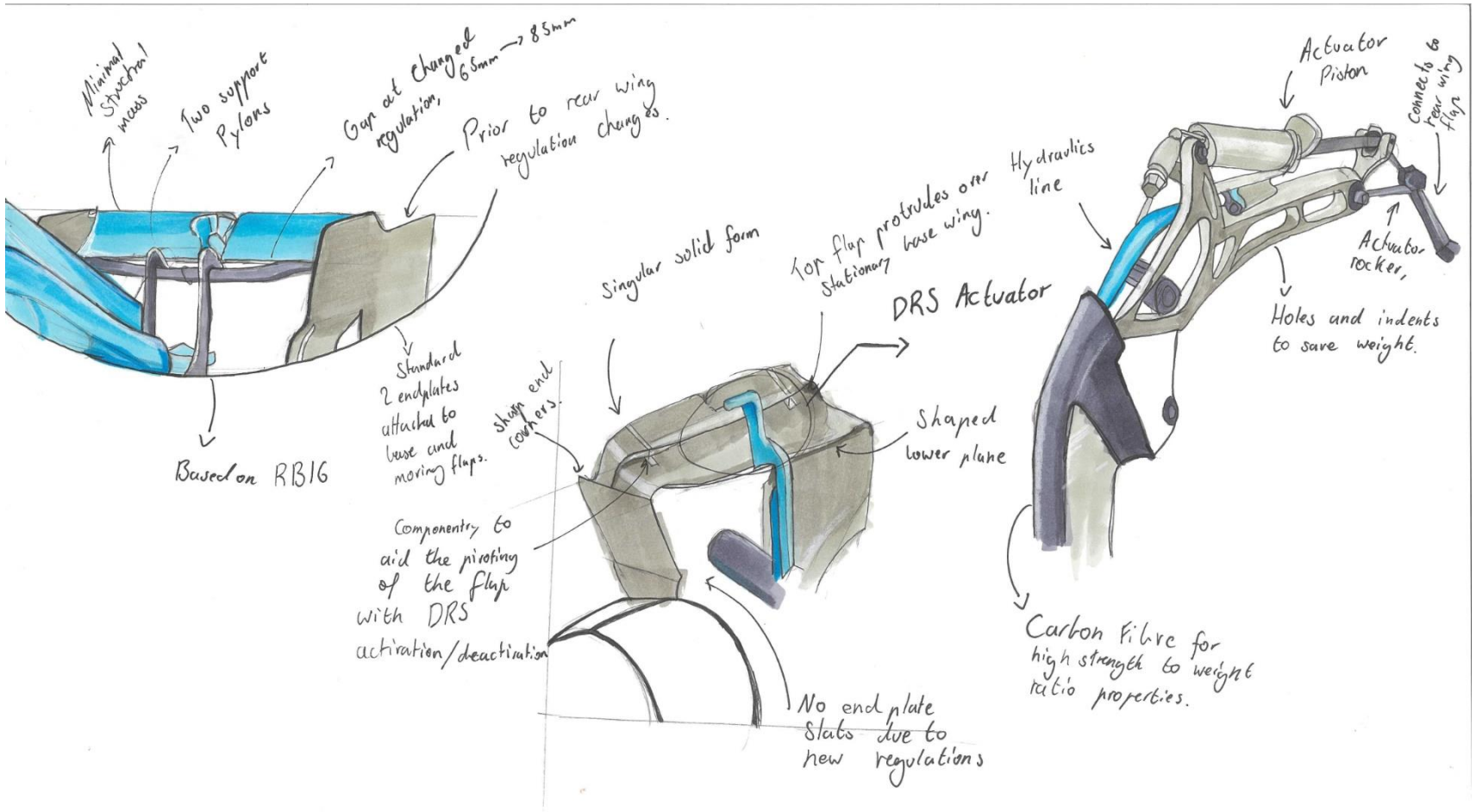


Shorter
Corners at
back end of
rear wing
Angled raised edges for
down-force generation
Direction
of travel
RBR
Close to regulation
shape, more angled
design. Less curves.

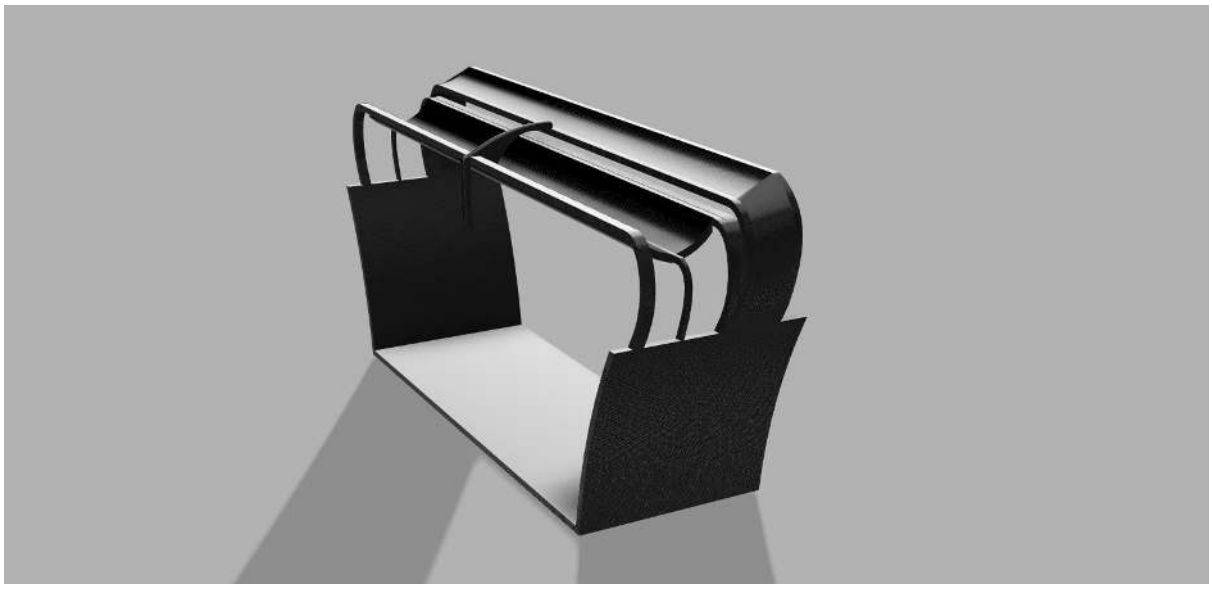
j

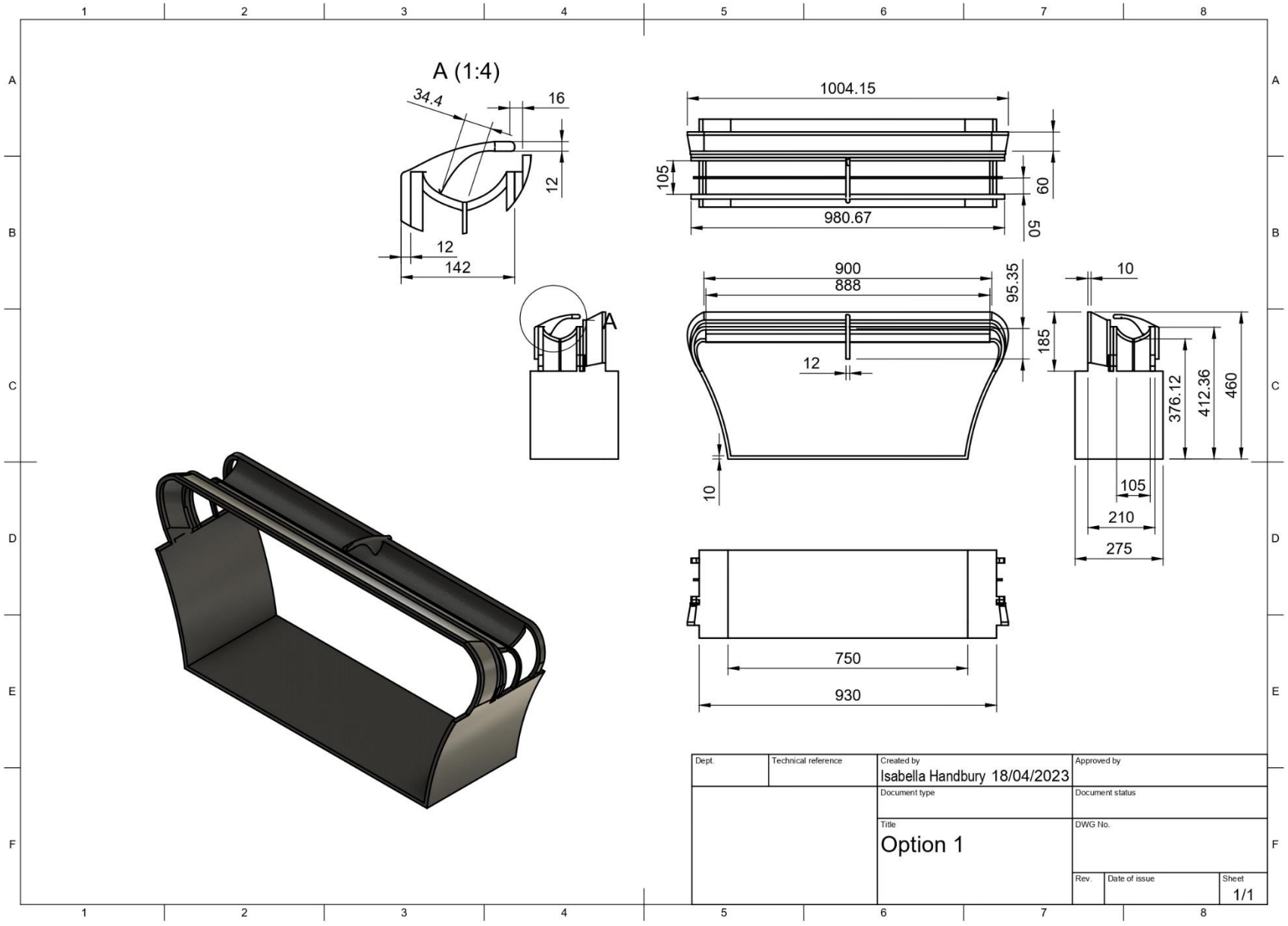

Housing for all electronics
Control board screen.
ACTIVATE
DE-ACTIVATE
Displays Sensor data along with selection decisions
Two colours for clearer selection of activation vs deactivation
Buttons for selections
Larger container to house all componetary
3D printed to account for specific component sizes
Screen for display and selection? Or just displaying sensor data?
Larger selection buttons
Single colour creates possible confusing selection tools for DRS activation
Press fit join
Two faces of the housing 'box'.
Systems Control


1230mm
960mm
910mm
'Moving section', pivoting rear wing flaps.
Dimensions from Giorgio Piola's designs
4 major Sections for Construction
· Beam wings
· Upper 'moving' wing
· Lower 'stationary' wing
· Main frame [endplates]
Beam wings
Sections of rear wing.
Continuous curve of endplates to wing base
Overlap of top and base wing


Minimal Structural mass
Two support Pylons
Gap at Changed regulation, 65mm → 85mm
Prior to rear wing regulation changes.
Standard 2 endplates attached to base and moving flaps.
Based on RB16
Singular solid form
Top flap protrudes over stationary base wing.
DRS Actuator
Sharp end corners.
Shaped lower plane
Componentry to aid the pivoting of the flap with DRS activation/deactivation
No end plate slats due to new regulations
Actuator Piston
Connects to rear wing flap
Hydraulics line
Actuator rocker,
Holes and indents to save weight.
Carbon Fibre for high strength to weight ratio properties.

Option 1




A (1:4)
34.4
16
12
12
142
A
1004.15
105
60
980.67
60
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888
95.35
12
10
185
10
376.12
412.36
460
105
210
275
750
930
Dept.
Technical reference
Created by
Isabella Handbury 18/04/2023
Approved by
Document type
Document status
Title
Option 1
DWG No.
Rev.
Date of issue
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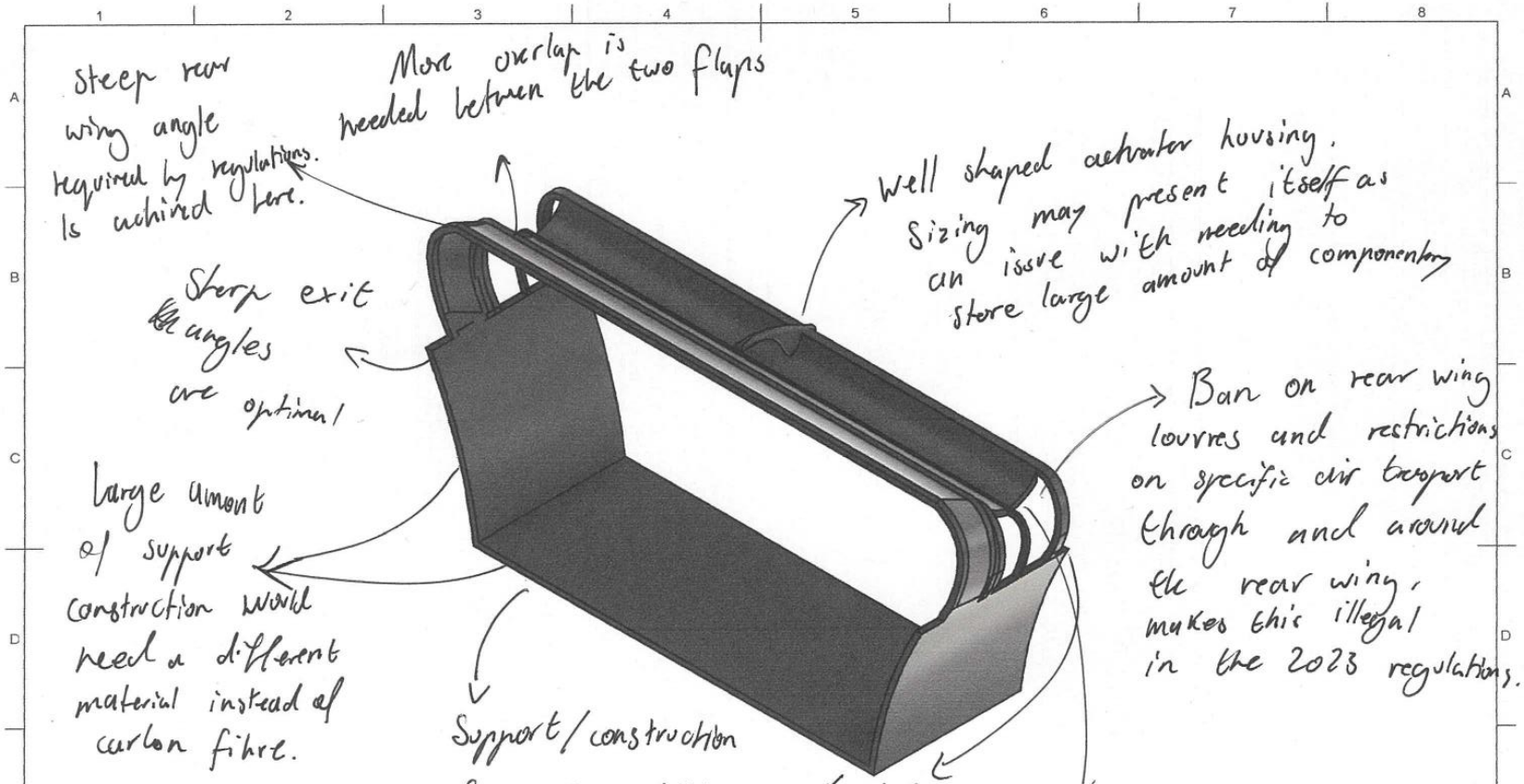

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		Isabella Handbury 18/04/2023	
		Document type	Document status
		Title: Option 1	DWG No.
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CUT

Relevant Criteria

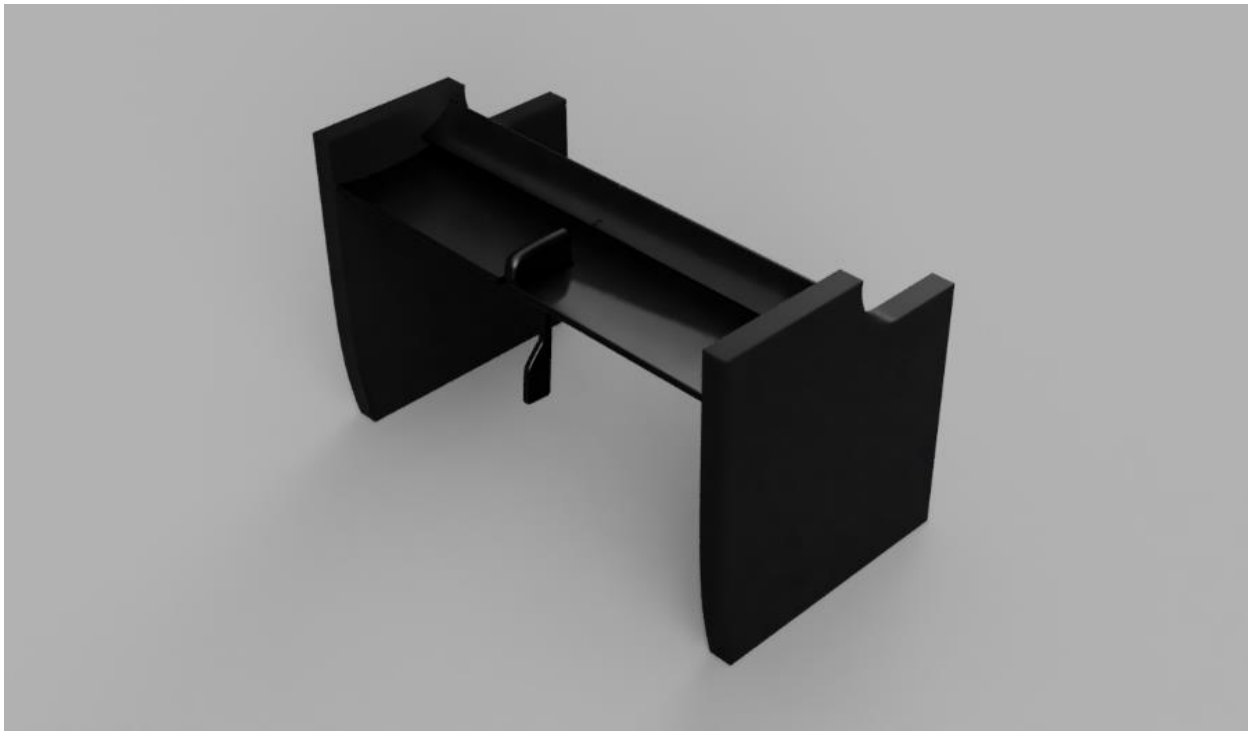
Criteria	Description	Rating	Justification
Functionality	Does the system include an efficient usage of the rear wing actuator to activate and deactivate the DRS?	4/10	While all the componentry for the actuator is not shown in the design, the location and housing for the actuator can be assessed to determine its perceived efficiency. The housing in this design option takes up minimal space reducing airflow impact and weight. However, this does affect the durability of the actuator housing and the space for componentry. Based on the design, it would be a challenge to fit all componentry into the housing and organise it well. The location of the housing means that extra connectors would be needed to change the actuators horizontal motion into vertical motion to move the rear wing flap. While it's doable it creates extra challenges. Another issue is that the actuator housing doesn't include holes for the power supply and pneumatic pipes and isn't connected to the support structure.
	Is the system aerodynamic and reduce drag?	8/10	The modern curvature of the rear wing suggest that the system will be aerodynamic and reduce drag. As well as having the correct angle of the rear wing flap to project the dirty air at the optimal angle, as stated by the FIA regulations. The front facing edges could be curved into the rear wing flaps to help with reducing drag as well as promoting positive airflow around the wing, in establishing this the rear edges are sharp which is optimal for aerodynamics.
	Does the system include a brake pedal to deactivate DRS?	0/10	With this design option a brake pedal was not included.
Materials and components	Were only materials permitted in the FIA 2023 technical regulations used?	10/10	Yes only materials permitted by the FIA in their 2023 regulations were used in this design. The design does not include the specifics of any fasteners which would

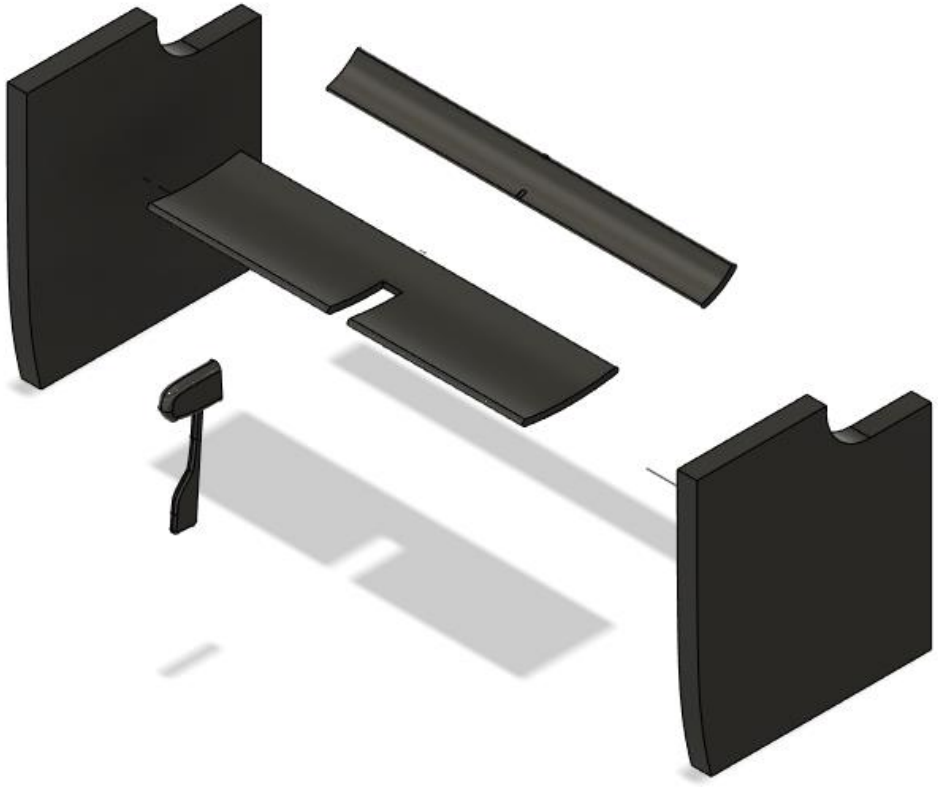
				need to be checked against the regulations. The only material in this design is carbon fibre which is allowed by the FIA.
		Are the materials chosen strong and suitable for the system? Are they assembled using methods to maximise strength?	8/10	The materials used in this design (carbon fibre) are suitable for the requirements of the system. Even though the process of constructing with carbon fibre is not simple it is the best material for the constraints of this system. Having one overall form with the absence of endplates does change the strong and weak points of the design. While the overall form is stronger and more aerodynamic, if not constructed properly could have some issues with needing to join at the curves which could create weak points.
Environmental use (use area)		Is the Model DRS able to withstand high aerodynamic forces through strength and durability?	8/10	Based on what can be assumed from the design and intended construction techniques the Model DRS will be able to withstand the aerodynamic forces and will have minimal weak points. While the durability also relies heavily on the construction of the Model DRS, some things can be inferred from the form in this design option. Unlike the older designs of the rear wings which relied heavily on end plates, this design option has the positive of less sharp joins that could present as weak points against aerodynamic forces. Having the structural support for the Model DRS also increases its strength and acts as the mounting point for the rear wing. A negative with this design option is the thin rear wing flap joiners to the support structure. They are an obvious weak point and have the potential to break easily under force.
Safety		Is all wiring for the system covered and safe?	9/10	The housing design for the actuator aims to contain all wires safely, protecting the users and manufacturers of the Model DRS. This is achieved in the design with the enclosed form. However, an issue with the housing is that it lacks the correct holes for the pneumatic pipes and power supply.

Quality standards Styling and appearance		Does the system have appealing and aesthetic qualities?	8/10	The curvature and overall shape of the rear wing is appealing and flows well with the support structure. Even though the rear wing louvres are banned by the FIA's regulations they add some negative space to the design stopping it from looking like one massive form. They also separate this design from the standard modern rear wing design, exploring some innovation.
Size		Does the Model DRS fit withing the FIA's dimension rules or a scaled alternative?	6/10	Yes this design fits within the depth, length, and height restriction of the FIA's 2023 technical regulations. However, it has issues in regards to the inclusion of the rear wing louvres which were banned by the FIA to stop their transport of air around the back of the car. This means that this design is illegal in terms of the FIA regulations.
Maintenance		Are the components organised and managed properly to allow for easy access and maintenance? This includes wires, electronics, and power supply.	4/10	The actuator housing in this design intends to house all the componentry associated with the pneumatics, an issue that the actuator housing could have is properly fitting all componentry. Not having adequate space could limit the ease of organising all wiring and electronics as well as making maintenance difficult. To improve the design specifics regarding how to access the components inside the housing could be included. This design lacks specifics for the inclusion of sensors which are vital in the system, the location and wiring or the sensors needs to be explored.
Production methods		Did the system utilise the most efficient production methods?	6/10	The design being one interconnecting form does pose some challenges with the construction especially with needing to either mould the overall form in sections or doing one mould. Both construction methods have pros and cons that effect the ease of manufacture and the quality of the final product. While it is clear to where each section would be individually formed there are still issues with joining the separate sections efficiently, as the process could take a very long time. The housing for the actuator is a very unique shape that needs to fit to the

				curvature of the rear wing precisely. This could pose some issues and would need to be completed and tested multiple times to get the best result. While this is the method to get the best result it is a lengthy process and also has the issue of wasting materials.

Option 2





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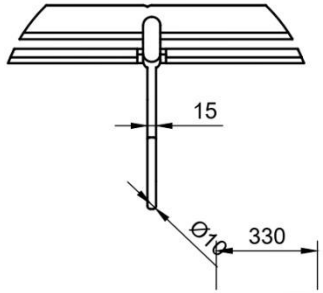

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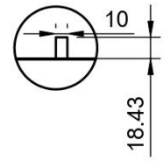

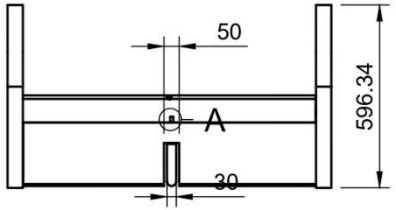

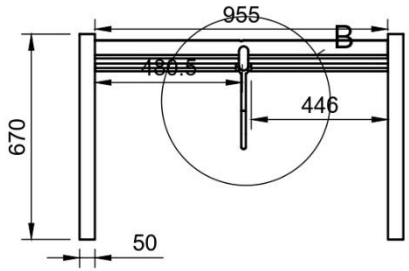

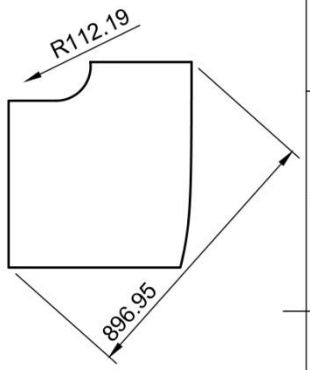

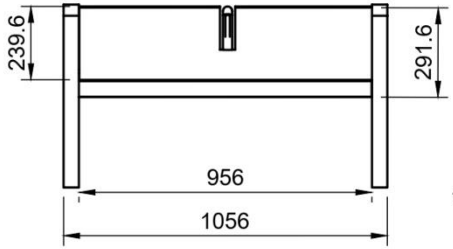

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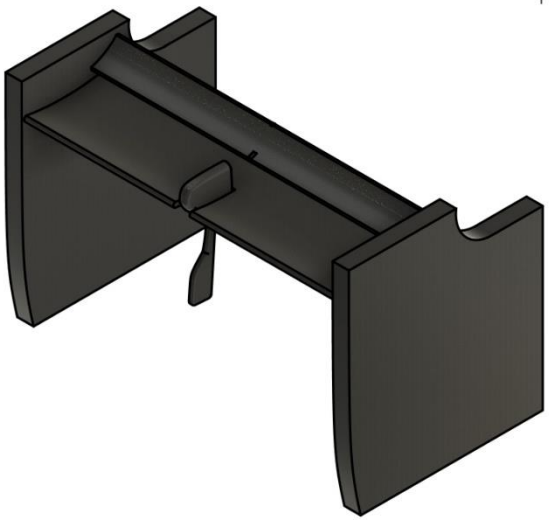

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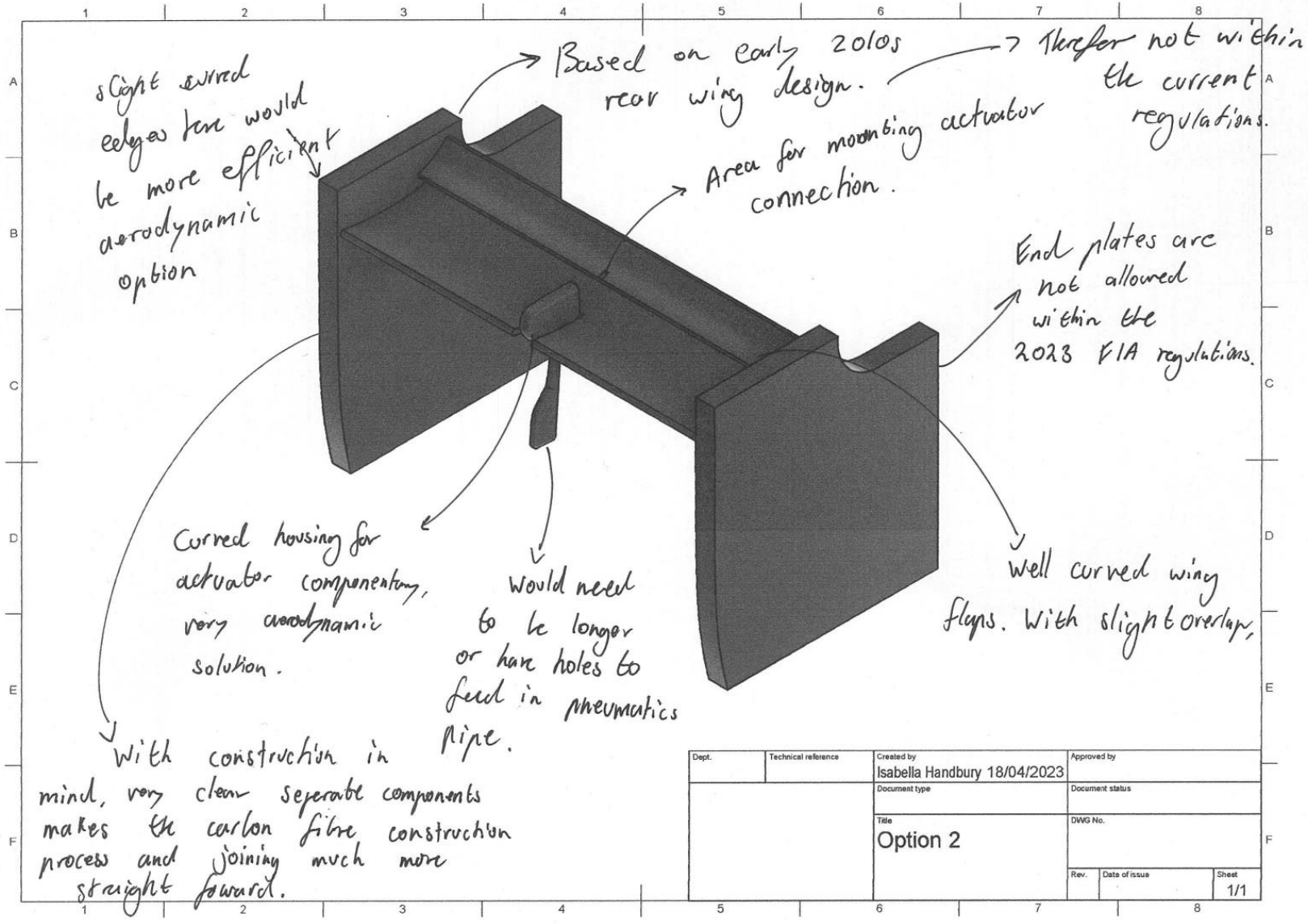
		Document type	Document status
		Title: Option 2	DWG No.
			Rev. / Date of issue / Sheet: 1/1


Based on early 2010s rear wing design.
Therefor not within the current regulations.
Slight curved edges here would be more efficient aerodynamic option
Area for mounting actuator connection.
End plates are not allowed within the 2023 FIA regulations.
Curved housing for actuator components, very aerodynamic solution.
Would need to be longer or have holes to feed in pneumatics pipe.
Well curved wing flaps. With slight overlap,
With construction in mind, very clean seperate components makes the carbon fibre construction process and joining much more straight fowards.
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Technical reference
Created by
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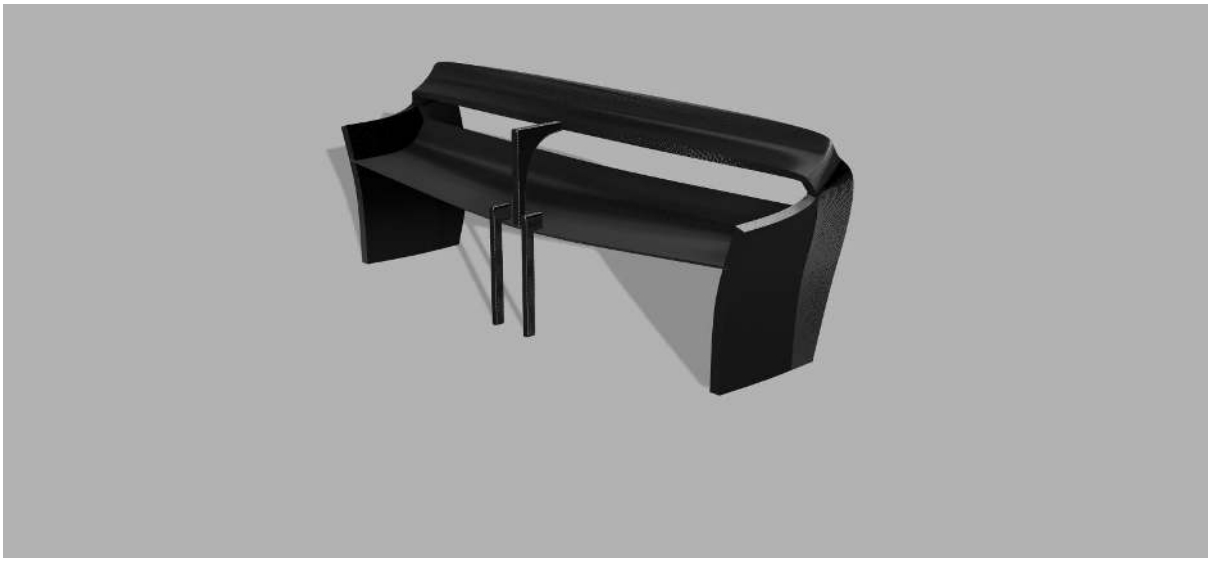
Relevant Criteria

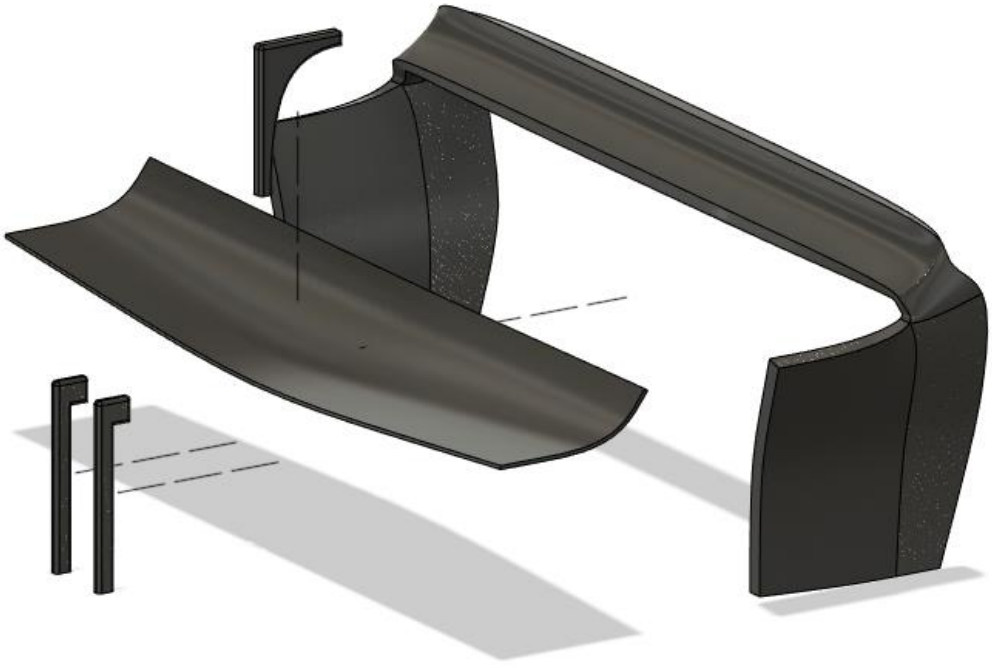
Criteria	Description	Rating	Justification
Functionality	Does the system include an efficient usage of the rear wing actuator to activate and deactivate the DRS?	8/10	While the design does not explicitly display the actuator, it does exhibit the location and housing of the actuator which contribute to its efficiency. The housing has great curvature for airflow which means that the actuator isn't interrupting the rear wing's airflow. The actuator is well aligned with the top 'moving' rear wing flap, which means that more simplistic actuator componentry can be utilised. Meaning that the actuator can work linearly without needing extra connectors to transfer the horizontal motion into vertical. The only issue with this actuator housing design is that it doesn't include holes for the pneumatic pipes and is 'floating'. The solution to this would be to design around the pipes and to add some structural support for the component.
	Is the system aerodynamic and reduce drag?	6/10	The curvature of the rear wing flaps suggests that without any testing the assumption can be made that, the specific portion of the rear wing is aerodynamic. As for the end plates, they are based on an older design of the rear wings and their sharp edges do affect the amount of drag the components produces. A more curved design for the front facing edges would produce a more drag efficient design.
	Does the system include a brake pedal to deactivate DRS?	0/10	A brake pedal was overlooked in the designing of this rear wing option, while it is a consideration it should be included to enhance user experience with the model.
Materials and components	Were only materials permitted in the FIA 2023 technical regulations used?	10/10	In this design a type of carbon fibre was used that is in compliance with the FIA technical regulations for 2023, the only other material used was the small alloy for the actuator connector. By cross checking the regulations it was found that for components that are not made by

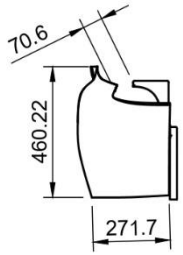
				additive manufacture, any steel alloy is allowed.
		Are the materials chosen strong and suitable for the system? Are they assembled using methods to maximise strength?	7/10	The chosen materials are fit for purpose and are selected for their durability within the specifics of the system. The only issue that is evident is the design's noticeable weak points. The connection of the endplates to the rear wing flaps creates a weak point at the join for breakage. Depending on the construction method used this has the potential to affect the strength of the system.
Environmental use (use area)		Is the Model DRS able to withstand high aerodynamic forces through strength and durability?	7/10	Based on the configuration of this design option in terms of minimising any weak points or potential short falls that lead to a lack of durability, there are definitely pros and cons in terms of structural stability. The major pros are the thickness of the endplates and wing flaps that add durability as well as the shaping that allow the components to work with the airflow not against it. The cons appear with the join points of the separate components, with the older rear wing design the sharp joins lead to weak points. For example, the joining of the endplates with the rear wing flaps creates a point that could weaken against strong aerodynamic forces.
Safety		Is all wiring for the system covered and safe?	9/10	For this actuator cover design, all wiring is covered and there is space to account for the componentry. This makes the system safe and user friendly. The only improvement that could be made is to have holes in the actuator cover for the pneumatic pipes and have structural supports for the housing.
Quality standards Styling and appearance		Does the system have appealing and aesthetic qualities?	8/10	While this design option is based on the older rear wings, around mid to early 2010s. I find the design has clear and simplistic lines and potentially nostalgic qualities. The endplates have a concise shape that complements that curved shaping of the rear wing flaps nicely.
Size		Does the Model DRS fit withing the FIA's dimension rules or a scaled alternative?	2/10	While this Model DRS design fits within the regulation's dimension restrictions it has other issues with the regulations. The inclusion of the end plates makes the

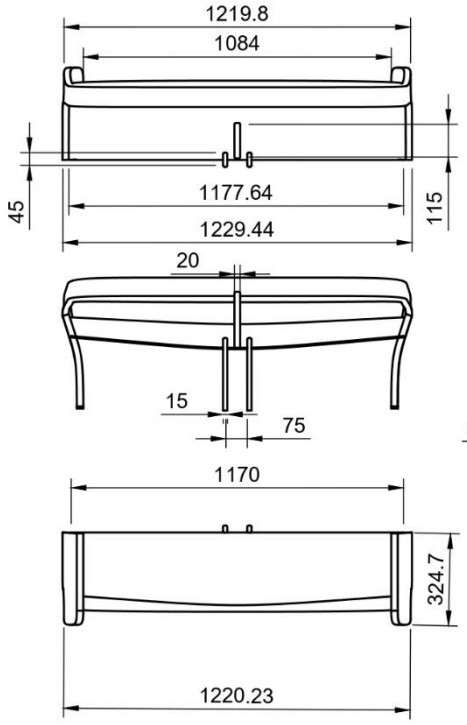
				design uncompliant with the 2023 regulations that worked to remove the inclusion of endplates for the more curved shape.
Maintenance		Are the components organised and managed properly to allow for easy access and maintenance? This includes wires, electronics, and power supply.	5/10	Based on the details that this design option provides it appears that the componentry associated with the pneumatic actuator is efficiently stored along with the wiring. However, the downfall of the design is not having a way to access the components in their housing, also not having a hole in the housing for power supply and the airline for the pneumatics. Something that has been neglected with this design is the location and wiring position for any sensors that will be attached to the rear wing.
Production methods		Did the system utilise the most efficient production methods?	7/10	In regards to designing for ease of manufacture, this design has aspects that aim to make production as efficient as possible. The clear separate components of the two rear wing flaps and two end plates, (while not allowed by the modern regulations) make construction much more efficient. Having clear separation of components would make the carbon fibre construction process much more simplistic, as each piece can be separately moulded and then created to join together. The only improvement that could be made is in regard to the actuator housing, its complex shape could present issues for creating a mould. This process could utilise 3D printing to make it more efficient, but its something that should be further considered.

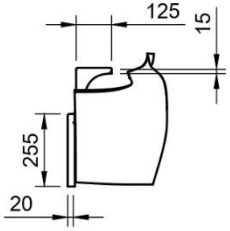
Option 3











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		Isabella Handbury 24/04/2023	
		Document type	Document status
		Title: Option 3	DWG No.
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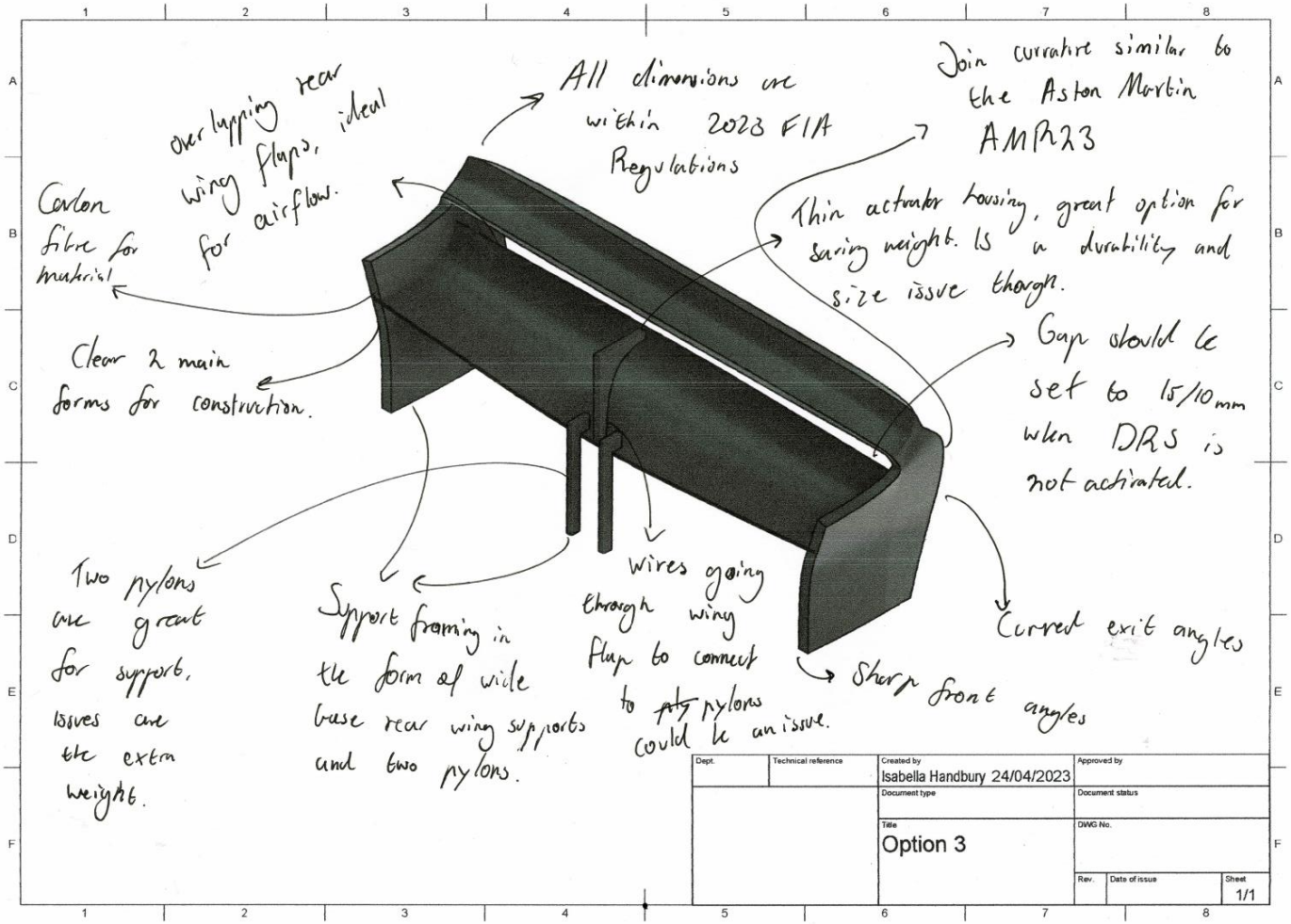
overlapping rear wing flaps, ideal for airflow.
All dimensions are within 2023 FIA Regulations
Join curvature similar to the Aston Martin AMR23
Carbon fibre for material
Thin actuator housing, great option for saving weight. Is a durability and size issue though.
Clear 2 main forms for construction.
Gap should be set to 15/10mm when DRS is not activated.
Two pylons are great for support, issues are the extra weight.
Support framing in the form of wide base rear wing supports and two pylons.
wires going through wing flap to connect to ~~fly~~ pylons could be an issue.
Curved exit angles
Sharp front angles
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Technical reference
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Isabella Handbury 24/04/2023
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Relevant Criteria

Criteria	Description	Rating	Justification
Functionality	Does the system include an efficient usage of the rear wing actuator to activate and deactivate the DRS?	6/10	Even though the model does not specifically display the actuator, a conclusion to this question can be gained from the actuator housing design and location. While the actuator housing has aerodynamic curvature it does have issues that impact its efficiency. The actuator being really thin is an issue for durability by also for housing all associated componentry that is required. Having the wiring and airline go from the actuator housing to the two pylons in this design is a good solution in terms of needing the pylons for extra support. However, it does create more complex issues with the wiring and airline having to go through the lower rear wing flap. This can create structural issues and makes the construction process much more difficult. The position of the actuator housing would require a connector to join the rear wing flap to the electronics. This is needed to change the direction of motion of the actuator into motion that would move the rear wing flap to activate DRS. This is a simple problem to solve however it does require extra time and materials.
	Is the system aerodynamic and reduce drag?	9/10	The well-shaped curvature of both the rear wing flaps and the main frame of the rear wing suggest a very aerodynamic solution. The angle of the rear wing flaps works to reduce drag, as well as minimising dirty air behind the car. The joining curvature on the corners of the main frame create added aerodynamic surfaces and are a smart use of space.

				The only improvements to be made in this criteria are having a sharper angled rear facing edges, this would further enhance these rear wing attributes.
		Does the system include a brake pedal to deactivate DRS?	2/10	A brake pedal was not included in this design which is an oversight that should be remedied. However, the inclusion of two rear wing pylons was completed with the requirement for wiring separation in mind. Not only do the two pylons add extra support but they create the ability to separate wiring. This means one of the pylons could have all pneumatic wiring and airlines, while the other have the wiring for the brake pedal.
Materials and components		Were only materials permitted in the FIA 2023 technical regulations used?	10/10	This design only included the carbon fibre components of the Model DRS. As such all materials were allowed within the FIA's technical regulations.
		Are the materials chosen strong and suitable for the system? Are they assembled using methods to maximise strength?	9/10	As carbon fibre is the chosen material for this design, it is specifically chosen for its attributes. It is durable, low in density and highly stable as a material. If the processes in construction with the carbon fibre are completed properly and producing a high-quality end result, then the design will be optimal in terms of strength. This is due to a minimisation of weak points through the shape of the parts and the way that they join. The only minor issue is the thickness of the stationary rear wing flap, while being thinner saves weight and creates less drag it is a risk in terms of durability.
Environmental use (use area)		Is the Model DRS able to withstand high aerodynamic forces through strength and durability?	9/10	The strong curvature and thickness of the base frame and supports make the design very capable at handling the large forces intended on being acted upon it. By having the pylons rear the ground and work in conjunction with the wide supports that extend off the rear wing curvature the design is set up to withstand high aerodynamic forces. The angle of the stationary rear wing flap stops force from being pushed directly down on it, instead guiding the air to flow through to the back of the wing. This puts less strain of the most ridged join.

				Comparatively the upper rear wing flap avoided this issue altogether by not only having the correct angle but also though the type of connection between the rear wing flap and the base form of the rear wing. The only issue that would arise in terms of the design's durability is with the actuator housing. This component is very thin and is only joined to the rear wing with minimal surface area. This has the potential to create a durability and reliability problem.
Safety		Is all wiring for the system covered and safe?	8/10	With this actuator cover design all wiring is covered either within the pylons or within the actuator housing. The only consideration is with the transfer of wiring or componentry between the two. The only joining location would have to be though the rear wing flap which creates lots of issues. Not only structurally and in terms of construction but also for wire management and safety.
Quality standards Styling and appearance		Does the system have appealing and aesthetic qualities?	9/10	As this design is within the 2023 Regulations and is inspired by the cars from this year, aesthetically it is very modern. The sleek joins between the upper rear wing flap and the main body form are very smooth, as well and the curvature of the main form being visually appealing. The only issue with the appearance of the design is the two pylons, they are distractive and unbalance the controlled appearance of the system.
Size		Does the Model DRS fit withing the FIA's dimension rules or a scaled alternative?	7/10	Almost all the dimensions are within the FIA's 2023 technical regulations. The only issue is the gap between the two rear wing flaps is larger than the 10-15mm gap that is allowed. For the design to be legal in terms of the regulations this would need to be altered.
Maintenance		Are the components organised and managed properly to allow for easy access and maintenance? This includes wires,	6/10	While there are strong pros and cons surrounding the two pylons, a major positive to their inclusion is the additional wire management. By having two pathways to access the actuator housing and componentry, wires and airlines can be sectioned based on function.

		electronics, and power supply.		Although this design is well thought out with its organisation of electronics and associated componentry, there is an issue with maintenance and access. As previously mentioned there is an issue with connecting components from the pylons to the actuator housing through the rear wing flap. But there is also an issue with accessing all componentry in the pylons as they are very integral to supporting the rear wing.
Production methods		Did the system utilise the most efficient production methods?	6/10	As this design is very specific with its shapes especially with the curvature, it is not the most efficient to manufacture. This could be considered the price to pay for a more effective and durable system, but steps would have to be taken to ensure that the project would be completed on time and to a high quality. The overall form is able to be separated into sections to be moulded for the carbon fibre construction process just like other designs have been intended to do. The only big difference is the joining of the sections is vastly more important with the design due to the complex curves. This creates a longer process in order to achieve the same high-quality result as other more simplistically shaped designs.

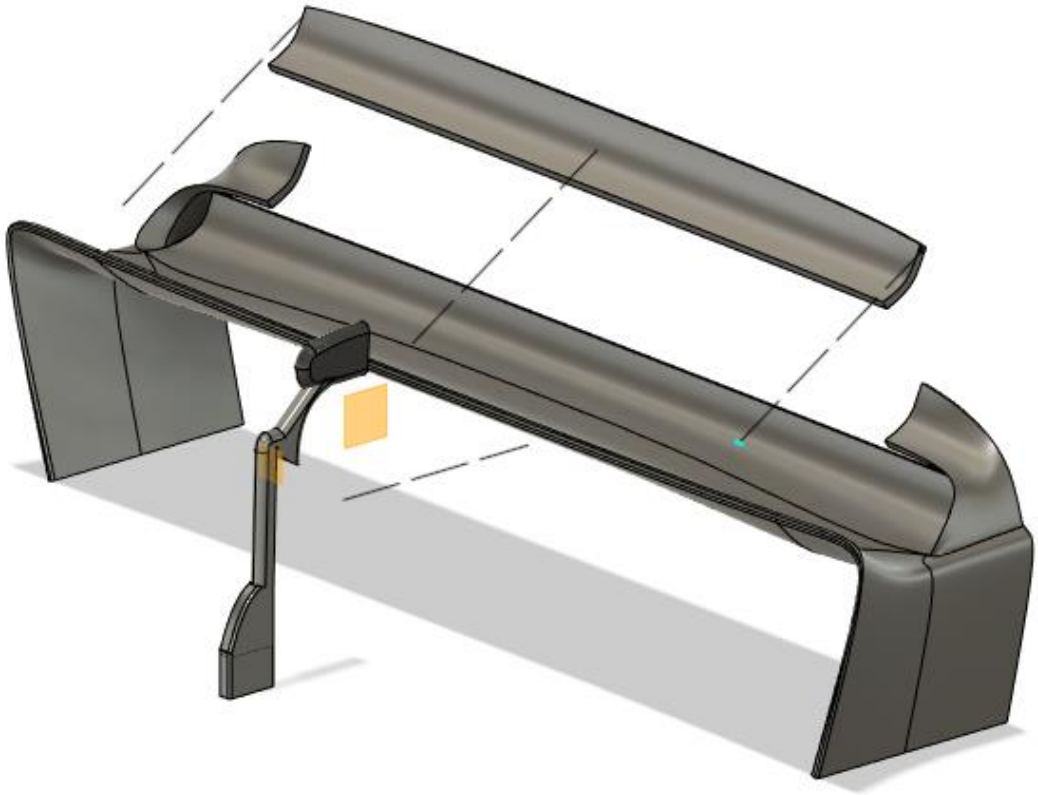
Creation of Preferred Design Option and Justification

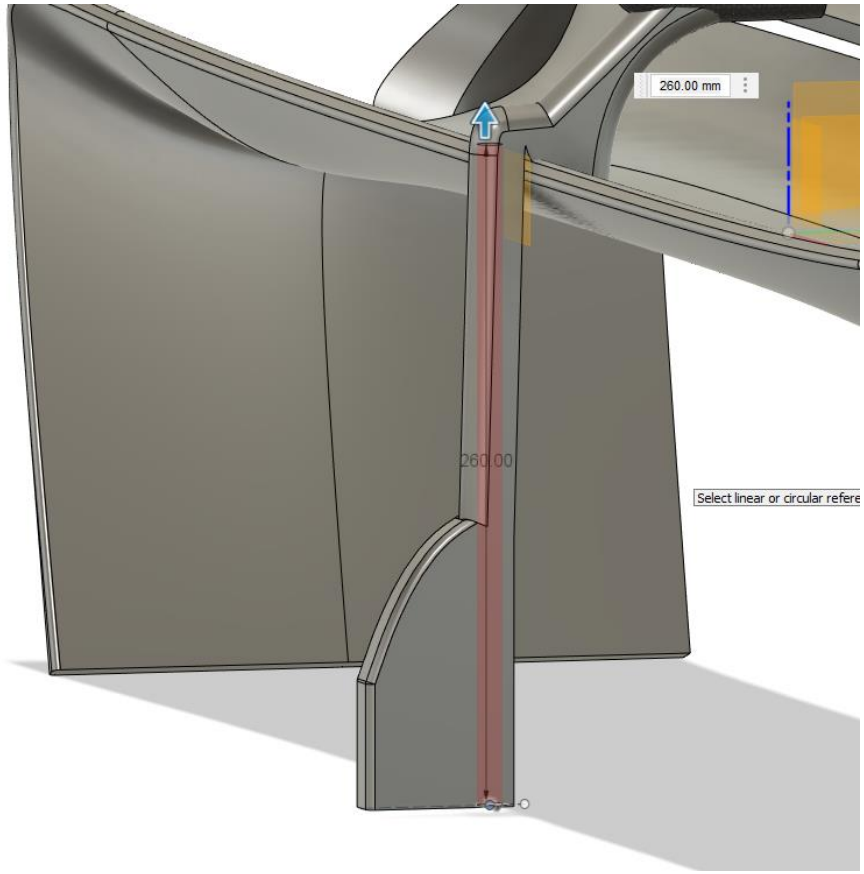
Design Option	Total Score
Option 1	71/110
Option 2	70/110
Option 3	81/110

After evaluating all three design options on the relevant criteria and collecting useful information on the pros and cons of each design a final design was generated. This design includes the main body of the Model DRS, along with the system control design and a CAD model for a brake pedal that could be altered to work with my project. The brake pedal is not my design and was chosen from thinkiverse as a suitable solution to the consideration of having a brake pedal deactivate DRS. As the brake pedal is not vital to the functionality of the system and has a main purpose of aesthetic and interactive purposes. It was logical to choose the most time effective solution to including it in the system, which is to utilise a premade model and fit it out to work within the constraints of my project.

Option 3 was the closest to the intended design of the system, however there were aspects of all design options that were suitable for the Model DRS. As such the new model was created to combine all the best attributes of each initial design to create the best outcome. For example while the curvature and sizing of the model are very much based off of options 1 and 3, the actuator housing is very similar to the one designed in option 2.

The only improvement that would need to be made to the design before it is tested or prototyped would be the accuracy of measurements. By using forms in Fusion 360 and also scaling the model the measurements have become overly specific and complex which is not ideal for constructing the design. When the prototypes and initial iterations are completed more parameters should be used to help control the dimensions of the models.

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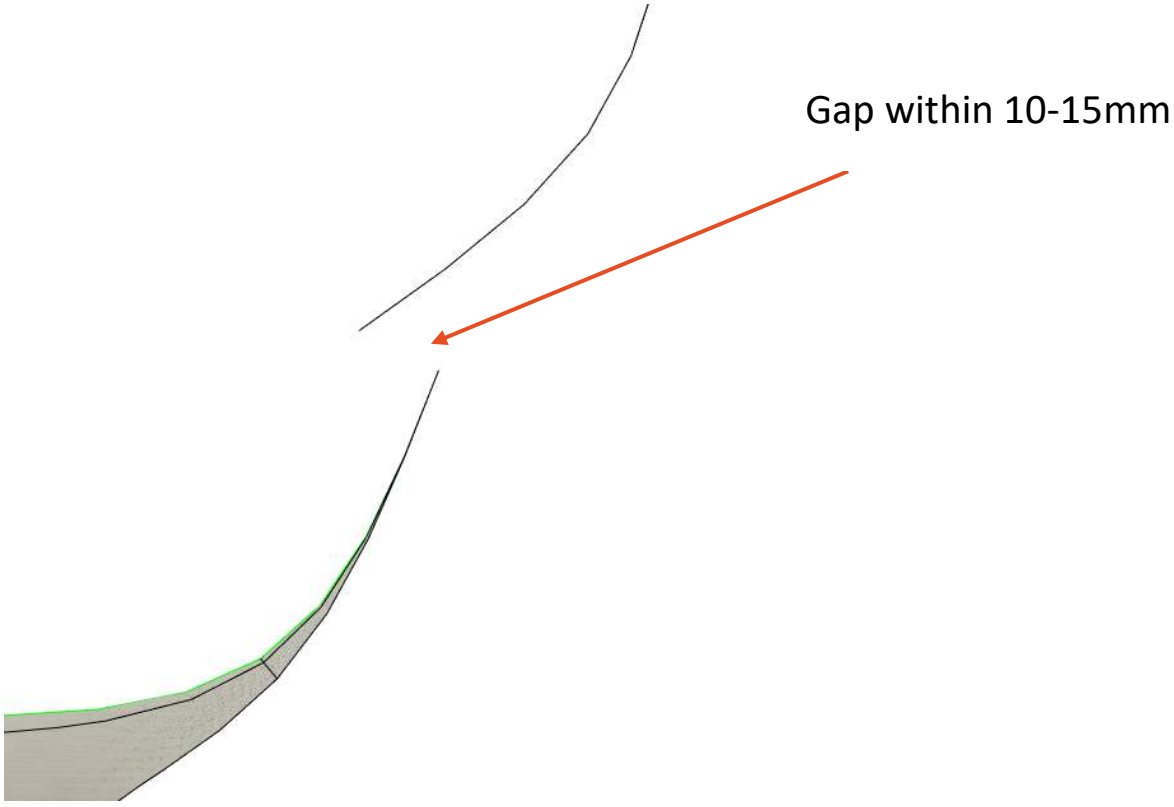
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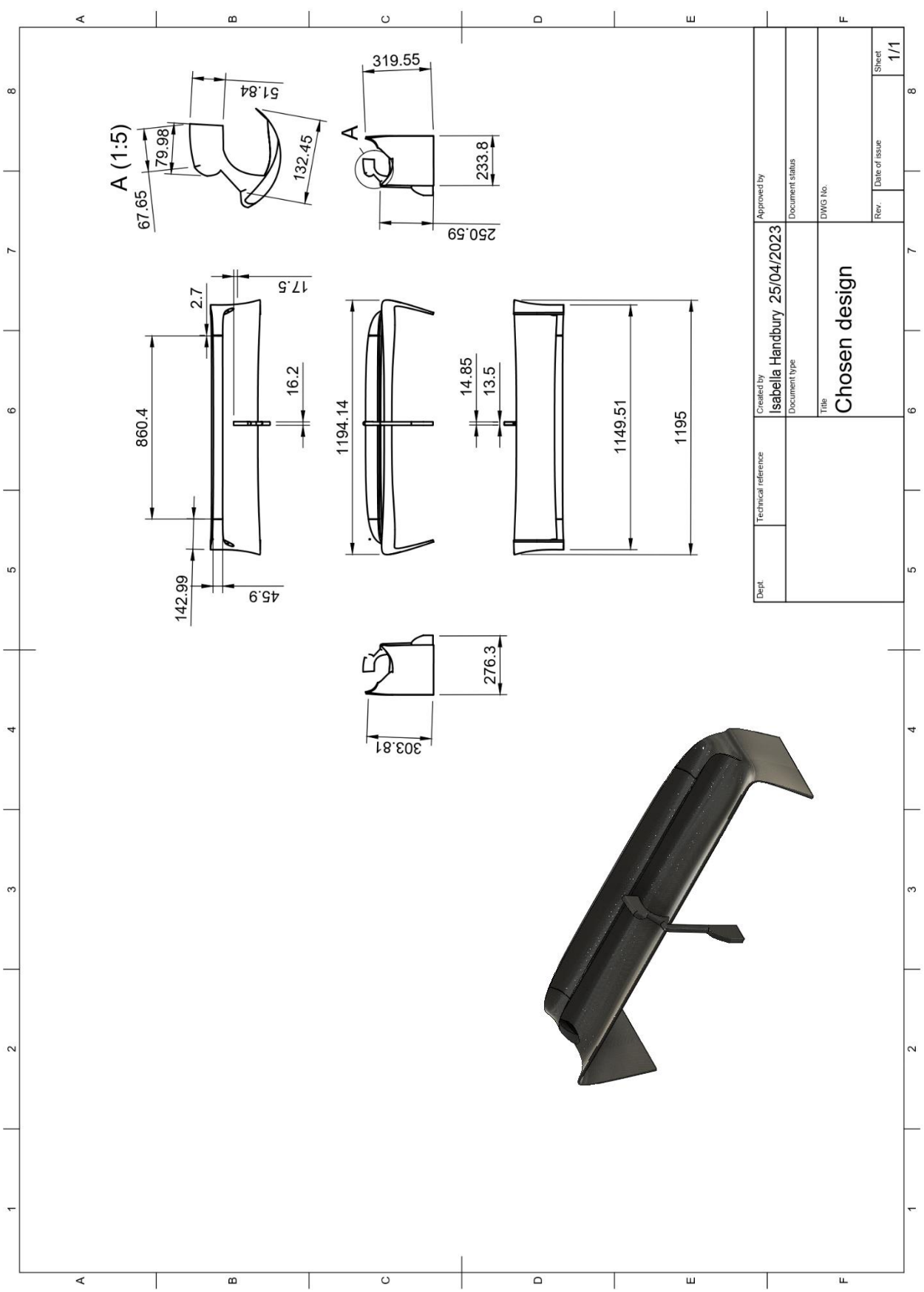
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A
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45.9
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13.5
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1195
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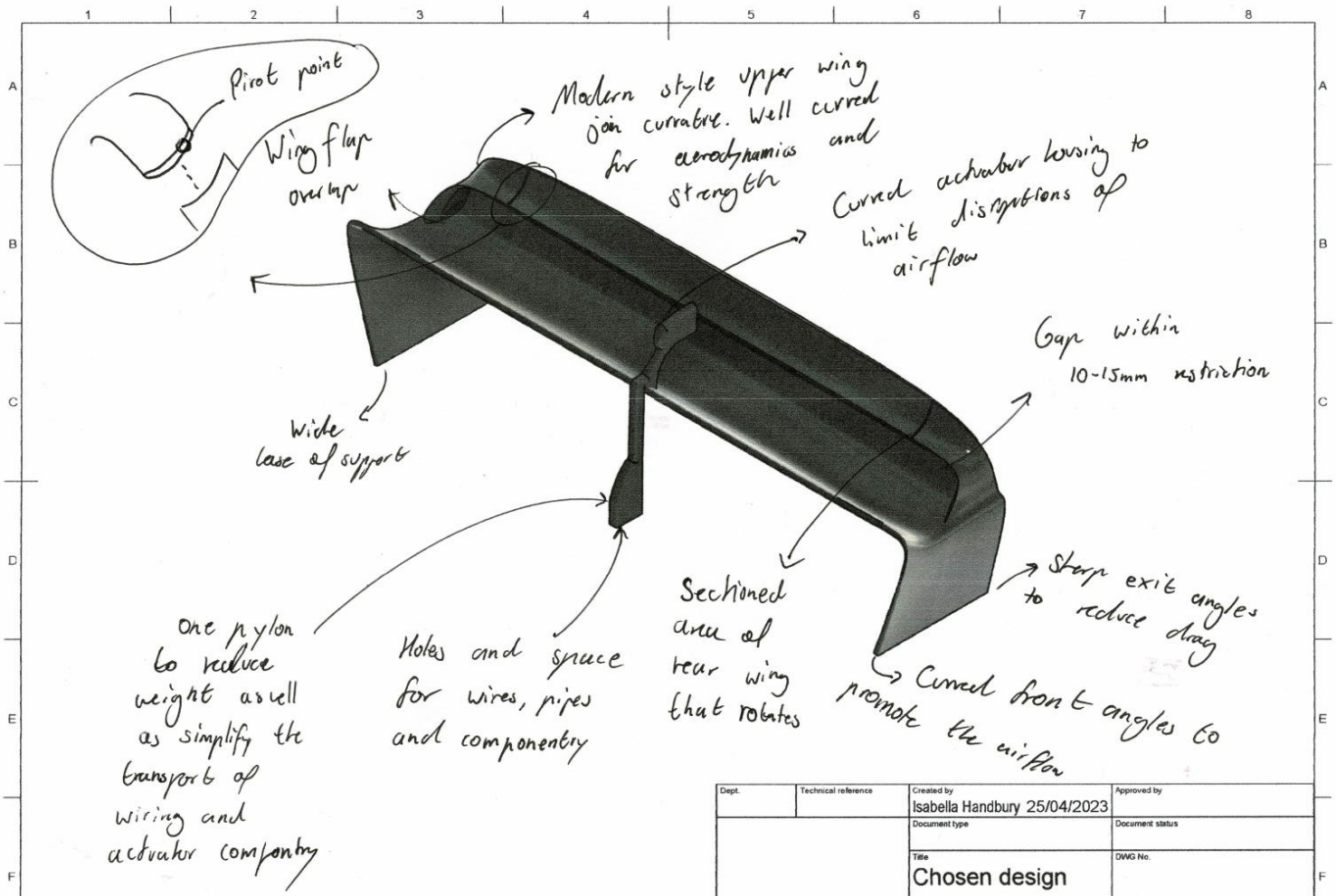
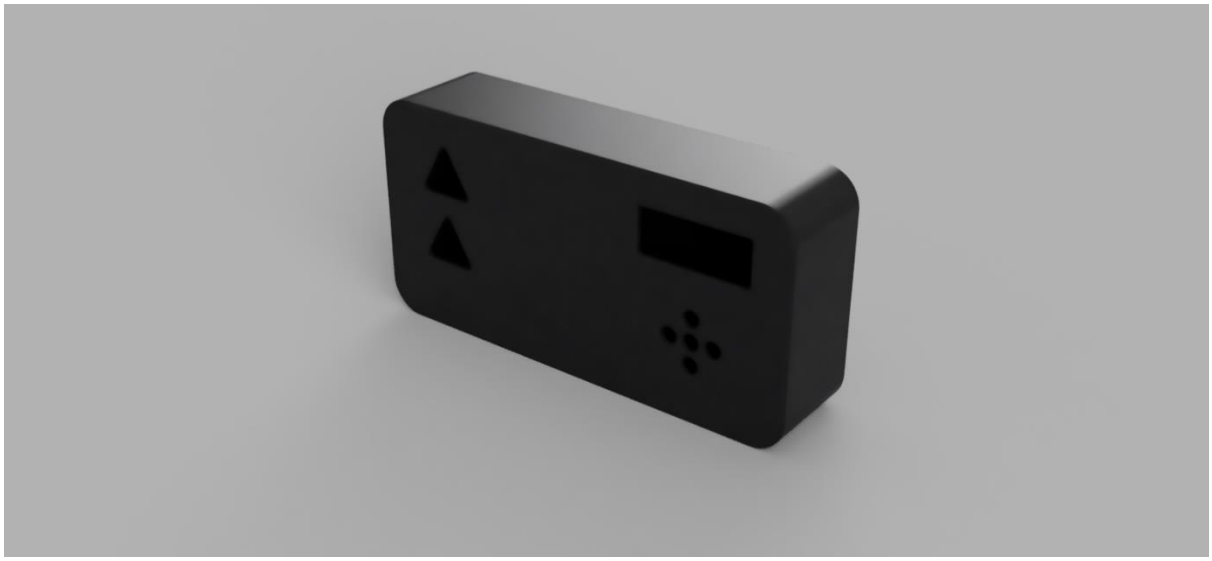
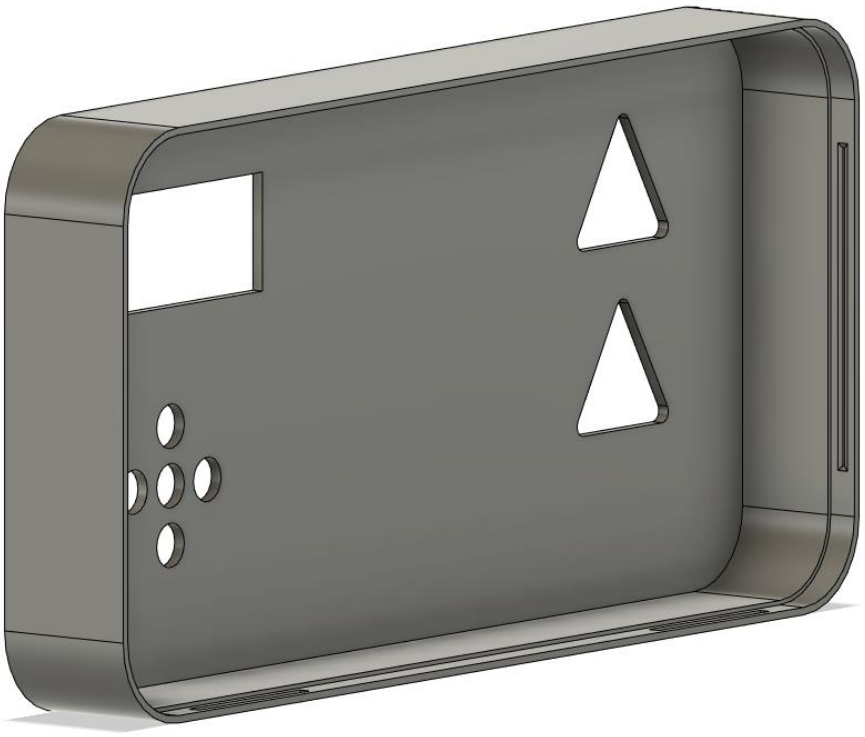
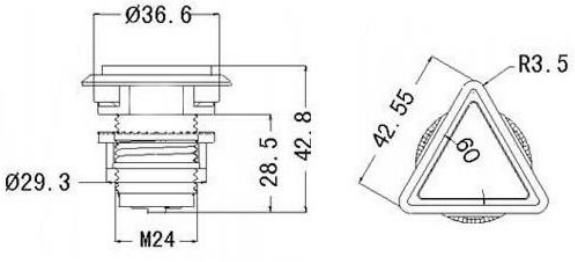
Dept.
Technical reference
Created by
Isabella Handbury 25/04/2023
Document type
Approved by
Document status
DWG No.
Title
Chosen design
Rev.
Date of issue
Sheet
1/1


Pivot point
Wing flap overlap
Modern style upper wing join curvature. Well curved for aerodynamics and strength
Curved actuator housing to limit disruptions of airflow
Gap within 10-15mm restriction
Wide base of support
One pylon to reduce weight as well as simplify the transport of wiring and actuator componentry
Holes and space for wires, pipes and componentry
Sectioned area of rear wing that rotates
Sharp exit angles to reduce drag
Curved front angles to promote the airflow
Dept.
Technical reference
Created by
Isabella Handbury 25/04/2023
Approved by
Document type
Document status
Title
Chosen design
DWG No.
Rev.
Date of issue
Sheet
1/1



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Ø29.3
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R3.5
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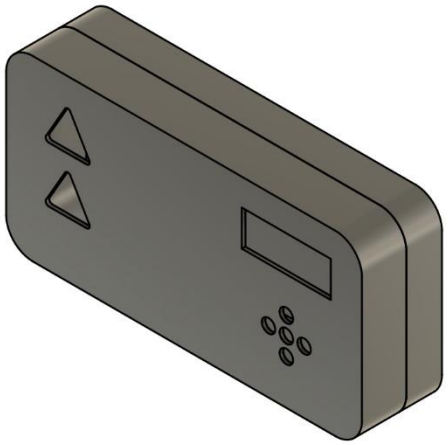
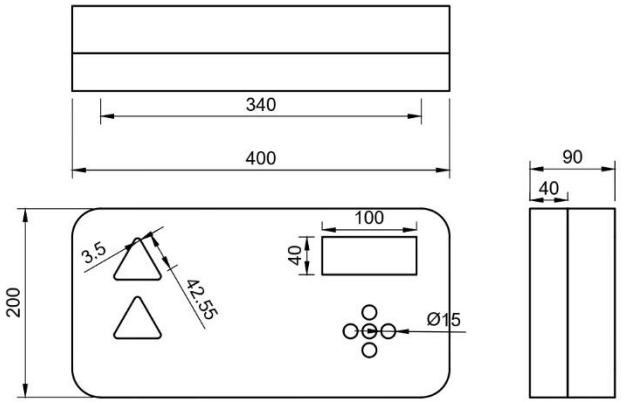
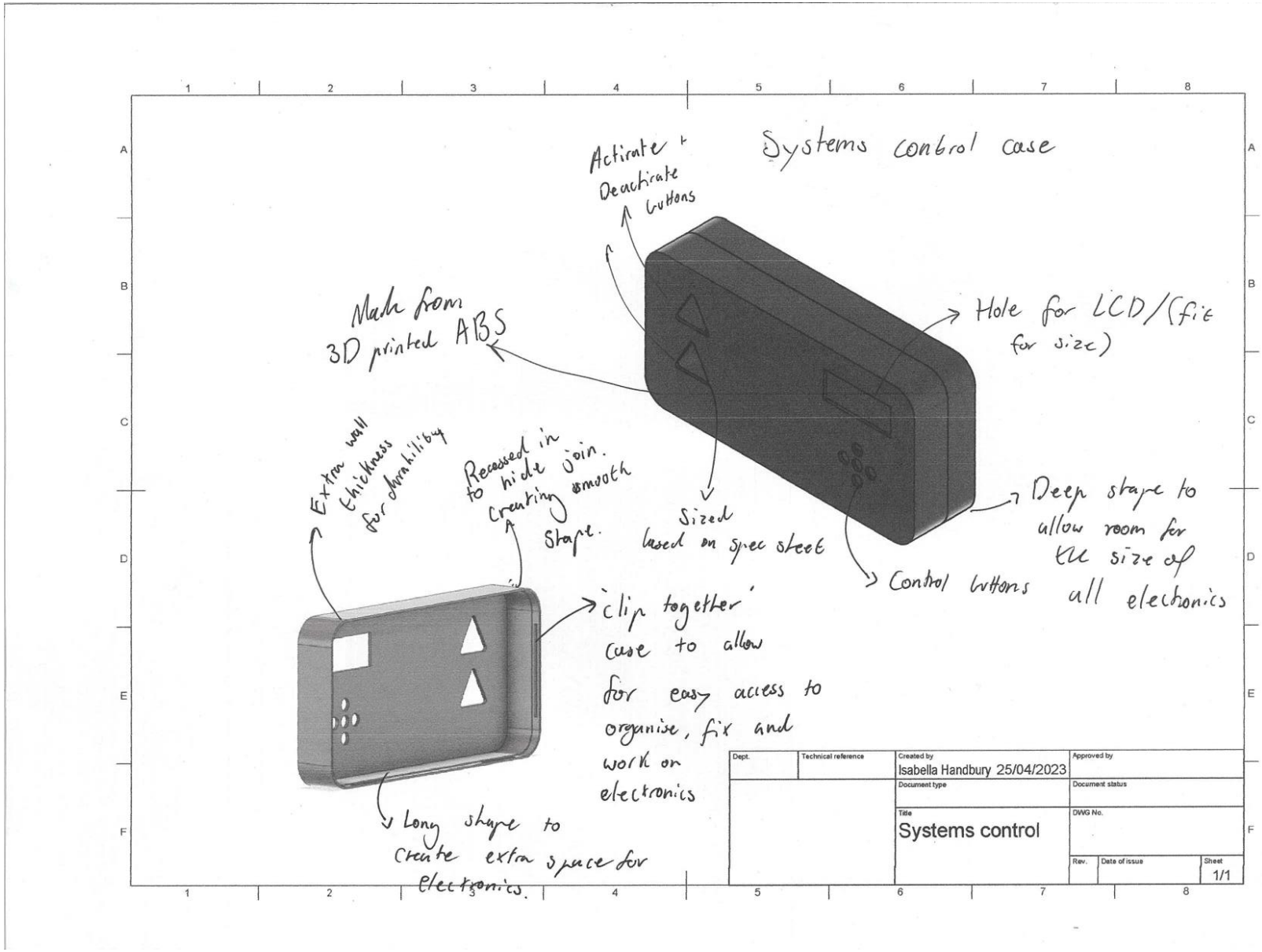
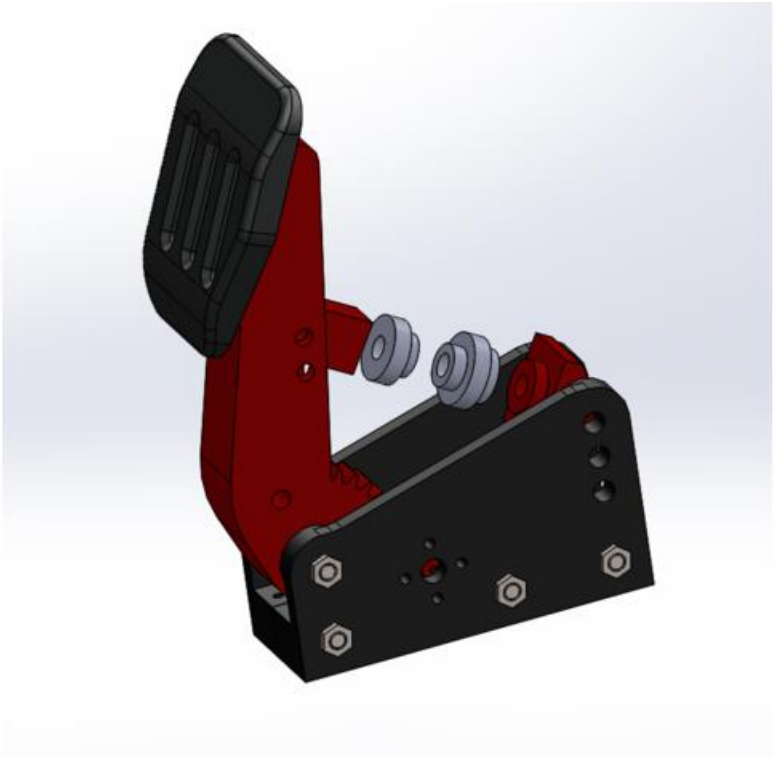




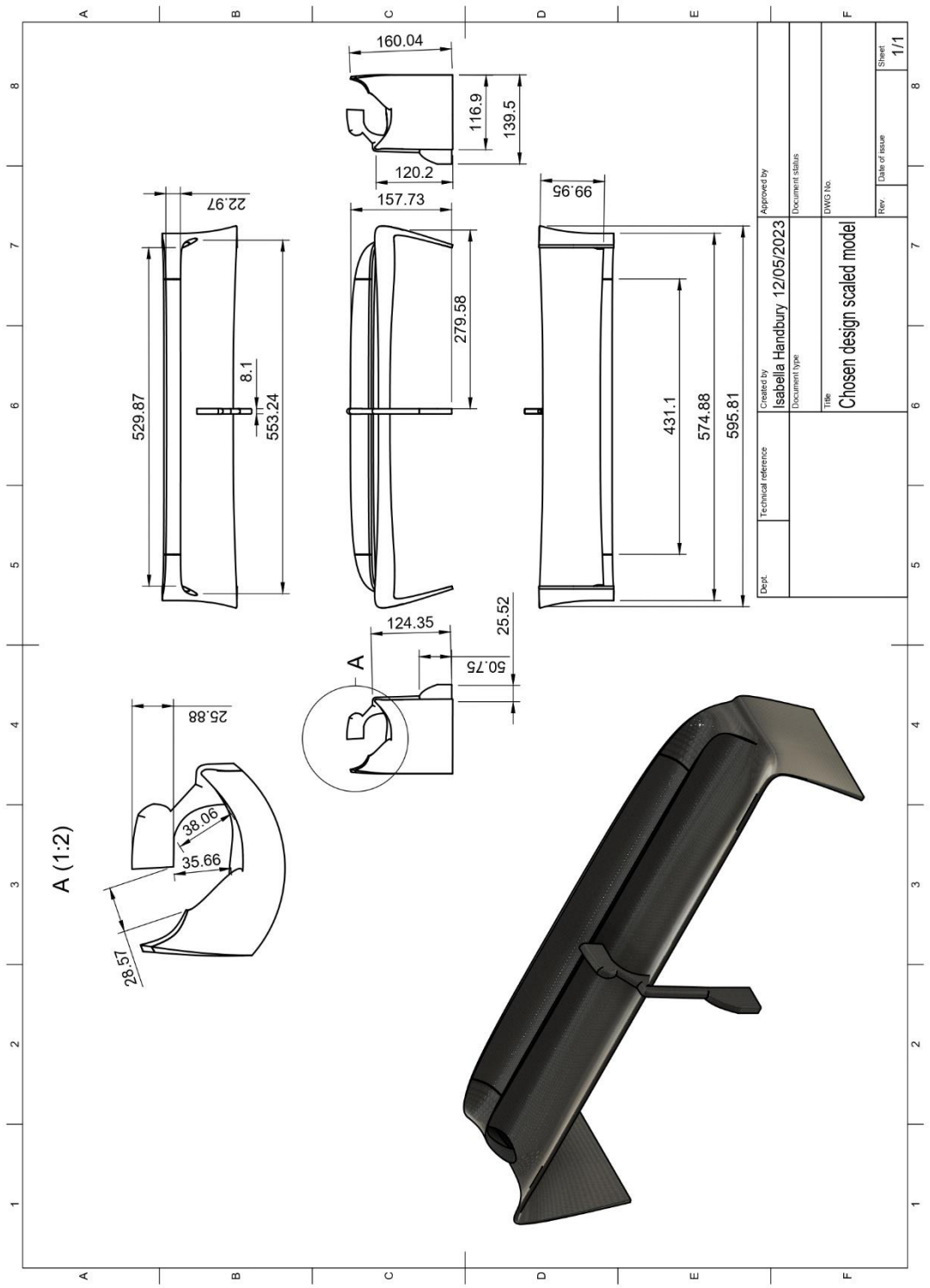
Dept.	Technical reference	Created by Isabella Handbury 25/04/2023	Approved by
		Document type	Document status
		Title Systems control	DWG No.
			Rev. / Date of issue / Sheet 1/1


Systems control case
Activate + Deactivate buttons
Made from 3D printed ABS
Hole for LCD/(fit for size)
Extra wall thickness for durability
Recessed in to hide join. creating smooth shape.
Sized based on spec sheet
Deep shape to allow room for the size of all electronics
Control buttons
'clip together' case to allow for easy access to organise, fix and work on electronics
Long shape to create extra space for electronics.
Dept.
Technical reference
Created by
Isabella Handbury 25/04/2023
Approved by
Document type
Document status
Title
Systems control
DWG No.
Rev.
Date of issue
Sheet
1/1

(Brake Pedal, 2023)



Scaled Dimensions



Criteria 3

Bill of Materials

School sourced	
Personally sourced	
Already available	

PART NUMBER	PART NAME	DESCRIPTION	QUANTITY	PART IMAGE (Link)	UNIT COST	AMOUNT
1	Arduino Uno Rev 3	The Arduino Uno Rev 3 's purpose within this project is to act as the microcontroller that is responsible for all the electronics' functionality. The microcontroller will be responsible for channelling the inputs from the three sensors into information that can be read from the LCD screen. The Arduino is also responsible for the activation and de-activation of the pneumatics through the control of the solenoid with the buttons.	1	https://core-electronics.com.au/arduino-uno-r3.html	$ 45.90	$ 45.90
2	DSN Double Acting Mini Cylinder	The pneumatic cylinder is the major part of the electromechanical system that will be made functional through the control of the pneumatics through the solenoid. The cylinder is double acting, which means that air can enter through both ends of the cylinder. This creates the efficient activation and deactivation of DRS. The actuator in conjunction with a fixed size connector will rotate the upper rear wing flap on its pivots, in order to increase and decrease the gap between the two rear wing flaps. As calculated the specifics of the part are; Bore= 16mm, Stroke=25mm	1	https://pneumaticsolutions.com.au/dsn-double-acting-mini-cylinder.html	$ 79.20	$ 79.20
3	DHT 11 Temperature and Humidity Sensor	The DHT11 sensor will be used to collect data from the surroundings of the Model DRS, simulating the information send between the car and pit wall during an F1 session. The sensor will be connected to the microcontroller which will send the data to the LCD screen.	1	https://core-electronics.com.au/dht11-temperature-and-relative-humidity-sensor-module.html		$ -

4	Gravity BMP388 Barometric Pressure Sensor	The Pressure sensor will be utilised to gain data around the Model DRS and send it to the microcontroller. It will then be displayed on the LCD screen and calculations can be completed based on the data.	1	https://core-electronics.com.au/gravity-bmp388-barometric-pressure-sensors.html	$ 11.85	$ 11.85
5	Load Cell	The Load cell will be used to collect data on the down force that the rear wing is producing. By mounting the sensor under the wing, the data of the change in weight when air is flowing over the wing, when it is open and closed can be gained. This can be displayed to explain the purpose of DRS, but it can also allow for accurate calculations to be completed regarding the efficiency of the DRS.	1	https://core-electronics.com.au/makerverse-20kg-load-cell.html	$ 7.80	$ 7.80
6	LCD	The purpose of the screen is to display the data collected from the three sensors to the user. The specifics of the screen are; 16x2 dimensions	1	https://core-electronics.com.au/1602-character-lcd-black-on-green-hd44780-16x2.html		$ -
7	5/2 Solenoid	The solenoid will be used to control the pneumatic section of the system through the information sent from the microcontroller. Because of the key job that the solenoid plays there are some key custom specification aspects for it to properly integrate into the system. The specifics of the Solenoid are; 4V210 5/2 Solenoid Valve with the 1/4 BSP port size, 24VDC voltage, plug with LED electrical connector.	1	https://pneumaticsolutions.com.au/4v210-5-2-solenoid-valve.html	$ 121.00	$ 121.00
8	Airline Regulator	The airline regulator will be used to monitor the air pressure being sent to the solenoid and limiting it to be within the limit that the parts are rated for. The only specific detail for this part is; 1/4" BSP size	1	https://www.valvesonline.com.au/airline-regulator	$ 24.20	$ 24.20

9	LED Illuminated Triangle Push Button	The push buttons will be used to activate and de-activate the DRS on the model. These specific buttons have been chosen due to their aesthetics and the ease of their wiring and coding. The specifics for ordering the parts are: 1 Blue, 1 Red	2	https://www.adafruit.com/product/4190	$ 3.95	$ 7.90
10	Airline pipe	The airline pipe will be used to connect all the pneumatic parts of the system, sending the air to its required location. The pipe is a Flexible Polyurethane Tube which is only available in 10 metre lots. This means that the specifics for this part are; 10 metre lot, 6mm tube size, colour black.	1	https://pneumaticsolutions.com.au/flexible-polyurethane tube.html	$ 20.46	$ 20.46
11	Carbon Fibre Nylon Filament	The carbon fibre filament that the school has readily available is Esun PA6+CF. This filament will be utilised for the prototype, the pneumatics housing and the systems control case.	1			$ -
12	Carbon Fibre Sheet	The carbon fibre sheet is being used to construct the final iteration of the rear wing bodywork. The parts will be created with two layers of 2mm twill carbon fibre, bonded with the epoxy resin. As determined by the CAD model the rough surface area of the scaled rear wing bodywork is 0.211 m^2 and the currently available dimensions of the carbon fibre sheet it 3.042m^2. This means that there is no need to order more carbon fibre sheet.	1			$ -
13	PVA Release Agent	The PVA release agent is to stop the carbon fibre pieces from sticking to the moulds. The release agent is to be sponged onto the mould to create a barrier for the epoxy composite (epoxy resin).	1	https://carbonparts.com.au/product/release-agent/	$ 11.00	$ 11.00

14	Epoxy Resin part A and B	Epoxy Resin is the ideal bonding agent for carbon fibre wet lay-up manufacturing. The resin comes in two parts, R105 Resin and H207 Clear hardener. These two parts are mixed together following the correct ratio to create the epoxy resin that will be used to bond the carbon fibre pieces together and set them in the shape of the mould. The ratio is 3:1 and the resin has a pot life of 20 minutes. The volume that will be appropriate for the size of the pieces being created is 1.33L of resin.	1	https://carbonparts.com.au/product/west-system-laminating-resin/	$ 99.00	$ 99.00
15	Expanding Foam	The expanding foam is intended to be used for the inside of the carbon fibre pieces. So that the parts are not hollow, expanding foam is an efficient and cost effective solution to this. This material is already available	1	https://www.bunnings.com.au/bostik-750ml-fill-it-expanda-foam_p0089098		
16	High Density Foam	For the carbon fibre moulds a high density foam is required to be able to CNC the shape of the rear wing body work. Clark Rubber have a suitable foam that can be ordered with specific dimensions to account for the size of the project. To account for the size of the carbon fibre parts the specifics should be : Width: 1000mm, Length: 200mm, Thickness: 150mm.	4	https://www.clarkrubber.com.au/products/25039p-high-density-foam-36-130?variant=31435078303855	$ 67.80	$ 271.20
17	Wires	A selection of wires will be required for the electronics of this system. M/M, F/F and F/M jumper wires which will be needed for the bread boarding of the components are already available. The electrical wire for the final assembly of the components is also already readily available. This speeds up the initial stages of production as acquiring specific components could be holding the processes up.		https://www.sparkfun.com/products/12794		$ -
18	Mufflers	The mufflers will be used to help aid the airflow with the solenoid as well as reducing the sound that the component produces. The specifics of the part are: Brass	2	https://pneumaticsolutions.com.au/bslm-brass-threaded-	$ 3.30	$ 6.60

		Treaded Button Silencer, thread size is 1/4" BSP.		button-silencer.html		
19	Solenoid Fittings	These fittings are intended to connect the airline to the solenoid so that air flows to it efficiently. Because these parts need to integrate together smoothly there are clear specifics for the parts. For this part they include; Female Bulkhead Connector, 6mm tube size, 1/4" BSP thread size.	3	https://pneumaticsolutions.com.au/3136-female-bulkhead-connector.html	$ 14.36	$ 43.08
20	Pneumatic Cylinder Elbow Fittings	This fitting is to connect the solenoid to the cylinder. The shape of the fitting is to ease the alignment of the components and to be able to mount the cylinder with ease. The specifics of the part are; Male Stud Elbow fitting, 6mm tube size, 1/4" BSP thread size.	2	https://pneumaticsolutions.com.au/3199-male-stud-elbow.html	$ 7.27	$ 14.54
21	Pivot Pins	Brass rod cut to the size to have as pivot pins. The pivots are to be used to rotate the upper rear wing flap. The brass rods are already available in the required sizing that is approximately 1/16"	2			$ -
22	5V Single Channel Relay Module 10A	The 5V relay module is required to be able to have a functioning solenoid. The part is needed as it acts as a switch to be able to turn the solenoid on and off, due to its large current. Due to the Arduino being a low voltage microcontroller this is required to handle the difference is current size between the solenoid and the microcontroller.	1	https://core-electronics.com.au/5v-single-channel-relay-module-10a.html?gclid=Cj0KCQjw6cKiBhD5ARIsAKXUdyY6Z1illzk3eqn0dTGCKA98q0zCz-xLdmqNF7OCgILHLi_Cjd82BvgaAssBEALw_wcB	$ 4.95	$ 4.95

23	Electrical tape	The electrical tape will be ordered with the intension of aiding the attachment and organisation of the sensors when locating them on the rear wing bodywork. The material will also be used as a precautionary measure in case any of the electronic componentry require it. The materials is also readily available which simplifies this.		https://www.bunnings.com.au/deta-20m-18mm-black-insulation-pvc-tape-5-pack_p0381007		$ -
24	I2C	The I2C is required to ease the wiring and coding of the LCD screen. The part reduces the amount of data pins needed to control the LCD screen. The part has the benefit of helping organise all the wiring and being incredibly compatible with the Arduino framework. This part is readily available so production processes requiring the I2C can begin.		https://core-electronics.com.au/i2c-lcd-backpack-for-1602-to-2004-lcds.html		$ -
25	Resistors	For the buttons to function well and not overwork the microcontroller, resistors are needed to reduce the power loss with the buttons. 2 10K Ohm resistors are already available so they do not need to be ordered.	2	https://www.jaycar.com.au/10k-ohm-0-5-watt-metal-film-resistors-pack-of-8/p/RR0596		
26	HX711 Amplifier	The amplifier is required to enhance the functionality of the load cell. Due to the output of the load cell being so small the HX711 is required to increase the output so that the microcontroller can easily read it. This part increases the accuracy of the data which is vital for the downforce calculations for this project.	1	https://core-electronics.com.au/sparkfun-load-cell-amplifier-hx711-38610.html	$ 18.25	$ 18.25
		TOTAL PARTS	33		**TOTAL**	$ 786.93

Production Plan

Step	Process	Materials	Equipment	Estimated Time
1. Gather materials	Attain all the materials required for the initial stages of production and ensure that all other required materials are ordered and on the way. With all materials have one location to store them to make sure nothing gets lost or misplaced.	All materials available in the workshop, check ordered components	A container	1.5 hours
2. 3D print prototype	Using the predesigned model 3D print a prototype. With the intension of having a physical object to run test and calculations with as well as providing a guide to the physically design. The prototype needs to be printed in two sections. One section for the main bodywork made up of the 'endplate' curves and lower rear wing flap. The second section is the upper rear wing flap that moves. The separation is not only for ease of printing but also to be able to create pivots for the upper wing to rotate on. The model will need additional scaling down to be able to fit on the print bed and not have to be completely sectioned. The filament being used will be Esun PA6+CF which is the high temperature Nylon6- Carbon Fibre that the school has in stock. The printer settings are; Print hot-end temperature: 260-300 C Print bed temperature: 60-80 C Print speed: 40-100 mm/s This stage also includes finishing the prototype properly by sanding it down to create a quality prototype.	Esun PA6+CF Sandpaper Glue	3D printer	1.5 hours (editing model and refining, not including print time)

3. Bread board and code all electronics	Bread board the electronics to get everything working in conjunction with all the other components. First testing that that the electronics are not faulty. This stage includes the setting up the bread board for; - Sensors: DHT11, BMP388, Load Cell (with the HX711 amplifier) - The solenoid (with the 5V relay module) - The LCD Screen (with the i2C) - The two push buttons (with 10K-Ohm resistors) Once all the components are bread boarded and functioning independent to the specific code, then coding should begin. The code should be set up as sheets of individual components code that needs to be working correctly before integrating all the code into one script. To be precise the solenoid, buttons, individual sensors, and LCD screen all need to be functioning independently of each other before combining them into one script. The code should be integrated in stages with the individual sensors being combined in one document prior to fully combining them into the one main script. This is the same for getting the buttons to work with the solenoid and getting the sensors working with the lcd screen. Before putting them all into one script.	Arduino Uno Rev 3, load cell, HD711 Amplifier, DHT11, BMP388, solenoid, I2C, HD44700, 5V Relay, buttons, wires, resistors	Bread board	5 hours
4. Test + calculations	Following the coding of all the components testing should be completed and calculations should be gained in reference to the performance of all the electronics. Tests should be completed on the sensors to check their accuracy and the speed they read and display changes. This also ties in the with calibration of the load cell. Whilst everything is functioning the current through the circuit should be monitored to account for power loss. Temperature increases from overheating should also be tested for and modifications should be made if required.	All electronics up to this stage of production	Thermometer Ammeter Voltmeter 20 kg weight	1 hour

5. Assemble all pneumatics	The pneumatics portion of the electromechanical system is the next stage of production. This stage includes cutting and fitting the airlines to size, connecting the components and the air compressor. The initial step is to measure and cut the airline to size so that the airline reaches from the solenoid to the actuator as well as to the air compressor. Once the airline is cut to size it should be connected to the solenoid, air compressor and double acting pneumatic actuator. The solenoid has 5 ports, one side with 3 ports one side with 2. On the side with 3 ports the solenoid is connected to the air compressor and two mufflers are connected to the other ports. On the other side, the solenoid is connected to the pneumatic actuator. The airline connects to the solenoid with specific solenoid fittings and is connected to the actuator with elbow fittings. Once this is all connected the airline regulator will be connected in series to the airline going from the compressor to the solenoid to monitor and control the flow of air. The pneumatic system should then be able to be controlled by the buttons connected to the microcontroller. As the actuator is double acting it is able to be open and close fast as air can enter at both ends of the cylinder.	Solenoid, actuator, solenoid fittings, mufflers, elbow fittings, airline, airline regulator	Scissors	2 hours
6. Attach pneumatics to 3D printed prototype	To be able to complete the next stage of tests, the pneumatic system needs to be loosely attached to the prototype so data can be collected. As well as being able to test for functionality. The prototype will need to have a variation of the pivot pins inserted into the upper rear wing flap for the it can open and close. The actuator will be loosely attached to a free-standing object instead of the actuator housings, as well as the electronics still being bread boarded. This saves time and allows any changes to be made before the components are printed and wired.	Pivot pins	clamps	1 hour

7. Test + calculations	The data being collected will be the speed the actuator responds to the input from the buttons to open and close. This will provide information on if the system needs improvements, and the specifics of those improvements. The other data will be the ratio of stroke length to the angle the actuator is at so that the gap at DRS activation increases from 15mm to 85mm. Even though the model is a reduced scale of the final iteration the ratio can be transferred across to ease the later construction. It will also be kept in consideration that the speed of activation and deactivation will be different due to the reduced size and weight of the upper rear wing flap.	All componentry up until this stage of production	Stopwatch Voltmeter Ammeter	1 hour
8. Do a trial carbon fibre mould and construction	As manufacturing with carbon fibre is such a complex process it should first be trialled to iron out any initial mistakes and misconceptions. A small mould should be constructed out of foam for the wet lay-up using the CNC router. The mould can be described and the 'negative' of the piece as it is the shape of the externals of the intended final piece. The mould will be two pieces, one for each side of the piece that will be joined together once finished. Once the mould has be created and prepped with the proper release agent a small amount of resin should be mixed with the correct ratio to the hardening agent as well. The resin is time sensitive so this process must be completed within the same time frame. Once ready the first sheet of carbon fibre will be laid over the mould and resin will be carefully applied to the first layer before another carbon fibre layer is added and the same resin process is completed. The shape of the practise piece should be curved so that the proper method to getting the carbon fibre laid over the curves is practised. Once the final layer of resin is added the carbon fibre needs to be left to cure. Once cured the pieces should be taken out of their moulds and any excess carbon fibre cut away. The two pieces need to be joined together using the epoxy resin. The edges of	Epoxy resin, PVA release agent, foam, carbon fibre sheets,	CNC Router Band saw Clamps Scissors Paint brush	3 hours

	the two pieces being joined together should be sanded back and degreased using acetone. The epoxy resin should be the mixed again using the correct ratio and applied to both edges of the pieces. The pieces should then be pressed together and clamped into place, and then left to bond together. Once bonded the join area of the whole piece should be cleaned up and then the practise piece will be completed.			
9. Create carbon fibre moulds	Following the completion of the practise carbon fibre piece, the process of creating the moulds for the final iteration's carbon fibre body can begin. The initial stage is to transfer the CAD file drawings and dimensions to an appropriate format for the CNC router to read the information and to set up the CNC router. The foam needs to divide into sections to be cut, the pieces need to be big enough that they have room around the indent in the foam for some rigidity. There will be 4 sections to create separately which means there will be 8 moulds. Each 'endplate' needs a mould for both sides of the form. The upper and lower rear wing flaps both need 2 moulds each. To speed up the process the automated stages of this production section should be set up to be completed outside of scheduled class time. This creates more time to do the final adjustments to the shaping of the mould if anything needs small minute changes. Once the moulds are created they can be prepped for the carbon fibre to be moulded around them.	Foam	CNC router Bandsaw Hot wires cutters	8 hours
10. Manufacture carbon fibre pieces	With the moulds completed they need to be coated in the proper release agent; the carbon fibre sheets need to be to the correct size for the moulds. Leaving some overhang to be able to properly mauver the carbon fibre sheets into the curves of the moulds. When everything is prepped and there is enough time to fully completed one piece, then the resin can be mixed with the correct ratio. With the epoxy resin being time sensitive, the construction process needs to be completed efficiently. The first layer of the carbon fibre sheet will be laid over the mould	Previously made moulds, carbon fibre sheets, PVA release agent, epoxy resin,	Paint brush Clamps Heat gun	8 hours

	and carefully pressed into the curves and corners of the mould. The resin will be dabbed onto the carbon fibre sheet not painted on in lines, as this effects the thread directions. The next piece will also be added but the twirl needs to be facing in the opposite direction to the previous sheet to increase strength in both directions. The same process is repeated on the second sheet. A heat gun can be used to remove any bubbles from the pieces as well. Then the carbon fibre is left to cure. The curing takes roughly 10 hours. Once the carbon fibre has cured it is released from the mould and then lightly sanded to achieve a smooth high-quality finish. This process is repeated for all moulds, ensuring that the space is left for the pivot point pins.			
11. Assemble Model DRS body including pivot points	Once all carbon fibre pieces are completed and sanded they will be ready for assembly. To join the sides of the 'endplates' they will need to be sanded back and the epoxy resin will need to be added to bond them together. The two sides will need to have expander foam in between them and then be clamped together to properly bond. This process of joining also applies for the upper and lower rear wing flaps, but it also applies for joining the 'endplates' to the lower rear wing flaps. To join the main stationary bodywork of the Model DRS, the edges of both the pieces will need to be sanded back and then joined with the epoxy resin. The pieces will then need to be clamped together and left to cure and bond. For the assembly of the rotating upper rear wing flap, the pivot pins will need to be added to connect the wing flap to the main bodywork. A backup plan of also adding a rod like support all the way through the upper rear wing flap, and the pivot pins should be considered.	Pivot pins, Previously made carbon fibre bodywork, epoxy resin,	Hot glue gun Handheld sander Clamps	1.5 hours
12. 3D print actuator housing	Taking the already designed actuator housing, make any changes that need to be made due to any new discovers throughout the production process. Or any improvements that can be made through the data collected during testing. An example of this could be	Esun PA6+CF filament	3D printer sandpaper	1.5 hours (editing model and refining, not

	changing the angle the actuator sits in the housing or changing how the airline travels through the housing. Using the Esun PA6+CF filament, the actuator housing should be printed lying flat so that 3D printer errors are minimised as well as, so the piece fits on the bed. If any alterations to the design means that the piece doesn't fit it can be printed in sections and joined together. Once the housing is printed it needs to be sanded back to achieve a high-quality finish.			including print time)
13. 3D print systems control case	Using the already prepared systems control case, changes can be made to account for any changes made with the configuration of the componentry following their testing and calculations. The control case will need to be printed in two pieces (one front, one back) to maximise the quality and functionality of the print. The print will be completed using Esun PA6+CF filament to achieve a cohesive look throughout the project. With the 3D printed pieces, they should be sanded once the print is completed to achieve a very high-quality finish. This includes sanding the cut-out holes for the LCD screen, buttons, and wiring.	Esun PA6+CF filament	3D printer sandpaper	1.5 hours (editing model and refining, not including print time)
14. Assemble all components	The assembly of all components will happen in 3 stages; 1. Assembly of all electronics 2. Assembly of pneumatics 3. Integrating electronics and pneumatics into the physical bodywork. 1.. The electronics will need to be wired to each other and properly connected to the microcontroller, essentially removing the bread board. The electronics will also need to be fitted into the systems control unit and properly organised. Heat shrink will need to be used to ensure the electronics are as neat and organised as possible. This stage also includes fitting the sensors to the correct locations on the physical bodywork and if needed covering the wiring with electrical tape. 2.. The assembly of the pneumatics will be very similar to the process of the electronics	All components worked on up until this stage, solder, heat shrink, electrical tape	Soldering iron Heat gun	3 hours

	assembly. The pneumatics will be properly fitted to the pneumatics housing and all wiring and airline will be properly organised. This also includes wiring the solenoid to the systems control case. 3.. With both the electronics and the pneumatics set up ,it will be time to begin the overarching assembly of all components. The main aspect of this is attaching the actuator to the upper rear wing flap and ensuring that the actuator housing stands in the correct locations relative to the body work. Once these three stages are complete and ensured that everything is finished to the highest of quality, the Model DRS will be completed.			
15. Test + calculations	With the completed project a wide range of tests will be run to determine its functionality and its completion relative to the evaluation criteria. The main test will include; - Speed of actuator opening and closing - Material flex (strength test) - The accuracy of sensor readings - The overall aerodynamics tested by mapping out the flow of air over and around the rear wing - The reduction in drag with the load cell results	Completed project	Thermometer Voltmeter Ammeter Stopwatch Weights Pressurised air	1.5 hours
16. Extra allowance of time	This extra time allows for missed production lessons and to a lot for production methods taking longer than expected. By scheduling the project to finished ahead of schedule a high quality end result will be produced by the due date.			

Gantt Chart

Steps	
1. Gather materials	2. 3D print prototype
3. Bread board and code all electronics	4. Test + calculations
5. Assemble all pneumatics	6. Attach pneumatics to 3D printed prototype
7. Test + calculations	8. Do a trial carbon fibre mould and construction
9. Create carbon fibre moulds	10. Manufacture carbon fibre piece
11. Assemble Model DRS body including pivot points	12. 3D print actuator housing
13. 3D print systems control case	14. Assemble all components
15. Test + calculations	16. Extra allowance of time

Steps	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12
1												
2												
3												
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Risk Assessment

Tool	Use (within the context of this project)	Risk	Hazard Consequence	Risk Probability	Risk Rating	Control Measure
3D Printer	The 3D printer will be used for a variety of purposes with this project. For a range of prototyping, tester mould creation and for the electronic housing. The 3D printer will be prototyping the body work of the rear wing for testing and to assess any changes that need to be made for the final model. The printer will also be used for some of the small carbon fibre moulds to speed up the testing process of manufacturing with carbon fibre. The main manufacturing usage for the 3D printer is to create the housing for the systems control unit and the pneumatic controls.	Entanglement in moving parts	Minor	Possible	Medium	All required safety guarding and constraints should be placed on the 3D printer to protect from the internal moving parts. The work and tool should not be touched when they are in use. Instead, they need to be turned off before removing the work from the bed.
		Inhalation of fumes/ particulate	Moderate	Likely	High	The 3D printers should be located in a well-ventilated room that is labelled with the required warnings for fume inhalation. Time in the 3D printer area should be minimised and if required a mask should be worn to protect from fumes.
		Electric shock	Catastrophic	Unlikely	High	No electrical tool should be used without checking it has been recently assessed for safe use. Prior to use tools should be visually inspected by the

						user to check for faulty wiring or noticeable damage. Any faulty tools should be reported and not be used. For the 3D printer the wires should be well organised and kept away from the moving mechanisms, along with keeping them away from any heat. This aims to minimise the probability of damaging the electric components.
		Burns	Insignificant	Likely	Medium	The proper guards should be placed on the 3D printer to stop contact with hot parts of the tool. Additionally, the work should not be removed from the bed until it has fully cooled.
Soldering iron	The soldering iron will be used for all the electronic wiring and joining, for all electronic componentry. The soldering iron will be used to heat up the solder to join the wires to all components and the microcontroller.	Inhalation of fumes/ particulate	Major	Likely	Extreme	A high quality and effective fume extractor should be used at all times when working with the soldering iron. Other people in the workshop should be removed from the area the fumes are located in to remove their risk of fume inhalation.

		Burns	Minor	Likely	Medium	To minimise the probability of burns the soldering iron should always be held correctly, with the silicon heat protector. Another method of controlling the risk is by implementing stands to hold the work. Removing the need for fingers to be near the soldering iron.
		Electric shock	Catastrophic	Unlikely	High	No electrical tool should be used without checking it has been recently assessed for safe use. Prior to use tools should be visually inspected by the user to check for faulty wiring or noticeable damage. Any faulty tools should be reported and not be used. In the case of the soldering iron, the heat should be kept away from all wiring to prevent damaging the wires leading to an increased risk of electric shock.
Heat gun	The heat gun will be used in the context of this project to heat up the heat shrink to aid in the organisation of the wires. It will also be used in the carbon	Electric shock	Catastrophic	Unlikely	High	No electrical tool should be used without checking it has been recently assessed for safe use. Prior to use tools

	fibre process to remove any air bubbles to enhance the quality of the final project.					should be visually inspected by the user to check for faulty wiring or noticeable damage. Any faulty tools should be reported and not be used. For the heat gun the heat should always be aimed away from any wiring and should always be turned off when set to rest. This limits the possibility of burning the power cord which could result in electric based issues.
		Burns	Minor	Likely	Medium	Set the temperature correctly and used caution when working with this tool. Keep hands clear of the work piece and the direction the gun is facing. Wear the appropriate PPE to minimise exposed skin.
		Inhalation of fumes/ particulate	Major	Likely	Extreme	Ensure the area of use is properly ventilated and that others are aware of the fumes being emitted. Wear the properly rated mask to protect against the specific fumes.

Epoxy resin	The epoxy resin is included in this list even through it is not a tool as there are factor that need to be considered to minimise risk. The epoxy resin will be used in the carbon fibre construction process in order to join the carbon fibre sheets together.	Skin contact (irritation)	Minor	Possible	Medium	Use an additional tool to mix or apply the resin e.g., a brush to keep hands clear of the substance. Wear the required PPE, in this case minimising exposed skin is the key to reducing the likelihood of skin irritation from the resin.
		Inhalation of fumes/ particulate	Minor	Possible	Medium	Mix and use the epoxy in a well-ventilated area, alert others to the hazard. Wear the correctly rated mask for the specific fumes that the resin emits.
		Burns	Minor	Unlikely	Low	Carefully read and closely adhere to the instructions of the mix ratios and times for the resin. Along with the warning time before the material becomes ineffective and hazardous.
Orbital sander	The orbital sander will be used to clean up the carbon fibre pieces. The sander will create a high-quality finish by removing any uneven resin layers or any rough spots created from removing the pieces from their moulds.	Cutting, laceration, entrapment	Major	Possible	High	Keep hands clear of the moving parts at all times, ensure the sander has fully stopped before packing it away or placing it down. Limit any loose clothing, including shirt sleeves and ties when using

						this tool to limit the risk of entrapment. This also includes tying up long hair.
		Inhalation of fumes/particulate	Major	Likely	Extreme	Ensure that the tool is used in a well-ventilated room with exhaust fans, and warn other people in the workspace. Wear the correctly rated mask for the fumes or particulate being expelled.
		Excessive noise	Minor	Almost certain	High	Warn others before using the tool and encourage the wearing of PPE in this case earmuffs. Limit the time spent on this specific tool to protect your ears.
		Kickback	Moderate	Possible	Medium	Utilise a correctly set up worktable or method to help hold the piece in one place. Wear safety glasses to protect eyes, but mainly just be aware of this risk and take care when operating this tool.
		Electric shock	Catastrophic	Unlikely	High	No electrical tool should be used without checking it has been recently assessed for safe use. Prior to use tools should be visually inspected by the user to check for

						faulty wiring or noticeable damage. Any faulty tools should be reported and not be used. With the orbital sander the power cable should be kept a safe distance away from the moving parts to reduce the probability of damaging the cord.
Band saw	The band saw will be utilised to shape the high-density foam that will be used for the moulds. The band saw will cut major pieces to size as well as gives the pieces a general shape.	Cuts and lacerations	Major	Possible	High	Keep hands and all body parts away from the moving saw blade during operation. When appropriate use a push stick or guide rail to minimise probability of the risk.
		Excessive noise	Minor	Almost certain	High	Wear the correct PPE and encourage others around the source of noise to either also wea the correct PPE or relocate. Where possible limit the time spent on the tool to minimise the risk.
		Inhalation of fumes/ particulate	Major	Possible	High	Ensure that the room is well ventilated and that the extraction systems are turned on. Where appropriate wear the correctly rated mask for the

						fumes or particulate.
CNC router	The CNC router has the intended purpose of shaping indentations in the foam. The CNC machine is accurate and efficient which makes it a very valuable tool in the manufacturing of the foam for the moulds.	Entanglement	Major	Possible	High	Make sure that hair, loose clothing, particularly ties or jewellery are kept away from the moving parts when in use. Aprons are an appropriate form of PPE to secure loose clothing and should be worn.
		Cuts and lacerations	Major	Possible	High	Ensure that hands are kept away from the moving parts during use. As well as checking that the work is correctly secured before commencing the operation.
		Electric shock	Catastrophic	Unlikely	High	No electrical tool should be used without checking it has been recently assessed for safe use. Prior to use tools should be visually inspected by the user to check for faulty wiring or noticeable damage. Any faulty tools should be reported and not be used. In the case of the CNC router the cables should be

						well organised and kept out of the way of the moving parts. This aims to limit the possibility of damaging the wires.
		Excessive noise	Minor	Almost certain	High	Wear the appropriate PPE of earmuffs and encourage others in the area near the noise to either relocate or also wear correct PPE. Limit time spent on this tool to minimise the risk.
		Inhalation of fumes/ particulate	Major	Likely	Extreme	Ensure that the extraction system is turned on and functioning correctly and that the tool is located in a well-ventilated room. Limit the amount of people around the tool when in operation and where appropriate wear a properly rate mask.
		Flying debris	Minor	Almost certain	High	Wear safety glasses to protect eyes from flying debris and ensure that the extractor is efficiently collecting the offcuts of the work.
Hot wire cutters	The hot wire cutters have the purpose of an additional shaping tool for the foam. Unlike the CNC and band saw the hot wire cutters can be used	Inhalation of fumes/ particulate	Major	Likely	Extreme	Ensure that the material being cut is safe within reason in terms of fume production. Ensure all cutting is done away from

	for small and specific foam shaping. Having a wide variety of tools for shaping the foam creates a variety of efficient methods for completing specific processes. This aids in the production of a high-quality end result.					others and in a well-ventilated room with and extraction system. A properly rated mask must be worn to minimize the risk.
		Burns	Moderate	Possible	Medium	Keep hands away from the hot wire at all times and ensure it is fully cooled down before packing away or adjusting the wire. If required wear a long sleeve shirt or jacket to limit exposed skin.
		Electric shock	Catastrophic	Unlikely	High	No electrical tool should be used without checking it has been recently assessed for safe use. Prior to use tools should be visually inspected by the user to check for faulty wiring or noticeable damage. Any faulty tools should be reported and not be used. For the hot wire cutters, the heat should always be kept away from any cord. As well always waiting for the tool to properly cool before storing it away.
		Eye irritant	Minor	Possible	Medium	Ensure that vapour is being correctly extracted and that

						the room is well ventilated. Wear the appropriate PPE of safety glasses to minimise the risk of eye irritation.
Handling materials	General transporting of materials around the workshops and handling materials while using the specified tools.	Muscle irritation from incorrect carrying of materials	Minor	Almost certain	High	Ensure that heavy materials are being carried with proper technique and assistance is gained when needed to carry materials.
		Cuts or abrasions	Insignificant	Likely	Medium	When handling abrasive or rough materials wear gloves. Ensure the materials are always finished to remove any sharp edges.
Workshop practises	General approach to being cautious around the workshop. Being sensible and safe.	Tripping or collision with others	Likely	Minor	Medium	Being careful and responsible in the workshop environment in conjunction with having a tidy workplace can reduce the risk of small workshop accidents.

MSDS

https://billanookviceduau-my.sharepoint.com/personal/isabhand_billanook_vic_edu_au/Documents/2023/Systems/MSDS

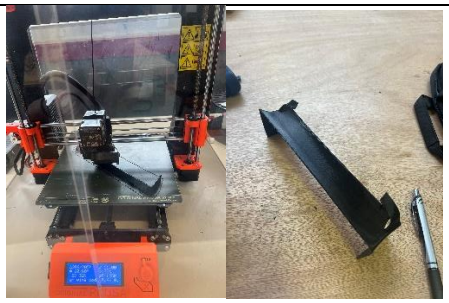
Criteria 4 - 8

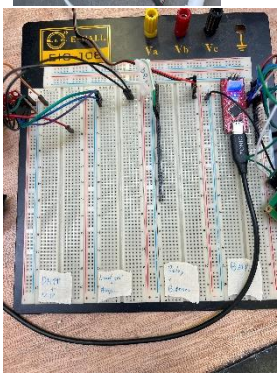
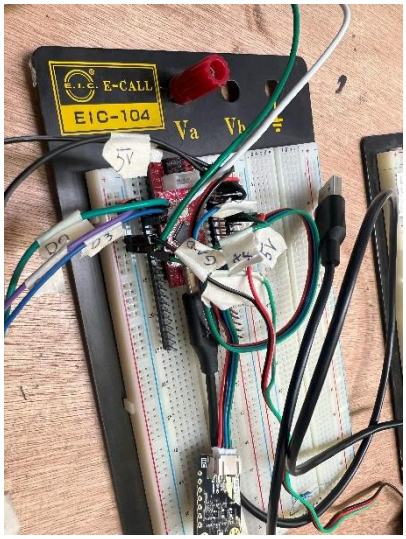
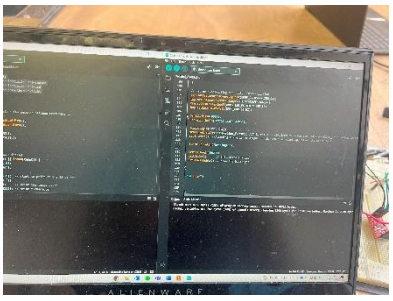
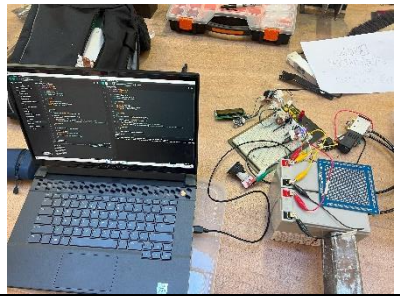
Production Journal

The actual progression of step within the construction of the system were significantly varied, however the journal remains in order based on the plan. The changes in order of occurrence can be reviewed in the Gantt chart and explanations for the alterations can be found in the "Design Iteration and Improvement (Progressive Testing)" section.

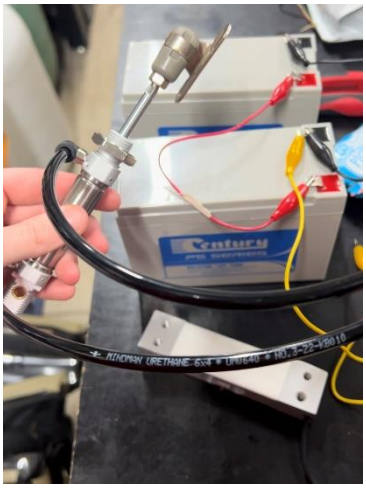
The removal of the testing and calculations stages was undertaken for two reasons. The first being that having rigid dates for when testing needed to occur was not conducive with random testing needing to occur to trouble shoot, or tests being conjoined or delayed providing more beneficial or accurate results. The second reason is due to the realisation of the time constant of the project brought about by large changes in the manufacturing process. By removing this step within the journal to save time and instead having testing be completed when relevant to the design as explored in the "Design Iteration and Improvement (Progressive Testing)" section as well as in the "Testing" section. More succinct and relevant testing was able to be completed.

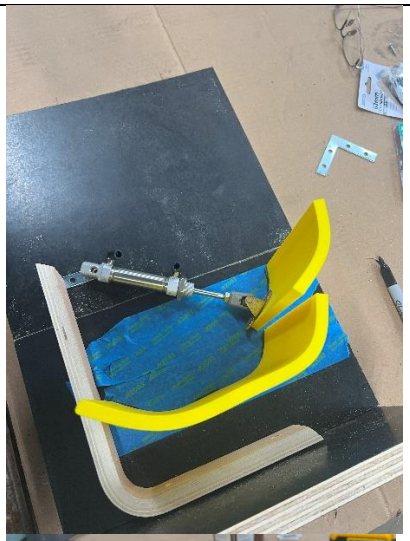
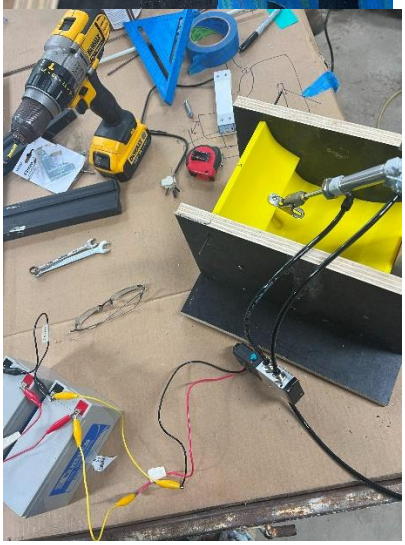
Step	Changes/ Altered Process	Image
1. Gather materials	This process took longer than expected due to issues with the parts ordered from Pneumatic Solutions. The solenoid and cylinder were correct however there were issues with the fittings. The solenoid didn't have a spec sheet which meant that the mufflers were ordered at 1/4 BSP instead of 1/8. The other fittings were also ordered incorrectly with the solenoid fittings needing to be changed for male 1/8 fittings and the cylinder fittings needing to be changed to 1/8 push fit male swivel elbow fittings. While this was easily resolved as highlighted in the correspondence following the production journal, it did slow down this stage of production.	

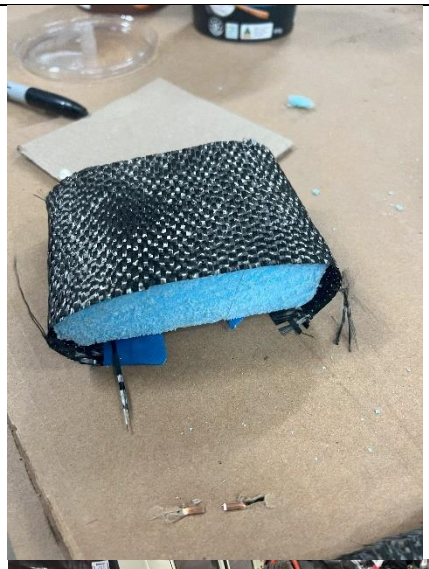
	The electronics were simple to acquire and just took time to be shipped which was accounted for.	
2. 3D print prototype	This process was delayed due to some small issues with the 3D printer. However, this did not majorly impact the timeline of the project due to the 3D printing being able to be completed in the background. To fit the print bed, the prototype was printed at 41.8% of the already scaled model. While this made the prototype too small to properly attached the pneumatics and electronics, it provided a useful visual aid for the following stages of production.	

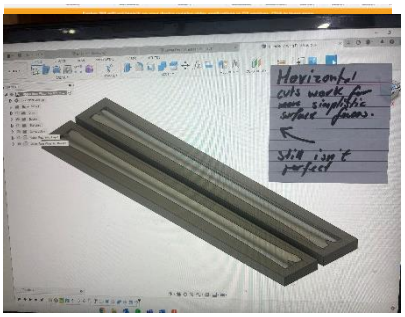
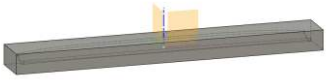
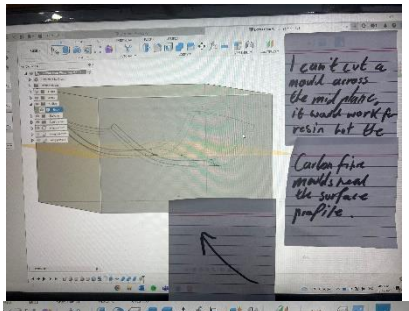
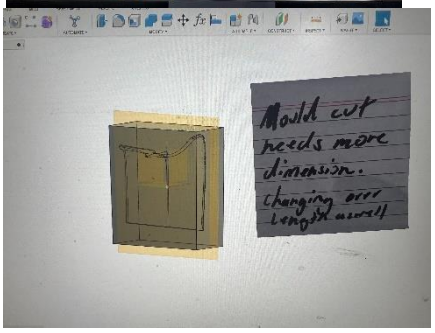
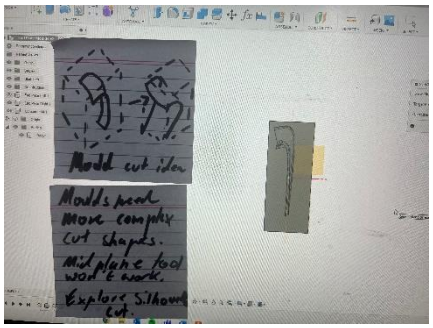
3. Bread board and code all electronics	Various alterations to this process were carried out, and the coding of all the electronics took a significant amount of time compared to what was allotted for. This resulted in other processes being delayed and alternative methods of manufacturing the project needing to be carried out. The initial step to breadboard and code all the electronics was as follows; Solder the headers pins to the microcontroller, connect the I2C to the LCD screen and allocate pins to each component. The first component that I breadboarded and coded was the LCD screen along with the DHT11 temperature and humidity sensor as they are components I am familiar with and was able to complete this step efficiently. Once this step was completed the next progression was to begin the wiring and coding for all the other components. To keep everything organised and to be able to effectively troubleshoot any issues that arose, I separated each component onto its own bread board and had individual sketches for each input. Once all the components were functioning independently the next step was to integrate them physically with wiring then test their functionality again. I worked through each individual step to ensure that the wiring was correct prior to integrating the code together. This step took a large amount of time as thoroughly checking each component's functionality and troubleshooting any small issues such as the relay being on the wrong pin and the load cell	    

	amplifier being connected with a faulty wire. Once I was sure that all the physical components were correctly wired and labelled to minimize confusion and optimise time management through limiting time spent relearning the wiring setup, I began slowly integrating the code through defined float return functions. This method allowed me to organise the code as well as having an easily readable setting out to easily trouble shoot. Following each new sketch that was integrated I ran the code to, test for any issues and resolve any problems. During this time I made the decision to remove the DHT11 temperature sensor from the electronics as the BMP388 sensor detects the same data points along with the addition of altitude. It was impractical to have two sensors collecting the same data as they were not being compared to check accuracy, so the DHT11 was removed. This decision also simplified the coding and later on would simplify the wiring process. Once I had integrated the sensor code into the compiled sketch to work with the LCD screen, intending on being able to display all 4 data types, I moved on to integrating the relay and button code. This is where many issues arose, due to the code within the void loop including the 4 data points from the sensors that rotated their time being displayed through a delay. The button could not be integrated into the code without being affected by the delays. A contingency plan was devised to utilise two micro controllers. One with the sensor	

	and LCD screen code, and the other with the relay and button code. Before falling back on this contingency plan, I explored using 'Millis' to create delays in the code without interrupting other function. While this was a feasible solution, I failed to allow the time to properly make it work and made the decision to fall back on my contingency plan to move on with other stages of construction.	
4. Assemble all pneumatics	Once the components are breadboarded and the solenoid is coded to work alongside the 5V relay the pneumatics could be assembled. The first stage of this was to connect two 12V batteries in series in order to produce the 24V DC that the solenoid was rated for. This then connected directly to the battery to check if it worked before later needing to be connected through the relay. With this set up connecting the solenoid and actuator with the air tube and associated connected was a simple process. The solenoid was then connected to the air compressor, joined through a valve that allows for the airflow to be turned on and off without disrupting the compressor.	

5. Attach pneumatics to 3D printed prototype	This stage of production was drastically altered compared to what the initial intension was. Initially the idea was to 3D print a large prototype at school of the rear wing that would be tested to ensure that all pneumatics were working according to the design. Due to the coding and electronics taking so long, along with the 3D printed prototype being very small due to the bed size limitations of the school printer, attaching the pneumatics to this protype would have proven to be pointless and would have provided no significant data or evidence of any alterations that needed to be made. As a result the small 3D printed prototype was used to visually assess the appearance of the wing and was a very useful tool in visualising the future assembly of the rear wing componentry. The intention behind this step regarding checking the fit of the actuator on an initial iteration of the design prior to connected it to the final carbon fibre version was later realised by testing the pneumatics on the 3D printed sections of the rear wings. With the design shift to have the pneumatics attached to a 150mm sectioned display of the rear wing this stage was able to be carried out on the first iteration of that piece. The actuator was visually and physically measured against the location of the rear wing flaps in order to correctly angle it within the restrictions of its max and min stroke lengths. The end of the actuator rod was connected to a ball and socket type of connector that following the actuators connection to the actuator stand,	 

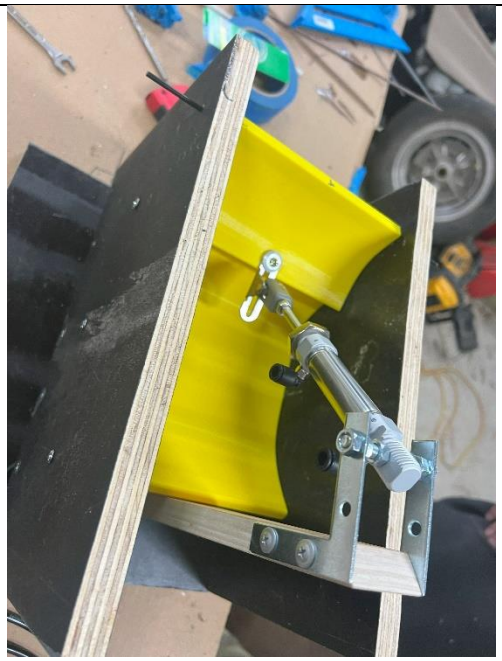
	was bolted to the upper rear wing flap. The actuator was connected to the stand with a bolt through the end opposite the actuator rod which went through the 90 degree steel brackets. To properly centre the actuator, 2 nuts were screwed either side of it. This allowed for it to be easily adjusted and then sure in position.	
6. Do a trial carbon fibre mould and construction	A trial carbon fibre piece was created and experimented on prior to the generation of the final 150mm carbon fibre wing sections. The piece utilised spare high density foam that was cut and shaped into a slightly complex wing shape to gain practise for the complex rear wing flap carbon fibre sections. The resin was then mixed at a 3.5:1 ratio by weight. For the test piece 100gram of the hardener was mixed with 350grams of the resin. This left excess resin after the test which provided a good margin for how much would be needed for the actual pieces. Prior to applying the resin, the carbon fibre sheet was cut to size and tapped to the underside of the foam piece to secure the material and also to minimise disturbing the twill. A coat of the resin mixture was then applied and the piece was left to cure for 24 hours. This piece was also used to test all construction methods prior to completing them on the actual carbon fibre pieces. This was to minimise the likelihood of damaging the final carbon fibre pieces and also to allow me to practise my technique on a control piece allowing me to	

	achieve a high quality result on the final pieces.	
7. Create carbon fibre moulds	[After reassessing the production plan and Gantt chart I concluded that outsourcing this stage of production would save a remarkable amount of time as well as produce a higher quality end project. The correspondence for the outsourcing of this stage began very early in the production process as reflected in the final Gantt chart. The process of outsourcing was more complex than I anticipated with an extensive volume of emails and phone calls with CADDEZINERS regarding the feasibility of the project. Following my initial email regarding the possibly of them designing the carbon fibre moulds based on my CAD model of the wing, there was a misunderstanding. The company sent me files and a quote for printing the entire wing. After some clarification phone calls and emails, I was informed that they would get back to me on whether this was a service they offered. Preparing for the possibility that they were unable to design the moulds themselves I began creating CAD models for them based on the already prepared model. While waiting for CADDEZINDERS to get back to me, I decided to make a start on modifying my CAD model to create the moulds. Once the wing had been sectioned into various components the process to create the moulds was quite simplistic. However, some issued arose with the angle of the cut for the moulds. While for some of the	    

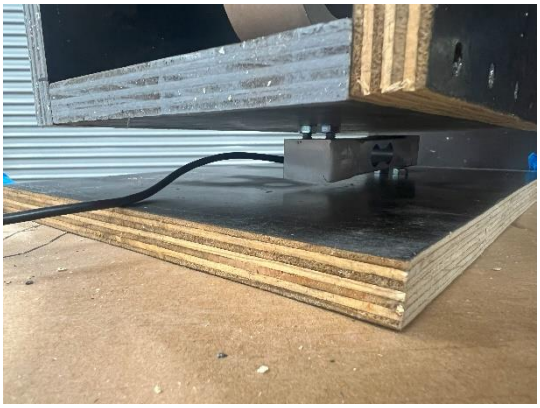
	more simply shaped pieces straight horizontal or vertical cuts worked there were issues with the more complex shapes especially with the endplates and connectors for the endplates to the wings. Following numerous talks with the CADDEZINERS company and through my own personal field research throughout the course of this length stage of production, I came to the conclusion that creating the moulds using this method was overly time consuming and would not produce the end result I am looking for. The revised method of creating the moulds for the project entails only creating the upper rear wing flap out of carbon fibre. This decision is explored further in the "Design Iteration and Improvement (Progressive Testing)" section of this folio. The revised method of creating a singular mould is a far more simplistic method that included designing and 3D printing a skeleton of the upper rear wing that will then be utilised along with foam to create the carbon fibre piece. This process had its own iterations and required to be changed but in the end proved to be an efficient and effective method of moulding the carbon fibre for the upper rear wing flap.] ********** The above iterative stages of progression of decision making with the carbon fibre moulds were heavily changed in the final revised method of creating the carbon fibre pieces for the 150mm sections.	

	While the creation of female moulds would have produced a high quality product if executed properly with adequate time; the budget, time and resource restrictions on the project limited this from being the method of creating the carbon fibre pieces. The final method of creating the moulds for the carbon fibre pieces was to 3D print male moulds of the wing sections that were a 0.98 scaled version of the prototyped pieces to allow for the thickness of the carbon fibre and resin. This process is explored more within the “manufacture carbon fibre pieces stage”.	
8. Manufacture carbon fibre pieces	Due to the time constraint on this project as well and the budget constraints, an altered method of producing the body and upper rear wing flap of the project was required. To complete further research into the process of carbon fibre componentry and mould creation I visited JPM Shipwrights in Cheltenham. Here I toured the workshop and learnt in more detail the process and cost of creating carbon fibre parts. The most notable details for production was within the specifics of mould creation, which focused on optimising the quality of the final part through creating run off area for easy separation of the two sides of the moulds. Whilst learning from the parts and moulds in house, I also discovered that the cost of creating the entirety of my project out of carbon fibre would cost upwards of $1000 to be professionally made. This resulted in the method of production of my project pivoting towards the new solution.	

The alteration from my initial design includes outsourcing the model of my design to be 3D printed. This piece acts as the educational aid regards of the aerodynamics of my design as well as being proof of my design concept. A separate 150mm section of a full scale upper and lower rear wing flap was created out of carbon fibre and attached to the pneumatics and electronics as the educational aid for the movement of the DRS and the representation of how the system actually works.



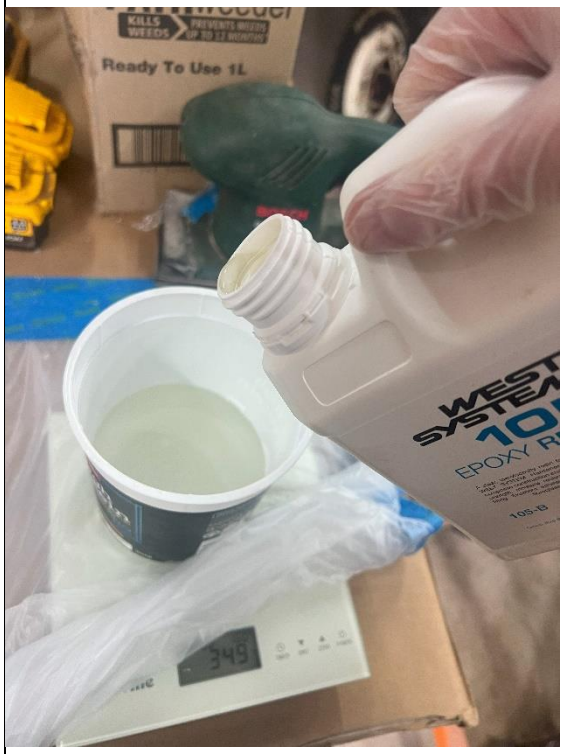
The outsourcing of the design came with many issues with are clearly outlined in the "correspondence" section. There were many misunderstandings surrounding what I was looking for and then unclear changes that needed to be addressed. In the end the piece was printed to a very high quality. The piece was SLA resin printed in ABS white and has a very smooth and professional finish.



The section part of the display of the DRS model was the actuating upper rear wing flap with both the upper and lower flap being made of carbon fibre. The first iteration of this has the wing sections being 3D printed to ensure everything was working prior to changing out the wings for carbon fibre pieces. the initial step was to cut the wooden boards to the correct size to support the 150mm wings sections, the actuator stand, the load cell and have a place for the batteries. With the pieces cut the locations of the wings and actuator componentry was measures and marked.The

support frame was then screwed together and the lower rear wing section was screwed into place as it is stationary. With this section in place the actuator stand could be secured and the actuator componentry was also connected to the stand. The actuator was then secured onto the wing at the front centre with too washer and a nut and bolt. The pieces had been measured to ensure that the wing was at the optimal angles when the actuator was at maximum and minimum stroke.

The next stage was to secure the wing within the support structure at the pivot points that had been located and holes drilled. Prior to securing the wing the sides needed to be sanded down to ensure that it moved without friction against the walls of the support structure.



Due to the 3D print being partially hollow because of the internal support fill it was printed with, it was easy to drive a carbon fibre rod through the top edge of the wing to act as its pivot axle. The rod was then passed through the side walls to secure the wing section.

To complete this initial iteration, the wing sections and actuator stand were removed in order to install the load cell under the centre of mass.

The bolts for the load cell were recessed into the wood and hidden underneath the location of the actuator stand in order to achieve an organised and nice looking end result.

The pieces were then reassembled and tested with the

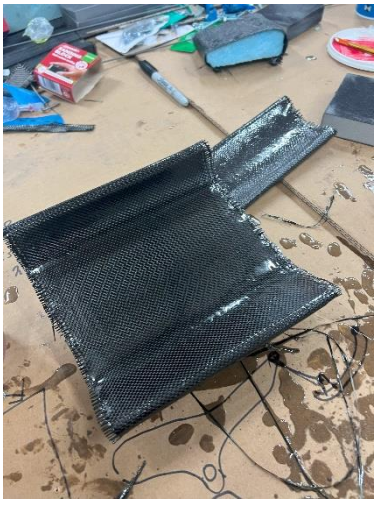
actuator before the wings and actuator components were removed in order to spray paint the support structure.

The support structure was spray painted black with multiple coats to achieve a high quality finish. To avoid removing the load cell I taped up the component and its wires.

The carbon fibre process was very similar to the test piece just on a larger scale and with a different mould material. While it would have been ideal to use the 3D printed sections to create foam moulds from, there was not enough time to complete this complex additional step. The best alternative was to use the 3D printed sections as the male moulds and to wrap them in carbon fibre. While this defeats the purpose of carbon fibre as it is meant to save weight through its properties that allow for light internal structures this process allowed me to create a finished product out of carbon fibre that works as an educational aid of a model DRS rear wing.

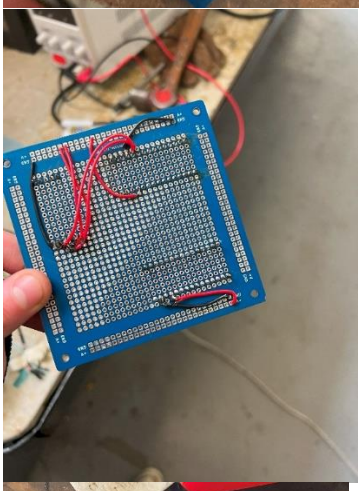
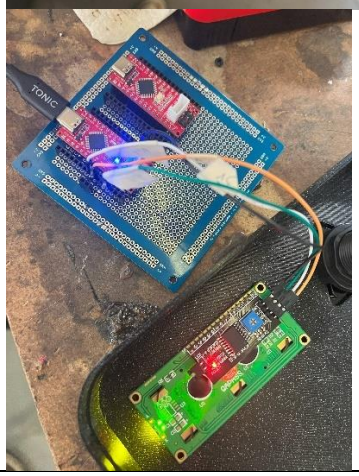
The process is as follows;

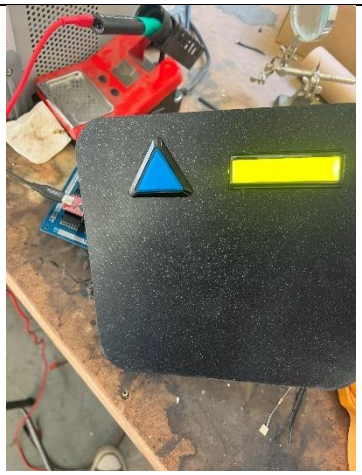
Mixing the resin and hardener at the 3.5:1 ratio, applying it generously to the carbon fibre that is wrapped around the 3D printed pieces making sure to hold it tight around the curved surfaces without disturbing the carbon fibre twill. For the upper rear wing flap piece this process was completed in one section while the lower rear wing flap was completed in two due to the more complex shape and bigger size. As the mixture begins to adhere to the mould it is easier to



	secure the carbon fibre to the more difficult corners. Before leaving the pieces to cure, I ensured that the brush strokes were going in the same direction so that the final result is visually appealing. After letting the pieces cure for 24 hours another batch of resin mixture was created to finish off the lower rear wing flap section and to touch up on any imperfections in the upper rear wing flap sections. The only noticeable sections that required touching up where the resin cured in uneven bumps that needed to be sanded back and then lightly covered in the mixture. Once the pieces had cured for another 24 hours they were ready to have the overhang carbon fibre sawed off and sanded down, ensuring that the pieces fit inside the support structure. While I am extremely happy with how the pieces turned out, the equipment limitations of not having a vacuum bag noticeably effected the final result of the carbon fibre wing sections. This is discussed further in the "evaluation" section. To complete this section of the DRS Model display the upper and lower carbon fibre pieces were fitted into the support structure and connected to the actuator using the exact same process as the first iteration with the 3D printed pieces. The only difference is the upper rear wing flap having two small pieces of carbon rod glued in to act as the pivot points instead of having the rod go all the way through the	

	piece. The intension of this was to have the wing are as level as possible.	
9. Assemble Model DRS body including pivot points	The majority of this step is covered in the previous step, however the externally 3D printed piece needed to have the upper rear wing flap attached with pivot points. Due to the very thin wall thickness of the model this step had some challenges. In order to drill a hole in the upper rear wing flap supports, extremely small drill bits were used and hand screwed into the 3D printed piece. This process was completed for both support and the pivot points on the upper rear wing flap itself. Very thing carbon fibre rod was then cut to size and epoxy glued into the upper rear wing flap. This means that the wing flap can still rotate and move on its pivot to simulate the DRS being open and closed.	
10. 3D print actuator housing	This step was removed due to the alteration in the process of how the actuator is supported in the model DRS display piece.	
11. 3D print systems control case	This stage required many design iterations which are outlined in the " Design Iteration and Improvement (Progressive Testing)" section. The case went through many redesign stages to account for improvements and changes within the intended method for organising all the electronics. The process of printing the case was relatively simple once the	

	designs correctly dimensioned the case to fit on the print bed. The piece was printed in a black marble PLA filament with a glittery sparkly which produced a very clean and aesthetic finish. Aside from the finish being of a really high quality, the final iteration fit all the components correctly and allowed for easy assembly of all the components.	
12. Assemble all components	The first step in assembling all the components was to connect all the electronics to a PCB board and wire them up. This step was very finicky and time consuming particularly with bridging the connections with the board and wires, as well as the time it took to fix any mistakes (see the image of one of the boards soldered in backwards). As discussed in the 'Code/wiring' section of the "Design Iteration and Improvement (Progressive Testing)" pages of this folio, the method behind completing this stage of assembling the components was altered in order to achieve a higher quality result that was easier to trouble shoot and looked a lot cleaner. To summarise, the alteration in process included; Not wiring everything directly onto the board, but instead using header pins to connect everything. This limited the amount of bridging of the gaps between connecters required, allowing for clear visibility to all wiring and simply looked nicer. The wiring of this stage included connected the GND and 5V pins to the outer allocated locations for those functions, with silicon wrapped wire. With the use of header pins all the sensors and button componentry were	  

	connected almost identically to how it was completed on the bread boards. With all the electronics connected the code was loaded onto the boards and tested for any issues. After some brief trouble shooting the electronics were ready for the next stage of assembly. The next stage entailed fitting all the components into the systems control case with press fittings. A key focus of this step was wire management, to achieve a well organised, safe and nice looking result. The LCD screen, PCB board, 5V regulator componentry, button and load cell amplifier were all internally fitted within the case. With the load cell, 5V relay, BMP sensor, and power source being externally fitted to the model and having cables organised and protected which go into the control case to be connected to their associated componentry. All wiring was organised and protected with heat shrink and braided cable sleeving. This produced a really high quality looking end result but also protected the components and people from harm from exposed wired. Once the assembly was completed and everything was fixed into the correct locations, all sections of the DRS model could be brought together to be displayed as one educational model. This included, the proof of design with the externally 3D printed half scale rear wing, the 150mm section of the full scale rear wing connected to the air compressor and the electronics within their housing. All sections were brought together to complete the project.	

13. Extra allowance of time		

Step	1	2	3	4	5	6	7	8	9	10	11	12
Time taken (estimated) (hours)	3	5	20	8	7	2	15	20	1		4	8

Correspondence

CADDEZINERS

From: Isabella Handbury
Sent: Thursday, May 25, 2023 4:36 PM
To: info@caddeziners.com.au
Subject: Feasibility of Mould Creation for School Project

Dear member of CADDEZINERS,

I'm writing to you in regard to feasibility of using your 3D printing services for a school project. For my Year 12 VCE Systems Engineering project I am designing, constructing, and testing a half scale model Formula 1 rear wing with DRS (Drag reduction system). The final iteration of the rear wing will be made using carbon fibre, which I had intended to make the moulds for myself but am currently having difficulties due to the size limitation of the school's 3D printer. Upon further research I have decided to outsource the creation of the moulds to save time and achieve a higher quality result.

The important questions I have for you is how feasible outsourcing the moulds to you is? In terms of cost and the actual creation of the moulds. As well as what further information you need from me to achieve this or determine its feasibility?

I already have a CAD model of the design and context in my design folio, which I can send through if it would help you answer any of my questions.

Thank you so much for your time.

Kind regards.

Isabella Handbury

Fwd: Feasibility of Mould Creation for School Project

CD Cad deziners <info@caddeziners.com.au>
To Isabella Handbury

Reply | Reply All | Forward

Tue 6/06/2023 12:34 PM

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image_2023_06_01T13_29_55_384Z.png 70 KB | image_2023_06_01T13_30_05_695Z.png 54 KB | ABS- SLA.png 1 MB

Hello Isabella,

Thank you for the follow up call. Please find the quote below:

--------- Forwarded message ---------
From: **Cad deziners** <info@caddeziners.com.au>
Date: Sat, Jun 3, 2023 at 11:45 AM
Subject: Re: Feasibility of Mould Creation for School Project
To: Isabella Handbury <isabhand@billanook.vic.edu.au>

Hello Isabella,

Thank you for your email. Please find the quote below:

Description	Material	Process	Qtys	Cost/Unit	Total Cost
Model Rear Wing	White ABS	3D Printing (SLA)	1	$310.0	$310
			10	$188.0	$1,880

Kindly note that the above quote is excluding **GST** and **Shipping Cost.** The cost per unit will decrease if the quantities are more.

White ABS (SLA) is not a strong material option but will have a smooth finish. Let us know if you are looking for a different material option.

Please confirm the dimensions (are in mm) which are attached below and feel free to contact us should you have any questions.

Shaby
Design Engineer
Engineering Department

Re: Feasibility of Mould Creation for School Project

CD Cad deziners <info@caddeziners.com.au>
To Isabella Handbury

Reply | Reply All | Forward

Wed 7/06/2023 4:24 PM

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Hello Isabella,

Apologies for the inconvenience. I will check with the team for costing and get back to you.

Please feel free to contact us should you have any questions.

On Wed, Jun 7, 2023 at 4:31 AM Isabella Handbury <isabhand@billanook.vic.edu.au> wrote:

> Hello,
>
> Thank you for responding to my email, however I'm not sure I was clear in regards to the questions I was asking. I am creating the wing out of carbon fibre and was asking if you offer a service where you could design and 3D print moulds using the CAD model of the wing I provided. If this is not something you offer I can model the moulds and send you a CAD file to quote them.
>
> Thank you for your assistance,
>
> Isabella

Re: Feasibility of Mould Creation for School Project

CD Cad deziners <info@caddeziners.com.au>
To Isabella Handbury

Reply | Reply All | Forward

Fri 23/06/2023 12:27 PM

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image_2023_06_21T09_18_18_824Z.png 160 KB | ABS- SLA.png 1 MB

Thank you for your email. Please find the quote below:

The cost of printing 1 set of your parts (6 parts) in White ABS (SLA) will be: **$268 + GST + Shipping cost**.

Kindly note that the cost per set will decrease if the quantities are more. I am afraid your part is too big to print in PETG and moreover petg will have rough finish compared to ABS.

ABS will have a smooth finish but will not be that strong. Carbon fiber is an expensive filament.

Please confirm the dimensions which are attached below (in mm) and
feel free to contact us should you have any questions.

On Wed, Jun 21, 2023 at 7:56 AM Isabella Handbury <isabhand@billanook.vic.edu.au> wrote:

> Hello,
>
> Following our phone call this morning I have sent through the CAD models for the separate components. Just to clarify from these component files I'm checking if it is feasible for you to design two-part moulds for each part, that I will check before they are printed.
>
> The best filament for the moulds so that when I will use them to create carbon fibre parts is PET-G filament and I wanted to ask what the approximate price would be for this filament.
>
> My final question is also to clarify what the cost of printing the previously quoted model out of Carbon Fibre filament would be? If this is a more simplistic and cost effective means of production, then this is possibly a better avenue to explore, since this is a one off creation for a school project.
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> Thank you for all your assistance,
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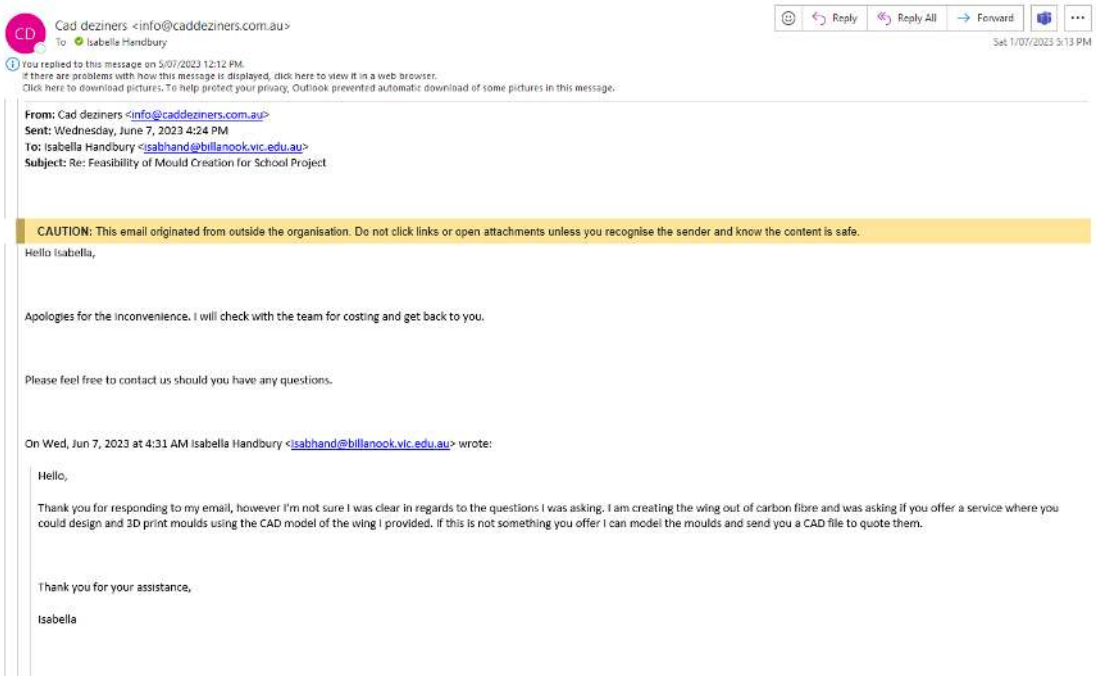

Cad deziners <info@caddeziners.com.au>
To Isabella Handbury

Reply | Reply All | Forward

Sat 1/07/2023 5:13 PM

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Sent: Wednesday, June 7, 2023 4:24 PM
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Subject: Re: Feasibility of Mould Creation for School Project

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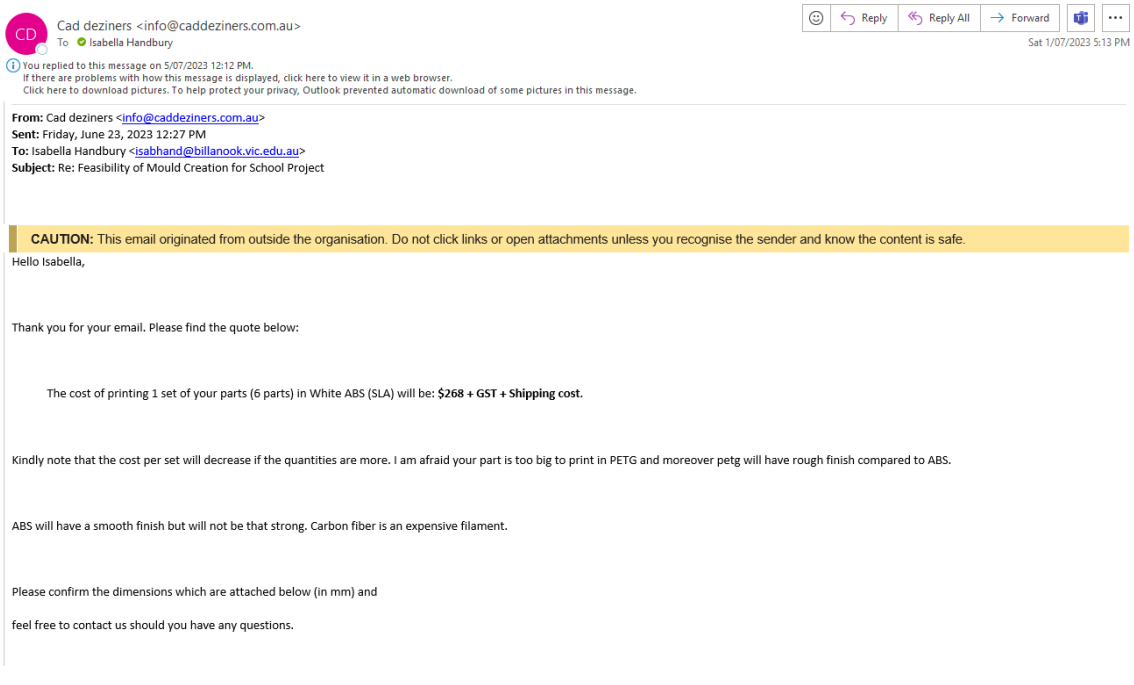

Cad deziners <info@caddeziners.com.au>
To Isabella Handbury

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From: Cad deziners <info@caddeziners.com.au>
Sent: Friday, June 23, 2023 12:27 PM
To: Isabella Handbury <isabhand@billanook.vic.edu.au>
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Hello Isabella,

Thank you for your email. Apologies for the inconvenience, the earlier quote was for printing the parts.

Can you confirm if you would like us to make a mold for only the Main Rear Wing? And just to update you that we can print your part in White ABS instead of Vero Black.

Please feel free to contact us should you have any questions.

On Wed, Jun 28, 2023 at 8:14 AM Isabella Handbury <isabhand@billanook.vic.edu.au> wrote:

> Hello,
>
> I have decided to go down the path of 3D printing the main body of the rear wing. Just checking for a final quote for the model attached, using Vero Black Plus filament.
>
> Thank you,
>
> Isabella

Re: Feasibility of Mould Creation for School Project

Cad deziners <info@caddeziners.com.au>
To Isabella Handbury

Reply | Reply All | Forward

Tue 25/07/2023 7:03 PM

If there are problems with how this message is displayed, click here to view it in a web browser.
Click here to download pictures. To help protect your privacy, Outlook prevented automatic download of some pictures in this message.

image_2023_07_25T08_12_06_880Z.png
40 KB

CAUTION: This email originated from outside the organisation. Do not click links or open attachments unless you recognise the sender and know the content is safe.

Hello Isabella,

Apologies for the late reply. Please find the quote below:

The cost of printing 1 set of your parts (Main rear wing body and mold of Upper rear wing flap) in White ABS (SLA) will be: **$590 + GST + Shipping cost**.

Kindly note that the cost per set will decrease if the quantities are more.

ABS will have a smooth finish and can be used as a mold but will not be that strong.

Please confirm the dimensions (in mm) which are attached below and feel free to contact us should you have any questions.

On Wed, Jul 5, 2023 at 7:42 AM Isabella Handbury <isabhand@billanook.vic.edu.au> wrote:

> Hello,
>
> I am confirming that I would like the main body of the rear wing printed in Vero Black, but would like to see a quote before finalising that process. In a previous email I asked for the feasibility of mould creation but did not specify which parts. The only part that needs a mould is the upper rear wing flap and I would like the mould to have location points and run off if that was possible. If the mould creation is not feasible printing the upper rear wing flap in the same material as the main rear wing body is also an option.
>
> Also if its not too much to ask would I be able to get a timeline on the completion of this? I have an upcoming deadline for the school project and just would like confirmation that everything is on schedule.
>
> Thank you for all your help,
>
> Isabella

Re: Feasibility of Mould Creation for School Project

CD Cad deziners <info@caddeziners.com.au>
To Isabella Handbury

Reply | Reply All | Forward

Tue 1/08/2023 2:13 PM

You replied to this message on 1/08/2023 7:39 PM.
If there are problems with how this message is displayed, click here to view it in a web browser.
Click here to download pictures. To help protect your privacy, Outlook prevented automatic download of some pictures in this message.

CAUTION: This email originated from outside the organisation. Do not click links or open attachments unless you recognise the sender and know the content is safe.

Hello Isabella,

Thank you for your email. Please find the reply to your questions below:

- **The timeline for this project, I need to confirm this can be completed within the next two weeks.**
  A: If you place an order within 2 days, then we will try to ship within 8 working days.

- **That the print will be completed in black ABS not white, as previously stated.**
  A: The quote which we have provided was for White ABS. Black ABS will be a little more expensive than White ABS and will have a rough finish. (Find the sample image below)

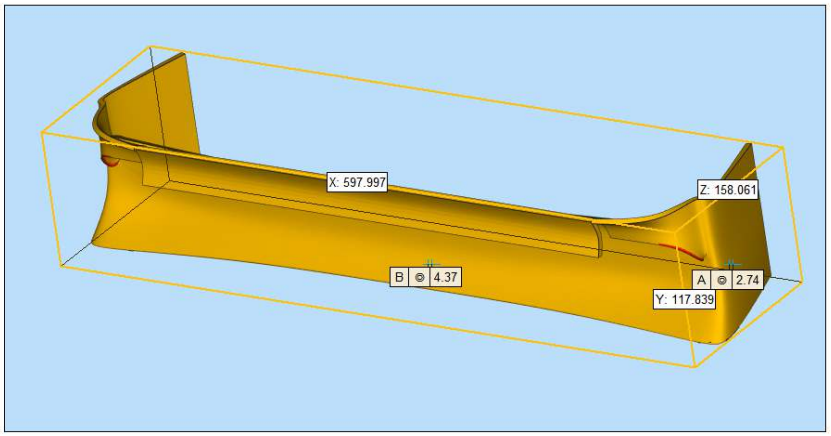
- An explanation in the significant price increase from previous quotes to the most recent. See email sent on the 6th of June.
  A: The quote which we have initially provided was directly printing your parts in ABS instead and the quote which we have provided recently was for printing 1 part directly and mold of another part in White ABS.

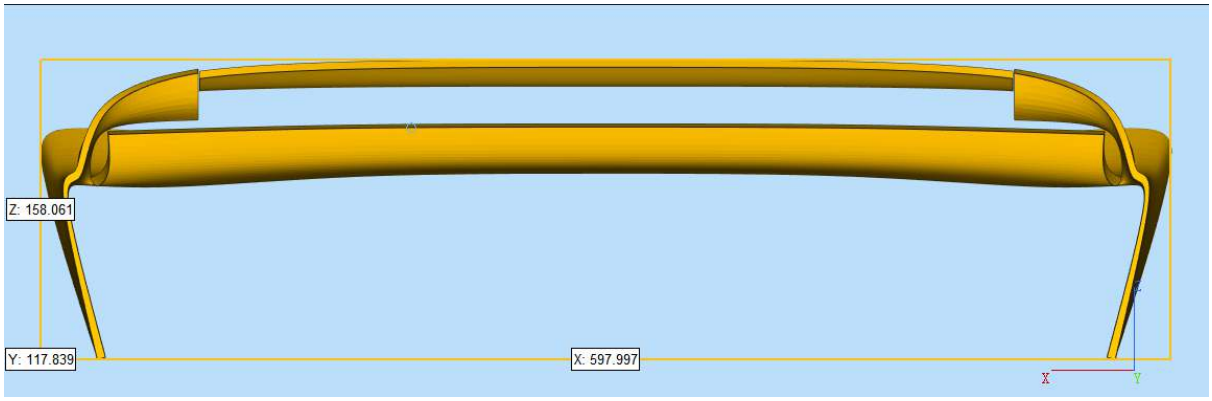
Please let us know how you would like to proceed and feel free to contact us should you have any questions.

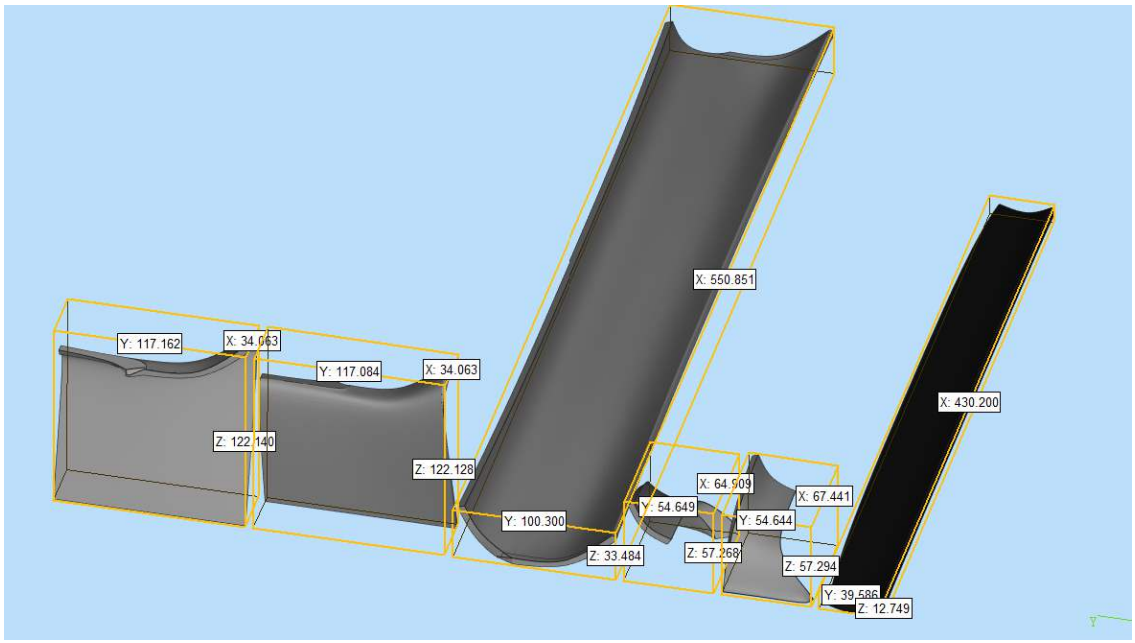
On Mon, Jul 31, 2023 at 5:41 PM Isabella Handbury <isabhand@billanook.vic.edu.au> wrote:

> Hi Shaby,
>
> Just following up your email with some things to confirm before paying for the project.
>
> The things I need confirmed are;
>
> - The timeline for this project, I need to confirm this can be completed within the next two weeks.
> - That the print will be completed in black ABS not white, as previously stated.
> - An explanation in the significant price increase from previous quotes to the most recent. See email sent on the 6th of June.
>
> I can confirm that the print dimensions are in mm.
>
> Kind regards,
> Isabella

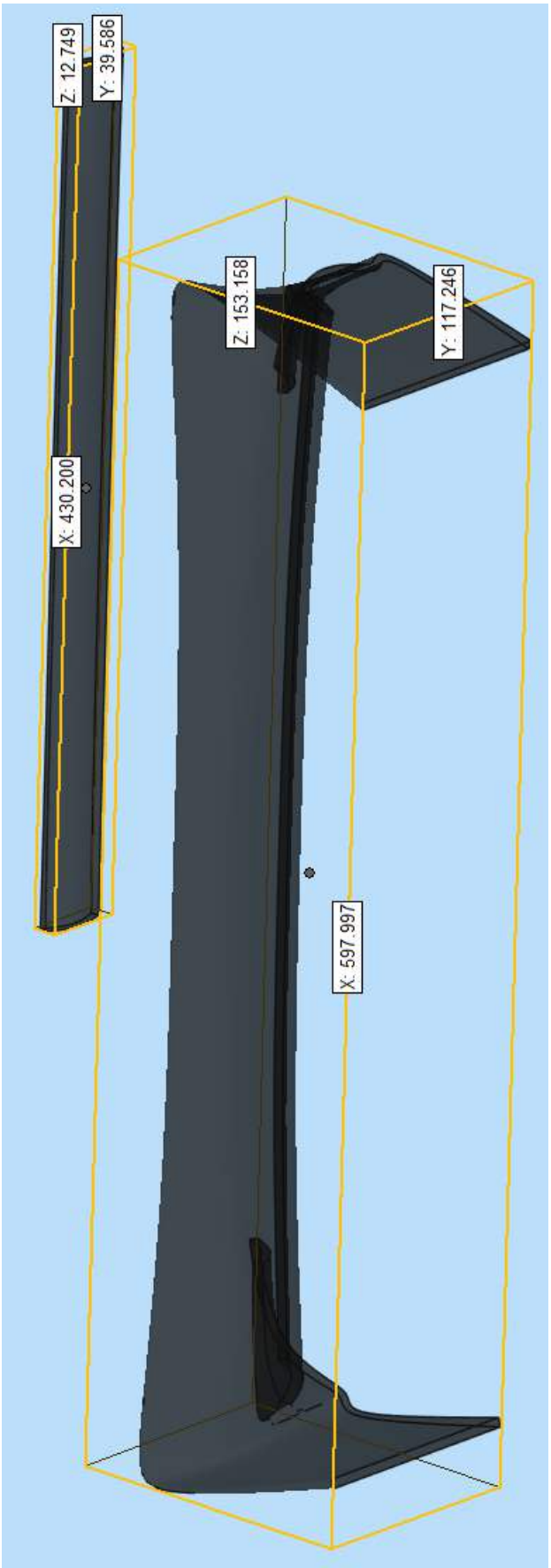
Attachments







Final Design Outsourced to Print



FIA

FIA Technical Regulations in Regards to a School Project

IH

Isabella Handbury
To press@fia.com

6/04/2023

Dear FIA member,

I am writing this email in regards to a school project I am completing, and am requesting assistance in order to produce the best outcome possible. For my Year 12 VCE Systems Engineering project I am designing, constructing, and testing a model Formula 1 rear wing with DRS. I have chosen to constrain my project within the regulations outlined in the 2023 Formula 1 Technical Regulations, to better simulate the engineering processes followed by the Formula 1 teams.

I have completed my initial stages of research and am in the midst of designing and digitally modelling the rear wing. Whilst attempting to attain the regulated dimensions for the rear wing and associated componentry I have run into problems with comprehending the dimensions of the rear wing profiles (3.10.1 of the regulations) and applying the information available to my designs.

Whist going through the regulations searching for any relevant information that could assist me in creating parameters in my CAD model for the rear wing, I came across a point in Appendix 1: Regulation Volumes. In 1.6 of said Appendix it states that;

"The CAD models of all the volumes and surfaces listed in this Appendix are available for Teams to download with a standard range of parameters. The FIA will provide a Team with a specific model on request."

Is there a possibility of gaining access to these models with associated parameters, or any other useful resources? If you are unable to assist me could you please direct me to the suitable pathway to gain this information, any assistance would be greatly appreciated. I am also more than happy to send through the initial stages of my folio which contains the relevant information for my project if that would be helpful.

Thank you for taking the time to read my email.

Kind regards,
Isabella Handbury

Pneumatic Solutions

Order return and queries

IH

Isabella Handbury
To salesvic@pneumaticsolutions.com.au

Reply | Reply All | Forward

Tue 30/05/2023 12:04 PM

Hello member of Pneumatic Solutions Australia,

I am contacting you in regards to your returns policy and with some questions to alter my order. I was unable to open the return policy section of your website due to an error code so I thought emailing you would be the best option. I recently ordered some parts for my year 12 Systems Engineering project and have found some issues with the fittings once my order arrived. My order included;

- 4V210 5/2 Solenoid Valve at ¼ BSP
- DSN Double Acting Mini Cylinder, 16mm bore and 25mm stroke
- 2 x BSLM Brass Threaded Button Silencer, ¼ BSP tread
- 3 x 3136 Female Bulkhead Connector, 6mm tube size, ¼ BSP tread
- 2 x 3199 Male Stud Elbow, 6mm tube size, ¼ BSP tread
- Flexible Polyurethane Tube, 6mm

While the Solenoid, Cylinder and Polyurethane tube are all correct there are some issues with the fittings. Based on your return policy I would like to send back the Silencers, Female bulkhead connectors and the Male stud elbow connectors and then reorder the correct parts.

To be able to order the correct parts I first have some questions that I hope you could help me with. Firstly I am unsure what the thread size of the silencers should be for the solenoid as there are no technical drawings that specified that the tread for them was different to the ¼ BSP size selected. My next issue is with connecting the DSN Double Acting Mini Cylinder, 16mm bore and 25mm stroke, to the 6mm Polyurethane tube. What connector would you recommend to fix this issue?

Thank you for your time,

Isabella Handbury

Field Research – JP ShipWrights

To gather a better understanding of the extensive process of creating moulds for carbon fibre components as well as the actual moulding of the carbon fibre, I visited JP ShipWrights in Cheltenham. They specialise in custom componentry, fit out work, boat maintenance and modification. The custom work completed there includes but is not limited to; bespoke fibreglass furniture, stunt plane work, competitive yacht componentry and many small custom projects. At the workshop there were many examples of carbon fibre moulds and the machinery that it takes to create the best quality finishes.

The images below are showing an example of a mould for some boat componentry and the fibreglass part is being left to properly cure in the mould before further steps are taken in the construction of this part. The icy pole sticks are they to assist with removing the part from the mould aiming to limit the risk of damaging the part.

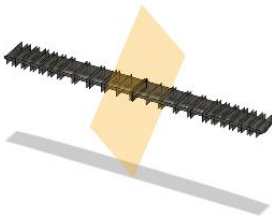
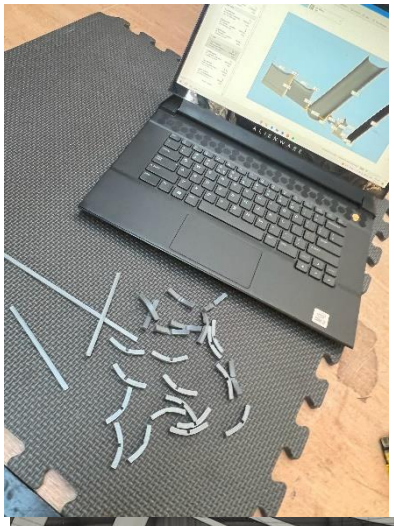
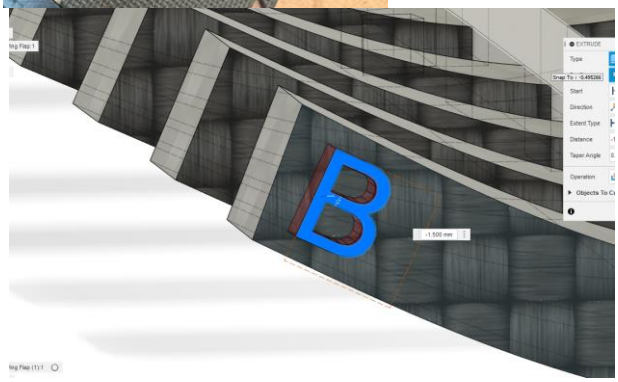
Key takeaways

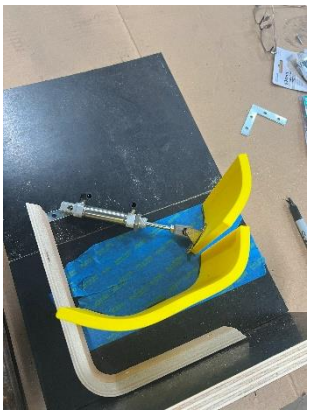
- In F1 a female mould autoclave would be utilised in order to achieve the best results with all carbon fibre parts. Owning, using, and preforming maintenance on this machinery is incredibly expensive, however it does produce the best possible result.
- When you don't have access then an autoclave, a way to simulate the process it completed on the carbon fibre is by utilising a vacuum bag with the wet layup technique.
- When creating and designing the moulds for any part to achieve the best results a flange around the mould is a must. This allows for any runoff of material to be easily cleared, provides something to apply leverage on when removing a part from the mould and overall make the mould easier to handle.
- The discussion was had about using CNC to shape polystyrene to create moulds for my project. The major talking points was how this process is very lengthy and requires keen precision, as well as the polystyrene needing to be sealed.

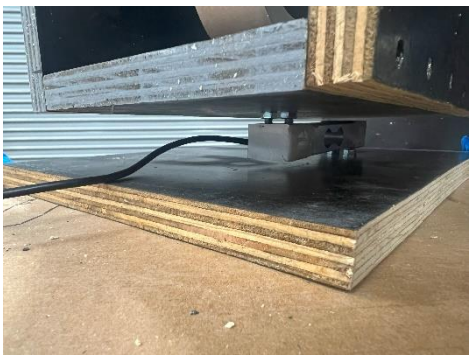
Design Iteration and Improvement (Progressive Testing)

Wing Main Body

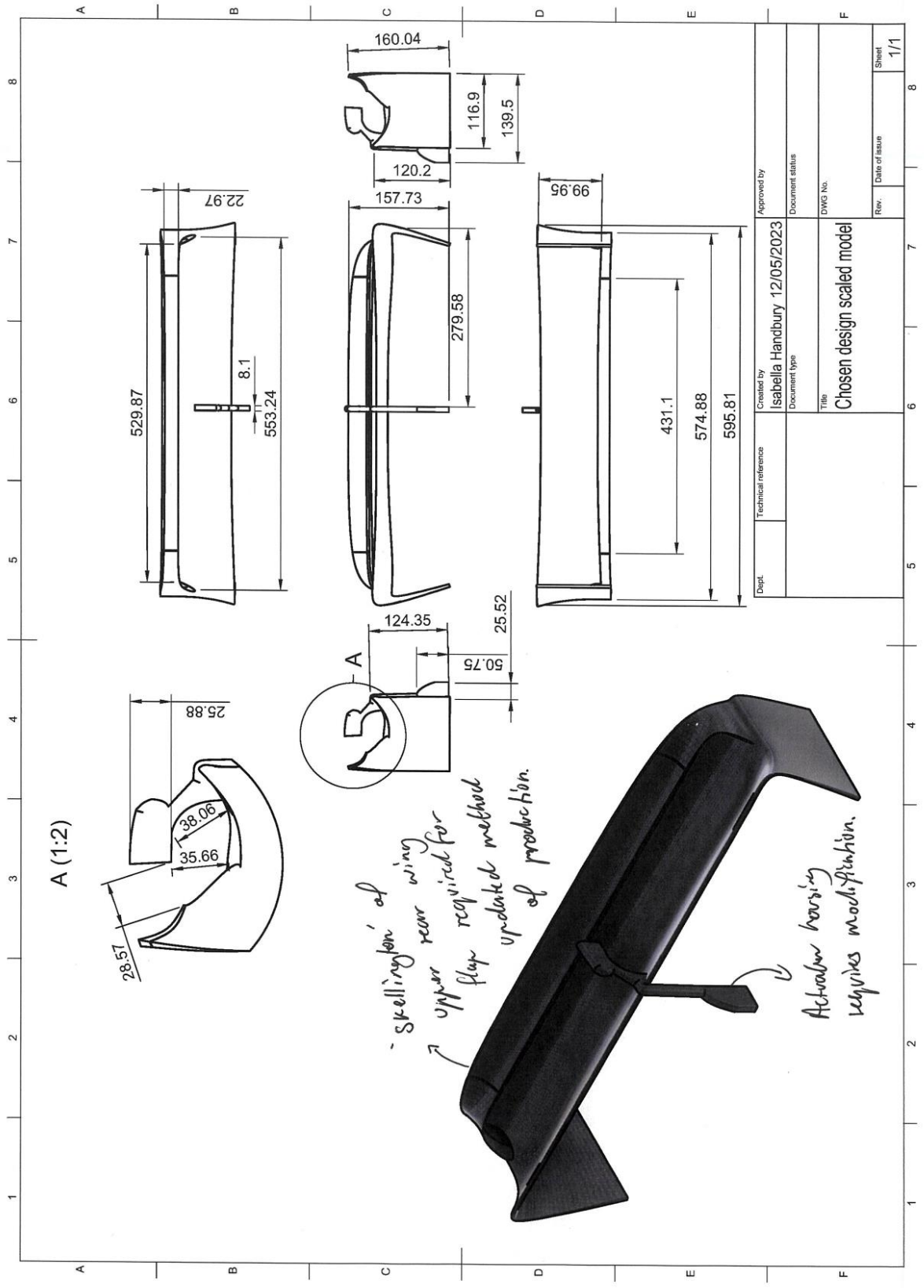
Initial Design Plan	
The initial plan for the wing main body was to generate moulds myself and to create the entire project out of carbon fibre. The CAD design for the wing was intended to be used to create the moulds to generate carbon fibre parts using a wet layup technique.	Image:

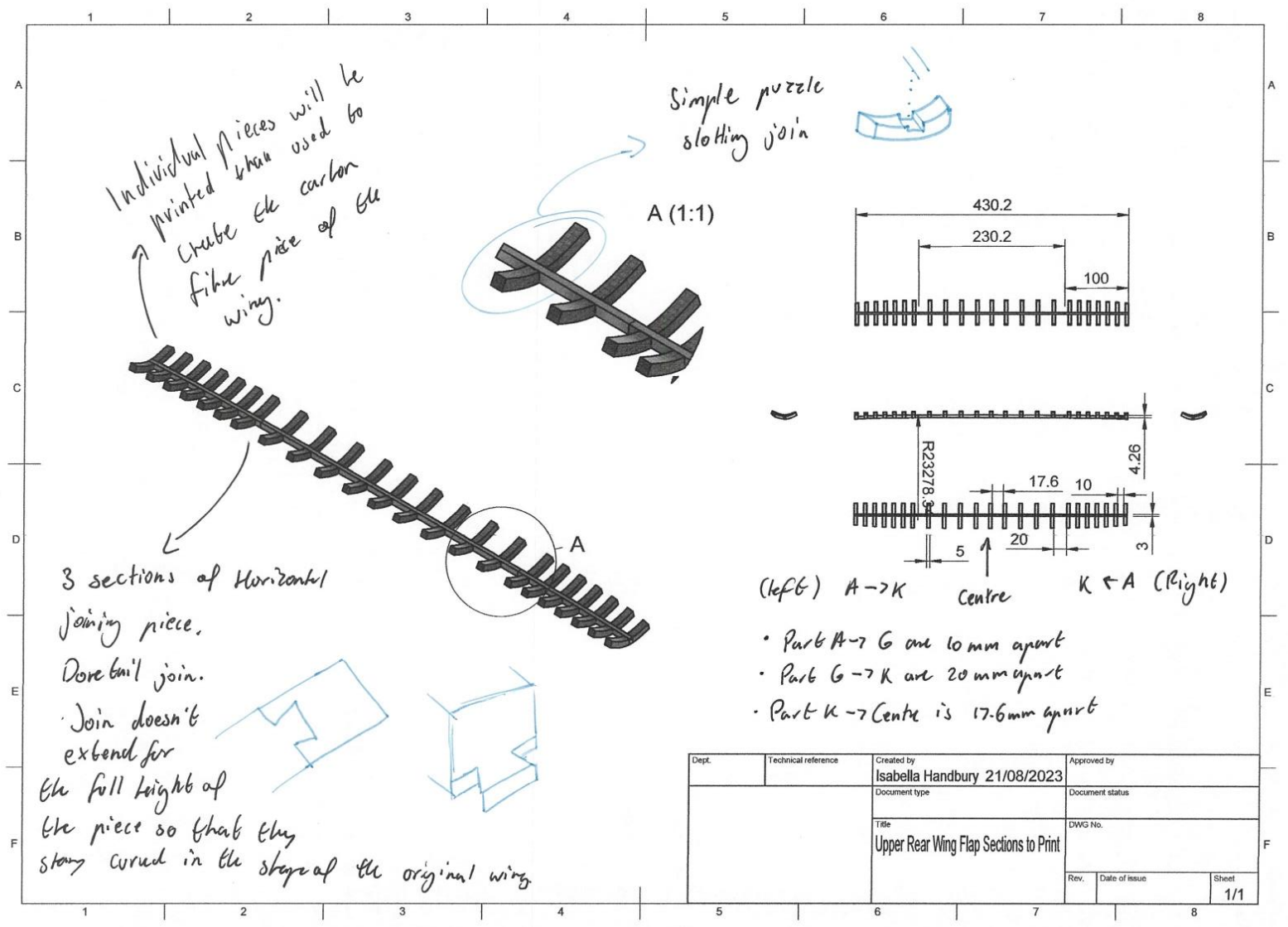
Design Iteration 1 (Upper rear wing flap)	
The design and idea development stage of this project had the wing main body go through copious amounts of design iterations. Following the initiation of construction, many issues arose that presented limitations on the feasibility of certain aspects of the wing main body when creating it out of carbon fibre. The idea of 3D printing the rear wing moulds was developed over a long period of time, with designs being created in CAD through generating the negatives of the already produced rear wing design. This idea was deserted following calculations into the high cost of getting the moulds 3D printed along with the high time requirement of creating all the parts out of carbon fibre using these moulds. There were also issues with the time it would take to print the moulds and the uncertainty that they would produce a high quality end result. This led to having the main rear wing body externally 3D printed and exploring the carbon fibre process with only the upper rear wing flap. With this change in direction of this construction process, issues arose with designing moulds for the upper rear wing flap. The solution was to design a Skellington of the upper rear wing flap in CAD with the intension being that it would be used in combination with foam to aid in shaping the carbon fibre into the ideal wing shape I had designed. Each piece is a slightly different shape and the clip to the horizontal connector like a puzzle piece. The horizonal connector piece is designed to slot together with a dovetail join. While the initial print of this design was successful there were some issues that required alterations to the design. The first issue was that once the parts were printed they appear so similar that they were indistinguishable from each other. Due to the complexity of the upper rear wing flap shape, it is vital that all the pieces are in their correct position. The solution to this was to alter the CAD model to have embossed labels on the small pieces to aid in telling them apart. Another issue was that because the pieces are so small the dovetail connector is affected by any tiny warp in the area around the connection point. While this is not something that required a design change it did prompt for more post print work to be completed on the horizontal pieces.	Images:   

Design Iteration 2 (Upper rear wing flap)	
The decision to switch from the 'Design iteration 1' to this second iteration was due to a number of factors; - Time constraints producing issues - Limitations of producing high quality end result - Problems arising with the scale of the externally 3D printed main rear wing body. When the externally 3D printed piece arrived it was realised that attempting to fit a carbon fibre wing into the connectors for the upper rear wing flap was not feasible. The wall thickness was too thin and the creation of actuator housing to fit with the main rear body and contain the actuator and associated componentry was unfeasible. As well as the actuator was too strong which would lead to damaging the 3D printed wing, along with requiring further implementation of support structures for the wing. All these factors lead to a large pivot in what the final product would look like. To properly display the functionality of the DRS of the rear wing as well as display proof of my rear wing design the project was technically split in two. The first bit was keeping the 3D printed rear wing model separate from any electronics and just adding small carbon fibre rod pieces into the rear connector structure and upper rear wing flap in order to have the upper rear wing flap pivot. This displays an example of what the DRS on my design would look like without risking damaging the rear wing. The second section of my project is taking a 150mm middle section of the upper and lower rear wing flaps and adding support structures, actuator supports and fitting all electromechanical componentry to the section. The first step was to cut the wooden boards to the correct size to fit the upper and lower rear wing flap sections so that it would join together whist leaving room for the batteries and being able to have the load cell under the wings. This was followed by 3D printing the first iteration of the rear wing flap sections that will later be replaced with a carbon fibre alternative that is moulded off of the 3D printed sections.	Image: 

The lower rear wing flap section was secured with screws in order to be stationary, while the upper rear wing flap had a carbon fibre rod inserted so that is able to rotate. The construction and fitting of the actuator housing is further explored within the 'Actuator Housing section." The next step was to fit the load cell under the wing flap section of the model. This required screws to be hidden under the actuator housing and recessed into the bottom of the board in order to hide the screws so they don't interfere with other components. More importantly, this allowed for the load cell to be correctly installed to get accurate readings.	

Design Iteration Annotations:

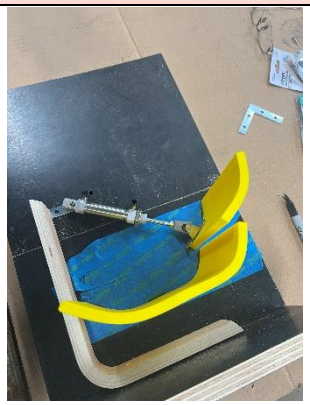
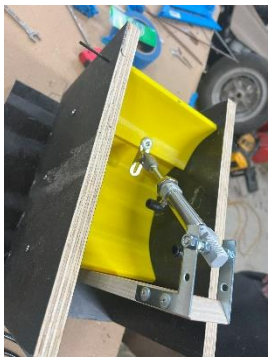



Individual pieces will be printed then used to create the carbon fibre skin of the wing.
Simple puzzle slotting join
A (1:1)
430.2
230.2
100
R232783
4.26
17.6
10
20
5
3
(left) A -> K
Centre
K <- A (Right)
A
3 sections of Horizontal joining piece.
Dovetail join.
Join doesn't extend for the full height of the piece so that they stay curved in the shape of the original wing
• Part A-> G are 10 mm apart
• Part G -> K are 20 mm apart
• Part K -> Centre is 17.6mm apart
Dept.
Technical reference
Created by
Isabella Handbury 21/08/2023
Approved by
Document type
Document status
Title
Upper Rear Wing Flap Sections to Print
DWG No.
Rev.
Date of issue
Sheet
1/1

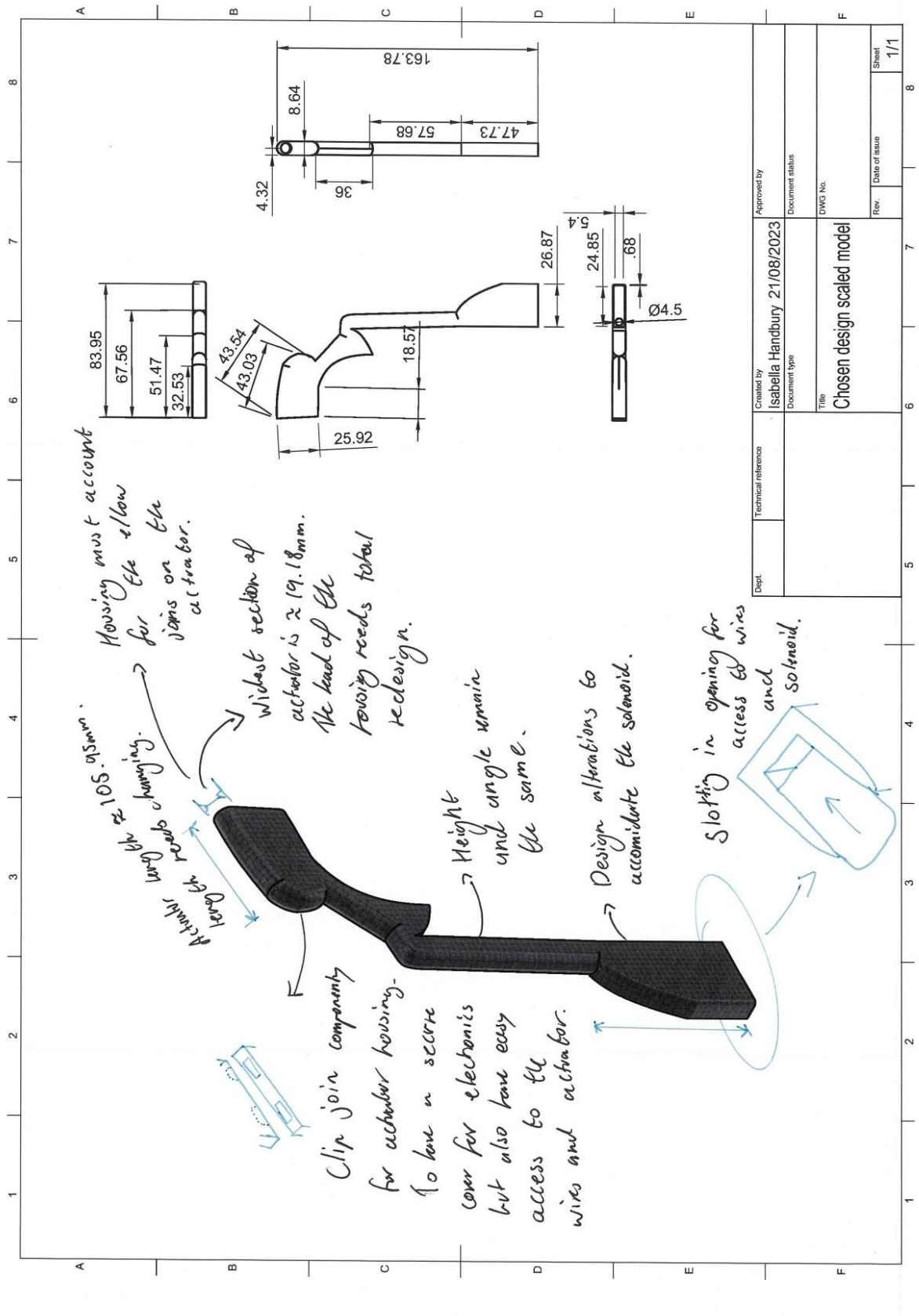
End Result	
The end result for the main boy of the rear wing is having the DRS model displayed with two sections. The first is the half scale externally 3D printed model of my initial design, which acts as a proof of design concept. The second part is the 150mm full scale version of the rear wing with operational DRS, made with carbon fibre wing sections.	Image:

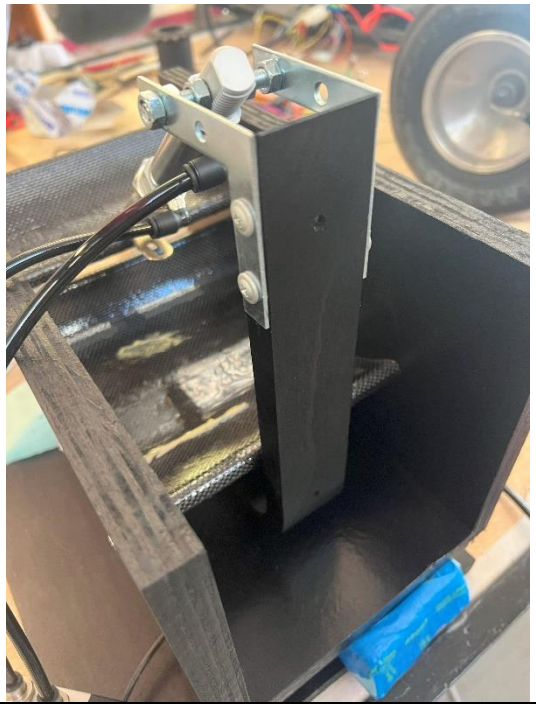
Actuator Housing

Initial Design Plan	
The initial plan for the actuator housing was to have a fully 3D printed housing design based on the model I had created during the design stage of the project. The housing was planned to store the actuator in the head of the housing as well as store the solenoid in the base, with all the air tube and wiring travelling though the housing.	Image:

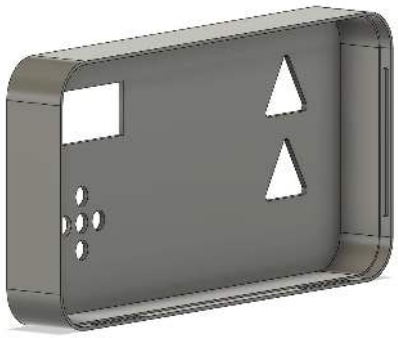
Design Iteration	
The design iteration for the actuator housing coincided with the changes made to not only the upper rear wing flap but to the way that the DRS is presented altogether. As addressed in 'Main Rear Wing Body' section, the externally 3D printed model of the rear wing prompted for changes to be made in order to save time and to be able to properly present the DRS movement and the sensors associated with it. The idea of 3D printing the housing was moved away from in order to save time and also to avoid the complex design process required to fit all the componentry properly. Instead, a more simplistic method was used which has the added benefit of having the componentry easily viewable. This enhances the project as an educational tool. The materials used to create the housing were a 90 degree wooden stand, two 90 degree joiners, a long bolt to hold the actuator and 6 nuts to hold the actuator in place. The first step was mapping out the location of the actuator in relation to the upper rear wing flap and the impact this has on the location of the supports. Once all the angles were determined with the actuator and the joins, they were screwed together and the nuts and bolts were used to centre the actuator and keep in place. The wooden support stand was centred between the two walls of the wing support and screwed into place.	 

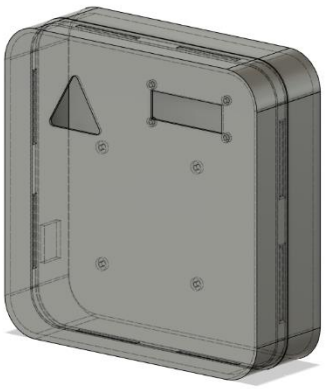
Design Iteration Annotations:



End Result	
The end result of this is the actuator stand instead of the planned 3D printed design. The actuator stand works in conjunction with the 150mm section of the rear wing to properly display the componentry behind the functional DRS. The change to having the exposed solenoid and actuator aids in the educational intensions behind this project.	Image: 

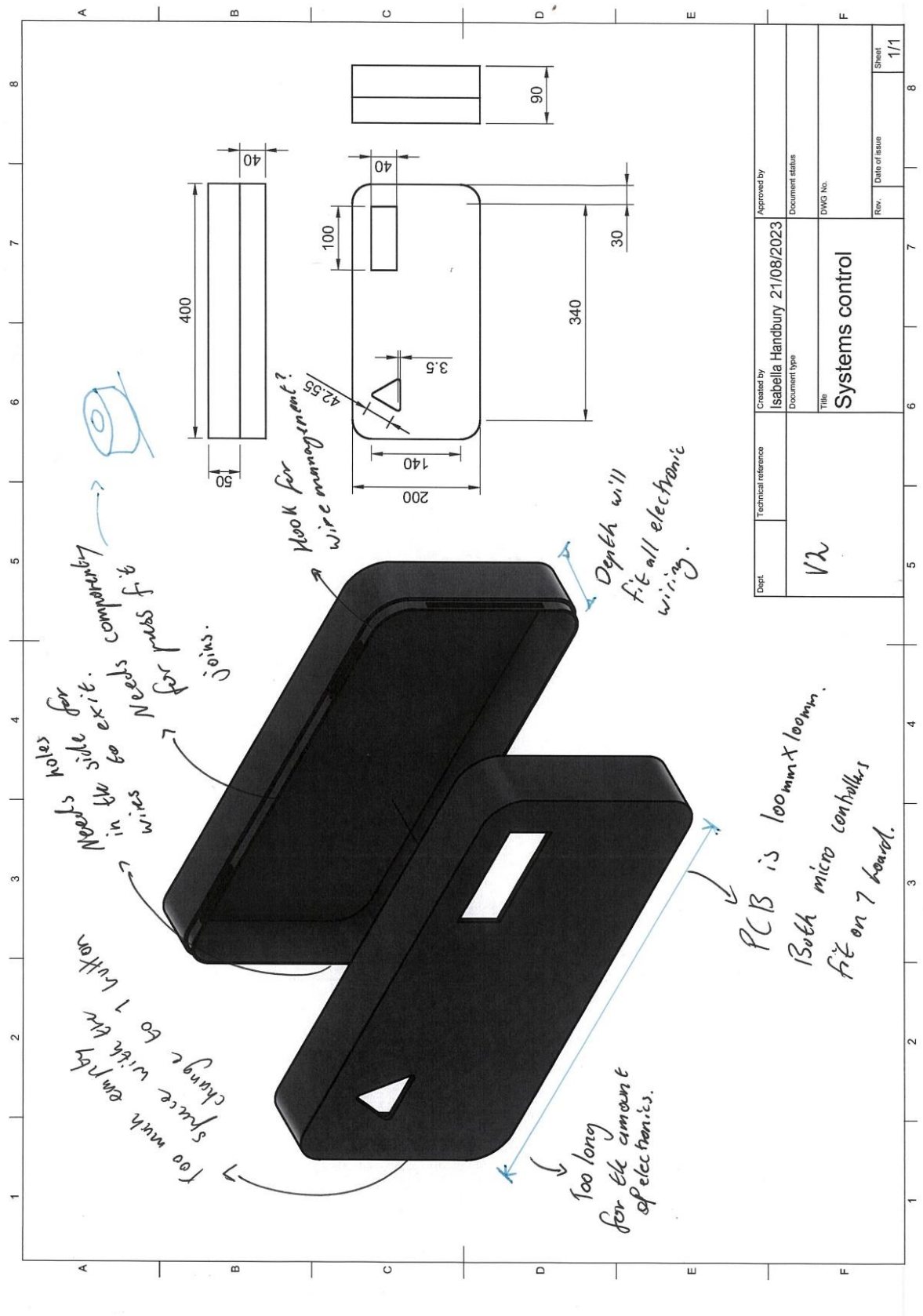
Systems Control Case

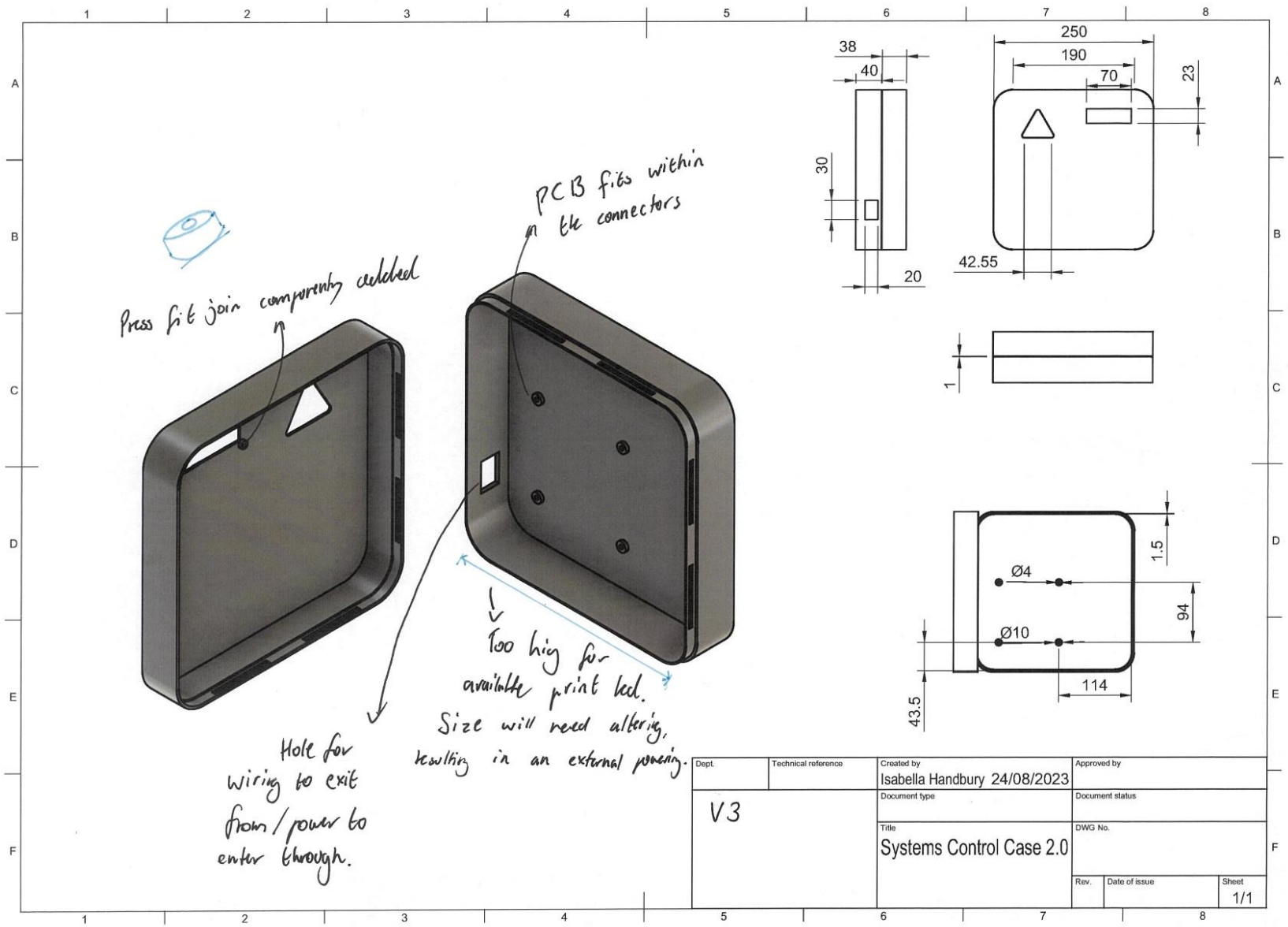
Initial Design Plan	
The planned design for the systems control case was to 3D print the design from the earlier criteria of this project. This design included two DRS on and off buttons, buttons to control the LCD screen and was specifically rectangular in shape for aesthetics.	Image: 

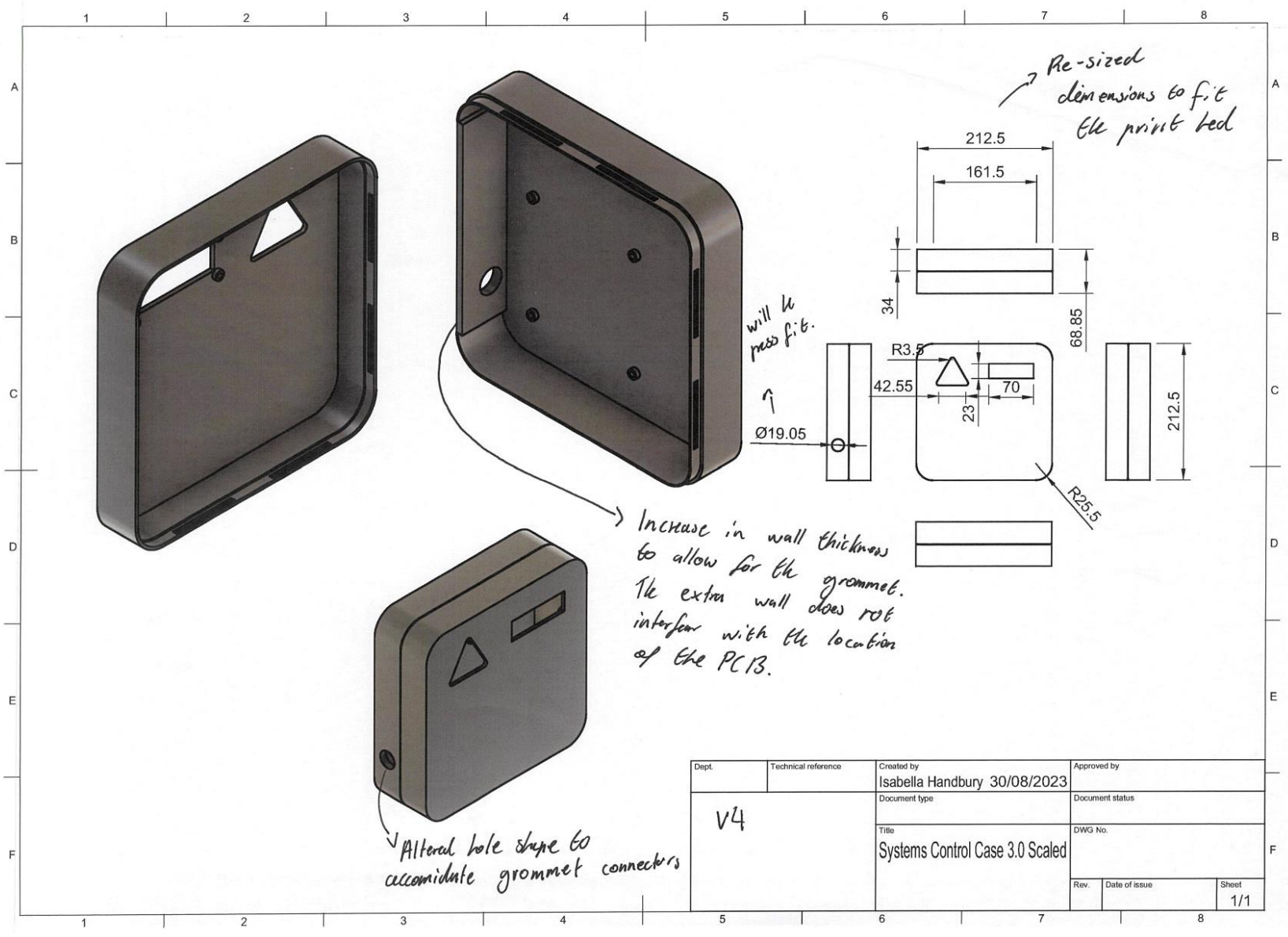
Design Iteration	
After the completion of all the wiring and coding of electronic componentry, it became apparent that the initially designed systems control case was insufficient for the changes made. The first change that was needed was to remove one of the triangle buttonholes and to remove the 5 circular button holes. These planned features were removed throughout the coding and wiring process when it was determined that they were either an unnecessary overcomplication of a simple process or were removed to simplify the code. The next big change was due to the usage of 2 seedling nano boards. The best way to organise the wiring inside the systems control case was with a PCB board. This generated the need for alterations in the sizing of the case and the configuration of the press fit join locations. Another change that needed to be made was with the appearance of the case. With the removal of the second button there needed to be a change to the front of the case to display the button and LCD screen better. These changes required for the systems control case to go through many design iterations, in order to create the best solution of wire management as well as producing a visually appealing case. Many iterations were required in order to get the location	Image: 

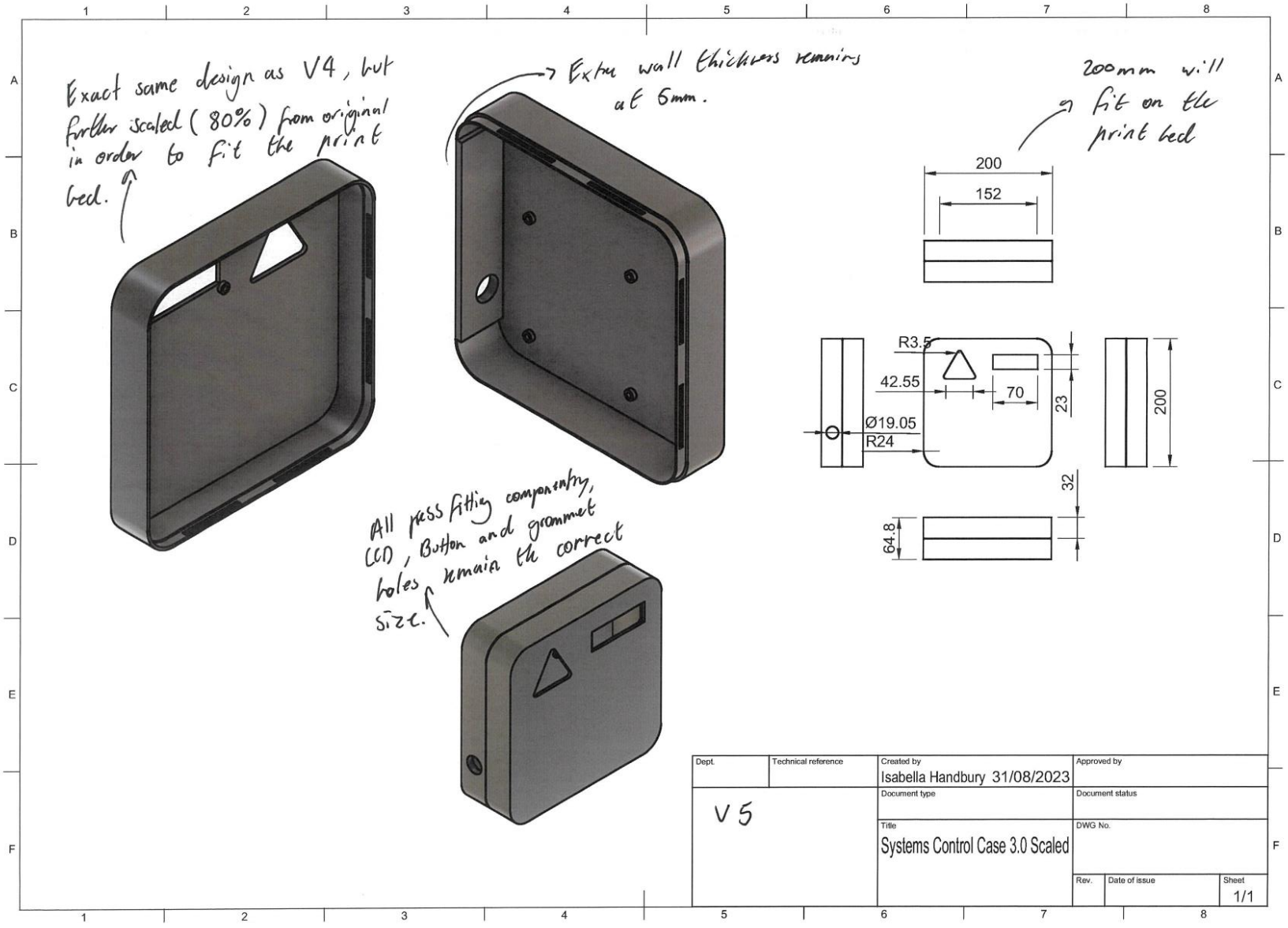
of all the components correct and investigate the ideal set up. While through the 5 different iterations (seen in annotated section below) the changes became much more minute, there was a lot of time spent changing the size of the case to fit the 3D printer bed. This required the other componentry to be rearranged as well.	

Design Iteration Annotations:



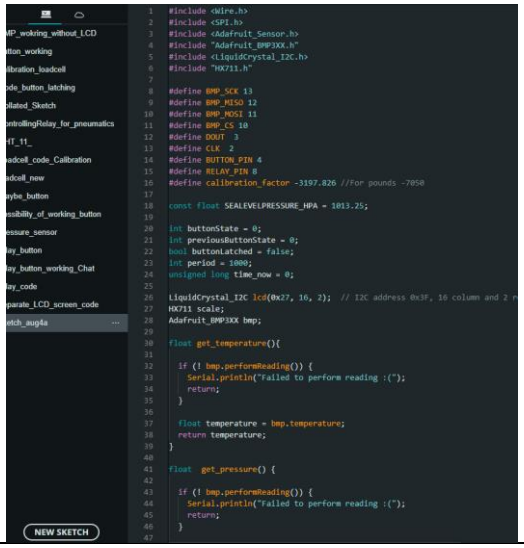

38
40
30
20
250
190
70
23
42.55
1
1.5
Ø4
94
Ø10
114
43.5
PCB fits within the connectors
Press fit join components soldered
Too big for available print bed. Size will need altering, resulting in an external housing.
Hole for wiring to exit from / power to enter through.
Dept.
Technical reference
Created by
Isabella Handbury 24/08/2023
Approved by
V3
Document type
Document status
Title
Systems Control Case 2.0
DWG No.
Rev.
Date of issue
Sheet
1/1

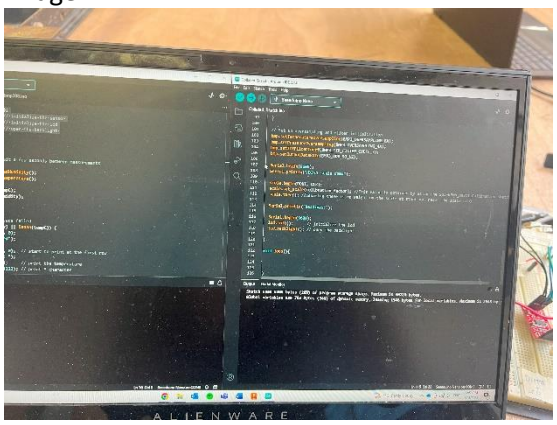
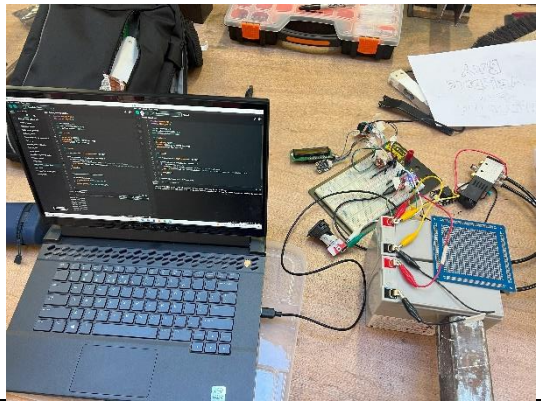

Re-sized dimensions to fit the print bed
212.5
161.5
34
68.85
R3.5
42.55
70
23
212.5
Ø19.05
R25.5
will it pass fit.
Increase in wall thickness to allow for the grommet. The extra wall does not interfere with the location of the PCB.
Altered hole shape to accomidate grommet connectors
Dept.
Technical reference
Created by
Isabella Handbury 30/08/2023
Approved by
V4
Document type
Document status
Title
Systems Control Case 3.0 Scaled
DWG No.
Rev.
Date of issue
Sheet
1/1


Exact same design as V4, but further scaled (80%) from original in order to fit the print bed.
Extra wall thickness remains at 6mm.
200mm will fit on the print bed
200
152
R3.5
42.55
70
23
Ø19.05
R24
200
32
64.8
All press fitting components, LCD, Button and grommet holes remain the correct size.
Dept.
Technical reference
Created by
Isabella Handbury 31/08/2023
Approved by
V5
Document type
Document status
Title
Systems Control Case 3.0 Scaled
DWG No.
Rev.
Date of issue
Sheet
1/1

End Result	
While the version 5 is the final result of the systems control case, during the final stages of production of the project, some minute but very instrumental problems were realised. When designing the holes for the button based on the data sheet, I generated the model sketch with the side lengths being measured from the internal centre of each curved triangle corner. Following printing the piece and noticing the hole was much bigger than required, it was realised that the data sheet had the side lengths being measured from the external centre of the triangle corners. This required for just the front piece to be reprinted, I also increased the hole from the LCD screen by 1mm in each direction to make it easier to fit the piece in without needing to sand anything back. Once the new front piece was printed, the case was completed and was ready to have everything fitted into it.	Image:

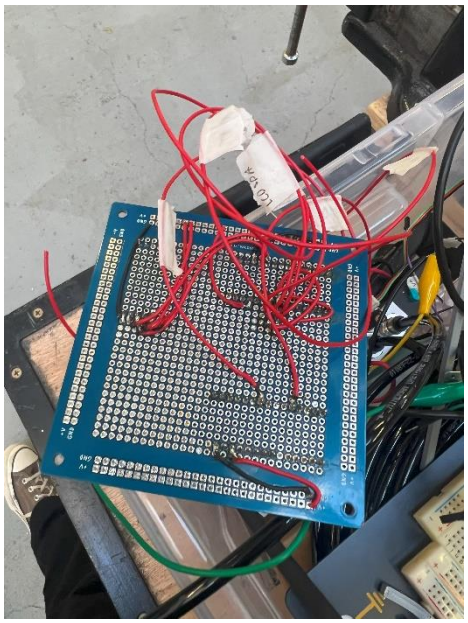
Code/Wiring

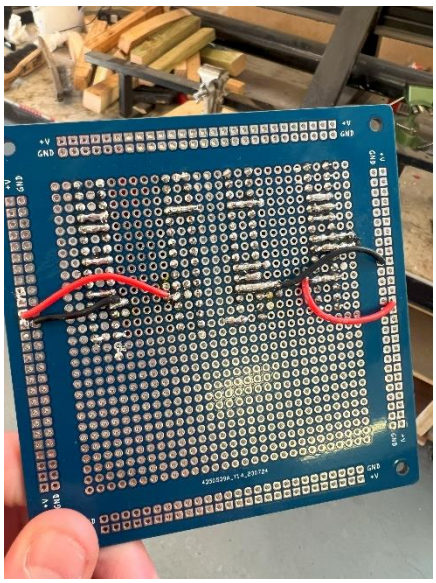
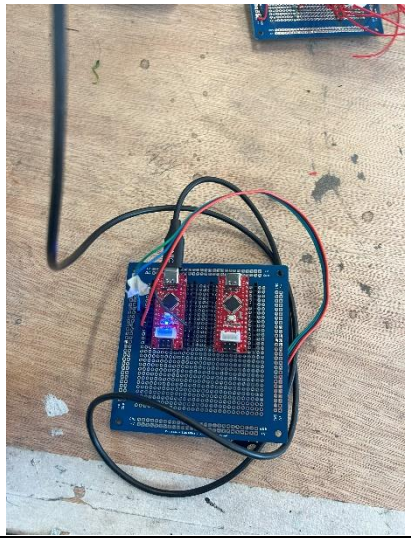
Initial Design Plan	
The initial plan for the code was to combine the sensor code through float functions alongside the button and relay code. This would require the use of only one microcontroller which was predicted to be easily wired to accommodate the needs of all componentry.	Image: 

Design Iteration	
Throughout the coding process many small iterations were made including; - Starting the process by isolation of each component into the own code script and own bread board, then testing each worked. - Slowly combining the code and wiring, starting with bringing the sensors into the LCD code. This had some issues that were later resolved with the use of float functions with return calls. - With the sensor code all set up and working with the LCD screen the next stage was to implement the relay and button code. The combination of code was where the majority of the issues and changes needed to be made. The sensors changed from all being in the void loop function to having their own float function definitions and then being called on in the void loop. This organised the code a lot better and also made integrating the sensors into the LCD screen a lot clearer. With this established, the process of integrating the relay and sensor code was predicted to be simplistic. The same process was used for defining	Image:  

the relay and button code within a float function then calling upon it in the void loop section. However, the delays in the sensor code generated lots of issues. Due to the LCD screen flipping between 2 different screens in order to display 4 different data points, there were delays in order to not have the data screens overlap each other. When implementing the relay and button code this meant that the button press would only be registered during a miniscule gap in the delay. When the relays were removed to test if the screen could display the code without it, the screen displayed the 4 data points over each other as predicted.

The initial solution to this problem was to explore adding millis into the code instead of delays. This is due to millis only pausing specified aspects of the code not the entire script. The investigation and attempts to implement this into the code was a lengthy process and was not achieving the results required in the time allocated. This caused the reversal to the contingency plan of splitting the code onto two microcontrollers which illuminated the issue altogether and saves time. While this solution worked very well for the code it generated the need to use a PCB board in order to organise all the wiring of the components. The initial iteration of the PCB board had the connections between the micro controller and the electronics being made with just wire. This produced a messy result that produced issues when tested with a power supply, so the wiring process was restarted with the use of header pins in order to properly organise everything. It also has the added benefit of making troubleshooting any issue that arise much easier.



End Result	
The end result of the code was to have two separate sketches running on two different PCB boards. One sketch for displaying the sensors and one to control the button and relay. The end wiring result was to abandon just using wire to connect all the componentry but instead to used header pins and jumper cables. Only using wire when necessary for connecting the GND and 5V as well as connecting the 5 volt regulator that was included in the final iteration made the process of powering the boards more organised. To enhance the organisation and quality of the wiring everything was encased in either heat shrink or in braided cable sleeving. The gromet was also connected into the systems housing case which further protected and organised the wiring.	Image:  

Testing

Explanation for change in testing format:

While progressive tests were run throughout the entirety of the production process (See: “Design Iteration and Improvement (Progressive Testing)”) the timeline for testing was significantly altered to allow for more production time and to have more succinct testing be performed at the conclusion of the production of the system. This results in the testing + calculations being removed from the production journal and altered within the Gantt chart.

Tests:

- Fusion 360 Testing
- Aerodynamics
- Downforce Testing
- Code Troubleshooting
- Power Testing
- Sensor Functionality
- Simulation of DRS Functionality

Fusion 360 Testing

Aim/Method:

Fusion 360 Digital Simulation Testing

- Purpose for testing: The general intent behind using the CAD model of the rear wing to achieve some test results, it is to gather and interpret data that isn't physically able to be recorded. These limitations are due to lack of professional testing equipment as well as time constraints.

  While the digital simulations do provide some general understanding of a prediction of the wing's performance, they do not account for real world factors which include the specifics of the construction techniques, any imperfections in the materials or changes in surrounding environment. This results in inaccurate results, however they do still hold some value in the potential results that could occur.

- Method of testing: • Using the Fusion 360 CAD file for the rear wing and in some cases isolating certain components. The simulation feature of the program was used. After dealing with some issues with the joining of the components causing errors with simulations, and an outdated version of the software halting testing; testing was possible.

  • Firstly creating a viewable mesh of the components was needed before adding fixed constraints to stop unmoving/mounted components to section from altering the results. Then the location and direction of forces was selected and specifically in the case of the wing flaps the angle of force was selected.

  • The different test categories are; • Structural buckling
  • Shape Optimisation
  • Static stress
  • Static stress displacement

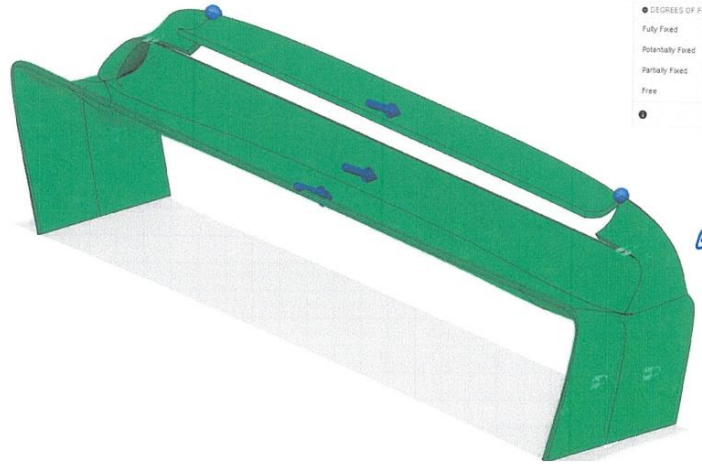


Image of connection contact error.

Predicted results:

My predictions for the results of the Fusion 360 testings are;

- The biggest structural issue will stem from the upper rear wing connector as well as the curves in the end plates. The shape of these forms should present issues with the structural integrity of the wing. As they are the also responsible for holding key components together (eg. Upper and lower rear wing flaps), they will have issues with structural buckling.
- With the shape optimisation I believe that the front face of the lower rear wing flap will require the most amount of alteration. This face is specifically shaped with the intentions of maximising aerodynamics, but I predict that this will generate areas that optimisation is recommended for strength and rigidity.
- Due to the locations that the loads will be applied to the rear wing I predict that majority of the buckling will be along the Z axis of the form. This logically makes sense due to the force pushing the wing back.

Results/Discussion:

Structural Buckling Simulation

Total Displacement

X Displacement

Y Displacement

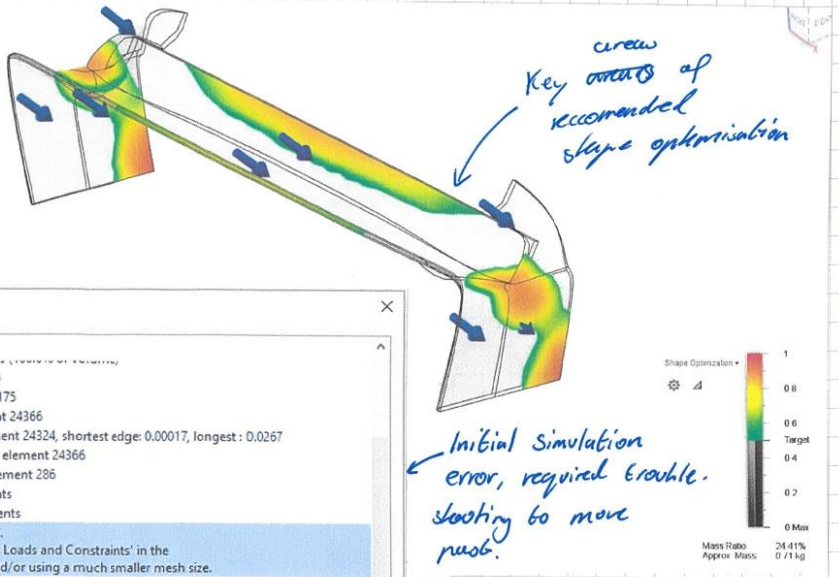
Z Displacement

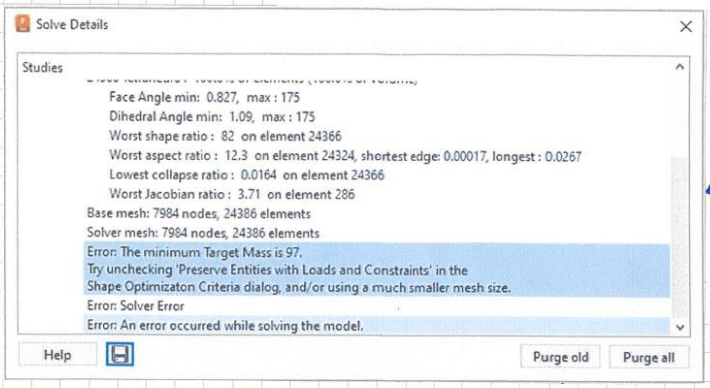
- Reason for test: The purpose of the structural buckling test is to analyse any buckling caused by specific axial loads. The data recorded above shows the results of direct force applied on the model vertically down. This provides results that can determain the models tendencies to shift/buckle under applied load. This helps to determain weak points in the model and can provide evidence to analyse the displacement of the rear wing. To summerise this data allows an analysis of the strength and predicted durability of the rear wing.

- Results/conclusions: The results of the structural buckling test exhibit some weaker points of the model as well as displacement tendancies.

  Key takeaways:
  - As seen in the 'total displacement' the overall main body of the wing has minimal displacement. Even though the stationary rear wing flap has some displacement it is minor.
  - The areas of the model that are 'weak points' and would require re-inforcing to minimise the displacement are the connectors to the upper rear wing flaps and the corners of the end plates.

Shape Optimisation Simulation





Initial Simulation error, required trouble shooting to move push.

- **Reason for test:** The purpose of gathering data on areas that are recommended for shape optimisation is to analyse where parts could require alteration or reinforcements to maximise strength based on the specified loads and their direction.

  In this simulation the load was directed at approximately a 30° angle (~~from~~ from vertically down) to represent the strong aerodynamic forces that would be applied to the rear wing of an F1 car. While the magnitude and direction of the load are not direct representations of the actual loads applied to an actual rear wing the produce an estimated representation of how the forces would effect my design.

- **Results/conclusions:** The areas highlighted in the image of results above, exhibit points in my design that are not as structurally sound. These areas are my joins between sections of the rear wing. The join between the endplates, stationary rear wing flap and the connector to the upper rear wing flap is a large joint that is recommended to recieve optimisation. For this particular point I would reinforce the join but also would create smoother joins between the components. The other two major points that shape optimisation is recommended are the front face and rear face of the stationary rear wing flap. These areas being highlighted is logical due to them being fixed at both ends without extra reinforcement. They are sections with large surface areas, recieving high force load loads. These sections would also require reinforcin

Static Stress Simulation

Image A

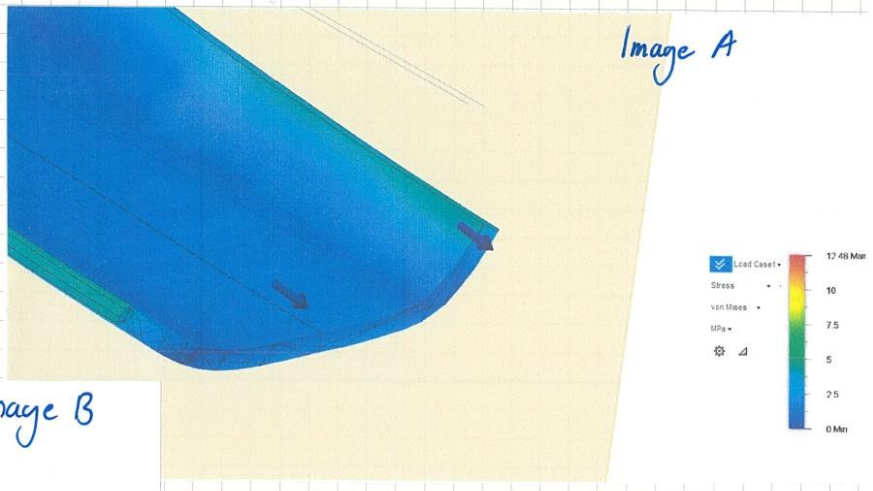


Image B

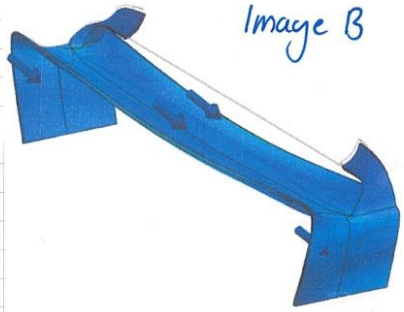


Image C

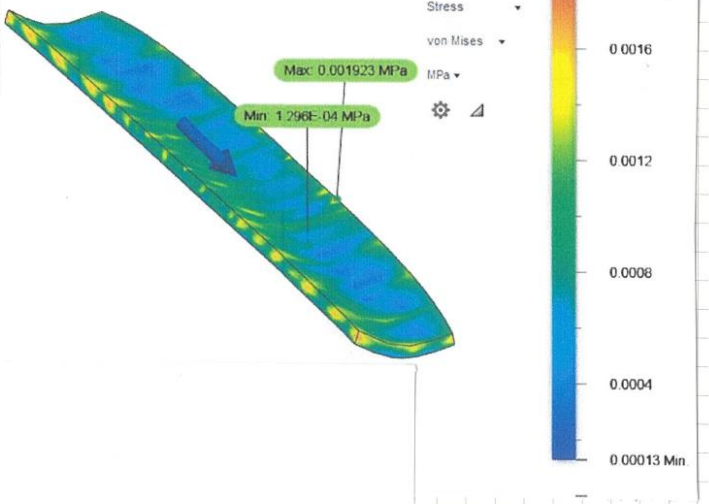
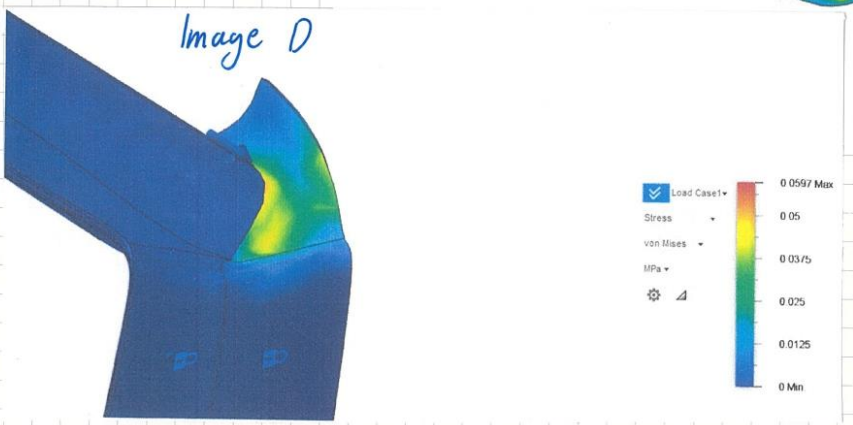


Image D



A: Material cross section stress simulation.
B: Static stress simulation of entire main rear wing body.
C: Static stress simulation of upper rear wing flap.
D: Static stress simulation of connecter to upper rear wing flap.

- Reason for test: The purpose of the static stress simulation tests is to ascertain information on likely areas of failure in the design based on the load conditions. In this simulation force was directed at approximately 30° (from horizontally down) towards the model. The magnitude of the force is not a representation of the exact real world forces applied to an F1 car, but is only an estimated value to exhibit usable data. In this simulation the strain, stress and deformation results are analysed to determine 'weak points' in the model!

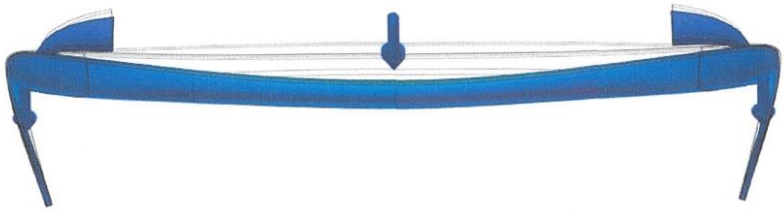
- Results/conclusions: A: The material cross section shows minimal stress caused by the load applied. The simulated material chosen was CFRP which was the closest material as it is Carbon-fiber-reinforced polymers. The only areas of minimal concern are the edges of the wing, overall a successful design.

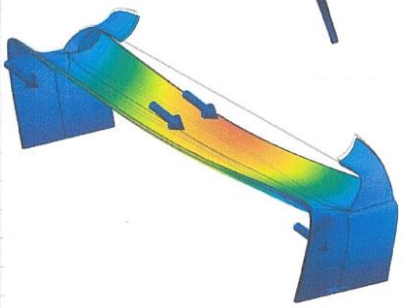
B: The overall analysis of the main rear wing body (when fixed by the bottom of the end plates) exhibits a successful design with minimal detections of stress caused by load. This can produce the conclusion that overall the design exhibits signs of being structurally strong and able to cope with force loads.

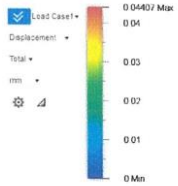
C: The isolated upper rear wing flap produces interesting results with the stress simulations. The key issue appears to be the edges of the wing. A solution to this could be to redesign the edges of the wing to curve them resulting in less load being forced upon one edge point. A conclusion can be drawn that the wing flap is relatively stable but could benefit from some redesigning of the edges of the wing.

D: When isolating the connectors of the upper rear wing flap to the main rear wing body and altering the fixed points of the design. Interesting results appeared with the weak points of the component. The weak points appearing on the curve makes logical sense especially with the angle of force applied. This result causes for consideration of inferring these 'weak curves' or a redesign of the component to minimise 'weak points' caused by stress.

Static Stress Displacement Simulation





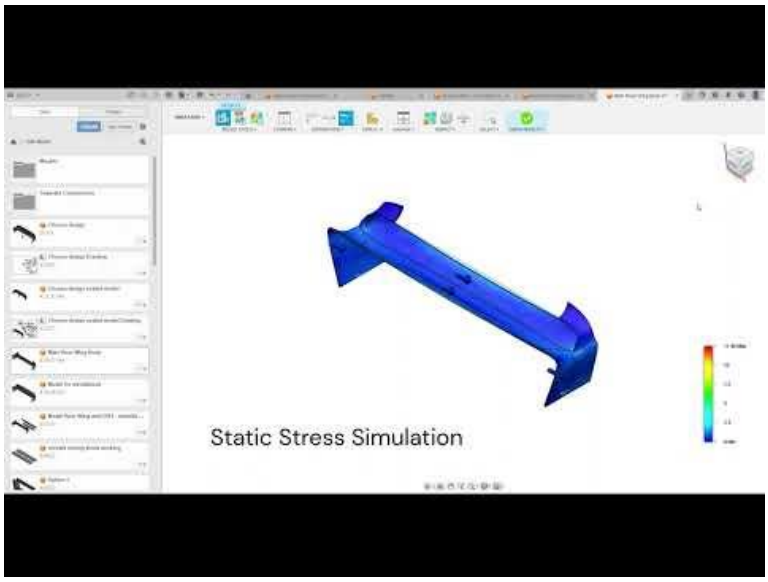


- Reason for test: The purpose of the displacement simulation in addition to the static stress simulation test is to understand how strain, stress and deformation also effects how the model moves when under stress. This simulation provides data on the comparison between the initial fixed position of the model ~~in~~ to and the displaced position caused or under load. This provides information on where the model needs to be altered/reinforced to reduce displacement.

- Results/conclusions: The major takeaway made obvious by the data from the simulations is the high amount of displacement of the stationary rear wing flap. This high level of displacement is a definite strength and durability issue and to improve the design strong reinforcements would need to be added. I am hesitant to suggest a redesign of this section of the model as I predict that this is a structal issue present with all rear wing designs. The clear solution would be to add internal reinforcements to the wing.

  Another more minor issue highlighted by the images above is the displacement of the connectors to the upper rear wing flap. As seen in the other simulations this particular component definitely requires either a redesign to improve its rigidity or reinforcements.

YouTube link to my results: https://www.youtube.com/watch?v=_FZaYW-uoVI



MicroCDF Virtual Wind Tunnel Testing

Aim:

The aim of this test is to gain an understanding of the aerodynamics of the rear wing and the effect that the DRS has on the airflow.

Predicted results:

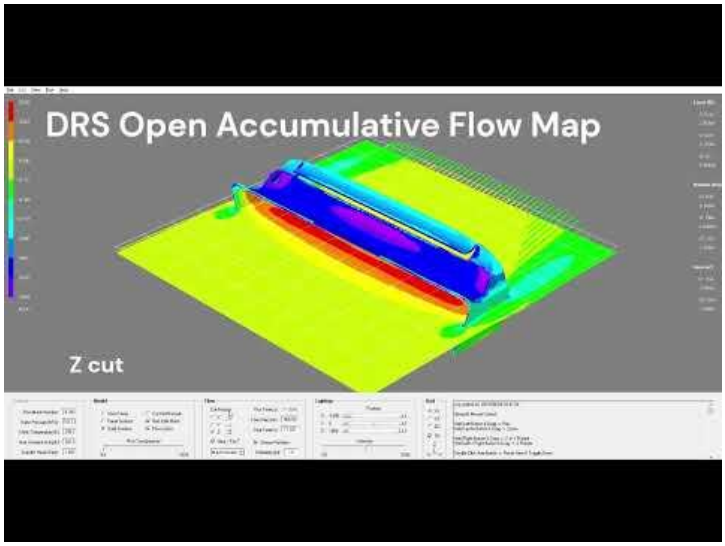
My predictions are that the DRS will have a significant effect on the airflow through the wing. A comparison between the open and closed upper rear wing flap will demonstrate the significant extra downforce production with the open rear wing. My prediction for a potential flaw with the aerodynamics of the design is with the connectors to the upper rear wing flap. I believe that this section of the design will interrupt airflow around the rear wing flap with and without DRS activated.

Method:

- Export a STL file from Fusion 360 (the program automatically exports a binary STL file), make sure that the components are facing the XZ axis plane.
- Open up the MicroCDF program and import the stl file, use the edit view function to light up the model with the direction of wind flow (X axis).
- Check that all normals of the stl file's vectors are outwardly pointed by selecting to view facet surfaces.
- Refine the surface of the model in the edit drop down tab. Then in the same location, create a mesh of the model.
- Set the stability of the model to medium and select the correct CPU to begin running the simulation.
- Select 'Run' in the drop down menu and wait approximately 5 hours for the simulation to run.

Results/Discussion:

YouTube link to my results: https://www.youtube.com/watch?v=ldS2Cs2ny8c



DRS Closed Accumulative Flow Map

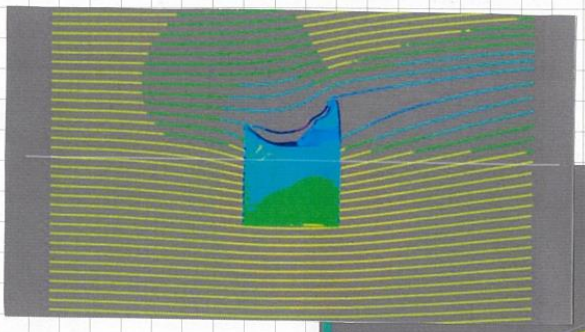


Image A

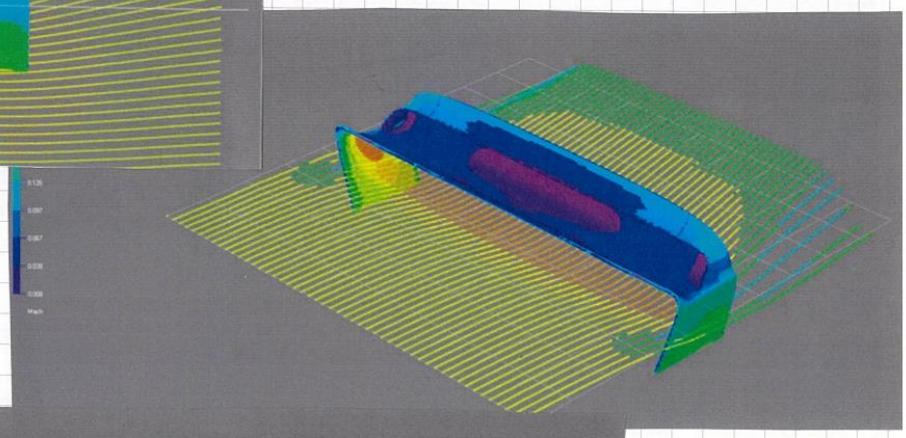


Image B

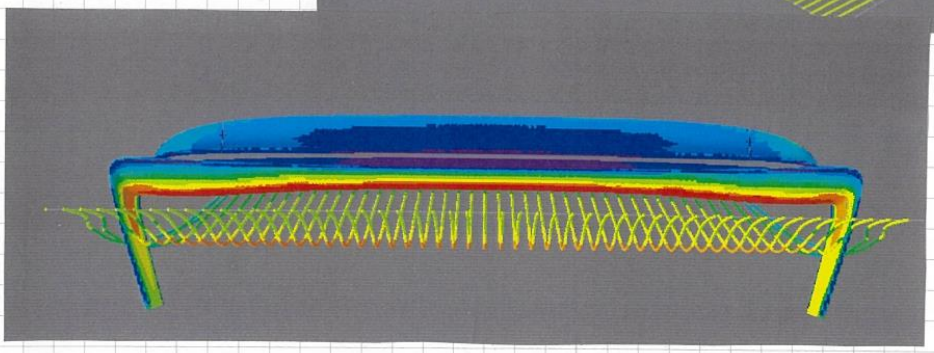


Image C

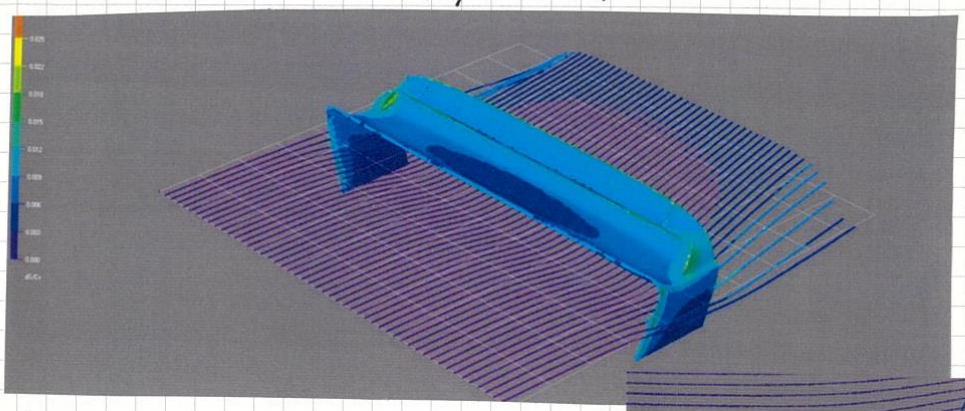
- Purpose for testing: The reasoning behind this specific test is to gather information on and analyse the cumulative flow of the wing and understand ~~and~~ any flaws in the design.

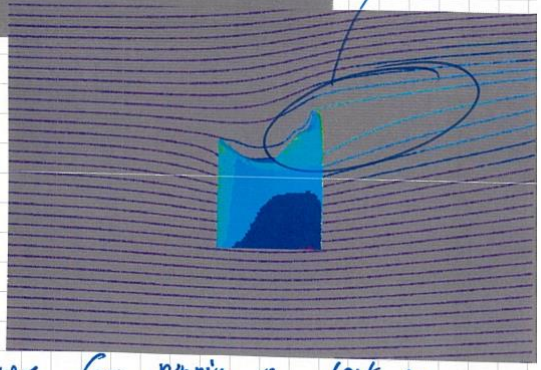
- Results/conclusions: A: From the side view of the wing we can interpretate the flow of air over the wing in the x cut direction. This supports the intension behind the wing to generate downforce through the depiction of the purple flow line over the upper rear wing flap.

The flow lines depicting turbulent air exiting the rear wing comparably suggests that the wing is interrupting the flow of air pushing force down.

B, C: The pictural and front view of the wing provides depth on the main points where down force is created. Having the central points of both the upper and lower rear wing flaps ~~presents flow~~ being high pressure points suggests that these areas are producing downforce from slowing the flow of air due to their slope and larger surface area.

DRS Closed Entropy Analysis





Entropy

- 'A thermodynamic quantity representing the unavailability of a system's thermal energy for conversion into mechanical work, often interpreted as the degree of disorder or randomness in the system.'

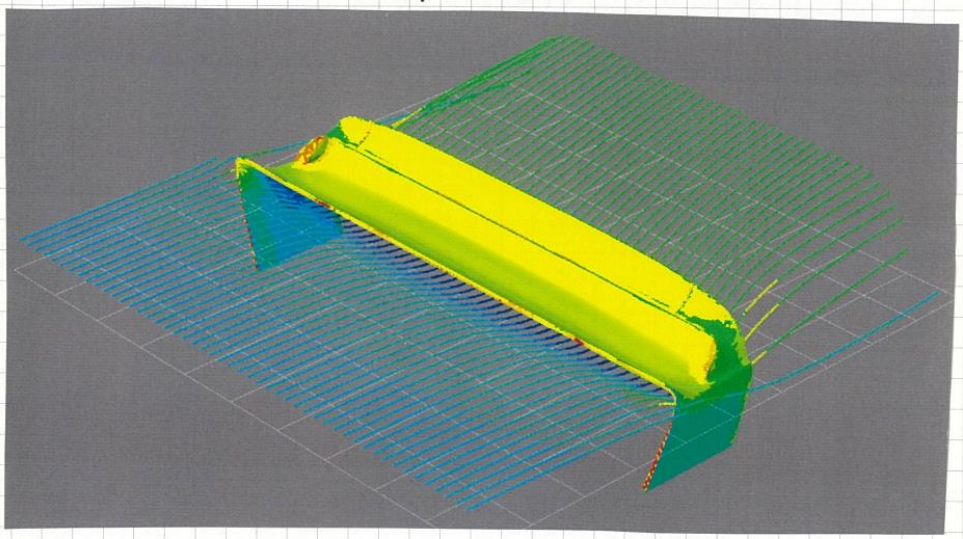
- Purpose for testing: The key purpose for running a test on the entropy of the rear wing is to gain greater access to the disruption of airflow over and around the wing.

- Results/conclusions: With the rear wing endplates protruding outside the shape of the wing due to their curved shape, it can be interpreted that this shape is negatively obstructing the airflow around the wing which could be causing unnecessary drag.

The exit path of the airflow over the wing and the flow separation is positive in terms of the production of downforce. The only concerning point of this section is the lack of air travelling through the small separation of the upper and lower rear wing flaps when it is closed. While this potentially could be due to the program regarding it as relevant, I think the produce the shape of the wing not producing maximum downforce for correct speed when DRS is closed.

Overall this suggests that even with minor flaws the wing is exceptionally productive at transporting air over and around its curved surfaces. The only changes to improve this design would be to reassess the shape of the upper rear wing flaps.

DRS Closed Temperature Analysis

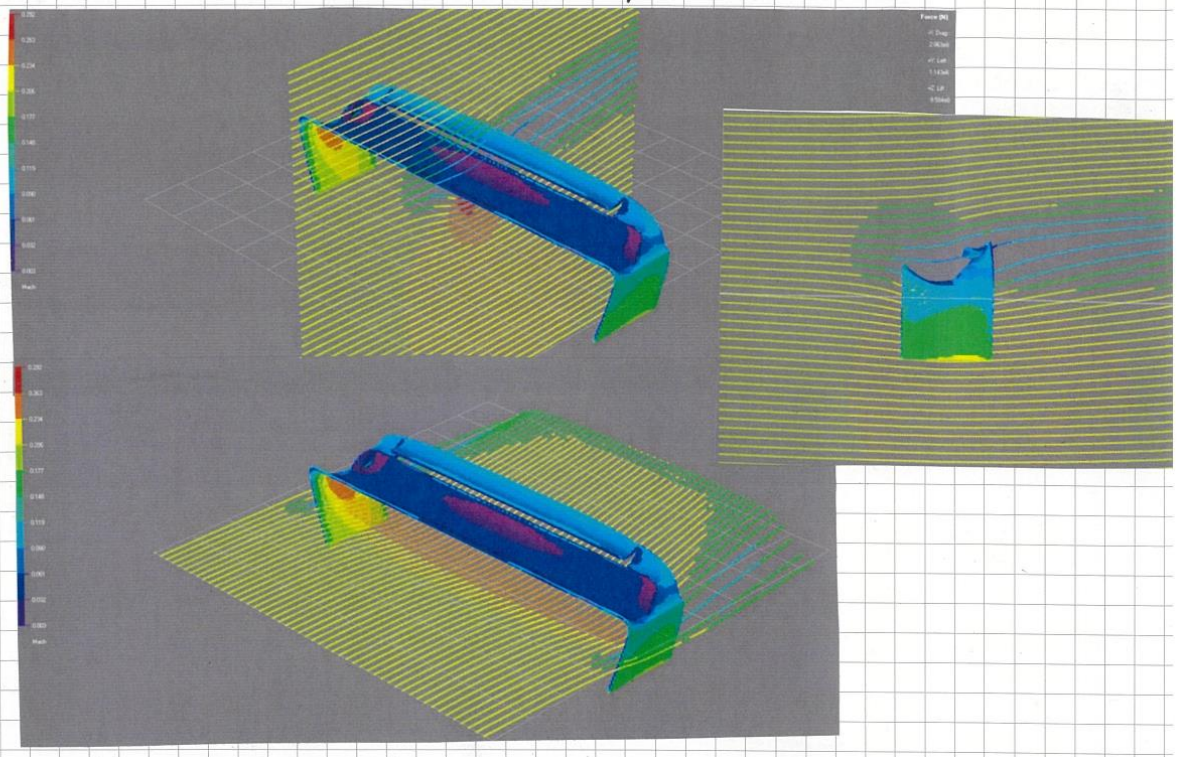


- Purpose for testing: The purpose for evaluating the temperature changes produced through the direction of airflow is to gain an understanding of any potential areas for concern. This is in terms of any weak points that may occur through continuous loading, but more importantly heat is produced from continuous friction which would suggest areas the could require aerodynamic improvements.

- Results/conclusions: The image of the temperature analysis suggest only a few areas but have a noticeable temperature difference. The main point to look into is the front facing edges of the wing which have majority of the airflow onto the front of the wing. Creating this impact but due to the location of this force the result is to be expected.

  Aside from the front facing edge there are no other noticable areas of concern temperature - wise. Which concludes that the wing is efficient at managing airflow limiting friction.

DRS Open Accumulative Flow Map

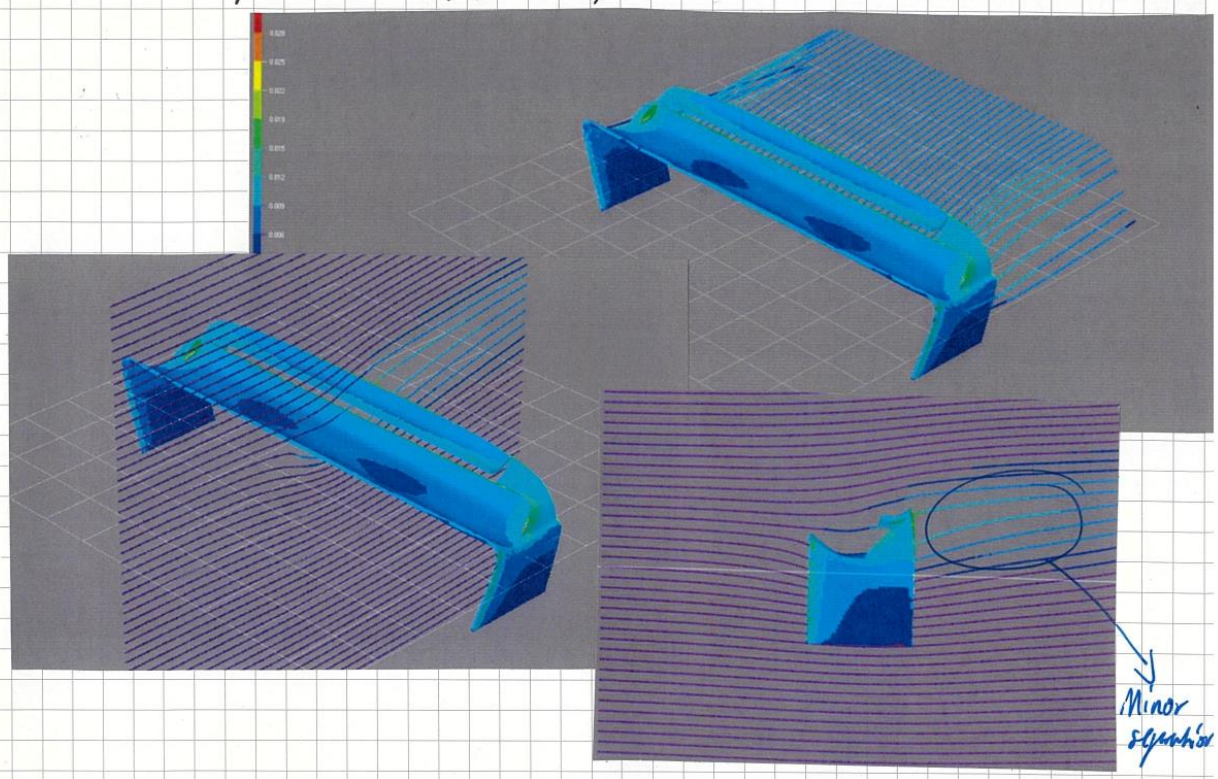


- Purpose for testing: The intention behind running the exact same test but instead having the upper rear wing flap in its open setting is to be able to compare results on airflow due to hopefully increased downforce reduction.

- Results/conclusions: The major difference between the two tests is being able to see the airflow lines going under the upper rear wing flap which means that in its open setting the wing reduces drag. This concludes that the DRS is functional within the components of my rear wing design.

  There is noticably less interruption in airflow behind the wing which further concludes that the slope and angle of the upper rear wing flap reduces drag.

DRS Open Entropy Analysis



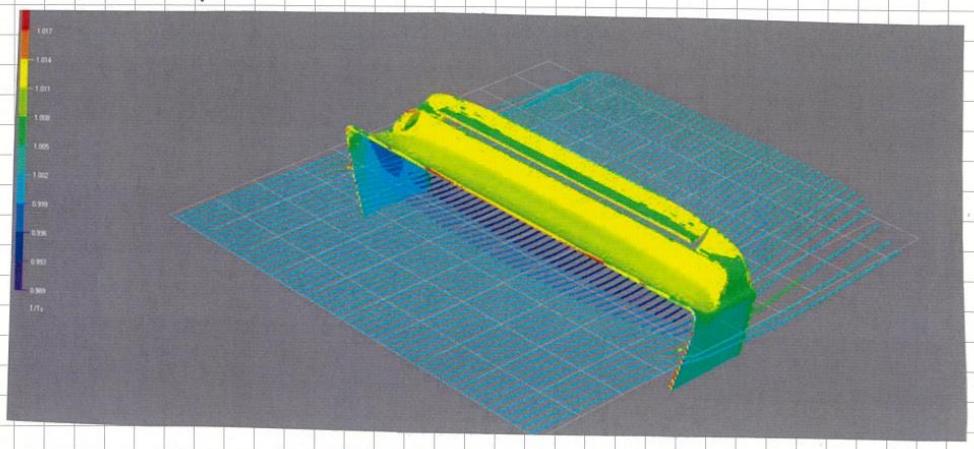
• Purpose for testing: The intension behind this test is to compare the results of the closed DRS entropy and the open DRS entropy. The main points of comparison are the flow separation behind the wing and the flow of air over the wing.

• Results/conclusions: While the flow of air around the main body of the wing remains the same the focus point for this is the flow over, around and into the open upper rear wing flaps.

The exit path of the wing interrupted air through the gap between the rear wing flaps suggest large drag reduction through minor flow separation. While the closed wing had large separation of flow due to it interrupting air and creating drag. The open wing only has minor separation which concludes that the wing is successful at reducing drag.

Overall this analysis shows that the design of my wing and the DRS components (angle of rear wing flap when open) is effective at reducing drag.

DRS Open Temperature Analysis



· Purpose for testing: The intension for this test is to compare the temperature differences between the open and closed DRS rear wing flap. The comparison allows for an analysis of temperature focus points that could allude to weak points and high friction/drag points.

· Results/conclusions: While both tests are almost completely identicle in terms of the effect on the main rear wing body. The differences arise with the centre of the upper rear wing flap and the exit airflow temperature under the main wing body.

Due to the changed angle of the upper rear wing flap it makes logical sense that it would be impacted by on-flowing front facing airflow. More airflow parallel facing surface area increases the level of direct airflow contact. Increasing temperature concentration. This is not a point for concern.

The interesting change observed is the change of temperature under the main body of the rear wing. My predictions for this is due to a smaller interruption in airflow higher up on the model has a run on impact on the lower airflow.

The differences between the two tests are minute and don't demonstate large temperature effects of the DRS being open.

Downforce testing

Aim:

The aim of this test is to gain an understanding of the effectiveness of the DRS on the rear wing model I have produced. The overall data that I am aiming to attain is the difference in down force when the wing has the DRS open and closed.

Predicted results:

The results that I predict is that there will be a reduction in the force the wing produces downwards when the upper rear wing flap in open. Through the weight data that is recorded I anticipate that the size of the difference will be very small due to the small size of the wing sections. However any consistent change in weight recorded being forced downwards is proof of the DRS working.

I also anticipate that the results are not going to the most accurate as the recording device being used is a kitchen scale instead of the preferred load cell as at the time of testing this part of the project the load cell was not wired up completely.

Required Material/Equipment:

- Leaf blower
- Cardboard box
- Scale
- Tape
- Stanley knife
- Painter's sheet

Method:

- Cut a cardboard box to size leaving the top a bottom open and secure the sides with tape in order to stop the box from collapsing in on itself.
- Wrap the scale in a thin protective painter's sheet to limit it getting damaged by the model.
- Place the model on the scale in the box and centre it.
- Cut a small hole in the top of the box, with the purpose of using it to have the phone camera record the results from the scale.
- Take the model off the scale and zero it before placing it back on and begin recording with your phone.
- Line up the leaf blower with the wings of the model centred in relation to the actuator stand, keeping it level.
- Keep the leaf blower at a consistence airflow rate for 20 seconds in order to get a range of data to analyse and find the peak lightest weight.
- Repeat the step above for 3 separate tests (each test contains 2 sub tests, one with DRS open and one with DRS closed), making sure to change the upper rear wing flap from open to closed between each sub test. Record the data.

Results:

Control weight of the wing section model: 2.890Kg

Peak Lowest Weight results (Kg)	Test 1	Test 2	Test 3	Average
DRS Open	2.194	2.153	2.154	2.167
DRS Closed	2.818	2.285	2.286	2.463

Discussion:

Before discussing the results in the table above I think its key to outline major aspects of the testing method that impact the accuracy of the test results. Some points are;

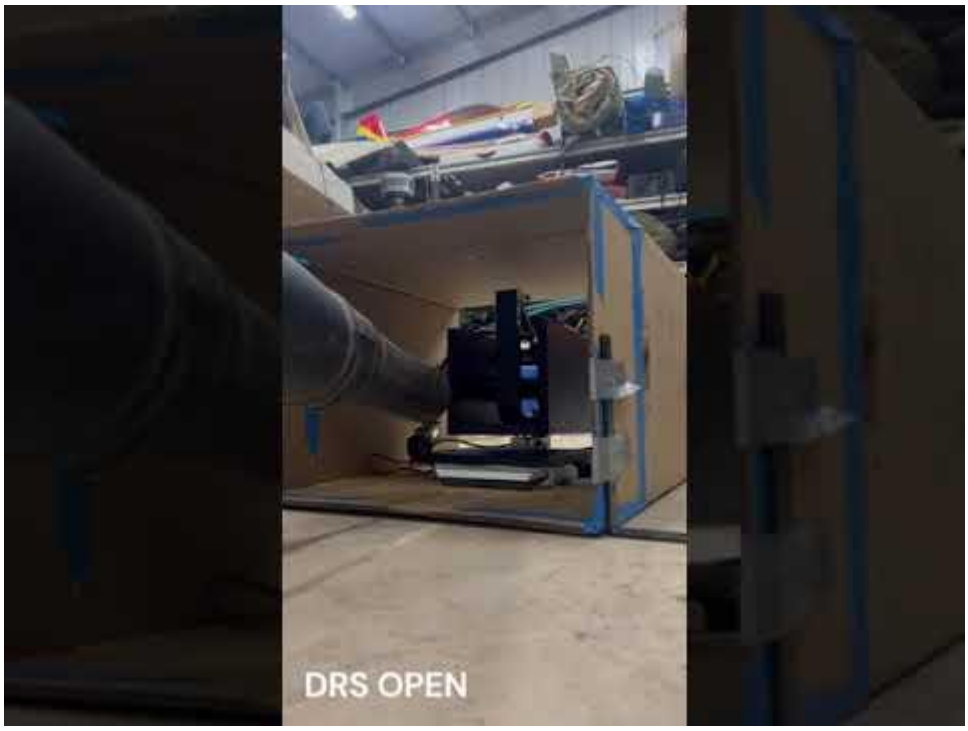
- The wind tunnel not being a professional setup with all optimised materials, wind distribution, measuring tools ect. plays a big role in how the results can be interpreted. Each aspect of the inaccuracies are outlined below.
- The scale used to measure the weight data is not optimised to read specific measurements fast. This impacts the validity of the results and the speed at which fluctuation in results is recorded.
- Using a leaf blower that does not widely distribute the flow of air effects the readings through more sections of the systems getting greater airflow. Also not having the wind blower fixed to a stationary point with a consistent flow of air skews the data. As each test does not have the exact same conditions, the data is opened up to being influenced by outside factors. If I was to redo this test I would place the air source in I fixed location and set it to a constant airflow.
- The outlier of 'Test 1 DRS Closed' does highlight the existence of inaccuracy in the testing, this outliner could be present due to all the reasons listed above.

The results presented suggest that the model is effective within the definition of a Drag Reduction System. The difference of 296 grams between the average of the open and closed DRS are evidence to support this claim. While the inaccuracies presented above do play a role in the accuracy of the results even if the specifics of the numbers are inaccurate, the major take away is that the average weight force when the DRS is open is less then when it is closed.

While the results above are not specific calculations of the downforce produced with consideration to the wing dimensions and lift coefficient, some basic calculations on the percentage reduction can be made. Real F1 rear wings have a maximum drag reduction of 80% from the rear wing with a average closer to 67% (Developmetn of a Drag Reduction System (DRS) For Multi-Element Race Car Wings, 2023). By calculating the percentage difference between the two weight values and the difference between them. The result is an approximately 12% drag reduction. Due to the testing model only being 150mm out of an entire F1 rear wing these results are still positive and provide evidence for effective DRS capabilities.

Images:

YouTube link to video of testing running: https://youtu.be/55LzpFW4G28?si=pKmCuXG56NE0NhnZ



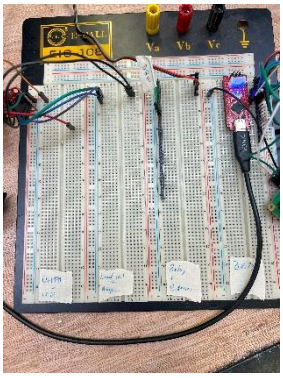
Code Troubleshooting

The code testing was a continuous iterative process, and as a result testing analysis is much broader and explores a wide range of code trouble shooting.

Aim:

The aim of every coding troubleshooting test is to isolate individual sections of the code to locate the problem and then troubleshoot it until it is solved.

Required Material/Equipment:



- Computer
- Bread board
- Jumper wires
- USB to USB-C cable
- Chat GPT

Method/Examples:

Method 1 of troubleshooting code:

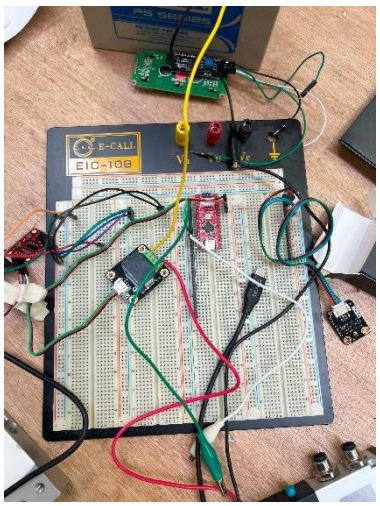
The first method is to physically isolate electronic componentry to either conclude that the issue is one involving wiring or pin locations. The best example of me following this method is with the individual breadboarding of every separate electronics component that was individually troubleshooted and tested prior to integrating them together.

Method 2 of troubleshooting code:

This method is the nest step after ruling out if the issue is a physical wiring or componentry issue. This method includes isolating the code into individual sketches or commenting out code, in order to narrow down the specific issue. An example of how I followed this process is with the initial integration of all of my individual sketches together. When everything was combined into one sketch there were a lot of initial issues. To simplify the process of reading and evaluation of the code to locate the issue, I commented out any code that I knew wasn't needed. This started with any Serial.print code and when I further couldn't find the issue I commented out entire sections of code to narrow down the possibilities.

Method 3 of troubleshooting code:

When all else fails a viable method of location errors in code and discover ways to simplify and condense code is by using Chat GPT. The AI is very adept at locating issues in code and then explaining them in an understandable way. The AI was very helpful when I was integrating all my code as it could quickly comb through my code and locate any repeated set up functions that clogged up the code as well as efficiently suggest potential issues with the code and provide different methods of finding a solution to the problem.



Power Testing

Test 1 (PCB):

Aim:

The aim for this test is to determine if the componentry for the powering of the two microcontrollers is connected properly, as well as if it's supplying the correct voltage level.

Predicted results:

Reviewing the clarity of the soldering of the capacitor, voltage regulator and terminal block I predict that there will be an issue with the voltage output. The solder that is bridging the connection between the components is messy and presents issues with components being incorrectly connected.

Required Material/Equipment:

- Voltmeter
- Voltmeter test probes
- DC Power Supply
- Wires with alligator clips
- The wired PBC boards with attached components

Method:

- Set up both the DC power supply and the voltmeter to correctly gather data from the voltage regulation circuit. The power supply needs to be set to 12V and the voltmeter set to the correct voltage setting with test probes attached.
- Using alligator clip connected wires with one end being exposed stripped wire, connect the power supple and the terminal block on the PCB board.
- With the voltmeter test probes, place the negative on the ground pin of the voltage regulator and the positive probe on the capacitor Vout pin.
- Collect the readings from the voltmeter.

Results:

The results from this test provided little conclusions, the voltmeter was receiving voltage so small it couldn't be displayed with the rare spike to a tiny readable voltage. Even when troubleshooting to changing the location of the voltmeter probes and switching out the power supply. The results remained the same.

Discussion:

Due to the inconclusive results from the initial test preformed a follow up test with isolated components is required. However, it can be determined that there are issues with the wiring/ soldering of the components that have led to the lack of a voltage recorded from the components.

Test 2 (Bread Board):

Aim:

The aim for this test is to isolate the voltage regulator componentry in order to determine if the wiring of the PCB is the issue or if it is an issue with the components.

Predicted results:

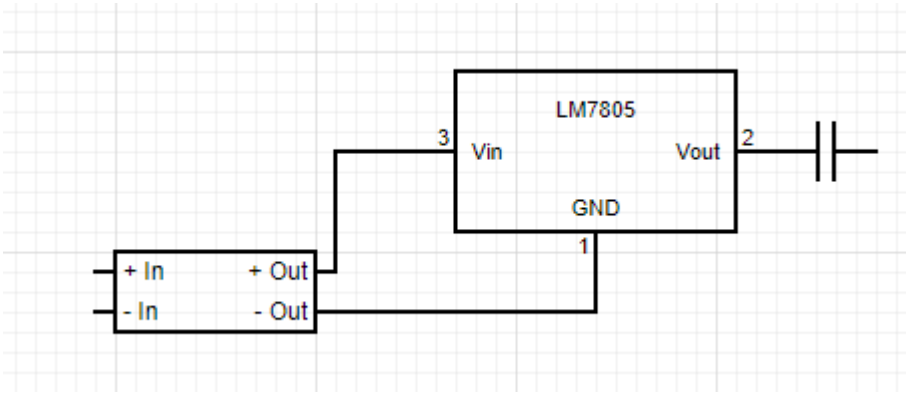
Due to the components being isolated I predict that there will be an expected voltage output of the required 5V. The removal of external factors including the soldering of the components to the PCB board increases the likelihood that this result occurs.

Required Material/Equipment:

- Bread board
- Jumper cables
- DC power supply
- Wires strippers
- Terminal block
- 5V regulator
- Electrolytic Capacitor
- Voltmeter
- Voltmeter test probes

Method:

- Set up the bread board with the terminal block, capacitor, and voltage regulator. The wiring of the setup is as shown in the circuit diagram I've created below.



- Once all wiring is set up and a jumper cable is connected to the positive side of the electrolytic capacitor, the DC power supply can be turned on and set to 12V.
- The DC power supply is connected to the + In and – In pins of the terminal block using alligator clip wires with stripper ends to connect the power supply and the terminal block.
- Connect the test probes to the voltmeter and set the voltmeter to the correct volt setting.
- Once this is all set up, using a voltmeter first test the voltage output from the negative capacitor pin and the positive jumper cable connection. To troubleshoot any issues with this data collection, carefully connect the voltmeter test probes to the GND and Vout pins of the capacity.
- Record the results from the voltmeter.

Results:

The initial results from the voltmeter were a wide spread of volts quickly flicking between values. The values ranged between -0.2 to 1.3 V. This occurred when taking the data from the jumper pins

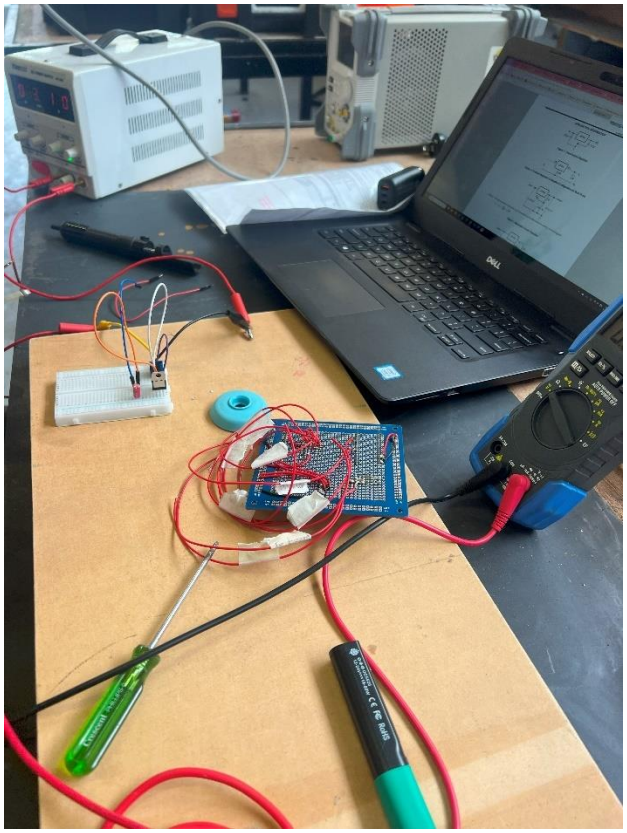
of the capacitor. To trouble shoot this a second reading was taken directly from the voltage regulator which produced the resulting voltage of 5.055V

Discussion:

The conclusion being drawn about the initial issue with the readings produced from connecting to the capacity and jumper cable is that the problem could have been either the pins from the jumper cable and capacitor intermittently touching the bread board terminals. There is also the possibility of a microfracture in the jumper cable.

Once the troubleshooting occurred of altering the data collection point, clear results that align with my predictions were achieved. The results of the 5V reading as well as a comparison between the issues with Test 1 and the cleaner and more simplified set up of Test 2, prompted design iteration of the wiring of the PCB board. The change is to use header pins instead of wiring everything directly to the board in order to achieve better wire management and be able to easily troubleshoot and swap out components as needed.

Images:



Sensor Functionality

Aim:

The aim of this test is to determine the functionality of the BMP sensor's temperature and pressure detection in different environments and understanding if it is changing the recorded data.

Predicted results:

I predict that the sensor will output data that is different in each environment, but for smaller increases in environmental changes I anticipate that the sensor will take longer to output the final reading for the data changes.

Required Material/Equipment:

- BMP388 sensor and associated wiring to the microcontroller
- USB-USBC cable
- Computer with the functioning code for the sensor

Method:

- Ensure that the wiring for the sensor is correct and connect it to the computer using the USB cable.
- Upload the code that works for the sensor that was written earlier.
- Change the environment of the sensors and wait for approximately 2 minutes in each location to observe the sensor's change in pressure and temperature readings.
- (The altitude reading was negated during this experiment as it is not impacted by the environmental changes being produced here and analysing those recording would bear no impact on the conclusions created).

Results:

Environment	Classroom	In path of heat gun	Outside
Peak temperature reading (degrees Celsius)	21.79	55.96	24.67
Peak pressure reading (Pa)	99437.07	99458.03	99410.97

- The sensor was able to rapidly change from a base room temperature value to the 55.96 degree value. When it reset to the room temperature value and was then exposed to the conditions outside, it took significantly longer for the data to reach its peak temperature.

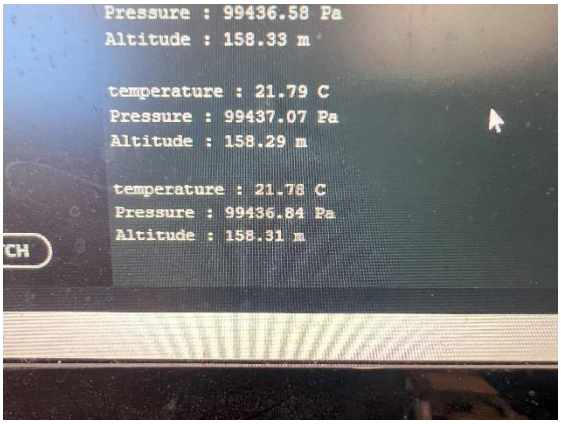
Discussion:

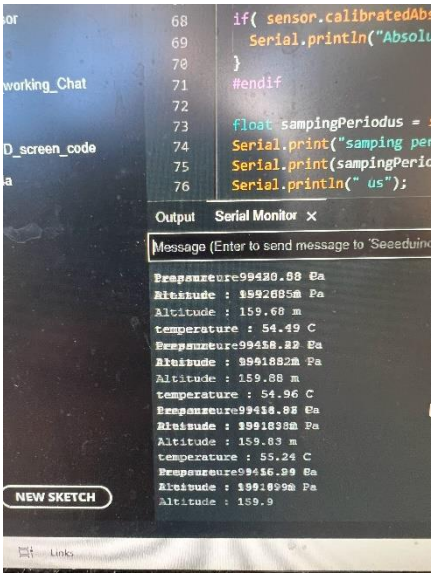
The above results allow for the conclusion to be made that the predicted results were correct. The sensor was functional in each environment and displayed accurate and succinct ranges of data before settling at a peak. Smaller changes in the temperature with a difference of 2.88 degrees from inside the classroom to outside took significantly longer to reach its peak reading compared to the 34.17 degree difference between the classroom temperature value and the heat gun value. This

prompts the resolution that the sensor is capable of reading a range of data points and can narrow down small changes in temperature event though this takes longer than reading larger changes.

The minute alterations in pressure is logically understandable as the environment change was not directly effecting those data reading. This conveys that the sensor is not sporadically outputting data and that the temperature recordings aren't inadvertently effecting the other data points. This provides the conclusion that the sensor is accurate in multiple areas of its functionality and is able to produce data outputs in a range of environments.

Images:





Simulation of DRS Functionality

Aim:

The aim of this test is to access the functionality of the sectioned rear wing model at simulating the actual functioning of DRS on an F1 car. The purpose of this is being able to determine if it fits its original purpose of aiding in the education of the technical aspects of the sport.

Predicted results:

I predict that the model will clearly demonstrate the opening and closing of the upper rear wing flap as a simulation of the actual DRS.

Required Material/Equipment:

- Air compressor
- Stopwatch
- The rear wing sectioned model with associated componentry

Method:

- Set up the air compressor to operate at 40 PSI which is within the safety and functionality ratings for the solenoid and actuator, while also generating fast actuation times for the DRS.
- Connect the solenoid to the air compressor and ensure the power supply to the solenoid is set up with two 12volt batteries in series.
- Run the stopwatch and actuate the DRS at the time intervals measured during the research stage of this system.
-

Results:

Here is a video of the comparison:
https://drive.google.com/file/d/1IbupxUFQvKqxvWi7U3nSr392LEsJSG-h/view?usp=sharing

Discussion:

The conclusions that can be drawn from this is that the system is functional in simulating the DRS of a Formula 1 car. While the model is not being subject to the same conditions as a on track rear wing, it is capable of simulating the process of initiating and deactivating DRS within the zones of a track.

Even without the planned functional aspect of a location sensor being a part of this to simulate when the wing had entered a DRS zone, just having the actuation occur by the button is a fair and understandable representation of what happens in Formula one cars when DRS in enabled by the drivers.

To sum up, the DRS model is an effective educational representation of the activation and deactivation of DRS on a real F1 car. This is displayed with the efficient opening and closing of the upper rear wing flap compared to a real Formula 1 lap time.

Evaluation with Criteria

Criteria	Description	How was this evaluated with the system? (Revised)	How well was this achieved? (1-10)	Evaluation feedback
Functionality	Does the system include an efficient usage of the rear wing actuator to activate and deactivate the DRS?	This was evaluated through the documentation and iterative review of producing a functional and efficient actuator. At the completion of the project the actuator was reviewed for its efficient activation and deactivation.	10	After assessing the actuator within its role in the DRS model and evaluating its efficiency I have concluded that it is highly efficient at activating and deactivating the DRS. Not only in terms on the speed at which it actuates but also through its durability and succinctness within the system. The actuator is well integrated into the system and consistently performs its role of controlling the opening and closing of the rear wing flap to a high standard.
	Is the system aerodynamic and reduce drag?	This section was evaluated in two different forms of testing. The first being the MicroCDF virtual wind tunnel analysis and the physical creation of a wing tunnel and testing the change in weight distributed down.	10	Following a review of the conclusions made from the two different tests completed it can be determined that the model DRS achieves being both aerodynamic and reducing drag. The virtual test demonstrates the efficient way the air flows over and around the wing, solidifying evidence for its aerodynamic capabilities and alluding to a reduction in drag through the depiction of air flowing through the gap in the rear wing when the DRS is activated. However the physical test solidifies proof of the system's ability to reduce drag through the data that was collected conveying an approximately 12% reduction in drag.
	Does the Model DRS include delayed activation of the	This section was evaluated by reviewing the production journal in comparison to a	0	After following the process for evaluating the system it can be concluded that the system does not include this

		DRS if it is attempted to be activated before the sensors indicated it is able to do so?	physical assessment of the final product.		consideration. The system does not include location sensors that indicate the ability to activate DRS based on location. The reasoning behind the exclusion of this consideration is to save time and to prioritise getting a high quality more simplistic system. This criteria is a consideration to allow for it to be a possible way to propel the system's complexity further if there is time a resources available. As conveyed within the Gantt chart there was not extra available time to expand towards completing considerations that didn't affect the simple functionality of the system.
		Does the system include a brake pedal to deactivate DRS?	This criteria was reviewed through physically accessing the completed system and referring specifically to the method of activation for the DRS model. An investigation of the production journal would also present the same results.	0	After reviewing the finished product and concluding that the method of deactivating DRS is through pressing the same button as activating DRS, it is realised that this criteria was not met. The intension behind excluding this criteria from the final product is evident in the production journal. Within the stage of product that covers all electronics including those in charge of activating and deactivating DRS it is evident that a copious abouts of design iteration and troubleshooting occurred that significantly pushed the time spent on the electronics then what was allocated. This is the reason why the brake pedal never became a realisation, spare time was not available to explore the considerations that did not drastically alter the final product.

User needs and requirements		Does the Model DRS gain input from a sensor to then allow a button to activate DRS?	This section was explored through reviewing the production journal comparing it to the final product. As well as reviewing the code sketches.	0	Following the investigation within the final product and the documented progression of the design it is concluded that this constraint was not met. The system does not include the correct componentry to facilitate this constraint and it is not located in the code. The reason behind this is again evident in the Gantt chart and the struggles with time management of the electronics presented in the production. Reviewing the Gantt chart it is obvious that an inadequate amount of time was allocated to coding, bread-boarding and wiring all the componentry. This resulted in the more complex methods of achieving the functionality of the project, for instance this criteria of having to gain sensor input prior to activating DRS, was opted out for a simple button that can be pressed at anytime. While it is unfortunate that this constraint was not met when reviewing the struggles with code in the production journal and the section on design iterations, it can be concluded that not overcomplicating the electronics was the best decision in order to complete the system.
		Is the system completed by the 1st of September?	This criteria was investigated by cross checking the production journal and the comparison between the actual and expected Gantt chart, this helped determine if the project was completed on time.	10	By reviewing the production journal and the Gantt chart it can be concluded that the project was completed by the 1st of September. While there were copious amounts of alterations to the production process and drastic reordering of steps the project was completed within the allocated time.

		Does the system have logically laid out and easily understandable controls?	This was accessed through a visual investigation of the systems controls and through gathering peer feedback to acquire an uninfluenced perspective on the ability to understand the controls. The design iterations also provided insight into the changes made to simplify the controls.	8	Progressing from the visual assessment of the end product it is obvious that through design iterations the systems control case has been simplified to include one button on the interface. I cannot fully conclude that the controls are easy to understand as there is only one control. The extreme simplification of the systems' components and casing alludes to very easy to understand controls. Peer reviewed feedback also supports the suggestions that the controls are very logically laid out but it is almost impossible to say otherwise as there is only one control.
Materials and components		Were only materials permitted in the FIA 2023 technical regulations used?	This was determined by comparing the explanations in the production journal to the FIA Technical Regulations (Appendix A). Following this comparison, a conclusion was drawn on if the system stayed within the regulations.	7	Following a comparison it can be determined that the majority of materials stayed within the regulations. Due to the alteration of the project to have the 150mm section model of the rear wing, some restricted materials were included in the support structure. The wooden actuator stand is not allowed within the FIA Technical Regulations however it is not technically part of the DRS functional system, so it is not majorly detrimental to this criteria. A material that is not allowed that was used is the 3D printed plastics. However, the externally 3D printed material is a different situation due to it being a proof of design and a representation of the full view of the sectioned model. The 3D printed male moulds that are within the core of the carbon

					fibre pieces are not allowed within the FIA regulations. Even though it clearly went against the constraint outlined here I proceeded to use this banned material as other options that were explored to negate the use of it were not feasible. Using 3D printed internal moulds for the carbon fibre pieces was the most cost effective, time efficient and was the production method that was predicted to produce the highest quality end result.
		Are the materials chosen strong and suitable for the system? Are they assembled using methods to maximise strength?	This criteria was accessed by visually reviewing the end result and determining its structural strength. This was also determined by running virtual CAD strength simulations.	8	Following a visual assessment of the end result it can be concluded that the 150mm section is constructed using suitable fastenings, intended to maximise strength and secure the moving aspects of the system. This includes all supports on the actuator stand and the way the load cell was bolted into the system. Using the results from the Fusion 360 Digital Simulation, some conclusions can be draw regarding the actual design of the entire wing. The majority of the wing handles the static stress simulations well when the force is applied at a 30 degree angle vertically down. The two areas of concern are the connectors for the upper rear wing flap and the centre of the lower rear wing flap. Both locations were flagged as high impact areas for static stress and were both also highlighted during structural buckling simulations as areas of high displacement. Overall the wing presented a being very durable with only a few areas for improvement.

					In regards to materials, while the internal structure of the rear wing sections is not the most durable option, the carbon fibre is the most suitable material for this system's wing sections. The support structure of the wing is produced with materials that are suitable for maximising the strength of the system and ensuring that all components stay together and are strong with repeated movement from the actuator.
Environmental use (use area)		Is the Model DRS able to withstand high aerodynamic forces through strength and durability?	This section of the evaluation criteria was reviewed using the results from the virtual wind tunnel, and by preforming a visual analysis of the durability of the system.	9	Taking the results from the MicroCDF Virtual Wind Tunnel it can be determined that the design for the wing is very durable in terms of wear caused by aerodynamic forces. The particular tests that outline this are the temperature analysis in comparison to the accumulative flow map. Both when the DRS is activated and deactivated, the only noticeable area that is being affected by the force of the aerodynamics is the front facing edge of the design. This area is highlighted to be a concentrated area for higher temperatures which alludes to more frictional wear on this portion of the wing. Aside from that, from the results of the virtual wind tunnel the design for the rear wing is very durable in terms of strength against aerodynamic forces. In terms a physical analysis of the end result, the materials chosen are ones that are durable against repeat aerodynamic forces. This was inadvertently tested with the physical wind tunnel created to

					test the down force of the wing. While this test was not targeted at the durability of the wing section, it was observed that the wing structurally showed no signs of wear following the multiple tests that were run.
Safety		Is all wiring for the system covered and safe?	This criteria was reviewed by visually accessing the system and checking the production journal for the methods taken to ensure a safely wired system.	10	Following a review of the production journal which outline processes including covering wire with heat shrink and braided cable sleeves, it can be determined that the system's wiring was safely covered. This is also proven by a visual assessment of the final system which also conveys clearly laid out and safe wiring.
Cost		Is the system completed within the budget of $500?	To evaluate this criteria, following the completion of the system all records of purchases were gathered and the total cost was calculated.	2	Following an assessment of the total costing of all materials and components it has been determined that the project did not stay within the allocated budget. The major contributing factors to this were the cost of outsourcing the 3D printing of the rear wing design along with large changes in the production process requiring materials that were not accounted for the be purchased. However, it can be seen by reviewing the use of materials that were already available, (seen in the design iteration and production journal). That steps were taken to reduce the total cost of the system. This justifies the score of 2/10 for this criteria.
Waste and energy		Did the system maximise sustainable processes and minimise environmental impact?	This was accessed by reviewing the production journal and evaluating the processes carried out, along with reviewing the design iteration stages to check	6	Following the assessment of the production journal and reviewing the design iterations it can been concluded that steps were taken to reduce the environmental impact and maximise sustainable processes. However it is

			for any wasted materials.		evident that a lot of waste was produced, increasing negative environmental impact. An example of both cases are; - The use of already available scrap wood for the support structure of the sectioned wing, reducing materials become waste. - The waste produced through unsuccessful 3D prints and test pieces. This is waste that can't be recycled and negatively impacts the environment by adding to landfill.
Quality standards Styling and appearance		Does the system have appealing and aesthetic qualities?	This was reviewed through visually accessing the end results and comparing my biased opinions to peer feedback.	9	Following a visual evaluation of the final system and a consideration of peer feedback I have concluded that the final system is completed to a high aesthetic quality. The organisation of wiring is not only highly valuable in terms of troubleshooting the system but also looks very clean and orderly. The time and care taken in regards to the finish of the system with the spraying of the support structure for the wing sections, the choice of filament for the control case and the high quality professional finish on the externally 3D printed rear wing, all contribute to the visual appeal of the final system and its aesthetic quality. The only area that I would like to improve with the appearance of this part of the system is a simple casing for the two 12V batteries.
Performance and durability		Does the Model DRS include	To draw a conclusion on this criteria the final system was run and	0	After the review of the final system and a consultation of the production journal it is

		failsafe backup sensors?	checked for this functionality, the production journal was also consulted to understand the implementation of this consideration.		concluded that this consideration did not make it into the final system. As displayed in the production journal and the expected vs actual Gantt chart, it is evident that the allotted for time for the electronics and coding had already been far surpassed at the generation of a base working electronic system. This is the reasoning behind negating to include failsafe backup sensors.
Size		Does the Model DRS fit within the FIA's dimension rules or a scaled alternative?	This criteria was evaluated by measuring the final end result of the system and comparing it to the FIA's regulations. The CAD models were also reviewed to determine the result for this criteria.	10	By reviewing the CAD model for both the half scale rear wing model that was externally 3D printed and the CAD model of the full scale 150mm section of the rear wing, along with actually measuring the end result it is determined that both fit within the FIA rear wing dimension rules. The wings are within the 1230mm maximum length, 960mm maximum length of upper rear wing flap and the 910mm total height dimensions. Or they are within the scaled alternative.
Maintenance		Are the components organised and managed properly to allow for easy access and maintenance? This includes wires, electronics, and power supply.	This criteria was concluded by reviewing the ease of completing some of the testing process and reviewing the troubleshooting mentioned in the production journal. The assessment of this criteria was also completed by visually reviewing the end result of the system.	7	By reviewing the physical wiring of the system and checking the production journal it is concluded that the components are all properly organised not only for safety and aesthetic purposes but also to have easy access to components for trouble shooting and maintenance. The prime example of the organisation of components is the systems control case which houses the majority of the electronics and has everything sectioned and all wires are grouped in braided cable sleeves and heat shrink. The case is very easy to access with

					a press fit 'snap' together case design. The only major downfall of this criteria is the housing of the 12V batteries and the organisation of the actuator airline. The batteries not having a case or specifically designed component to slot into for housing is something that lets down the organisation of the system. The same thing applies for the airline of the actuator and solenoid which both don't have housing due to the design changes of the actuator housing. However even though these components aren't well organised they are easy to access for maintenance and to troubleshoot any issues.
Production methods		Did the system utilise the most efficient production methods?	This consideration was reviewed by checking the predicted the actual Gantt chart and cross checking it with the production journal.	7	By cross checking the production journal to the predicted vs actual Gantt chart, conclusions can be made regarding the efficiency of production methods. The common trend seen is that closer to the beginning of production time was lost due to overly complex methods of production being pursued. The best example of this is the mould creation of the carbon fibre pieces which was far from the more efficient method and cost the project lots of time, only to be abandoned for the more efficient alternative of the male 3D printed moulds. While the common trend is that the initial production steps were overly complex and risked impacting the efficiency, this trend leads into the production steps in the latter portion of completing this system being much more efficient. This was due to

					needing to make up for any time lost due to inefficient production process that occurred in the early stages of the project.
Regulations		Construction of the Model DRS must not include processes that use 240V. "Students are not permitted to carry out any electrical work on electrical products or equipment that operate above 50 volts AC or 120 volts ripple-free DC" (Victorian Certificate of Education , 2023)	This criteria was reviewed by assessing the productional journal and ensuring that no processed breeched this VCE regulation.	10	By reviewing the production journal it is evident that no processes occurred that required the use of 240V. Therefor there was no breech of the VCE regulation.

How well was this achieved?	Total	With the removal of unachieved criteria (score of 0)
Constraints	100/130	100/120
Consideration	23/60	23/30
Total	123/210	123/150

The above table demonstrates the scores attributed to each criteria with an exploration into the difference between constraints and considerations. This table conveys how constraints were heavily prioritised over considerations which is logically how the results should end up. The column with the data depicting the removal of any unachieved scores is there in order to present how well the completed stages were achieved without uncompleted tasked skewing the data.

Overall key takeaways from the data;

- Constraints took higher precedent over considerations. As evident with half of the considerations scores being unachieved criteria, while only one constraint was not achieved.
- With the removal of the unachieved criteria skewing the data the project was completed to a very high standard in relation to the constraints and considerations.

Gantt Chart (predicted vs actual)

Steps	
1. Gather materials	2. 3D print prototype
3. Bread board and code all electronics	4. Test + calculations
5. Assemble all pneumatics	6. Attach pneumatics to 3D printed prototype
7. Test + calculations	8. Do a trial carbon fibre mould and construction
9. Create carbon fibre moulds	10. Manufacture carbon fibre piece
11. Assemble Model DRS body including pivot points	12. 3D print actuator housing
13. 3D print systems control case	14. Assemble all components
15. Test + calculations	16. Extra allowance of time

The greyed out rows of the Gantt chart are any steps that were removed from production either through being replaced and conjoined with another step or through being removed due to time constraints.

All test + calculations stages are greyed out due to their removal from the production journal. They still occurred but not within the rigidly scheduled times allocated.

Steps	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12
1												
2												
3												
4												
5												
6												
7												
8												
9												
10												
11												
12												
13												
14												
15												
16												

Comparison

Expected	
Actual	
Overlap	
Removed	

Steps	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12
1												
2												
3												
4												
5												
6												
7												
8												
9												
10												
11												
12												
13												
14												
15												
16												

The Gantt chart depicts a very noticeable change in timeline from the predicted progression of the manufacturing process in comparison to what actually occurred.

The most notable comparisons between the predicted and actual charts occur with the bread boarding and coding of electronics, along with the mould creation. By accessing the chart conclusions can be made of the flow on effect of these steps taking longer than the allotted time. The removal of strict scheduled testing and calculations along with the extra time accounted for being used to assemble the components. Is evidence for how these stages of production effected the timeline and caused a lack of time to explore some of the constraints.

More detailed explanations of the processes that altered the trajectory of the stages of production are found within the production journal and the design iteration stages of the folio.

Key Areas for Improvement

Even though I am very pleased with the high quality end result of my DRS Model, by reviewing the evaluation criteria and through learning during creating this system there are definitely some areas in which the system could be improved.

Appearance

While both the externally printed rear wing and the functional sectioned wing are visually appealing and have high quality finishes there are some points that improvements could be made. Some of which are;

- Creating a casing or organisation unit for the 12 V batteries to slot into. While their location on the side of the functional sectioned wing is purposefully located to maximise their organisation by being close to both the air compressor and the solenoid, it does not achieve the same organised appearance as the rest of the system. To improve this I would design a case that connected onto either that air compressor or the sectioned rear wing support structure. The case would have the upper section of the batteries exposed so they are easy to access but would provide a more organised appearing power supply section to match with the already well organised system.
- Having a more organised and cleaner looking method of housing the airlines that connect the actuator and solenoid. This works alongside having an improved method of housing the solenoid. Similar to the battery, the solenoid's location on the side of the support structure for the sectioned rear wing struct allows it to be easily connected to the actuator and the air compressor but compromises the organised appearance of the system. This has a simple fix to improve the appearance of the model whilst also improving organisation. While a case for the solenoid could easily be designed and fitted to the side of the support structure, the organisation of the airlines is a bit more complex. My initial idea for improving their organisation would be to somehow fix them to the actuator stand so that they aren't floating in the air.

Functionality

While the system is functional and in some aspects functions to an extremely high quality there are some sections of the final system that could be improve to further increase the standard of the functionality of the system. Some improvements that could be made are;

- As outlined in the MicroCDF virtual wind tunnel testing, the front facing edge of the wing has higher temperature and wear characteristics due to their location in relation to airflow around and over the wing. Due to the results of these tests, the suggested method of improving not only the amount of aerodynamic stress on the front facing edges of the wing but also the aerodynamic flow of air over the wing would be to create a more curved profile of all the front facing edges of the wing. This has the intention of creating less drag and reducing the impact of air hitting the straight planes of the wing's edge.
- As presented through the Fusion 360 static buckling test, a weak point of the wing is the connectors to the upper rear wing flaps. Due to the lack of structural support on this point of the wing design, they have a high level of displacement in all planes when force in applied. To

improve the strength and durability of my wing design I would redesign these connectors to have more structural support by how they are connected to the endplates of the rear wing. Alternatively, I would add internal supports to aid in the rigidity of this portion of the design.

- Similarly to the connectors to the upper rear wing flap, the centre of the lower rear wing flap had large levers of vertical displacement seen in the structural buckling simulations. Due to the importance of this part of the wing for aerodynamics and the direction of the flow of air, there is not a way to add additional external supports to stop the buckling. The alternative way to improve its durability is through internal supports.
- Results from the MicroCDF virtual wind tunnel also suggest that a key area of improvement in the flow of air through the gap between the upper and lower rear wing flap when DRS is disabled. The graphing of the flow lines did not present lots of air travelling though this section of the wing. This would need further investigation on how to redesign the shapes and angles of the rear wing componentry to improve the flow of air through this crucial location for drag reduction.
- A simple area of improvement for the functionality of the activation and deactivation of DRS that I would have liked to include is the use of two different buttons for activation and deactivation of DRS, instead of the one button used. Having two buttons more closely simulate the actual process in an F1 car enhancing the DRS model as a representation of these systems on a real world scale.
- While having two sections of the design, the externally 3D printed proof of design (1/2 scale) and the 150mm section of the upper and lower rear wing flaps (1/1 scale), was a viable solution to all of the issues that occurred during the exploration of the creation of carbon fibre moulds and allowed for a high quality end result to be produced in the allotted time. A further improvement to this solution would be being able to see the DRS working on an entire wing (could be 1/2 scale) as was originally intended.

Methods of Production

While the methods of production completed where all intended to efficiently produce a high quality end result and the majority of them did, there are a few that improvements can be acknowledged for. Some of these methods are;

- While the carbon fibre process and the end result that was achieved were at a high quality considering the resources available, my lack of experience with this production method, time pressure and available equipment. The carbon fibre construction process is still an area that has many sections that could be improved. While it is not reasonable to suggest the use of a professional grade autoclave, I do suggest research into and the implementation of vacuum bagging the carbon fibre pieces as an area of improvement within methods of production.

  The carbon fibre pieces turned out exceptionally well considering their method of production with the biggest proposed area for improvement evident with the final result having resin not seeping into the carbon fibre material in large spaces across the wing. This flaw in the final system, highlighted with the thick layer of resin seen over the wing surface, which defeats the purpose of using the carbon fibre in the first place as the excess resin adds weight and impedes on the properties of the carbon fibre. Is an area that I recommend improvements we made in, with the purposed method of using a vacuum bagging technique with the carbon fibre pieces, the resin would get integrated into the carbon fibre producing a much better quality end result.

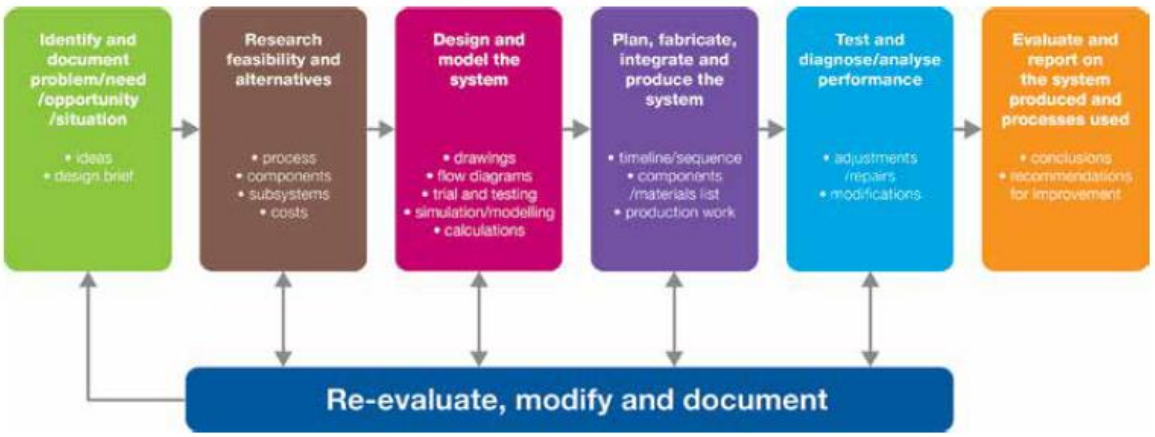
- Another point of the carbon fibre process that I could improve on is the joining of the two ends of the carbon fibre material at the underside of the wing. While the join is hidden in the final results of the project, it is an area of the carbon fibre process that could be significantly improved to enhance the appearance of the entire carbon fibre pieces. Where the joins occur with the carbon fibre piece the twill has been disturbed which leaves uneven layers of carbon fibre material which subtracts from the quality of the pieces. Ways to limit the disturbance of twill at the joins is a production method that could be further explored.
- An area of improvement within the mould creation of the carbon fibre pieces is to utilise high density foam (which was the initial intention of the design) instead of 3D printed pieces. while the 3D printed pieces were a great solution to the time constraint on the project and the vast complications during the design of the moulds they defeat the purpose of using carbon fibre for its beneficial strength to weight ratio. The use of foam for the internal moulds of the carbon fibre pieces would improve their simulation of rear F1 car rear wings and produce a better quality end result.
- A method of production that was improved in the later stages of production is the use of header pins and jumper cables with the wiring of the PCB board. While this process is how all the wiring of the electronics of the final system was completed, I'm addressing that this method of production should have been implemented earlier in the process of creating the final system to save time and make trouble shooting easier.

Further Progression

Even through the system is completed to a standard that I am proud of, there are areas that could be improved through further progression of ideas. A lot of these progressions are derived from considerations that did not get achieved.

- One way in which the completed system could be progressed is through the realisation of having location sensors on the rear wing that stop the DRS from being activated unless they are at a specified point. This would properly simulate the Formula 1 Sporting Regulations where DRS can only be activated at certain locations of the track. The next step within this progression is to have the activation of DRS delayed if an attempt was made to active it prior to a activation point. These progressions would require the addition of specific location sensors that worked in conjunction with each other to inform the code if it can allow the activation button to work, adding a delay into the code if the location sensors didn't provide the output that the system was at the correct location before attempted activation.
- Another progression that is the realisation of a consideration is to have a brake pedal instead of a button to deactivate DRS. This could simply be a 3D printed brake pedal that fits over a button to simulate the way that DRS is deactivated in F1 cars through braking into corners out of the DRS zones. This progression would further create an interactive capability of the DRS model.
- A progression to the final system would be to include failsafe backup sensors to the design which simulated the extra precautions placed on an F1 car so that the pit wall is always able to gain data from the car to analyse.
- With all these enhancements to the control of the system and the way in which it is all managed within the code, housing and wiring would also require progression.

Evaluation of Progression Through the Systems Engineering Process



Even though I am confident in my progression through the stages of the System Engineering process with each stage completed to a high quality, it is evident that the majority of the evaluation of my navigation through the process is linked to a lack of interconnection of processes at some points and areas where re-evaluation, modification and documentation could be improved.

The key areas in which I utilised the Systems Engineering Process well where with the progressing through the identification, research and design stages. During these stage there were a lot of evaluation of the project and modifications made following research into the feasibility of certain aspects of the project. A key point in this was referring to the evaluation criteria closely when designing the system. These stages flowed together well and the progression through them was far from linear, an area of my navigation through these stages that could be improved would be the re-evaluation of constraints and considerations. While the further research was targeted at the different aspects of the evaluation criteria, amendments were not made when information was discovered that would make a constraint unfeasible.

Further along the Systems Engineering process a large area for improvement would be to refer back to the evaluation criteria more throughout the production of the system and document it. This would have assisted in keeping the project on track and would have ensured that more of the considerations could have been explored. Even when considering the improvements that could be made to the process, the planning and journaling were areas which were completed very thoroughly and documented extensively during the production of my system.

With many iterations being made to the design of my system, the design and model portion of the Systems Engineering Process became very well integrated within the production of the system. This

was heavily documented and a lot of modifications were made based on the re-evaluation of the initial designs.

A large area of improvement that I believe would have made the production of my system significantly more conducive to the initial brief of the project would have been to more frequently cross check where I was in the stage of production compared to what the overall intensions were of my project. This would have aided in keeping everything on track and would have assisted in generating more testing throughout the production of the system. By having a clear idea of the requirements of the final system, more targeted tests could have been implemented throughout the production of my system to ensure that everything was working and being improved to satisfy the initial constraints and considerations of the system. The addition of more progressive tests would have fuelled more re-evaluation to be made earlier in the production process, saving time spend on pointless tasks that would eventually prove to be time wasting.

Overall I utilised the Systems Engineering Process thoroughly with my real strengths within the interconnection of the process being more succinct earlier on in the process. With my key area of improvement being not losing sight of the scope of the system by neglecting to consistently refer back to the initial documentation on the constraints and considerations of the project.

Final Evaluation

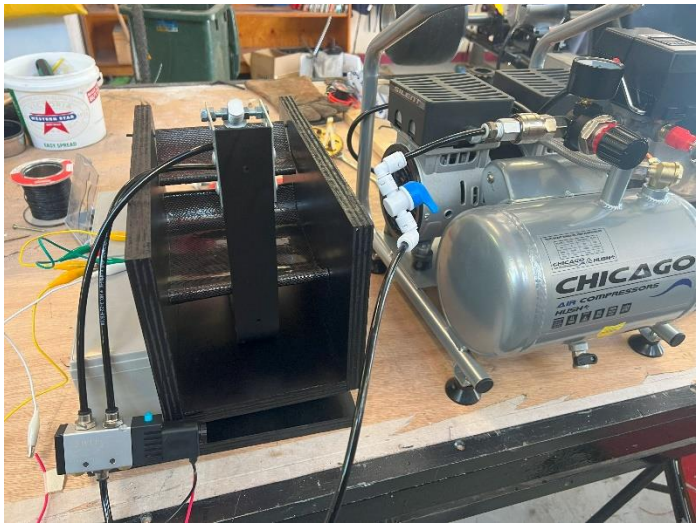
The final evaluation's intention is to review the end result of the system in comparison to the intended result as established early in the System Engineering Process.

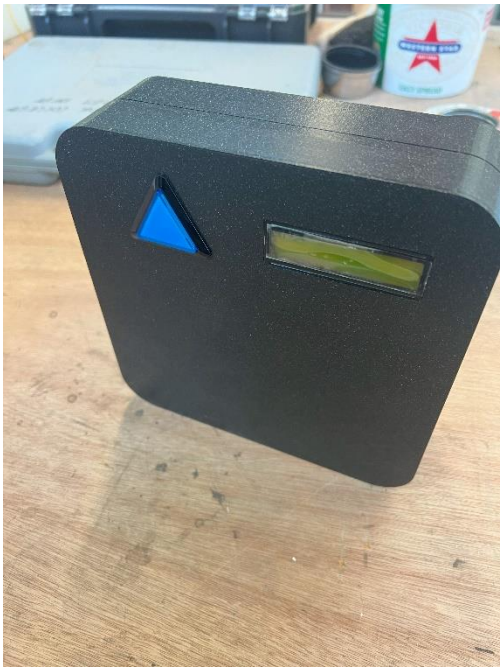
While the initial intension of my project was to produce an entirely carbon fibre half scale model of a Formular 1 car rear wing, with location sensors to only have DRS activate when at specific locations. Due to time restrictions, cost, complexity of the manufacturing processes and a list of other complications, the initial intension was not carried out to the exact specification imagined.

The high cost of producing carbon fibre pieces and the extreme cost and complexity of generating moulds for carbon fibre altered the realisation of the carbon fibre pieces. This resulted in the alternative method of realising the design of my rear wing through externally 3D printing my initial design. To be able to still properly display the actuator and all the components associated with it, the design had to pivot to be able to achieve the intention of displaying the technical aspects of my design. The addition of the section rear wing flap allowed a version of the carbon fibre plan to be realised and clearly exhibits the functionality of the actuator in activating and deactivating DRS. The complexity of the code and an underestimate of the time that would be needed to generate functional code of the DRS model limited the location sensors from becoming a reality. However the project is still fully functional without them.

My design brief states that I would be aiming to solve the issue of "The non-existent platform for testing and evaluating inhibits an influx of future development within the sport,". With my solution to the lack of information regarding the more technical aspects of Formula 1 limiting the possibility of inspiring future engineers being to create "a realistic and industry spec teaching aid," in the form of " the Model Drag Reduction System (Model DRS), the system would work to educate engineering students who are unfamiliar with the more technical aspects of the sport,".

While my final product looks and functions a lot differently to what was designed and planned for, the design brief was still met. I produced an educational aid that simulates the technical aspects of Formula 1, aiming to educate engineering students.





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Appendix

FIA Technical Regulations (Appendix A)

All information for these regulations are from the FIA publicly published ' FIA 2023 Formula 1 Technical Regulations – Issue 3 – 2022-10-19 '

Electronics

8.3 Control electronics

8.3.1 All components of the power unit, fuel system, transmission systems, brake system, tyre pressure monitoring system and DRS, in addition to all associated actuators, must be controlled by the FIA Standard ECU. The FIA Standard ECU may only be used with FIA approved software and may only be connected to the control system wiring loom, sensors and actuators in a manner specified by the FIA. Additional information regarding the FIA Standard ECU software versions and setup may be found in the Appendix to the Technical and Sporting Regulations.

Materials

15.3 Permitted Materials

With the exception of materials specifically permitted for certain components, as listed in Article 15.5, only the following materials may be used:

15.3.1 Metallic Materials – Not Used for Additive Manufacture

a. Iron Alloys: any

b. Aluminium Alloys:

i. Aluminium based alloys from the 1xxx, 2xxx, 3xxx, 4xxx, 5xxx, 6xxx and 7xxx series containing less than 1% Lithium.

ii. In addition, the specific alloys 2099, 2199, 2050 and 2055 are permitted.

c. Ni or Co based Superalloys:

i. UNS N06625; W.Nr.2.4856 Inconel 625

ii. UNS N07718; W.Nr.2.4668 Inconel 718

iii. UNS N07001; W.Nr.2.4654 Waspaloy

iv. UNS R30035; W.Nr2.4999 MP35N

v. UNS R30159 MP159

d. Titanium Alloys:

i. Low-alloy titanium alloys containing at least 97.5% Ti and less than 1% of any other element.

ii. Grade 9 Ti3Al2.5V

iii. Grade 5; grade 23 TiAl6V4, Ti64

iv. UNS R54620, Ti6242, Ti6242Si

v. UNS R56260 Ti6246

vi. UNS R56410 Ti10-2-3

vii. UNS R58153 Ti15-3-3-3

viii. UNS R58640 Allvac 38-644

ix. Ti-5Al-5Mo-5V-3Cr Ti5553

x. Ti-15Mo-3Nb-3Al-0.2Si Timetal Ti-21S

xi. Timetal Ti1100

xii. Ti 6Cr-5Mo-5V-4Al

e. Copper Alloys containing less than 2.5% Beryllium

f. Tungsten Alloys: any

DRS

3.10.10 Drag Reduction System (DRS)

The entire RW Flap described in Article 3.10.1, including any gurney if fitted, part of the rearward most section from any regions of the Rear Wing Tip described in Article 3.10.5 where there are exactly two sections in an intersection with any plane that contains the Xaligned axis [Y,Z]= [480, 670], and the portions of the Rear Wing Separators defined in 3.10.6 that are attached to these components, may be rotated about a fixed axis whilst the car is in motion. All bodywork to be rotated shall be known as "DRS Bodywork"

Furthermore:

a. No part of the "DRS Bodywork" may lie outboard of Y=490.

b. The axis of rotation of the RW Flap must be fixed and located no more than 20mm below the upper extremity and no more than 20mm forward of the rear extremity of RV-RW-PROFILES at all times.

c. There must be no relative movement between the constituent parts of the DRS Bodywork.

d. Any such variation of incidence maintains compliance with all of the bodywork regulations with the exception of Article 3.10.1 and 3.10.5.

e. It cannot be used to change the geometry of any duct, either directly or indirectly, other than the change to the distance between adjacent sections permitted by Article 3.10.1 and 3.10.5.

f. The design is such that failure of the system will result in the uppermost closed section returning to the normal high incidence position.

g. Any alteration of the incidence of the uppermost closed section may only be commanded by direct driver input and controlled using the control electronics specified in Article 8.3.

h. At any Y plane, the distance between the two sections of the Rear Wing Profiles at their closest position must lie between 10mm and 85mm.

In order to achieve the above adjustment, the actuator mechanism and any associated minimal fairings must lie inboard of Y=25, below Z=940 and not extend beyond RV-RW-PROFILES in the forward or rearward direction, when viewed from above. Furthermore, any additional minimal bodywork associated with the adjustment of the DRS Bodywork must be located outboard of Y= 465 and in side-view be below the top surface of RV-RW-PROFILES, must not be visible from below and must fit within a cuboid which is 30mm wide, 60mm long and 30mm high. This cuboid may have a free orientation in space, but its width must be aligned with the axis of rotation of the RW flap.

FIA Sporting Regulations (Appendix B)

All information for these regulations are from the FIA publicly published ' FIA 2023 Formula 1 Sporting Regulations – Issue 9 – 2022-10-19

22) DRIVER ADJUSTABLE BODYWORK

22.1 Driver adjustable bodywork permitted by Article 3.10.10 of the Technical Regulations.

a) The driver adjustable bodywork may only be activated by the driver in any of the pre-determined activation zones around each circuit. In conditions of poor visibility 2022 Formula 1 Sporting Regulations 20/107 19 October 2022 ©2022 Fédération Internationale de l'Automobile Issue 9 however the Race Director may, at his absolute discretion, disable all such systems until conditions improve.

b) If the adjustable bodywork is disabled in this way at any time during any of the three periods of the qualifying practice session (Q1, Q2 or Q3) it will remain disabled for the remainder of the relevant period.

c) For the sole purpose of improving overtaking opportunities during the sprint session and the race the adjustable bodywork may be activated by the driver after he has completed two (2) laps after the start or following a safety car period during each sprint session or a race.

d) The driver may only activate the adjustable bodywork in a sprint session or the race when he has been notified via the control electronics (see Article 8.2 of the Technical Regulations) that it is enabled. It will be enabled, and may only be used by the driver, if he is less than one (1) second behind another at any of the predetermined detection points around each track.

e) The system will be disabled by the control electronics the first time the driver uses the brakes after he has activated the system.

f) In conditions of poor visibility, or if yellow flags are being shown in the activation zone, the Race Director may, at his absolute discretion, disable all such systems until conditions improve or yellow flags are withdrawn.

g) The FIA may, after consulting all Competitors, adjust the above time proximity in order to ensure the stated purpose of the adjustable bodywork is met.

h) In the event of a failure in the system which notifies the driver that he was within one (1) second of the car in front, and is hence authorised to use the adjustable bodywork, the Competitor concerned may ask the Race Director for permission to override the system. If permission is given in this way the onus will be upon the Competitor concerned to ensure that their driver only uses the adjustable bodywork if he is within one (1) second of the car in front of him.

i) If the failure in the system is rectified the driver may no longer use this override, the Race Director will notify the Competitor if and when the fault has been remedied